# Vedic Maths

## Zero to Hero

From counting to competitive exams — a colourful self-study textbook
Foundations · 16 Sutras & 13 Sub-sutras · Mental maths · Children's course
Government-exam aptitude · 1,000+ verified practice questions with solutions

$97 \times 96 = 93|12$

$1000 - 357 = 643$

$35^2 = 12|25$

$43 \times 11 = 473$

Original practice material. Exam names are used only to describe the syllabus; no question is an official previous-year question.

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# Welcome

How to use this book

# Before You Begin

## Who this book is for

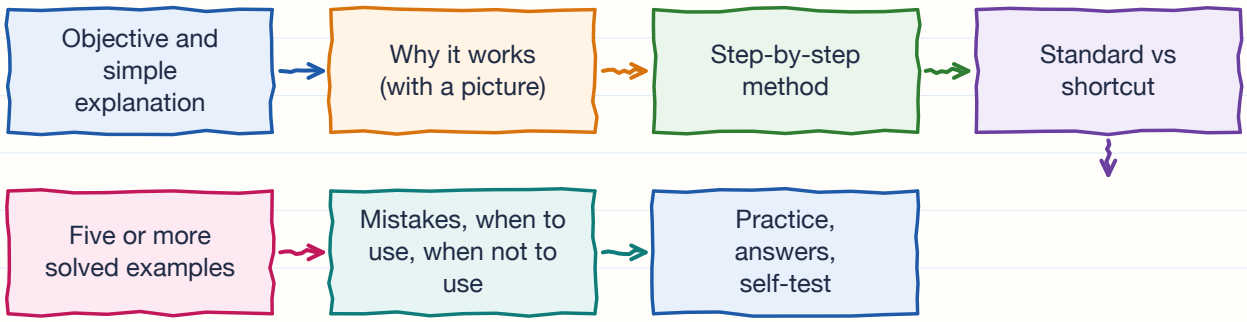
This book starts at **zero**. If you can count to ten, you can begin on page one. By the end you will be able to calculate quickly in your head, check your own answers, and solve the arithmetic and aptitude questions that appear in school tests and in competitive examinations such as SSC, Banking, Railways (RRB NTPC, RRB JE), state-level exams and the CSAT paper.

The book has separate paths for:

- **Children (ages 5–12)** — Part X is a gentle, game-based course. Parents and teachers will find instructions in the purple boxes.
- **School students** — Parts I–VII build strong calculation and algebra skills.
- **Adults** who want speed and confidence with numbers — Parts III, IV, VI and IX.
- **Government-exam aspirants** — Parts VIII, XI, XII and XIII, plus the roadmap in Part XIV.

## How every lesson is organised

Every major technique follows the same pattern, so you always know where to look:




The coloured boxes have fixed meanings:

### Formula boxes are blue

The rule or identity you should remember.

### Worked examples are orange

A fully solved problem with every step shown.

### Tips are green

A faster or safer way to think about the problem.

### Common mistakes are red

Errors that learners often make — and how to avoid them.

### Memory tricks are pink

A short phrase or picture that helps you recall the method.

### Approximate methods are yellow

Anything that gives an **estimate** rather than an exact answer is marked like this. Never confuse an estimate with an exact answer.

## How to study

1. Read the explanation slowly. Say the steps aloud.
2. Cover the solution of each worked example and try it yourself first.
3. Do the practice questions **on paper** before looking at the answer key.
4. Check every answer. If you got it wrong, find the exact step where you went wrong.
5. Revise with the quick-revision boxes the next day, and again after a week.

### Understanding first, speed second

Speed comes naturally after accuracy. A fast wrong answer scores nothing — and in exams with negative marking it costs marks. Practise slowly and correctly; speed will follow.

# What is Vedic Mathematics?

## The short answer

**Vedic Mathematics** is the name given to a collection of mental calculation techniques published by **Sri Bharati Krishna Tirthaji** (1884–1960), a scholar and the Shankaracharya of Puri. His book *Vedic Mathematics* was published in 1965, after his death. It presents the methods through **16 sutras** (short "word-formulae") and **13 sub-sutras**.

Examples of what these methods do:

- Multiply numbers close to 100 or 1,000 in one line (*Nikhilam*).
- Multiply any two numbers by a neat "vertically and crosswise" pattern (*Urdhva-Tiryagbhyam*).
- Square numbers ending in 5 instantly (*Ekadhikena Purvena*).
- Divide by 9, 99, or by numbers such as 12 and 112 with very little writing.
- Solve some special equations by inspection.

### What is known, and what is claimed

Tirthaji stated that the sutras came from an appendix (*parishishta*) of the *Atharvaveda*. Historians and Sanskrit scholars have **not found these sutras in any known Vedic text**, and many regard them as Tirthaji's own composition, written in Sanskrit style. This book therefore uses the phrase "traditional sutras" to mean *the sutras as presented in Tirthaji's book* — not a claim about ancient origin.

What is beyond dispute is that **the techniques themselves are correct mathematics**. Each one can be proved with ordinary algebra, and this book shows you the proof (the "Why it works" boxes). India also has a genuine and rich mathematical history — Aryabhata, Brahmagupta, Bhaskara II, the Kerala school — that is separate from this particular system.

## What Vedic methods are — and are not

| Vedic methods are… | Vedic methods are not… |
|---|---|
| Clever, correct shortcuts for particular kinds of numbers | A replacement for understanding place value, fractions and algebra |
| Very fast when the numbers suit the method (near a base, ending in 5, etc.) | Faster for *every* calculation |
| A great way to build number sense and confidence | Magic — every step can be explained |
| Useful for checking answers in a second way | Proof of any historical claim |

### Do not force a shortcut

If the numbers do not fit a method, use the standard method. Choosing the right method is itself a skill. Throughout this book, every technique has a "When **not** to use it" section.

## Notation used in this book

- Numbers are written with **Indian grouping**: 1,00,000 is one lakh; 1,00,00,000 is one crore. Part I explains how this compares with the international system (100,000 and 10,000,000).
- The **vertical bar** `|` separates the left and right parts of a Vedic answer. For example `93|12` means 9312.
- A **minus sign** is written –. Negative deviations are shown in brackets, e.g. (–3).
- A **bar over a digit** (vinculum), such as $\bar{1}$, means "minus 1" in that place. It is introduced in Part II.
- "≈" means *approximately equal to*. Answers with ≈ are estimates.
- Money is written in rupees, e.g. ₹2,450.

## About the practice questions

All practice questions in this book are **original**. They were produced by a program that also computes each answer, and every answer was checked a second time by an independent calculation (see the `tests/` folder in the source files). Where a topic is commonly tested in SSC, Banking or Railway exams this is mentioned, but **no question is presented as an official previous-year question**.

# Part I

## Mathematics Foundations

Every shortcut in this book rests on four simple ideas: **place value**, the **four operations**, **fractions**, and **powers**. This part builds them from zero. If you already feel confident, skim the quick-revision boxes and do the practice questions — they will show you quickly whether anything needs attention.

# Chapter 1: Numbers from Zero

> **Learning objective**
> By the end of this chapter you will be able to read and write large numbers in the Indian and international systems, find the place value of any digit, recognise odd, even, prime and composite numbers, work with negative numbers on a number line, and use basic divisibility tests.

## 1.1 What is a number?

A **number** tells us *how many* or *how much*. When you count five mangoes you are using the number 5. We write numbers with just ten symbols called **digits**:

**0, 1, 2, 3, 4, 5, 6, 7, 8, 9**

The digit **0 (zero)** means "nothing here". It was one of the greatest inventions in mathematics: Indian mathematicians such as Brahmagupta (7th century CE) wrote rules for calculating with zero, and the place-value system using zero spread from India to the rest of the world.

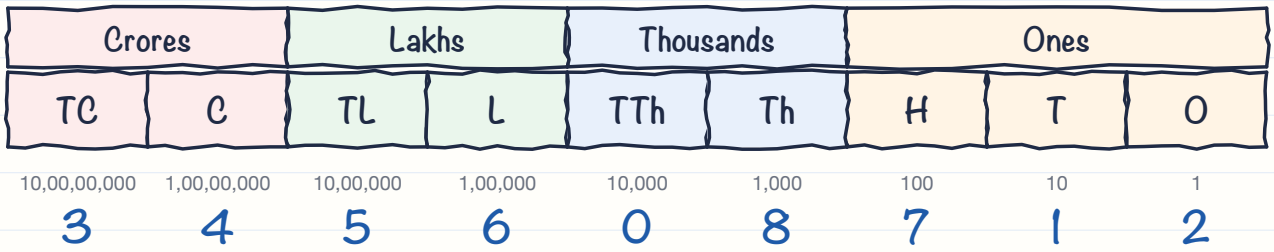
- **Natural numbers**: 1, 2, 3, 4, … (the counting numbers)
- **Whole numbers**: 0, 1, 2, 3, … (natural numbers plus zero)
- **Integers**: … –3, –2, –1, 0, 1, 2, 3, … (whole numbers and their negatives)

## 1.2 Place value

The *same* digit can mean different amounts depending on **where** it stands. In 555, the first 5 means five hundreds, the second means five tens, and the last means five ones.

> **Place value and face value**
> **Face value** = the digit itself. **Place value** = digit × value of its place. In 4,725 the digit 7 has face value 7 and place value 7 × 100 = 700.

Every place is **ten times** the place on its right. This is why our system is called the **decimal** (base-10) system — and it is exactly why so many Vedic shortcuts use bases such as 10, 100 and 1,000.




34,56,08,712 = thirty-four crore fifty-six lakh eight thousand seven hundred twelve

## 1.3 The Indian and international systems

Both systems use the same places; they only **group** and **name** them differently.

| Number | Indian system | International system |
|---|---|---|
| 1,000 | one thousand | one thousand (1,000) |
| 1,00,000 | one **lakh** | one hundred thousand (100,000) |
| 10,00,000 | ten lakh | one **million** (1,000,000) |
| 1,00,00,000 | one **crore** | ten million (10,000,000) |
| 10,00,00,000 | ten crore | one hundred million (100,000,000) |
| 1,00,00,00,000 | one hundred crore (one arab) | one **billion** (1,000,000,000) |

### Comma rules

**Indian:** first comma after 3 digits from the right, then after every **2** digits: 12,34,56,789.
**International:** a comma after every **3** digits: 123,456,789.

### Reading a large number

**Read 5,07,30,046 in words.** Groups from the left: 5 | 07 | 30 | 046 → five crore, seven lakh, thirty thousand, forty-six. In the international system: 50,730,046 = fifty million, seven hundred thirty thousand, forty-six.

### Lakh and crore

"**L**akh has **5** zeros, **C**rore has **7**." Ten lakh make a million; ten crore make a hundred million.

### Zeros in the middle

When writing "three crore four lakh nine", put zeros in every empty place: 3,04,00,009. Count the digits: a crore number has 8 digits.

## 1.4 Odd and even numbers

An **even** number can be split into two equal groups: it ends in 0, 2, 4, 6 or 8. An **odd** number ends in 1, 3, 5, 7 or 9.

| Rule | Example |
|---|---|
| even + even = even | 14 + 6 = 20 |
| odd + odd = even | 7 + 9 = 16 |
| even + odd = odd | 8 + 5 = 13 |
| odd × odd = odd | 7 × 9 = 63 |
| even × anything = even | 6 × 7 = 42 |

These rules give you a **free check**: if you add two odd numbers and get an odd answer, you have made a mistake.

## 1.5 Prime and composite numbers

A **prime number** has exactly two factors: 1 and itself (2, 3, 5, 7, 11, 13, …). A **composite number** has more than two factors (4, 6, 8, 9, 10, 12, …). The number **1 is neither** prime nor composite, and **2 is the only even prime**.

### Testing whether a number is prime

To test n, divide by the primes 2, 3, 5, 7, 11, … up to √n. If none divides n exactly, n is prime.

### Is 221 prime?

√221 is a little less than 15 (since $15^2 = 225$). Test 2, 3, 5, 7, 11, 13: 221 is odd; digit sum 5 is not divisible by 3; it does not end in 0 or 5; 221 ÷ 7 leaves remainder 4; 221 ÷ 11 leaves remainder 1; **221 ÷ 13 = 17**. So 221 = 13 × 17 is **composite**.

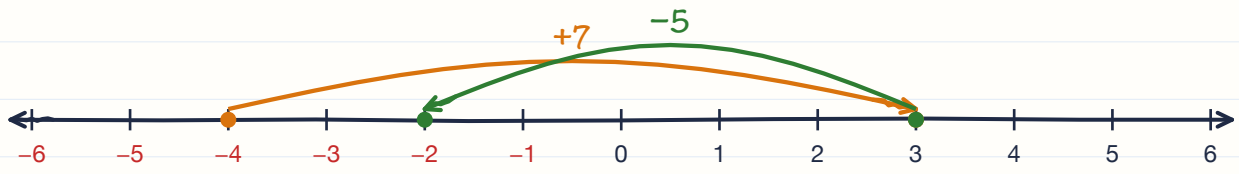
| Primes up to 100 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 5 | 7 | 11 | 13 | 17 | 19 | 23 | 29 |
| 31 | 37 | 41 | 43 | 47 | 53 | 59 | 61 | 67 | 71 |
| 73 | 79 | 83 | 89 | 97 | | | | | |

### 51, 57, 87 and 91 look prime — but are not

51 = 3 × 17, 57 = 3 × 19, 87 = 3 × 29, 91 = 7 × 13. Always test before you decide.

## 1.6 Positive and negative numbers

Numbers less than zero are **negative**. A temperature of −4 °C is four degrees below zero; a bank balance of −₹500 means you owe ₹500. On a number line, numbers increase to the right.




The picture shows −4 + 7 = 3 (seven steps right) and 3 − 5 = −2 (five steps left).

### Rules for signs

**Adding a negative** is the same as subtracting: 8 + (−3) = 5. **Subtracting a negative** is the same as adding: 8 − (−3) = 11. **Multiplying or dividing:** same signs → positive; different signs → negative. (−4) × (−5) = 20, (−4) × 5 = −20.

### Comparing negatives

The further left a number is, the smaller it is. So −15 < −2, even though 15 is bigger than 2.

## 1.7 Factors and multiples

A **factor** divides a number exactly. A **multiple** is the result of multiplying a number by 1, 2, 3, …

- Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36 (they come in pairs: 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6).
- Multiples of 7: 7, 14, 21, 28, 35, …

### Find factors in pairs

Start at 1 and move upward. Stop when the pair numbers meet — you have then found every factor.

## 1.8 Divisibility basics

A divisibility rule tells you whether a number divides exactly **without doing the division**. You will use these constantly for simplifying fractions and checking answers.

| Divisor | Rule | Example |
|---|---|---|
| 2 | Last digit even (0, 2, 4, 6, 8) | 3,574 ✓ |
| 3 | Digit sum divisible by 3 | 4,512 → 12 ✓ |
| 4 | Last two digits divisible by 4 | 7,316 → 16 ✓ |
| 5 | Last digit 0 or 5 | 8,435 ✓ |
| 6 | Divisible by 2 and by 3 | 2,154 ✓ |
| 7 | Double the last digit, subtract from the rest; repeat | 203 → 20 − 6 = 14 ✓ |
| 8 | Last three digits divisible by 8 | 51,128 → 128 ✓ |
| 9 | Digit sum divisible by 9 | 6,723 → 18 ✓ |
| 10 | Last digit 0 | 4,560 ✓ |
| 11 | Alternating digit sum divisible by 11 | 9,537 → (9+3) − (5+7) = 0 ✓ |
| 12 | Divisible by 3 and by 4 | 1,524 ✓ |
| 13 | Add 4 × last digit to the rest; repeat | 247 → 24 + 28 = 52 ✓ |
| 25 | Last two digits 00, 25, 50, 75 | 3,475 ✓ |

### Why the rule for 3 and 9 works

Every power of 10 is "a multiple of 9, plus 1": 10 = 9 + 1, 100 = 99 + 1, 1000 = 999 + 1. So 4,512 = 4 × 1000 + 5 × 100 + 1 × 10 + 2 leaves the same remainder on division by 9 (or by 3) as 4 + 5 + 1 + 2 = 12. This idea — the **digit sum** — returns in Part II as a powerful checking method.

## 1.9 Comparing and ordering numbers

1. The number with **more digits** is larger (for positive whole numbers).
2. If the digit counts are equal, compare from the **left**; the first different digit decides.
3. For negative numbers, the order is reversed.

### Arrange in ascending order: 4,05,612; 40,561; 4,50,126; −45,061

−45,061 is negative, so it is smallest. 40,561 has 5 digits. The two 6-digit numbers start 4,0… and 4,5… so 4,05,612 < 4,50,126. **Order:** −45,061 < 40,561 < 4,05,612 < 4,50,126.

### Chapter 1 in six lines

- Place value = digit × value of its place; each place is 10 × the one to its right.
- Indian commas: 3 then 2s (lakh, crore). International: 3s (million, billion).
- Even ends in 0/2/4/6/8. Odd + odd = even.
- Prime: exactly two factors. Test primes up to √n. 1 is neither; 2 is the only even prime.
- Same signs multiply to positive; subtracting a negative means adding.
- Digit sum tests divisibility by 3 and 9.

## ✎ Chapter 1 practice

**1.** In 7,94,45,911, what is the place value of the digit in the thousands place?

**2.** Rewrite 89,59,93,246 using international commas (millions).

**3.** Is 73 prime or composite?

**4.** Is 24,062 divisible by 4? Use the divisibility rule.

**5.** Calculate 14 − 5 + (−29).

**6.** How many factors does 48 have? List them.

**7.** In 8,82,59,382, what is the place value of the digit in the thousands place?

**8.** Rewrite 7,04,75,177 using international commas (millions).

**9.** Is 373 prime or composite?

**10.** Is 52,416 divisible by 3? Use the divisibility rule.

**11.** Calculate 11 − (−30) + 6.

**12.** How many factors does 120 have? List them.

**13.** In 6,56,45,349, what is the place value of the digit in the crores place?

**14.** Rewrite 58,79,34,175 using international commas (millions).

**15.** Is 349 prime or composite?

**16.** Is 86,258 divisible by 11? Use the divisibility rule.

**17.** Calculate (−18) − (−5) + (−9).

**18.** In 8,72,72,198, what is the place value of the digit in the crores place?

*Answers: see the answer key (questions 1–18).*

### Short self-test

1. How many zeros are there in ten crore? 2. Is 1 a prime number? 3. What is (−7) − (−10)? 4. Is 7,352 divisible by 4? *(Answers: 8; no; 3; yes, because 52 ÷ 4 = 13.)*

## Chapter 1 answer key

**1.** **5,000** — The digit is 5; thousands = 1,000; 5 × 1,000 = 5,000.

**2.** **895,993,246** — International grouping uses threes from the right: thousand, million, billion.

**3.** **Prime** — √73 ≈ 8. No prime up to 8 divides 73, so it is prime.

**4.** **No** — Rule for 4: last two digits divisible by 4. 24,062 ÷ 4 leaves remainder 2.

**5.** **−20** — Subtracting 5 is the same as adding −5. 14 + −5 + −29 = −20.

**6.** **10: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48** — Pair factors from the outside in: 1×48, 2×24, 3×16, 4×12, 6×8.

**7.** **9,000** — The digit is 9; thousands = 1,000; 9 × 1,000 = 9,000.

8. **70,475,177** — International grouping uses threes from the right: thousand, million, billion.
9. **Prime** — √373 ≈ 19. No prime up to 19 divides 373, so it is prime.
10. **Yes** — Rule for 3: digit sum divisible by 3. 52,416 ÷ 3 leaves remainder 0.
11. **47** — Subtracting (−30) is the same as adding 30. 11 + 30 + 6 = 47.
12. **16: 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120** — Pair factors from the outside in: 1×120, 2×60, 3×40, 4×30, 5×24, 6×20, 8×15, 10×12.
13. **6,00,00,000** — The digit is 6; crores = 1,00,00,000; 6 × 1,00,00,000 = 6,00,00,000.
14. **587,934,175** — International grouping uses threes from the right: thousand, million, billion.
15. **Prime** — √349 ≈ 18. No prime up to 18 divides 349, so it is prime.
16. **No** — Rule for 11: (sum of odd-place digits − sum of even-place digits) divisible by 11. 86,258 ÷ 11 leaves remainder 7.
17. **−22** — Subtracting (−5) is the same as adding 5. −18 + 5 + −9 = −22.
18. **8,00,00,000** — The digit is 8; crores = 1,00,00,000; 8 × 1,00,00,000 = 8,00,00,000.

# Chapter 2: Arithmetic Fundamentals

**Learning objective**

Add, subtract, multiply and divide with confidence; understand carrying and borrowing; use the commutative, associative and distributive properties to split numbers into convenient parts; estimate and check answers.

## 2.1 Addition and carrying

When a column adds to 10 or more, we **carry** the tens digit to the next column on the left.

```
  1 1        ← carries
  4 7 8
+ 3 6 5
-------
  8 4 3
```

Units: 8 + 5 = 13 → write 3, carry 1. Tens: 7 + 6 + 1 = 14 → write 4, carry 1. Hundreds: 4 + 3 + 1 = 8.

## 2.2 Subtraction and borrowing

When the top digit is smaller, we **borrow** 1 from the next column (worth 10 in the current column).

```
  6 ¹2 ¹3     (borrow: 3 becomes 13, then 2 becomes 12)
  7  3  3
- 2  8  6
---------
  4  4  7
```

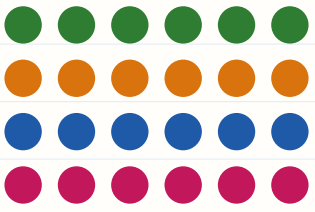
Check by adding: 447 + 286 = 733 ✓. **Every subtraction can be checked by addition.**

**Borrowing across zeros**

In 5,003 − 1,267 you must borrow through the zeros: 5,003 becomes 4 | 9 | 9 | 13. A faster way is the Vedic "all from 9 and the last from 10" method in Part III.

## 2.3 Multiplication and division

**Multiplication** is repeated addition: 4 × 6 means four groups of six.



4 rows × 6 = 24

**Division** is sharing into equal groups, or asking "how many groups?". 24 ÷ 6 = 4 because 6 × 4 = 24. Every division fact is a multiplication fact read backwards.

### Division vocabulary

Dividend = Divisor × Quotient + Remainder, and the remainder is always smaller than the divisor. Example: 47 ÷ 5 → 47 = 5 × 9 + 2, so quotient 9 and remainder 2.

## 2.4 Multiplication tables

Knowing tables up to 10 × 10 (better, up to 20 × 10) is the single most useful arithmetic skill. The full grid is reprinted in Part XV.

| × | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 |
| 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 | 30 |
| 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 | 40 |
| 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 |
| 6 | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 | 60 |
| 7 | 14 | 21 | 28 | 35 | 42 | 49 | 56 | 63 | 70 |
| 8 | 16 | 24 | 32 | 40 | 48 | 56 | 64 | 72 | 80 |
| 9 | 18 | 27 | 36 | 45 | 54 | 63 | 72 | 81 | 90 |
| 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |

### Patterns that make tables easy

- **×9:** the digits of each answer add to 9 (18, 27, 36, …), and the tens digit is one less than the number multiplied (9 × 7 = 63).
- **×5:** half of ×10. 5 × 14 = 140 ÷ 2 = 70.
- **×4:** double, then double again. 4 × 17 = 34 doubled = 68.
- **×11 (single digits):** repeat the digit: 11 × 7 = 77.
- **Turn-around facts:** 7 × 8 = 8 × 7, so there are only 45 facts to learn in the 10 × 10 table, not 100.

## 2.5 The three properties that power mental maths

### The properties

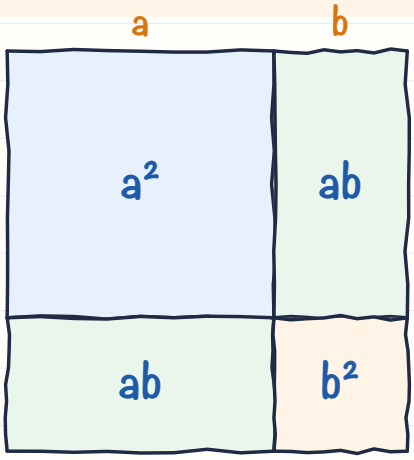
**Commutative:** a + b = b + a and a × b = b × a. (Order does not matter.) **Associative:** (a + b) + c = a + (b + c) and (a × b) × c = a × (b × c). (Grouping does not matter.) **Distributive:** a × (b + c) = a × b + a × c. (Multiplication spreads over addition.)

The distributive property is the secret behind almost every multiplication shortcut — including the Vedic ones.

### Using the properties

- 25 × 37 × 4 = (25 × 4) × 37 = 100 × 37 = 3,700. *(commutative + associative)*
- 47 + 38 + 53 = (47 + 53) + 38 = 100 + 38 = 138. *(find pairs that make 100)*

- $8 \times 46 = 8 \times 40 + 8 \times 6 = 320 + 48 = 368$. *(distributive)*
- $9 \times 58 = 10 \times 58 - 58 = 580 - 58 = 522$. *(distributive with subtraction)*
- $16 \times 25 = 4 \times 4 \times 25 = 4 \times 100 = 400$. *(break into factors)*




$$(a + b)^2 = a^2 + 2ab + b^2$$

The picture shows the distributive property as an area: a big square of side (a + b) is made of four rectangles. You will use this picture again for algebraic identities in Part VII.

## 2.6 Breaking numbers into convenient parts

Mental calculation is mostly about **re-writing a hard problem as easy ones**.

| Problem | Convenient split | Working |
|---|---|---|
| 68 + 57 | 57 = 60 − 3 | 68 + 60 − 3 = 125 |
| 342 − 198 | 198 = 200 − 2 | 342 − 200 + 2 = 144 |
| 7 × 98 | 98 = 100 − 2 | 700 − 14 = 686 |
| 15 × 24 | 15 × 24 = 30 × 12 | 360 (double one, halve the other) |
| 648 ÷ 8 | 648 = 640 + 8 | 80 + 1 = 81 |

## 2.7 Estimation and checking

### Estimating before you calculate

Round each number to one or two significant digits and calculate. $49 \times 21 \approx 50 \times 20 = 1{,}000$. The exact answer (1,029) should be close to this. Estimation catches misplaced digits and wrong decimal points. It does **not** give the exact answer.

Three quick checks you can always use:

1. **Size check** — is the answer roughly the estimate?
2. **Last-digit check** — 37 × 48 must end in 6, because 7 × 8 = 56.
3. **Digit-sum check** (casting out nines) — explained fully in Part II, section on *Gunitasamuccayah*.

## 2.8 Standard method versus mental method

### Standard written method

**Problem: 48 × 25**

```
   48
 × 25
 ----
  240
 960
 ----
 1200
```

### Mental method

**Problem: 48 × 25** 25 = 100 ÷ 4, so 48 × 25 = 48 × 100 ÷ 4 = 4,800 ÷ 4 = **1,200**

One line, no carrying.

### When to use / when not to use

**Use a mental method** when the numbers have a friendly structure (near 10/100, multiples of 5 or 25, pairs that make 100). **Use the written method** when the numbers are "ugly" and you need certainty, e.g. 7,384 × 2,917. A good calculator of numbers chooses the method *after* looking at the numbers.

### Chapter 2 in five lines

- Carry when a column reaches 10; borrow when the top digit is smaller.
- Dividend = divisor × quotient + remainder.
- Commutative and associative: re-order and re-group freely for + and ×.
- Distributive: a(b + c) = ab + ac — the engine of all multiplication shortcuts.
- Always estimate first and check last.

### ✎ Chapter 2 practice

**19.** Add mentally from left to right: 867 + 794

**20.** 2,250 − 2,008

**21.** 48 × 3

**22.** 2,175 ÷ 25

**23.** 7 × 3

**24.** Estimate 66 × 972 by rounding (approximate answer). *(approximate)*

**25.** Add mentally from left to right: 935 + 539

**26.** 4,217 − 1,578

**27.** 19 × 6

**28.** 7,479 ÷ 29

**29.** 3 × 3

**30.** Estimate 81 × 858 by rounding (approximate answer). *(approximate)*

**31.** Add mentally from left to right: 298 + 953

**32.** 662 − 318

**33.** 24 × 8

**34.** 7,847 ÷ 28

*Answers: see the answer key (questions 19–34).*

## Chapter 2 answer key

**19.** **1,661** — Hundreds 1500, tens 150, units 11: 1500 + 150 + 11 = 1661.

**20.** **242** — Subtract 2,010 instead: 2,250 − 2,010 = 240; add back 2 → 242.

**21.** **144** — (40 + 8) × 3 = 120 + 24 = 144.

**22.** **Q = 87, R = 0** — Check: 25 × 87 + 0 = 2,175.

**23.** **21** — 7 groups of 3 = 21.

**24.** **≈ 70,000 (exact 64,152)** — 66 ≈ 70, 972 ≈ 1000; 70 × 1000 = 70,000. This is an estimate, not the exact product.

**25.** **1,474** — Hundreds 1400, tens 60, units 14: 1400 + 60 + 14 = 1474.

**26.** **2,639** — Subtract 1,580 instead: 4,217 − 1,580 = 2,637; add back 2 → 2,639.

**27.** **114** — (10 + 9) × 6 = 60 + 54 = 114.

**28.** **Q = 257, R = 26** — Check: 29 × 257 + 26 = 7,479.

**29.** **9** — 3 groups of 3 = 9.

**30.** **≈ 72,000 (exact 69,498)** — 81 ≈ 80, 858 ≈ 900; 80 × 900 = 72,000. This is an estimate, not the exact product.

**31.** **1,251** — Hundreds 1100, tens 140, units 11: 1100 + 140 + 11 = 1251.

**32.** **344** — Subtract 320 instead: 662 − 320 = 342; add back 2 → 344.

**33.** **192** — (20 + 4) × 8 = 160 + 32 = 192.

**34.** **Q = 280, R = 7** — Check: 28 × 280 + 7 = 7,847.

# Chapter 3: Fractions, Decimals and Percentages

### Learning objective

Understand what a fraction means; convert between fractions, decimals and percentages; simplify and compare fractions; calculate percentages, percentage increase and decrease; memorise the fraction–percentage values that appear constantly in exams.

## 3.1 What is a fraction?

A fraction describes **part of a whole**. In 3/4, the whole is cut into **4 equal parts** (the *denominator*) and we take **3** of them (the *numerator*).




3 of 4 equal parts are shaded

| Type | Meaning | Example |
|---|---|---|
| Proper fraction | numerator < denominator | 3/8 |
| Improper fraction | numerator ≥ denominator | 11/4 |
| Mixed number | whole number + proper fraction | 2 3/4 (= 11/4) |
| Unit fraction | numerator 1 | 1/7 |

### Converting mixed ↔ improper

2 3/4 = (2 × 4 + 3)/4 = 11/4. 17/5 = 3 remainder 2 = 3 2/5.

## 3.2 Equivalent fractions and simplification

Multiplying or dividing the numerator **and** the denominator by the same non-zero number does not change the value: 1/2 = 2/4 = 50/100.

To **simplify**, divide both by their **HCF** (highest common factor).

### Simplify 84/126

HCF(84, 126) = 42. 84 ÷ 42 = 2 and 126 ÷ 42 = 3, so 84/126 = **2/3**. If you cannot see the HCF, divide step by step: 84/126 → (÷2) 42/63 → (÷3) 14/21 → (÷7) 2/3.

## 3.3 Adding, subtracting, multiplying and dividing fractions

### The four operations

**Add/subtract:** make the denominators equal (use the LCM), then add/subtract numerators. **Multiply:** numerator × numerator, denominator × denominator. Cancel first to keep numbers small. **Divide:** multiply by the reciprocal (flip the second fraction).

### Worked examples

- 2/3 + 3/4 = 8/12 + 9/12 = 17/12 = 1 5/12
- 5/6 − 1/4 = 10/12 − 3/12 = 7/12
- 4/9 × 3/8 = (cancel 4 with 8, 3 with 9) 1/3 × 1/2 = 1/6
- 3/5 ÷ 9/10 = 3/5 × 10/9 = 30/45 = 2/3
- 1 1/2 × 2 2/3 = 3/2 × 8/3 = 8/2 = 4

### Vertically and crosswise for fractions

For a/b + c/d, the answer is (ad + bc)/bd — "cross multiply for the top, multiply the bottoms". 2/3 + 3/4 = (2×4 + 3×3)/(3×4) = 17/12. This is the same crosswise pattern you will meet in *Urdhva-Tiryagbhyam* (Part II). Simplify at the end.

## 3.4 Decimals

A decimal is a fraction whose denominator is a power of 10: 0.7 = 7/10, 0.25 = 25/100, 0.125 = 125/1000.

- **Adding/subtracting:** line up the decimal points. 12.5 + 3.75 = 16.25
- **Multiplying:** multiply as whole numbers, then count the total decimal places. 1.2 × 0.03 → 12 × 3 = 36, three decimal places → 0.036
- **Dividing:** make the divisor a whole number by moving both points. 4.8 ÷ 0.12 = 480 ÷ 12 = 40

### Decimal places

0.2 × 0.3 = 0.06, **not** 0.6. Count: one place + one place = two places.

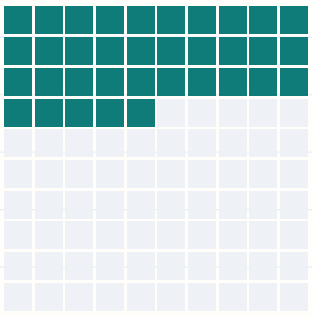
## 3.5 Fraction → decimal

Divide the numerator by the denominator. If the denominator has only the prime factors 2 and 5, the decimal **terminates** (3/8 = 0.375). Otherwise it **recurs**: 1/3 = 0.333…, 1/7 = 0.142857142857…

Part II shows two Vedic ways to find recurring decimals such as 1/19 and 1/7 quickly.

## 3.6 Percentages

**Per cent** means "per hundred". 35% means 35 out of 100 = 35/100 = 0.35.



35 out of 100
= 35% = 0.35

### Conversions

Fraction → % : multiply by 100. 3/8 × 100 = 37.5% % → fraction : divide by 100 and simplify. 45% = 45/100 = 9/20 Decimal → % : move the point two places right. 0.072 = 7.2%

## 3.7 Finding a percentage of a quantity

### Build any percentage from 10%, 5% and 1%

**Find 35% of 640.** 10% = 64, so 30% = 192; 5% = 32. Total 35% = 192 + 32 = **224**. **Find 18% of 450.** 10% = 45; 1% = 4.5 so 8% = 36. Total = 45 + 36 = **81**. **Find 12.5% of 488.** 12.5% = 1/8. 488 ÷ 8 = **61**. **Find 120% of 75.** 100% = 75 and 20% = 15. Total = **90**. **Find 0.5% of 3,000.** 1% = 30, so 0.5% = **15**.

### x% of y = y% of x

8% of 25 is hard? Swap: 25% of 8 = 2. This always works because x/100 × y = y/100 × x.

## 3.8 Percentage increase and decrease

### Percentage change

Percentage change = (change ÷ original value) × 100. New value after an increase of r% = original × (100 + r)/100. New value after a decrease of r% = original × (100 − r)/100.

### Price rise

A book's price rises from ₹240 to ₹276. Change = 36. Percentage increase = 36/240 × 100 = **15%**.

### Up 20% then down 20% is not "no change"

₹100 → up 20% → ₹120 → down 20% → ₹96. The second percentage is taken of a *bigger* number. Net change = −4%. Formula for successive changes a% and b%: net = a + b + ab/100 (use negative values for decreases): 20 − 20 − 400/100 = −4%.

## 3.9 The exam fraction–percentage chart

Learn this chart by heart. It turns many percentage questions into simple division.

| Fraction | Percentage | Decimal |
|---|---|---|
| 1/2 | 50% | 0.5 |
| 1/3 | 33 1/3% | 0.3333… (recurring) |
| 2/3 | 66 2/3% | 0.6667… (recurring) |
| 1/4 | 25% | 0.25 |
| 3/4 | 75% | 0.75 |
| 1/5 | 20% | 0.2 |
| 2/5 | 40% | 0.4 |
| 3/5 | 60% | 0.6 |
| 4/5 | 80% | 0.8 |

| Fraction | Percentage | Decimal |
|---|---|---|
| 1/6 | 16 2/3% | 0.1667… (recurring) |
| 5/6 | 83 1/3% | 0.8333… (recurring) |
| 1/7 | 14 2/7% | 0.1429… (recurring) |
| 1/8 | 12 1/2% | 0.125 |
| 3/8 | 37 1/2% | 0.375 |
| 5/8 | 62 1/2% | 0.625 |
| 7/8 | 87 1/2% | 0.875 |
| 1/9 | 11 1/9% | 0.1111… (recurring) |
| 1/10 | 10% | 0.1 |
| 1/11 | 9 1/11% | 0.0909… (recurring) |
| 1/12 | 8 1/3% | 0.0833… (recurring) |
| 1/15 | 6 2/3% | 0.0667… (recurring) |
| 1/16 | 6 1/4% | 0.0625 |
| 1/20 | 5% | 0.05 |
| 1/25 | 4% | 0.04 |
| 1/40 | 2 1/2% | 0.025 |
| 1/50 | 2% | 0.02 |

### The "eighths" family

1/8 = 12.5%, and every eighth adds 12.5%: 3/8 = 37.5%, 5/8 = 62.5%, 7/8 = 87.5%. The "sixths and thirds" family: 1/6 = 16 2/3%, 1/3 = 33 1/3%, 2/3 = 66 2/3%, 5/6 = 83 1/3%.

### Chapter 3 in five lines

- Fraction = parts taken / equal parts in the whole.
- Simplify by dividing top and bottom by the HCF.
- Add with a common denominator; multiply straight across; divide by flipping.
- Percent = per hundred; build any percentage from 10%, 5% and 1%.
- Percentage change is always measured against the **original** value.

## ✎ Chapter 3 practice

**35.** Simplify 117/18.

**36.** 1/5 – 3/7

**37.** Write 1/7 as a percentage.

**38.** Find 120% of 640.

**39.** A price changes from ₹400 to ₹360. Find the percentage decrease.

**40.** 2.1 × 7.6

**41.** Find 25% of 432 using a fraction.

**42.** Simplify 130/90.

**43.** 3/7 – 5/11

**44.** Write 1/11 as a percentage.

**45.** Find 40% of 480.

**46.** A price changes from ₹4,000 to ₹4,480. Find the percentage increase.

**47.** 3.7 × 0.11

**48.** Find 9 1/11% of 803 using a fraction.

**49.** Simplify 45/63.

**50.** 3/10 + 2/3

**51.** Write 5/6 as a percentage.

**52.** Find 150% of 80.

*Answers: see the answer key (questions 35–52).*

## Chapter 3 answer key

**35.** **13/2** — HCF of 117 and 18 is 9. Divide both by 9.

**36.** **–8/35** — LCM of denominators = 35. Convert, then subtract numerators and simplify: –8/35.

**37.** **14 2/7%** — 1/7 × 100 = 14 2/7%.

**38.** **768** — Split 120% into easy parts (10%, 5%, 1% …) or use fractions: 120% = 6/5. Result 768.

**39.** **10%** — Change = 40. % change = change ÷ original × 100 = 40/400 × 100 = 10%.

**40.** **15.96** — Multiply without points: 21 × 76 = 1596. Decimal places: 1 + 1 = 2, so the answer is 15.96.

**41.** **108** — 25% = 1/4. 1/4 × 432 = 108.

**42.** **13/9** — HCF of 130 and 90 is 10. Divide both by 10.

**43.** **–2/77** — LCM of denominators = 77. Convert, then subtract numerators and simplify: –2/77.

**44.** **9 1/11%** — 1/11 × 100 = 9 1/11%.

**45.** **192** — Split 40% into easy parts (10%, 5%, 1% …) or use fractions: 40% = 2/5. Result 192.

**46.** **12%** — Change = 480. % change = change ÷ original × 100 = 480/4000 × 100 = 12%.

**47.** **0.407** — Multiply without points: 37 × 11 = 407. Decimal places: 1 + 2 = 3, so the answer is 0.407.

**48.** **73** — 9 1/11% = 1/11. 1/11 × 803 = 73.

**49.** **5/7** — HCF of 45 and 63 is 9. Divide both by 9.

**50.** **29/30** — LCM of denominators = 30. Convert, then add numerators and simplify: 29/30.

**51.** **83 1/3%** — 5/6 × 100 = 83 1/3%.

**52.** **120** — Split 150% into easy parts (10%, 5%, 1% …) or use fractions: 150% = 3/2. Result 120.

# Chapter 4: Powers, Roots and Number Properties

### Learning objective

Use squares, cubes and exponents; find square and cube roots of perfect powers; write numbers as products of primes; find HCF and LCM; work with remainders; estimate sensibly.

## 4.1 Squares and cubes

The **square** of a number is the number times itself: $7^2 = 7 \times 7 = 49$. It is the area of a square with side 7. The **cube** is the number used three times: $4^3 = 4 \times 4 \times 4 = 64$, the volume of a cube with side 4.

| n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 7 | 49 | 13 | 169 | 19 | 361 | 25 | 625 |
| 2 | 4 | 8 | 64 | 14 | 196 | 20 | 400 | 26 | 676 |
| 3 | 9 | 9 | 81 | 15 | 225 | 21 | 441 | 27 | 729 |
| 4 | 16 | 10 | 100 | 16 | 256 | 22 | 484 | 28 | 784 |
| 5 | 25 | 11 | 121 | 17 | 289 | 23 | 529 | 29 | 841 |
| 6 | 36 | 12 | 144 | 18 | 324 | 24 | 576 | 30 | 900 |

| n | $n^3$ | n | $n^3$ | n | $n^3$ |
|---|---|---|---|---|---|
| 1 | 1 | 6 | 216 | 11 | 1,331 |
| 2 | 8 | 7 | 343 | 12 | 1,728 |
| 3 | 27 | 8 | 512 | 13 | 2,197 |
| 4 | 64 | 9 | 729 | 14 | 2,744 |
| 5 | 125 | 10 | 1,000 | 15 | 3,375 |

### Squares go up by odd numbers

1, 4, 9, 16, 25, … The gaps are 3, 5, 7, 9, … So $31^2 = 30^2 + 30 + 31 = 900 + 61 = 961$. In general $(n + 1)^2 = n^2 + n + (n + 1)$.

## 4.2 Exponents and powers

In $2^5$, 2 is the **base** and 5 is the **exponent** (or index). $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

### Laws of exponents ($a \neq 0$)

$a^m \times a^n = a^{m+n} \cdot a^m \div a^n = a^{m-n} \cdot (a^m)^n = a^{mn} \cdot (ab)^n = a^n b^n \; a^0 = 1 \cdot a^{-n} = 1/a^n \cdot a^{1/2} = \sqrt{a} \cdot a^{1/3} = \sqrt[3]{a}$

### Using the laws

- $2^3 \times 2^4 = 2^7 = 128$
- $5^6 \div 5^4 = 5^2 = 25$
- $(3^2)^3 = 3^6 = 729$
- $10^{-3} = 1/1000 = 0.001$
- $2^{10} = 1{,}024 \approx 1{,}000$ (useful for estimation)

| n | $2^n$ | $3^n$ | $5^n$ |
|---|---|---|---|
| 1 | 2 | 3 | 5 |
| 2 | 4 | 9 | 25 |
| 3 | 8 | 27 | 125 |
| 4 | 16 | 81 | 625 |
| 5 | 32 | 243 | 3,125 |
| 6 | 64 | 729 | 15,625 |
| 7 | 128 | 2,187 | 78,125 |
| 8 | 256 | 6,561 | 3,90,625 |
| 9 | 512 | 19,683 | |
| 10 | 1,024 | 59,049 | |
| 11 | 2,048 | | |
| 12 | 4,096 | | |

## 4.3 Square roots and cube roots

The **square root** $\sqrt{n}$ is the number which, squared, gives n: $\sqrt{144} = 12$. The **cube root** $\sqrt[3]{n}$ is the number which, cubed, gives n: $\sqrt[3]{343} = 7$.

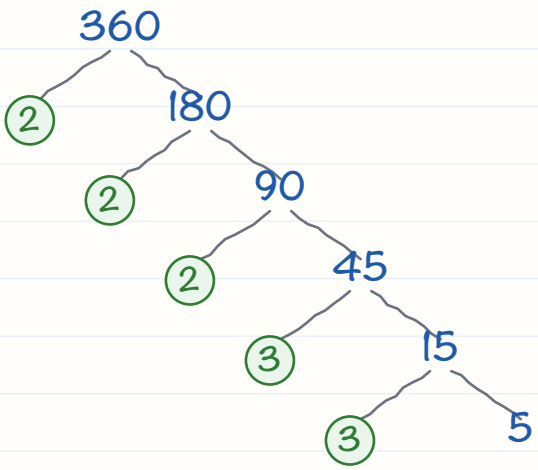
| Number ends in | Square ends in | Cube ends in |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 1 | 1 |
| 2 | 4 | 8 |
| 3 | 9 | 7 |
| 4 | 6 | 4 |
| 5 | 5 | 5 |
| 6 | 6 | 6 |
| 7 | 9 | 3 |
| 8 | 4 | 2 |
| 9 | 1 | 9 |

### Last digits identify roots

A perfect square can only end in 0, 1, 4, 5, 6 or 9. A number ending in 2, 3, 7 or 8 is **never** a perfect square. Every digit 0–9 appears exactly once as the last digit of a cube — so the last digit of a perfect cube tells you the last digit of its cube root. Part VI uses this to find cube roots of 4–6 digit perfect cubes in seconds.

## 4.4 Prime factorisation

Every whole number greater than 1 can be written as a product of primes in exactly one way (ignoring order). This is called the **Fundamental Theorem of Arithmetic**.




$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = 2^3 \times 3^2 \times 5$.

### Number of factors

If $n = p^a \times q^b \times r^c \ldots$, the number of factors is $(a + 1)(b + 1)(c + 1)\ldots$ $360 = 2^3 \times 3^2 \times 5^1 \rightarrow (3 + 1)(2 + 1)(1 + 1) = 24$ factors.

## 4.5 HCF and LCM

- **HCF** (highest common factor, also GCD): the biggest number dividing all the given numbers.
- **LCM** (lowest common multiple): the smallest number that all the given numbers divide.

### Using prime factors

HCF = product of common primes with the **smallest** powers. LCM = product of all primes with the **largest** powers. For two numbers: HCF × LCM = product of the numbers.

### HCF and LCM of 72 and 120

$72 = 2^3 \times 3^2$, $120 = 2^3 \times 3 \times 5$. HCF = $2^3 \times 3$ = **24**. LCM = $2^3 \times 3^2 \times 5$ = **360**. Check: $24 \times 360 = 8{,}640 = 72 \times 120$ ✓.

### When to use which

"Largest size of equal pieces / groups" → **HCF**. "When will they next happen together / smallest common length" → **LCM**. Bells ringing every 12 and 18 minutes ring together again after LCM(12, 18) = 36 minutes.

## 4.6 Remainders and divisibility

### Remainder arithmetic

The remainder of a sum (or product) equals the remainder of the sum (or product) of the remainders. (47 × 53) ÷ 5 → remainders 2 and 3 → 2 × 3 = 6 → and 6 leaves remainder **1** on division by 5. Check: 47 × 53 = 2,491, and 2,491 ÷ 5 leaves 1 ✓.

This simple idea lets you find remainders of huge numbers without calculating them, and it is the basis of digit-sum checking.

## 4.7 Approximation and estimation

### Rounding rules

To round to the nearest 10, 100, …: look at the next digit to the right. 5 or more → round up; less than 5 → round down. 4,768 ≈ 4,800 (nearest hundred) ≈ 5,000 (nearest thousand). For square roots: $\sqrt{50}$ is a bit more than 7 because $7^2 = 49$. A good estimate is $\sqrt{n} \approx a + (n - a^2)/(2a)$, where $a^2$ is the nearest perfect square: $\sqrt{50} \approx 7 + 1/14 \approx 7.07$ (true value 7.0711…).

### Chapter 4 in six lines

- $a^m \times a^n = a^{m+n}$; $a^0 = 1$; $a^{-n} = 1/a^n$.
- Perfect squares end only in 0, 1, 4, 5, 6, 9.
- Cube last digits: 2↔8 and 3↔7 swap; the others stay the same.
- Prime factorisation is unique; number of factors = product of (power + 1).
- HCF: smallest powers of common primes. LCM: largest powers of all primes.
- Remainders can be multiplied and added.

### ✎ Chapter 4 practice

**53.** $45^2$

**54.** $16^3$

**55.** $\sqrt{3,481}$

**56.** $\sqrt[3]{1,48,877}$

**57.** $2^{-1}$ as a fraction

**58.** Write 252 as a product of primes.

**59.** Find the HCF and LCM of 14 and 70.

**60.** Find the remainder when 66,136 is divided by 13.

**61.** Estimate $\sqrt{244}$ to 2 decimal places (approximate). *(approximate)*

**62.** $16^2$

**63.** $6^3$

**64.** $\sqrt{4,225}$

**65.** $\sqrt[3]{50,653}$

**66.** $2^6 \times 2^1$ (give the value)

**67.** Write 210 as a product of primes.

**68.** Find the HCF and LCM of 91 and 39.

**69.** Find the remainder when 71,676 is divided by 7.

**70.** Estimate √92 to 2 decimal places (approximate). *(approximate)*

**71.** $26^2$

**72.** $5^3$

*Answers: see the answer key (questions 53–72).*

### Part I mastery checklist

Tick each item only when you can do it without help: ☐ read a 9-digit number in both systems ☐ test a 3-digit number for primality ☐ use the distributive property to multiply mentally ☐ convert any fraction in the chart to a percentage instantly ☐ find HCF and LCM by prime factors ☐ estimate a product to within 10%.

## Chapter 4 answer key

**53.** **2,025** — 45 × 45 = 2025.

**54.** **4,096** — 16 × 16 × 16 = 4,096.

**55.** **59** — Last digit 1 → root ends in 1 or 9. Leading part 34 lies between $5^2$ and $6^2$, so tens digit is 5. Check $55^2 = 3025$ to choose → 59.

**56.** **53** — Last digit 7 → cube root ends in 3 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 148 lies between $5^3$ and $6^3$ → tens digit 5.

**57.** **1/2** — A negative exponent means reciprocal: $1/2^1$.

**58.** **$2^2 \times 3^2 \times 7$** — Divide repeatedly by the smallest prime that works (factor tree).

**59.** **HCF 14, LCM 70** — HCF × LCM = product: 14 × 70 = 980 = 14 × 70.

**60.** **5** — 66,136 = 13 × 5,087 + 5.

**61.** **≈ 15.62 (true value 15.620)** — Nearest perfect square 256 = $16^2$. $\sqrt{n} \approx 16 + (244 - 256) \div (2 \times 16) = 15.625$.

**62.** **256** — 16 × 16 = 256.

**63.** **216** — 6 × 6 × 6 = 216.

**64.** **65** — Last digit 5 → root ends in 5. Leading part 42 lies between $6^2$ and $7^2$, so tens digit is 6. Check $65^2 = 4225$ to choose → 65.

**65.** **37** — Last digit 3 → cube root ends in 7 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 50 lies between $3^3$ and $4^3$ → tens digit 3.

**66.** **128** — Add exponents: $2^7 = 128$.

**67.** **$2 \times 3 \times 5 \times 7$** — Divide repeatedly by the smallest prime that works (factor tree).

**68.** **HCF 13, LCM 273** — HCF × LCM = product: 13 × 273 = 3549 = 91 × 39.

**69.** **3** — 71,676 = 7 × 10,239 + 3.

**70.** **≈ 9.60 (true value 9.592)** — Nearest perfect square 100 = $10^2$. $\sqrt{n} \approx 10 + (92 - 100) \div (2 \times 10) = 9.600$.

**71.** **676** — 26 × 26 = 676.

**72.** **125** — 5 × 5 × 5 = 125.

# Part II

## Vedic Mathematics Core Techniques

This part introduces the **16 sutras** in the order used by Bharati Krishna Tirthaji, followed by the **13 sub-sutras**. For each sutra you get the traditional name, a plain-English meaning, the mathematics behind it, and its best uses. The most important sutras (1–4, 10 and 14) are developed in full lessons here and practised heavily in Parts III–VII. The more specialised ones are explained with worked examples and honest notes about their limits.

# The 16 Sutras at a Glance

### How to read the sutra names

Each sutra is a short Sanskrit phrase. The English meanings below are the ones commonly given in Tirthaji's book and in later teaching. Several sutras are applied in more than one way, and different teachers sometimes link the same technique to different sutras. Where that happens, this book says so rather than presenting one interpretation as the only correct one.

| # | Sutra | Common English meaning | Main uses in this book |
|---|---|---|---|
| 1 | Ekadhikena Purvena | By one more than the previous one | Squares ending in 5; 1/19, 1/29 … as decimals |
| 2 | Nikhilam Navatashcaramam Dashatah | All from 9 and the last from 10 | Subtraction from $10^n$; multiplication and division near a base |
| 3 | Urdhva-Tiryagbhyam | Vertically and crosswise | General multiplication; algebra |
| 4 | Paravartya Yojayet | Transpose and adjust | Division by 12, 112 …; solving equations |
| 5 | Shunyam Saamyasamuccaye | When the total is the same, that total is zero | Special equations solved by inspection |
| 6 | (Anurupye) Shunyamanyat | If one is in ratio, the other is zero | Special simultaneous equations |
| 7 | Sankalana-Vyavakal anabhyam | By addition and by subtraction | Equations with swapped coefficients |
| 8 | Puranapuranabhyam | By completion or non-completion | Completing the square / cube |
| 9 | Chalana-Kalanabhya m | Differences and similarities (sequential motion) | Quadratic equations via 2x + b |
| 10 | Yavadunam | Whatever the extent of its deficiency | Squaring numbers near a base |
| 11 | Vyashtisamanstih | Part and whole | Using an average or a split to simplify |
| 12 | Shesanyankena Charamena | The remainders by the last digit | Decimals of 1/7, 1/13, 1/17 … |
| 13 | Sopantyadvayamant yam | The ultimate and twice the penultimate | A special family of fraction equations |
| 14 | Ekanyunena Purvena | By one less than the previous one | Multiplying by 9, 99, 999 … |
| 15 | Gunitasamuccayah | The product of the sum is equal to the sum of the product | Checking products and factorisations |
| 16 | Gunakasamuccayah | The factors of the sum is equal to the sum of the factors | Checking factorisations; derivatives |

## Two ideas used throughout

**1. Base and deviation.** Many methods compare a number with a convenient **base** (10, 100, 1,000) and work with the **deviation** (how far above or below the base it is). 97 is 100 − 3; 104 is 100 + 4.

**2. Digit sums (Beejank).** Add the digits of a number repeatedly until one digit remains (treat 9 as 0). This is the remainder on division by 9. If a calculation is correct, the digit sums obey the same calculation. See Sutra 15.

# Sutra 1: Ekadhikena Purvena — "By One More than the Previous One"

### Learning objective

Square any number ending in 5 in one step; multiply pairs such as 43 × 47; write 1/19 (and similar fractions) as recurring decimals without long division.

**Simple explanation.** *Ekadhika* means "one more". The sutra tells you to take the digit (or number) that comes before, and use **one more than it**.

**Prerequisites:** multiplication tables, place value.

## 1A. Squaring numbers ending in 5

### The rule

For a number ending in 5, write it as (a)5. Then **(a)5² = a × (a + 1) | 25**.

$65^2$ take 6 6 × 7 = 42 attach 25 4,225

### Why it works

A number ending in 5 is 10a + 5. Its square is $(10a + 5)^2 = 100a^2 + 100a + 25 = 100 \times a(a + 1) + 25$. So the hundreds part is a × (a + 1) and the last two digits are always 25.

**Step-by-step algorithm**

1. Remove the final 5. Call what is left *a*.
2. Multiply *a* by the next number, *a* + 1.
3. Write 25 after the result.

### Standard method

**$75^2$**

```
  75
× 75
----
 375
525
----
5625
```

### Ekadhikena

**$75^2$** a = 7, one more = 8 7 × 8 = 56 attach 25 → **5,625**

### Five solved examples (increasing difficulty)

1. $25^2$ → 2 × 3 = 6 → **625**

2. $45^2 \rightarrow 4 \times 5 = 20 \rightarrow$ **2,025**
3. $95^2 \rightarrow 9 \times 10 = 90 \rightarrow$ **9,025**
4. $115^2 \rightarrow 11 \times 12 = 132 \rightarrow$ **13,225**
5. $7.5^2 \rightarrow 75^2 = 5{,}625$; one decimal place in 7.5, so two in the square → **56.25**

**Real-life use.** Area of a 35 m × 35 m plot = $35^2 = 3 \times 4 \mid 25 = 1{,}225$ m².

### Common mistakes

- Writing a × a instead of a × (a + 1): $65^2$ is **not** 3,625. It is 6 × 7 = 42 → 4,225.
- Forgetting that a can have two digits: $125^2 \rightarrow 12 \times 13 = 156 \rightarrow 15{,}625$.

## 1B. Same leading digits, last digits adding to 10

The same idea works for products like 43 × 47: the tens digits are equal (4) and the units add to 10 (3 + 7). This is the sub-sutra **Antyayordasake'pi** ("when the last digits total 10"), used together with Ekadhikena.

### The rule

If two numbers have the same leading part *a* and their last digits *u* and *v* satisfy u + v = 10, then the product is **a × (a + 1) | u × v**, where u × v is written as **two digits**.

### Solved examples

1. $43 \times 47 \rightarrow 4 \times 5 = 20 \mid 3 \times 7 = 21 \rightarrow$ **2,021**
2. $62 \times 68 \rightarrow 6 \times 7 = 42 \mid 2 \times 8 = 16 \rightarrow$ **4,216**
3. $81 \times 89 \rightarrow 8 \times 9 = 72 \mid 1 \times 9 = 09 \rightarrow$ **7,209** (note the 0)
4. $104 \times 106 \rightarrow 10 \times 11 = 110 \mid 4 \times 6 = 24 \rightarrow$ **11,024**
5. 7.3 × 7.7 → for 73 × 77 write 7 × 8 | 3 × 7, i.e. 56 | 21 → 5,621 → **56.21**

### Why it works

$(10a + u)(10a + v) = 100a^2 + 10a(u + v) + uv$. When u + v = 10 this is $100a^2 + 100a + uv = 100 \cdot a(a + 1) + uv$.

### When to use / when not to use

**Use it** only when *both* conditions hold: the same leading part, and last digits adding to 10. **Do not** use it for 43 × 57 (different tens digits) or 46 × 47 (units add to 13). Then use vertically and crosswise (Sutra 3).

## 1C. Recurring decimals for 1/19, 1/29, 1/39 ...

For a denominator ending in 9, the *ekadhika* is one more than the digits before the 9: for 19 it is **2**, for 29 it is **3**, for 49 it is **5**.

**Method for 1/19 (multiply from the right by 2):** start with 1 as the last digit and keep doubling, writing each new digit to the **left** and carrying any tens.

```
digits written right to left:  1, 2, 4, 8, 6 (16, carry 1), 3 (12+1=
13, carry 1),
7 (6+1), 4 (14, carry 1), 9 (8+1), 8 (18, carry 1), 7 (16+1=17, carr
y 1), 5 (14+1=15, carry 1),
1 (10+1=11, carry 1), 3 (2+1), 6, 2 (12, carry 1), 5 (4+1), 0 (10, c
arry 1) … stop after 18 digits

1/19 = 0.052631578947368421 052631…   (18-digit recurring block)
```

### How many digits?

For 1/19 the block has 18 digits (one less than 19). You can stop at 9 digits and use a beautiful shortcut: the second half is the **nines-complement** of the first half. 052631578 + 947368421 = 999999999.

### Ekadhikena

"**Five squares: one more, then 25.**"

# Sutra 2: Nikhilam Navatashcaramam Dashatah — "All from 9 and the Last from 10"

### Learning objective

Subtract any number from 100, 1,000, 10,000 … instantly; multiply numbers close to a base in one line; divide by 9, 8, 89, 98, 997 and similar divisors.

**Simple explanation.** *Nikhilam* means "all". To find how far a number is from the next power of 10, subtract **every digit from 9** and the **last digit from 10**.

## 2A. Subtracting from a power of 10

### Solved examples

1. 100 – 37: 9 – 3 = 6, 10 – 7 = 3 → **63**
2. 1,000 – 468: 9–4, 9–6, 10–8 → **532**
3. 10,000 – 2,791 → 7,2,0,9 → **7,209**
4. 1,000 – 46: think of 46 as 046 → 9–0, 9–4, 10–6 → **954**
5. 10,000 – 3,580: the last digit is 0, so apply the rule to 358 (from 1,000) → 642 and keep the 0 → **6,420**

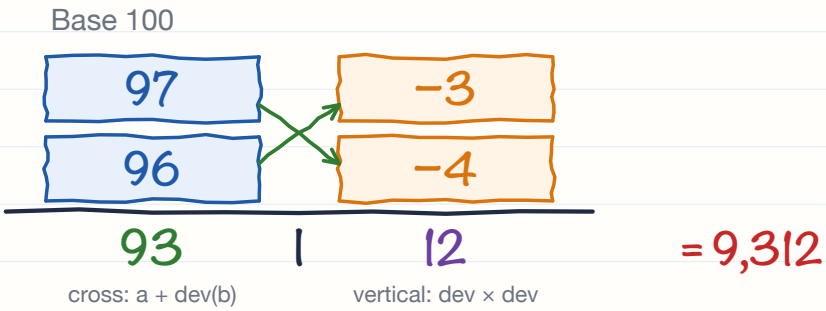
### Why it works

1,000 = 999 + 1. Subtracting from 999 never needs borrowing (each digit is subtracted from 9), and the extra 1 is added to the last digit, which is the same as subtracting that digit from 10.

### Trailing zeros

If the number ends in zero, the "last digit" is the last **non-zero** digit. 1,000 – 470: apply the rule to 47 (from 100) to get 53, then attach the 0 → 530.

## 2B. Multiplying numbers near a base




### Nikhilam multiplication

Write each number with its deviation from the base. **Right part** = product of the deviations (it must have as many digits as the base has zeros). **Left part** = either number plus the *other* number's deviation (cross addition).

**Step-by-step (97 × 96, base 100):**

1. Deviations: 97 = 100 − 3, 96 = 100 − 4.
2. Right part: (−3) × (−4) = 12 → two digits: 12.
3. Left part: 97 − 4 = 93 (or 96 − 3 = 93 — the same!).
4. Answer: 93 | 12 = **9,312**.

### Why it works

Let the numbers be B + a and B + b, where B is the base. Their product is $B^2 + Ba + Bb + ab = B \times (B + a + b) + ab$. The left part (B + a + b) is "one number plus the other's deviation"; it is multiplied by B, which is why the right part ab must fill exactly the number of zeros in B.

### Five solved examples

1. **88 × 94** (base 100; −12, −6): right = 72; left = 88 − 6 = 82 → **8,272**
2. **103 × 107** (+3, +7): right = 21; left = 103 + 7 = 110 → **11,021**
3. **112 × 95** (+12, −5): right = −60; left = 112 − 5 = 107 → 107 | −60 = 10,700 − 60 = **10,640**
4. **996 × 988** (base 1,000; −4, −12): right = 048 (three digits); left = 996 − 12 = 984 → **9,84,048**
5. **1,012 × 1,003** (+12, +3): right = 036; left = 1,015 → **10,15,036**

### Common mistakes

- **Too few digits in the right part.** With base 1,000 the right part needs three digits: 4 × 12 = 48 must be written 048.
- **Too many digits.** For 88 × 88 (base 100, −12 each), right = 144 — that is three digits. Carry the 1: left 76 + 1 → 77 | 44 = **7,744**.
- **Negative right part.** When one deviation is positive and the other negative, subtract: 107 | −60 means 10,700 − 60.

## 2C. Working bases (numbers near 50, 200, 300 ...)

When numbers are near 50 or 300, use a **working base** = (a convenient multiple) × (a power of 10).

### Working base 50 = 100 ÷ 2

**48 × 47** (deviations −2, −3 from 50): right = 6; left = 48 − 3 = 45. Because 50 = 100 ÷ 2, halve the left part: 45 ÷ 2 = 22.5. The .5 means "50 into the right part", so the right part becomes 56 and the answer is 22 | 56 = **2,256**. Alternatively use 50 = 5 × 10: left 45 × 5 = 225, right 6 → 225 | 6 = **2,256**.

### Working base 200 = 2 × 100

**212 × 207** (+12, +7): right = 84; left = 212 + 7 = 219 → × 2 = 438 → **43,884**.

## 2D. Nikhilam division (divisors just below a base)

To divide by 9, 8, 89, 98, 997 …, use the **complement** of the divisor (base − divisor), and repeatedly multiply and add.

### 1,235 ÷ 88

Base 100, complement 100 − 88 = **12**. Split the dividend: the right part has as many digits as the base has zeros.

```
   88 | 1   2 |  3   5
   12 |     1 |  2          ← 1 × 12
      |       |  3   6      ← (2 + 1) = 3, and 3 × 12 = 36
      |-------|--------
        1   3 |  8  11      → quotient 13, remainder 8×10 + 11 = 91
```

91 is bigger than 88, so take one more 88 out: quotient **14**, remainder 91 − 88 = **3**. Check: 88 × 14 + 3 = 1,235 ✓.

**Division by 9** is especially easy because the complement is 1: each quotient digit is the running total of the dividend's digits.

### 12,345 ÷ 9

Running totals: 1, 1+2 = 3, 3+3 = 6, 6+4 = 10 → quotient columns 1 | 3 | 6 | 10; remainder column 10 + 5 = 15. Resolve: 1,3,6,10 → 1,370 (carry the 1 from 10). Remainder 15 = 9 + 6 → add 1 to the quotient. **Quotient 1,371, remainder 6.** Check: 9 × 1,371 + 6 = 12,345 ✓.

### When to use / when not to use

**Use Nikhilam** when the numbers are within about 10–15 % of a base (88–112 for base 100). **Do not use it** for 67 × 58: the deviations (−33, −42) are larger than the numbers' distance to other bases, and the right part 1,386 overflows badly. Use vertically and crosswise instead.

### Nikhilam

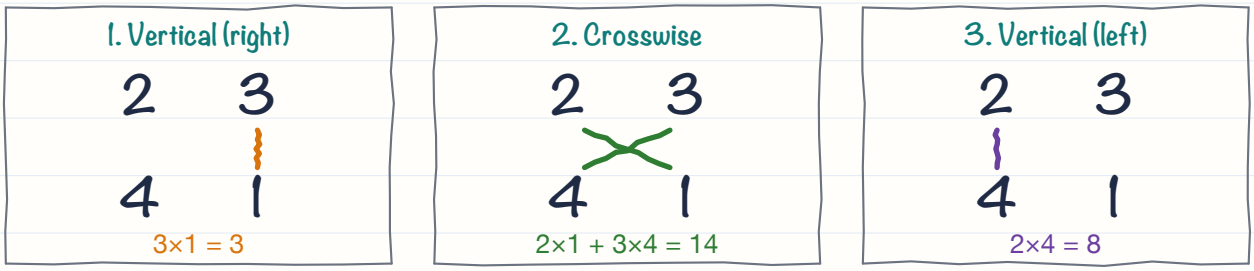
"**Cross-add on the left, multiply deviations on the right, fill the zeros.**"

# Sutra 3: Urdhva-Tiryagbhyam — "Vertically and Crosswise"

### Learning objective

Multiply any two numbers — of any size — in a single line, from right to left; multiply algebraic expressions with the same pattern.

**Simple explanation.** Instead of writing several partial products and adding them, you calculate each **column** of the answer directly: multiply vertically, then crosswise, then vertically again, adding as you go.




### 2-digit × 2-digit: (ab) × (cd)

Units: b × d. Tens: a × d + b × c (crosswise). Hundreds: a × c. Carry from right to left.

### 23 × 41

Step 1 (vertical right): 3 × 1 = 3 → write 3. Step 2 (crosswise): 2 × 1 + 3 × 4 = 2 + 12 = 14 → write 4, carry 1. Step 3 (vertical left): 2 × 4 = 8, + 1 = 9. **23 × 41 = 943.**

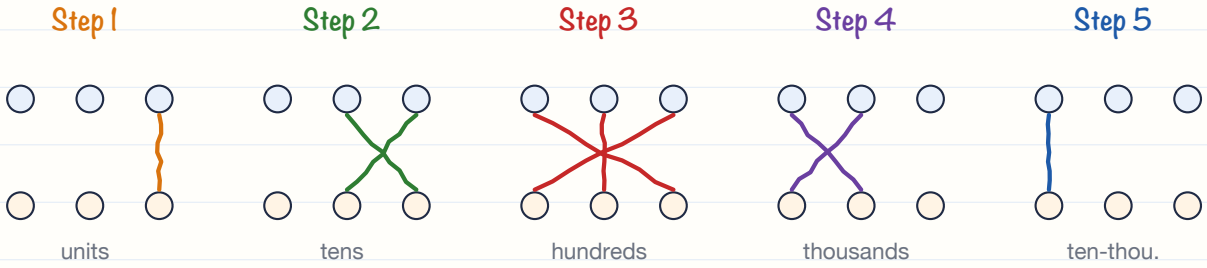
### Why it works

(10a + b)(10c + d) = 100ac + 10(ad + bc) + bd. The three steps are exactly the coefficients of 100, 10 and 1. It is the distributive property, organised by place value.

### Solved examples (increasing difficulty)

1. **32 × 21:** 2×1 = 2; 3×1 + 2×2 = 7; 3×2 = 6 → **672**
2. **46 × 37:** 6×7 = 42 → 2 carry 4; 4×7 + 6×3 = 46, +4 = 50 → 0 carry 5; 4×3 = 12, +5 = 17 → **1,702**
3. **89 × 76:** 9×6 = 54 → 4 c5; 8×6 + 9×7 = 111, +5 = 116 → 6 c11; 8×7 = 56, +11 = 67 → **6,764**
4. **123 × 456** (five steps, see below) → **56,088**
5. **(2x + 3)(4x + 5):** 2·4 = $8x^2$; 2·5 + 3·4 = 22x; 3·5 = 15 → **$8x^2$ + 22x + 15**

## 3-digit × 3-digit




```
  1 2 3
× 4 5 6
Step 1: 3×6                 = 18   → write 8, carry 1
Step 2: 2×6 + 3×5           = 27   +1 = 28 → write 8, carry 2
Step 3: 1×6 + 3×4 + 2×5     = 28   +2 = 30 → write 0, carry 3
Step 4: 1×5 + 2×4           = 13   +3 = 16 → write 6, carry 1
Step 5: 1×4                 =  4   +1 =  5
Answer: 56,088
```

### Common mistakes

- Forgetting one of the crosswise pairs in the middle step (there are **three** products in step 3 for 3-digit numbers).
- Adding the carry to the wrong column. Always carry to the **next column on the left**.
- Mixing up the number of digits: pad the shorter number with zeros (23 × 405 → 023 × 405).

### When to use / when not to use

**Use it** for any multiplication — it is the universal method. It shines for 2-digit × 2-digit products done mentally. For numbers near a base, Nikhilam is usually faster; for multiplying by 11, 25 or 99, the special rules in Part IV are faster still.

### Urdhva

"**Vertical, cross, vertical.** For 3 digits: 1 – 2 – 3 – 2 – 1 products in the five steps."

# Sutra 4: Paravartya Yojayet — "Transpose and Adjust"

### Learning objective

Divide by numbers slightly **above** a base (11, 12, 13, 112, 1,012 …); solve linear equations by transposing terms; understand the idea of "change the sign and apply".

**Simple explanation.** *Paravartya* means "transpose" (move across and change sign). When a divisor is a little more than a base, write its excess digits with **minus signs** and use them as multipliers.

## 4A. Paravartya division

### 1,452 ÷ 12

Base 10, excess +2 → transposed flag **–2**. The right part has one digit (base 10 has one zero).

```
  12 | 1    4    5 | 2
  -2 |     -2   -4 | -2
     |-------------|----
       1    2    1 | 0      → quotient 121, remainder 0
```

Bring down 1. 1 × (–2) = –2 → 4 – 2 = 2. 2 × (–2) = –4 → 5 – 4 = 1. 1 × (–2) = –2 → 2 – 2 = 0. **1,452 ÷ 12 = 121.** Check: 12 × 121 = 1,452 ✓.

### 13,789 ÷ 112

Base 100, excess 12 → flags **–1, –2**. Right part: 2 digits.

```
  112 | 1   3   7 |  8   9
  -1-2|    -1  -2 |
      |        -2 | -4          (2 × -1, 2 × -2)
      |           | -3  -6      (3 × -1, 3 × -2)
      |-----------|--------
        1   2   3 |  1   3      → quotient 123, remainder 13
```

**Quotient 123, remainder 13.** Check: 112 × 123 + 13 = 13,789 ✓.

### Negative remainder

Sometimes the remainder comes out negative (e.g. 2,489 ÷ 13 gives column values 2, –2, 14 | –33). Resolve the columns into an ordinary number, then **borrow one divisor** from the quotient until the remainder is between 0 and the divisor. 2,489 ÷ 13 → quotient 191, remainder 6.

## 4B. Solving linear equations by transposition

"Transpose and adjust" is exactly what you do when you move a term across the equals sign and change its sign.

### $ax + b = cx + d$

Transpose: $x = (d - b) \div (a - c)$.

### Solved examples

1. $5x + 3 = 28 \rightarrow 5x = 25 \rightarrow \mathbf{x = 5}$
2. $7x - 4 = 3x + 20 \rightarrow x = (20 + 4)/(7 - 3) = 24/4 = \mathbf{6}$
3. $2x + 9 = 5x - 6 \rightarrow x = (-6 - 9)/(2 - 5) = -15/-3 = \mathbf{5}$
4. $(x + 3)/4 = (x - 2)/3 \rightarrow 3(x + 3) = 4(x - 2) \rightarrow 3x + 9 = 4x - 8 \rightarrow \mathbf{x = 17}$
5. $0.5x + 1.2 = 0.2x + 3 \rightarrow x = (3 - 1.2)/(0.5 - 0.2) = 1.8/0.3 = \mathbf{6}$

### When to use / when not to use

**Paravartya division** is neat for divisors like 11, 12, 13, 102, 112, 1,012. For other divisors (e.g. 47, 73) use ordinary long division, or the more advanced "flag" (*dhvajanka*) method found in specialist books. **Do not force it**: dividing by 19 with Paravartya creates large negative columns and is slower than long division.

### Paravartya

"**Above the base? Flip the sign of the excess and multiply along.**"

### ✎ Practice: Sutras 1–4

**73.** $85^2$

**74.** $18 \times 12$

**75.** $10{,}000 - 1{,}631$

**76.** $94 \times 96$ (Nikhilam, base 100)

**77.** $104 \times 113$ (base 100)

**78.** $112 \times 92$

**79.** $82 \times 81$ (vertically and crosswise)

**80.** $118 \times 588$

**81.** $91{,}868 \div 9$ (quotient and remainder)

**82.** $39{,}099 \div 996$

**83.** $3{,}357 \div 113$

**84.** Solve $12x - 11 = 9x - 41$

**85.** Find the recurring block of $1/59$.

**86.** $185^2$

**87.** $82 \times 88$

**88.** $10{,}000 - 8{,}799$

**89.** $93 \times 92$ (Nikhilam, base 100)

**90.** 1,055 × 1,046 (base 1,000)

**91.** 109 × 98

**92.** 43 × 54 (vertically and crosswise)

**93.** 169 × 151

**94.** 60,082 ÷ 9 (quotient and remainder)

**95.** 34,046 ÷ 997

**96.** 6,884 ÷ 113

**97.** Solve 12x + 15 = 9x + 9

**98.** Find the recurring block of 1/39.

*Answers: see the answer key (questions 73–98).*

## Sutras 1–4 answer key

**73.** **7,225** — Digits before 5: 8 8 × (one more = 9) = 72 Attach 25 → 7,225

**74.** **216** — Same leading part 1; last digits 8 + 2 = 10 ✓ Left: 1 × 2 = 2 Right: 8 × 2 = 16 (always two digits) Answer = 216

**75.** **8,369** — All from 9, last from 10: 9–1=8 9–6=3 9–3=6, 10–1=9 → 8369.

**76.** **9,024** — Base = 100 Deviations: 94 → –6, 96 → –4 Left part (cross): 94 – 4 = 90 Right part: (–6) × (–4) = 24 Right part padded to 2 digit(s): 24 Answer = 9,024

**77.** **11,752** — Base = 100 Deviations: 104 → +4, 113 → +13 Left part (cross): 104 + 13 = 117 Right part: (+4) × (+13) = 52 Right part padded to 2 digit(s): 52 Answer = 11,752

**78.** **10,304** — Base = 100 Deviations: 112 → +12, 92 → –8 Left part (cross): 112 – 8 = 104 Right part: (+12) × (–8) = –96 Right part is negative: borrow 1 from left → 104 × 100 – 96 Answer = 10,304

**79.** **6,642** — Column 1 (from right): 2×1 = 2 → write 2, carry 0 Column 2 (from right): 2×8 + 8×1 = 24 → write 4, carry 2 Column 3 (from right): 8×8 = 64, + carry 2 = 66 → write 6, carry 6 Answer = 6,642

**80.** **69,384** — Column 1 (from right): 8×8 = 64 → write 4, carry 6 Column 2 (from right): 8×8 + 1×8 = 72, + carry 6 = 78 → write 8, carry 7 Column 3 (from right): 8×5 + 1×8 + 1×8 = 56, + carry 7 = 63 → write 3, carry 6 Column 4 (from right): 1×5 + 1×8 = 13, + carry 6 = 19 → write 9, carry 1 Column 5 (from right): 1×5 = 5, + carry 1 = 6 → write 6, carry 0 Answer = 69,384

**81.** **Q = 10,207, R = 5** — Running digit sums (quotient columns): 9, 10, 18, 24 Last running sum + last digit = remainder column 32 Adjust carries/remainder ≥ 9 → quotient 10,207, remainder 5

**82.** **Q = 39, R = 255** — Divisor 996 = 1,000 – 4; write complement digits 004 Split dividend: 39 | 099 Column totals after multiplying by the complement: [3, 9] | [0, 21, 45] Resolve carries and remainder → quotient 39, remainder 255

**83.** **Q = 29, R = 80** — Divisor 113 = 100 + 13; transposed flags: –1, –3 Column values after applying flags: [3, 0] | [-4, 7] Resolve negatives/carries → quotient 29, remainder 80

**84.** **x = –10** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (–41 – –11)/(12 – 9) = –30/3 = –10.

**85.** **0.(016949152542372881355932…)** — Ekadhika (one more than 5) = 6 Start with 1 on the right; repeatedly multiply by 6, write the unit digit, carry the rest Recurring block (58 digits): 016949152542372881355932…

**86.** **34,225** — Digits before 5: 18 18 × (one more = 19) = 342 Attach 25 → 34,225

**87.** **7,216** — Same leading part 8; last digits 2 + 8 = 10 ✓ Left: 8 × 9 = 72 Right: 2 × 8 = 16 (always two digits) Answer = 7,216

**88.** **1,201** — All from 9, last from 10: 9–8=1 9–7=2 9–9=0, 10–9=1 → 1201.

**89.** **8,556** — Base = 100 Deviations: 93 → –7, 92 → –8 Left part (cross): 93 – 8 = 85 Right part: (–7) × (–8) = 56 Right part padded to 2 digit(s): 56 Answer = 8,556

**90.** **11,03,530** — Base = 1,000 Deviations: 1055 → +55, 1046 → +46 Left part (cross): 1055 + 46 = 1101 Right part: (+55) × (+46) = 2530 Right part has more than 3 digit(s): carry 2 to the left Answer = 11,03,530

**91.** **10,682** — Base = 100 Deviations: 109 → +9, 98 → –2 Left part (cross): 109 – 2 = 107 Right part: (+9) × (–2) = –18 Right part is negative: borrow 1 from left → 107 × 100 – 18 Answer = 10,682

**92.** **2,322** — Column 1 (from right): 3×4 = 12 → write 2, carry 1 Column 2 (from right): 3×5 + 4×4 = 31, + carry 1 = 32 → write 2, carry 3 Column 3 (from right): 4×5 = 20, + carry 3 = 23 → write 3, carry 2 Answer = 2,322

**93.** **25,519** — Column 1 (from right): 9×1 = 9 → write 9, carry 0 Column 2 (from right): 9×5 + 6×1 = 51 → write 1, carry 5 Column 3 (from right): 9×1 + 6×5 + 1×1 = 40, + carry 5 = 45 → write 5, carry 4 Column 4 (from right): 6×1 + 1×5 = 11, + carry 4 = 15 → write 5, carry 1 Column 5 (from right): 1×1 = 1, + carry 1 = 2 → write 2, carry 0 Answer = 25,519

**94.** **Q = 6,675, R = 7** — Running digit sums (quotient columns): 6, 6, 6, 14 Last running sum + last digit = remainder column 16 Adjust carries/remainder ≥ 9 → quotient 6,675, remainder 7

**95.** **Q = 34, R = 148** — Divisor 997 = 1,000 − 3; write complement digits 003 Split dividend: 34 | 046 Column totals after multiplying by the complement: [3, 4] | [0, 13, 18] Resolve carries and remainder → quotient 34, remainder 148

**96.** **Q = 60, R = 104** — Divisor 113 = 100 + 13; transposed flags: −1, −3 Column values after applying flags: [6, 2] | [-12, -2] Resolve negatives/carries → quotient 60, remainder 104

**97.** **x = −2** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (9 − 15)/(12 − 9) = −6/3 = −2.

**98.** **0.(025641)** — Ekadhika (one more than 3) = 4 Start with 1 on the right; repeatedly multiply by 4, write the unit digit, carry the rest Recurring block (6 digits): 025641

# Sutras 5–9: Equations by Inspection

Sutras 5 to 9 are mainly about **solving equations quickly** when they have a special structure. They are clever and worth knowing, but remember: each one only applies to its special case. The ordinary algebra in Part VII always works.

## Sutra 5: Shunyam Saamyasamuccaye — "When the Total is the Same, that Total is Zero"

*Samuccaya* means a "combination" or "total" that appears on both sides of an equation. The sutra says: when the same samuccaya occurs throughout, it must equal **zero**. Tirthaji applies this in several situations; four common ones are below.

### Case 1 — a common factor

If the same expression is a factor of every term, it can be zero. $12x + 3x = 4x + 5x$ gives $15x = 9x$, which forces **x = 0**.

### Case 2 — equal independent terms

$(x + a)(x + b) = (x + c)(x + d)$ with **ab = cd** gives **x = 0**. Example: $(x + 7)(x + 9) = (x + 3)(x + 21)$. Since $7 \times 9 = 63 = 3 \times 21$, **x = 0**. In general ($ab \neq cd$): $x = (cd - ab) \div (a + b - c - d)$.

### Case 3 — equal numerators, equal sums of denominators

$1/(x - 4) + 1/(x - 6) = 1/(x - 2) + 1/(x - 8)$. The denominators on each side add to the same total, $2x - 10$. Set it to zero: $2x - 10 = 0$, **x = 5**. Check: $1/1 + 1/(-1) = 0$ and $1/3 + 1/(-3) = 0$ ✓.

### Case 4 — sum of numerators equals sum of denominators

$(2x + 3)/(2x + 5) = (2x + 5)/(2x + 3)$. Here $N_1 + N_2 = 4x + 8$ and $D_1 + D_2 = 4x + 8$ — the same. Set $4x + 8 = 0$: **x = –2**. Check: $(-1)/1 = 1/(-1)$ ✓.

### Use the exact pattern only

These rules are **pattern-matching**. If the pattern does not match exactly, the rule gives a wrong answer. Always substitute your answer back into the original equation to check, and watch for values that make a denominator zero.

## Sutra 6: (Anurupye) Shunyamanyat — "If One is in Ratio, the Other is Zero"

For simultaneous equations, if the coefficients of one variable are in the **same ratio** as the constants, then the **other** variable is zero.

### $3x + 7y = 2$ and $4x + 14y = 4$

The y-coefficients 7 : 14 = 1 : 2 and the constants 2 : 4 = 1 : 2 — the same ratio. So **x = 0**. Then $7y = 2$, giving **y = 2/7**. Check: $4(0) + 14(2/7) = 4$ ✓.

### When to use / when not to use

Look for this before starting elimination. It saves time in exam questions that are designed with this pattern, but it applies **only** when the ratios match exactly.

## Sutra 7: Sankalana-Vyavakalanabhyam — "By Addition and by Subtraction"

When the coefficients of x and y are **swapped** between two equations, add the equations and subtract them.

### $45x - 23y = 113$ and $23x - 45y = 91$

**Add:** $68x - 68y = 204 \rightarrow x - y = 3$. **Subtract:** $22x + 22y = 22 \rightarrow x + y = 1$. So $x = (3 + 1)/2 = \mathbf{2}$ and $y = (1 - 3)/2 = \mathbf{-1}$. Check: $45(2) - 23(-1) = 90 + 23 = 113$ ✓; $23(2) - 45(-1) = 46 + 45 = 91$ ✓.

### Why it works

For $ax + by = p$ and $bx + ay = q$, adding gives $(a + b)(x + y) = p + q$ and subtracting gives $(a - b)(x - y) = p - q$. Two easy equations replace two awkward ones.

## Sutra 8: Puranapuranabhyam — "By Completion or Non-completion"

This sutra means **completing** an expression into a perfect square or cube. Completing the square is a standard school technique; Tirthaji extends it to cubes.

### $x^3 + 6x^2 + 11x + 6 = 0$

$(x + 2)^3 = x^3 + 6x^2 + 12x + 8$. Our cubic is $(x + 2)^3 - (x + 2)$, because subtracting $x + 2$ takes $12x$ to $11x$ and $8$ to $6$. Let $y = x + 2$: $y^3 - y = 0 \rightarrow y(y - 1)(y + 1) = 0 \rightarrow y = 0, 1, -1 \rightarrow \mathbf{x = -2, -1, -3}$.

### Completing the square: $x^2 + 6x - 7 = 0$

$x^2 + 6x = 7 \rightarrow x^2 + 6x + 9 = 16 \rightarrow (x + 3)^2 = 16 \rightarrow x + 3 = \pm 4 \rightarrow \mathbf{x = 1 \text{ or } x = -7}$.

## Sutra 9: Chalana-Kalanabhyam — "Differences and Similarities"

Tirthaji uses this sutra for ideas from calculus. Its most practical use is a quick way to solve quadratic equations.

### For $ax^2 + bx + c = 0$

The "first differential" $2ax + b$ equals $\mathbf{\pm\sqrt{(b^2 - 4ac)}}$. This is the quadratic formula rearranged: $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div 2a$.

### $x^2 - 5x + 6 = 0$

$2x - 5 = \pm\sqrt{(25 - 24)} = \pm 1 \rightarrow 2x = 6 \text{ or } 4 \rightarrow \mathbf{x = 3 \text{ or } 2}$.

### $2x^2 + 3x - 9 = 0$

$4x + 3 = \pm\sqrt{(9 + 72)} = \pm 9 \rightarrow 4x = 6 \text{ or } -12 \rightarrow \mathbf{x = 3/2 \text{ or } -3}$.

### Interpretations vary

> Some teaching sources describe Chalana-Kalanabhyam differently (for example, as finding factors by comparing differences). The quadratic method above is the application given in Tirthaji's book and is mathematically identical to the quadratic formula.

# Sutras 10–16: Squares, Decimals, Nines and Checking

## Sutra 10: Yavadunam — "Whatever the Extent of its Deficiency"

*Yavadunam* (with the sub-sutra *Yavadunam Tavadunikritya Varganca Yojayet*: "lessen by the deficiency and set up the square of the deficiency") squares numbers close to a base.

### Squaring near a base

$n^2$ = (n + deviation) | (deviation)², where the right part has as many digits as the base has zeros.

### Solved examples

1. **$97^2$** (base 100, deficiency 3): 97 − 3 = 94 | $3^2$ = 09 → **9,409**
2. **$108^2$** (excess 8): 108 + 8 = 116 | 64 → **11,664**
3. **$88^2$** (deficiency 12): 88 − 12 = 76 | 144 → carry 1 → **7,744**
4. **$994^2$** (base 1,000, −6): 988 | 036 → **9,88,036**
5. **$1{,}025^2$** (+25): 1,050 | 625 → **10,50,625**

### Why it works

This is Nikhilam multiplication with both numbers equal: $(B + d)^2 = B(B + 2d) + d^2 = B \times (n + d) + d^2$.

## Sutra 11: Vyashtisamanstih — "Part and Whole"

The idea is to treat a quantity either as a **whole** or as **parts**, whichever is easier. Teaching sources apply this in different ways; two widely taught uses are:

### Multiplication by the average (product of numbers equidistant from a middle value)

47 × 53: the whole (average) is 50; each part differs by 3. So $47 \times 53 = 50^2 - 3^2 = 2{,}500 - 9 =$ **2,491**. $68 \times 72 = 70^2 - 2^2 = 4{,}900 - 4 =$ **4,896**.

### Averages by deviations

Average of 62, 67, 58, 71, 64: take 60 as the "whole" and average only the deviations (2, 7, −2, 11, 4 → total 22 → 22/5 = 4.4). Average = **64.4**.

### Interpretations vary

Tirthaji's own uses of this sutra include factorising expressions by splitting them into parts. The two methods above are the most common in modern teaching. Both are standard algebra $(a + b)(a - b) = a^2 - b^2$ and the "assumed mean" method.

## Sutra 12: Shesanyankena Charamena — "The Remainders by the Last Digit"

This sutra converts fractions such as 1/7, 1/13 and 1/17 into recurring decimals.

### 1/7

Successive remainders when 10, 100, 1,000 … are divided by 7 are **3, 2, 6, 4, 5, 1** (then they repeat). Multiply each remainder by the last digit of 7 (which is 7) and keep only the units digit: 3×7 = 21 → 1; 2×7 = 14 → 4; 6×7 = 42 → 2; 4×7 = 28 → 8; 5×7 = 35 → 5; 1×7 = 7 → 7. **1/7 = 0.142857 142857…**

### 1/13

Remainders: 10, 9, 12, 3, 4, 1. Multiply by 3 and keep units: 0, 7, 6, 9, 2, 3. **1/13 = 0.076923 076923…**

### Multiples of 1/7 are rotations

2/7 = 0.285714…, 3/7 = 0.428571…, 6/7 = 0.857142… — always the same six digits 1, 4, 2, 8, 5, 7, starting at a different place.

## Sutra 13: Sopantyadvayamantyam — "The Ultimate and Twice the Penultimate"

This sutra solves a special family of equations: when the four factors in the denominators form an **arithmetic progression** (like x + 2, x + 3, x + 4, x + 5) and are arranged as 1/(AB) + 1/(AC) = 1/(AD) + 1/(BC), then **ultimate + 2 × penultimate = 0**, i.e. D + 2C = 0.

### 1/((x+2)(x+3)) + 1/((x+2)(x+4)) = 1/((x+2)(x+5)) + 1/((x+3)(x+4))

Factors x + 2, x + 3, x + 4, x + 5 (an arithmetic progression). Ultimate D = x + 5, penultimate C = x + 4. (x + 5) + 2(x + 4) = 0 → 3x + 13 = 0 → **x = −13/3**. (Verified by substitution with exact fractions.)

### Very special case

This only works for exactly this arrangement. It is included for completeness; you will rarely need it outside Vedic Maths competitions.

## Sutra 14: Ekanyunena Purvena — "By One Less than the Previous One"

This is the partner of Sutra 1. Use it to multiply by **9, 99, 999 …** when the other number has the **same number of digits** as the nines (or fewer).

### n × 99…9 (same number of digits)

Left part: n − 1. Right part: the nines minus the left part (i.e. "all from 9 and last from 10" applied to n).

### Solved examples

1. 7 × 9 → 6 | 3 → **63**

2. 43 × 99 → 42 | 57 → **4,257**
3. 768 × 999 → 767 | 232 → **7,67,232**
4. 46 × 999 (fewer digits): think 046 → 045 | 954 → **45,954**
5. 1,234 × 9,999 → 1,233 | 8,766 → **1,23,38,766**

### When to use / when not to use

**Not directly** when the multiplicand has *more* digits than the nines (e.g. 456 × 99). Then use 456 × 100 − 456 = 45,600 − 456 = **45,144**, which is always valid.

## Sutra 15: Gunitasamuccayah — "The Product of the Sum is the Sum of the Product"

In algebra: the sum of the coefficients of a product equals the product of the sums of the coefficients of its factors. This gives a quick **check** for multiplication and factorisation.

### Checking $(x + 2)(x + 3) = x^2 + 5x + 6$

Coefficient sums of the factors: (1 + 2) = 3 and (1 + 3) = 4 → 3 × 4 = 12. Coefficient sum of the product: 1 + 5 + 6 = 12 ✓.

The same idea works for numbers using **digit sums** (Beejank). This is often called *casting out nines*.

### The digit-sum check

Find the digit sum (repeat until one digit; 9 counts as 0) of each number. Do the same operation on the digit sums. The result must match the digit sum of your answer.

### Checking 467 × 38 = 17,746

467 → 4 + 6 + 7 = 17 → 8. 38 → 11 → 2. 8 × 2 = 16 → 7. 17,746 → 1 + 7 + 7 + 4 + 6 = 25 → 7 ✓.

### What the check cannot catch

If two digits are swapped (17,746 written as 17,476) the digit sum is unchanged, so the check passes even though the answer is wrong. The digit-sum check catches most errors, not all. Combine it with an estimate.

## Sutra 16: Gunakasamuccayah — "The Factors of the Sum is Equal to the Sum of the Factors"

Tirthaji uses this sutra in connection with derivatives of factorised expressions: if a quadratic factorises as (x + a)(x + b), its first differential 2x + (a + b) equals the **sum of the factors** (x + a) + (x + b).

### $x^2 + 7x + 12 = (x + 3)(x + 4)$

Differential: 2x + 7. Sum of factors: (x + 3) + (x + 4) = 2x + 7 ✓. This confirms the factorisation.

### Interpretations vary

> This sutra has fewer standard school applications than the others. Treat it as a checking idea.

# The 13 Sub-sutras

| # | Sub-sutra | Common English meaning | Typical application |
|---|---|---|---|
| 1 | Anurupyena | Proportionately | Working bases (50, 200 …) — scale the left part |
| 2 | Sisyate Sesasamjnah | The remainder remains constant | Remainders in division |
| 3 | Adyamadyenantyamantyena | The first by the first and the last by the last | Factorising quadratics: first terms and last terms |
| 4 | Kevalaih Saptakam Gunyat | For 7, the multiplier is 143 | 7 × 143 = 1,001, used for division by 7 and divisibility |
| 5 | Vestanam | By osculation | Divisibility tests for 7, 13, 17, 19 … |
| 6 | Yavadunam Tavadunam | Lessen by the deficiency | Squares and cubes near a base |
| 7 | Yavadunam Tavadunikritya Varganca Yojayet | Lessen by the deficiency and set up the square | Squaring near a base (Sutra 10) |
| 8 | Antyayordasake'pi | Last digits totalling 10 | 43 × 47 type products (Sutra 1B) |
| 9 | Antyayoreva | Only the last terms | Some fraction equations |
| 10 | Samuccayagunitah | The sum of the coefficients in the product | Checking products (with Sutra 15) |
| 11 | Lopanasthapanabhyam | By alternate elimination and retention | Factorising expressions in several variables |
| 12 | Vilokanam | By mere observation | Solving equations by inspection |
| 13 | Gunitasamuccayah Samuccayagunitah | The product of the sum is the sum of the product | Checking factorisations |

### Sub-sutra 4: why 143 helps with 7

7 × 143 = 1,001, and 1,001 = 7 × 11 × 13. So a number like abc,abc (e.g. 365,365) is divisible by 7, 11 and 13: 365,365 = 365 × 1,001.

### Sub-sutra 5: osculation test for 7

For 7, the "osculator" is 2 (subtract twice the last digit from the rest). Is 343 divisible by 7? 34 − 2 × 3 = 28, and 28 = 7 × 4 ✓. Is 1,001 divisible by 7? 100 − 2 = 98 → 9 − 16 = −7 ✓.

### Sub-sutra 3: first by first and last by last

Factorise $6x^2 + 11x + 3$. Split the first term $6x^2 = 2x \times 3x$ and the last $3 = 3 \times 1$, then arrange so that the crosswise products add to the middle term: $(2x + 3)(3x + 1) \rightarrow 2x + 9x = 11x$ ✓.

### A note on the sub-sutras

Lists of sub-sutras in different books vary slightly in spelling and order, and some applications are attributed differently. The meanings above follow the commonly reproduced list from Tirthaji's book.

### Part II quick revision

- **Ekadhikena:** $(a)5^2 = a(a + 1) \mid 25$. Same tens, units add to 10: $a(a + 1) \mid u \times v$.

- **Nikhilam:** all from 9, last from 10. Near a base: cross-add | multiply deviations.
- **Urdhva:** vertical – cross – vertical (2 digits); 1–2–3–2–1 products (3 digits).
- **Paravartya:** above the base → negative flags; $x = (d - b)/(a - c)$.
- **Yavadunam:** $n^2 = (n + d) \mid d^2$.
- **Ekanyunena:** $n \times 99\ldots9 = (n - 1) \mid$ complement.
- **Gunitasamuccayah:** digit-sum and coefficient-sum checks.
- Sutras 5–9, 11–13, 16: special patterns — check by substitution.

## ✎ Practice: Sutras 5–16

**99.** Solve $(x + 10)(x + 3) = (x + 7)(x + 3)$

**100.** Solve: $39x + 32y = 380$, $32x + 39y = 401$

**101.** Solve $x^3 + 15x^2 + 74x + 120 = 0$ by completing the cube

**102.** Solve $x^2 + 3x - 10 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**103.** $110^2$

**104.** $4{,}272 \times 9{,}999$

**105.** Factorise $x^2 + 8x - 9$

**106.** Solve $(x + 1)(x + 1) = (x + 3)(x + 7)$

**107.** Solve: $11x + 27y = 12$, $27x + 11y = 140$

**108.** Solve $x^3 + 9x^2 + 26x + 24 = 0$ by completing the cube

**109.** Solve $x^2 + 5x + 0 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**110.** $111^2$

**111.** $69 \times 99$

**112.** Factorise $x^2 + 3x - 28$

**113.** Find the recurring block of 1/29.

**114.** Solve $(x + 2)(x + 12) = (x + 3)(x + 7)$

*Answers: see the answer key (questions 99–114).*

## Sutras 5–16 answer key

**99.** **x = –3** — $x = (cd - ab)/(a + b - c - d) = (21 - 30)/(13 - 10)$.

**100.** **x = 4, y = 7** — Add: $71(x + y) = 781 \rightarrow x + y = 11$. Subtract: $7(x - y) = -21 \rightarrow x - y = -3$. Solve: $x = 4$, $y = 7$.

**101.** **x = –6, –5, –4** — $(x + 5)^3 = x^3 + 15x^2 + 75x + 125$. The given cubic $= (x + 5)^3 - (x + 5)$. Let $y = x + 5$: $y^3 - y = 0 \rightarrow y = 0, \pm1$.

**102.** **x = 2 or –5** — Differential: $2x + 3$. Discriminant $= 9 - -40 = 49$, $\sqrt{} = 7$. $2x + 3 = \pm7$.

**103.** **12,100** — Base 100, deviation +10 Left: 110 + 10 = 120 Right: $10^2 = 100$ (needs 2 digit(s), carry 1) Answer = 12,100

**104.** **4,27,15,728** — Left: 4272 – 1 = 4271 Right: 9,999 – 4271 = 5728 Answer = 4,27,15,728

**105.** **(x + 9)(x – 1)** — Find two numbers with product –9 and sum 8: 9 and –1. Check with Gunita-samuccaya (put $x = 1$): $(1 + 9)(1 - 1) = 0$ and $1 + 8 - 9 = 0$.

**106.** **x = –2 1/2** — $x = (cd - ab)/(a + b - c - d) = (21 - 1)/(2 - 10)$.

**107.** **x = 6, y = –2** — Add: $38(x + y) = 152 \rightarrow x + y = 4$. Subtract: $-16(x - y) = -128 \rightarrow x - y = 8$. Solve: $x = 6$, $y = -2$.

**108.** **x = –4, –3, –2** — $(x + 3)^3 = x^3 + 9x^2 + 27x + 27$. The given cubic $= (x + 3)^3 - (x + 3)$. Let $y = x + 3$: $y^3 - y = 0 \rightarrow y = 0, \pm1$.

**109.** **x = 0 or –5** — Differential: $2x + 5$. Discriminant $= 25 - 0 = 25$, $\sqrt{} = 5$. $2x + 5 = \pm5$.

**110.** **12,321** — Base 100, deviation +11 Left: 111 + 11 = 122 Right: $11^2 = 121$ (needs 2 digit(s), carry 1) Answer = 12,321

**111.** **6,831** — Left: 69 – 1 = 68 Right: 99 – 68 = 31 Answer = 6,831

**112.** **(x – 4)(x + 7)** — Find two numbers with product –28 and sum 3: –4 and 7. Check with Gunita-samuccaya (put x = 1): (1 – 4)(1 + 7) = –24 and 1 + 3 – 28 = –24.

**113.** **0.(034482758620689655172413…)** — Ekadhika (one more than 2) = 3 Start with 1 on the right; repeatedly multiply by 3, write the unit digit, carry the rest Recurring block (28 digits): 034482758620689655172413…

**114.** **x = –3/4** — x = (cd – ab)/(a + b – c – d) = (21 – 24)/(14 – 10).

# Part III

## Mental Addition and Subtraction

Fast, accurate addition and subtraction are the foundation of every other calculation — including multiplication, percentages and data interpretation. This part teaches eight methods, with **more than 60 worked examples**. The goal is to do two- and three-digit sums in your head, reliably.

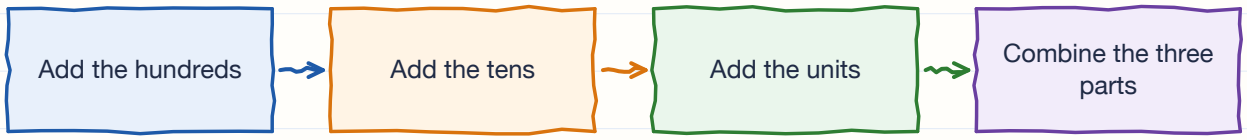
# Chapter 5: Mental Addition

## 5.1 Adding from left to right

### Learning objective

Add two- and three-digit numbers mentally by working from the largest place value first.

At school we add from right to left because of carrying. In the head it is easier to go **left to right**: you hear the most important part of the answer first, and you can stop early if an estimate is enough.




### Why it works

Addition is commutative and associative, so we may add the place values in any order: 456 + 378 = (400 + 300) + (50 + 70) + (6 + 8).

| # | Problem | Left-to-right working | Answer |
|---|---|---|---|
| 1 | 34 + 45 | 30 + 40 = 70; 4 + 5 = 9 | 79 |
| 2 | 58 + 27 | 50 + 20 = 70; 8 + 7 = 15; 70 + 15 | 85 |
| 3 | 76 + 69 | 70 + 60 = 130; 6 + 9 = 15 | 145 |
| 4 | 243 + 516 | 700; 50; 9 | 759 |
| 5 | 456 + 378 | 700; 120; 14 → 700 + 120 + 14 | 834 |
| 6 | 689 + 547 | 1,100; 120; 16 | 1,236 |
| 7 | 1,245 + 3,682 | 4,000; 800; 120; 7 | 4,927 |
| 8 | 8,764 + 5,398 | 13,000; 1,000; 150; 12 | 14,162 |

### Running total

Keep only one number in your head: 456 + 378 → 456 + 300 = 756 → + 70 = 826 → + 8 = 834. This "add the parts of the second number one at a time" style is the most popular mental method.

## 5.2 Breaking numbers into hundreds, tens and units

| # | Problem | Break the second number | Answer |
|---|---|---|---|
| 9 | 67 + 48 | 67 + 40 = 107; + 8 | 115 |
| 10 | 125 + 67 | 125 + 60 = 185; + 7 | 192 |
| 11 | 348 + 235 | 548; 578; 583 | 583 |
| 12 | 2,456 + 1,387 | 3,456; 3,756; 3,836; 3,843 | 3,843 |

## 5.3 Compensation: round, then adjust

### Compensation for addition

a + b = (a + rounded b) − (amount added in rounding)

| # | Problem | Round and adjust | Answer |
|---|---|---|---|
| 13 | 46 + 9 | 46 + 10 − 1 | 55 |
| 14 | 73 + 19 | 73 + 20 − 1 | 92 |
| 15 | 158 + 29 | 158 + 30 − 1 | 187 |
| 16 | 264 + 39 | 264 + 40 − 1 | 303 |
| 17 | 345 + 99 | 345 + 100 − 1 | 444 |
| 18 | 1,276 + 199 | 1,276 + 200 − 1 | 1,475 |
| 19 | 4,538 + 299 | 4,538 + 300 − 1 | 4,837 |
| 20 | 2,817 + 998 | 2,817 + 1,000 − 2 | 3,815 |
| 21 | 657 + 197 | 657 + 200 − 3 | 854 |
| 22 | 3,450 + 1,996 | 3,450 + 2,000 − 4 | 5,446 |

### Adjust in the right direction

You added **too much** when rounding up, so you must **subtract** the extra. A common slip is 345 + 99 = 345 + 100 + 1 = 446 (wrong). The correct answer is 444.

## 5.4 Making tens and hundreds (number bonds)

Look for pairs that make 10, 100 or 1,000.

| # | Problem | Pairing | Answer |
|---|---|---|---|
| 23 | 7 + 8 + 3 + 2 | (7 + 3) + (8 + 2) = 10 + 10 | 20 |
| 24 | 36 + 45 + 64 | (36 + 64) + 45 = 100 + 45 | 145 |
| 25 | 125 + 380 + 875 | (125 + 875) + 380 | 1,380 |
| 26 | 47 + 28 + 53 + 72 | (47 + 53) + (28 + 72) | 200 |
| 27 | 2.6 + 3.9 + 7.4 | (2.6 + 7.4) + 3.9 = 10 + 3.9 | 13.9 |

## 5.5 Fast addition of many numbers

### Adding a column — "drop the tens" method

**Add 47, 86, 59, 34, 78.** Units first: 7 + 6 = 13, + 9 = 22, + 4 = 26, + 8 = 34 → write 4, carry 3. Tens: 4 + 8 + 5 + 3 + 7 = 27, + 3 = 30. **Total = 304.**

### Adding by grouping near a common value

**Add 98, 102, 97, 105, 99, 101.** All are near 100. Deviations: −2, +2, −3, +5, −1, +1 → total +2. Sum = 6 × 100 + 2 = **602**.

| # | Problem | Method | Answer |
|---|---|---|---|
| 28 | 23 + 45 + 67 + 89 | units 24 → 4 c2; tens 2+4+6+8+2 = 22 | 224 |
| 29 | 312 + 288 + 456 | (312 + 288) + 456 = 600 + 456 | 1,056 |
| 30 | 49 + 51 + 48 + 52 + 50 | five numbers near 50, deviations total 0 | 250 |
| 31 | 1,999 + 2,001 + 3,500 | (1,999 + 2,001) + 3,500 | 7,500 |
| 32 | 18 + 27 + 36 + 45 + 54 | pairs 18 + 54 = 72, 27 + 45 = 72, + 36 | 180 |

## 5.6 Column totals and the digit-sum check

When you add a long column of numbers (bills, marks, data tables), check the total with digit sums:

### Checking 4,567 + 2,986 + 1,348 = 8,901

Digit sums: 4,567 → 22 → 4; 2,986 → 25 → 7; 1,348 → 16 → 7. 4 + 7 + 7 = 18 → 9 → 0. 8,901 → 18 → 9 → 0 ✓.

### Mental addition

Left to right · add the parts of the second number one at a time · round and adjust · pair to make 10s and 100s · use deviations for numbers near a common value · check with digit sums.

# Chapter 6: Mental Subtraction

## 6.1 Subtracting from powers of 10 (Nikhilam)

### All from 9 and the last from 10

To subtract a number from 10, 100, 1,000 …: subtract each digit from 9, and the last non-zero digit from 10. Pad with leading zeros to match the number of zeros in the base.

| # | Problem | Working | Answer |
|---|---|---|---|
| 33 | 100 – 64 | 9–6 = 3, 10–4 = 6 | 36 |
| 34 | 100 – 7 | 07 → 9–0, 10–7 | 93 |
| 35 | 1,000 – 273 | 7, 2, 7 | 727 |
| 36 | 1,000 – 89 | 089 → 9, 1, 1 | 911 |
| 37 | 10,000 – 4,826 | 5, 1, 7, 4 | 5,174 |
| 38 | 10,000 – 307 | 0307 → 9, 6, 9, 3 | 9,693 |
| 39 | 1,00,000 – 36,452 | 6, 3, 5, 4, 8 | 63,548 |
| 40 | 1,000 – 640 | apply to 64 (from 100) → 36, attach 0 | 360 |

### Making change

"₹500 note for a ₹347 bill" is 500 – 347 = 400 + (100 – 47) – 0 … even simpler: 500 – 347 = (499 – 347) + 1 = 152 + 1 = ₹**153**. Subtracting from 499 never needs borrowing.

## 6.2 Subtracting near a round number (compensation)

### Compensation for subtraction

a – b = (a – rounded b) + (amount added in rounding)

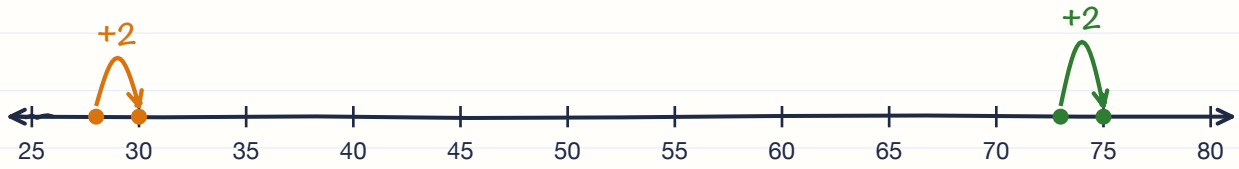
| # | Problem | Round and adjust | Answer |
|---|---|---|---|
| 41 | 83 – 9 | 83 – 10 + 1 | 74 |
| 42 | 156 – 38 | 156 – 40 + 2 | 118 |
| 43 | 432 – 99 | 432 – 100 + 1 | 333 |
| 44 | 725 – 198 | 725 – 200 + 2 | 527 |
| 45 | 1,264 – 497 | 1,264 – 500 + 3 | 767 |
| 46 | 5,000 – 1,998 | 5,000 – 2,000 + 2 | 3,002 |
| 47 | 8,345 – 2,989 | 8,345 – 3,000 + 11 | 5,356 |

## 6.3 Equal adjustment (shift both numbers)

Adding or subtracting the **same amount** to both numbers does not change the difference. Make the second number round.

| # | Problem | Shift both | Answer |
|---|---|---|---|
| 48 | 73 – 28 | (73 + 2) – (28 + 2) = 75 – 30 | 45 |
| 49 | 512 – 196 | 516 – 200 | 316 |
| 50 | 1,003 – 678 | 1,025 – 700 | 325 |
| 51 | 6,020 – 3,985 | 6,035 – 4,000 | 2,035 |
| 52 | 82.4 – 39.7 | 82.7 – 40 | 42.7 |

> **Why equal adjustment works**
> On a number line, the difference is the **distance** between two points. Sliding both points the same distance keeps the gap the same.




## 6.4 Counting up (the shopkeeper's method)

Find the difference by counting **up** from the smaller number to the larger, in friendly jumps.

| # | Problem | Count up | Answer |
|---|---|---|---|
| 53 | 92 – 57 | 57 → 60 (3) → 92 (32) | 35 |
| 54 | 403 – 168 | 168 → 200 (32) → 403 (203) | 235 |
| 55 | 2,005 – 1,387 | 1,387 → 1,400 (13) → 2,000 (600) → 2,005 (5) | 618 |
| 56 | 10,000 – 6,742 | Nikhilam: 3, 2, 5, 8 | 3,258 |

## 6.5 Subtracting close numbers

When two numbers are close, the difference is small — find it by looking at the **last digits** and adjusting.

| # | Problem | Working | Answer |
|---|---|---|---|
| 57 | 503 – 497 | 497 → 500 (3) → 503 (3) | 6 |
| 58 | 1,002 – 989 | 989 → 1,000 (11) → 1,002 (2) | 13 |
| 59 | 7,014 – 6,958 | 6,958 → 7,000 (42) → 7,014 (14) | 56 |

## 6.6 The complement method for subtraction

To subtract b from a, you can **add the complement** of b (from the next power of 10) and then remove that power of 10.

### 742 − 358

Complement of 358 from 1,000 is 642 (all from 9, last from 10). 742 + 642 = 1,384. Remove 1,000 → **384**.

This is how calculators and computers subtract internally, and it turns every subtraction into an addition.

| # | Problem | Complement working | Answer |
|---|---|---|---|
| 60 | 623 − 287 | 623 + 713 = 1,336 → remove 1,000 | 336 |
| 61 | 5,104 − 2,768 | 5,104 + 7,232 = 12,336 → remove 10,000 | 2,336 |

## 6.7 Cross-checking results

### Three checks for subtraction

1. **Add back:** answer + number subtracted = original number.
2. **Digit sums:** digit sum of (a − b) equals digit sum of a minus digit sum of b (add 9 if negative).
3. **Estimate:** 8,345 − 2,989 ≈ 8,300 − 3,000 = 5,300; the answer 5,356 is close ✓.

### Mistakes to watch for

- Subtracting the smaller digit from the larger in every column (writing 2 for "7 minus 9") — this is the most common error in written subtraction.
- Forgetting the padding zero in Nikhilam: 1,000 − 89 is 911, not 11.

# Chapter 7: Mental Arithmetic Games for Children

These games build the same skills in a playful way. Play for 5–10 minutes a day. Part X has a full children's course.

### How to play

Use real objects at first (coins, buttons, blocks). Let the child explain their thinking aloud. Praise strategies ("you made a ten — smart!") more than speed.

1. **Make Ten Snap.** Shuffle cards 1–9. Turn two over; if they make 10 (3 + 7, 4 + 6 …), shout "Ten!" and keep the pair.
2. **Bus Stop.** "There are 12 people on the bus. 5 get off, 3 get on. How many now?" Change numbers each round.
3. **Shop Keeper.** Give ₹50 in toy notes; buy items priced ₹18, ₹23, ₹37. The child counts up to give change.
4. **Number-line Jumps.** Draw a line from 0 to 30 on the floor with chalk. Call out "start at 8, jump forward 7" — the child jumps and calls the landing number.
5. **Doubles Race.** Call a number; the child doubles it. Then "near doubles": 7 + 8 is double 7 plus 1.
6. **Target 100.** Two players take turns adding 1–10 to a running total. Whoever says exactly 100 wins. (Hidden lesson: work backwards from 100 in steps of 11.)

### Part III summary

- Add left to right, one part at a time.
- Round and adjust: add 99 = add 100, take 1.
- Subtract from $10^n$: all from 9, last from 10.
- Shift both numbers to make subtraction easy.
- Count up for differences; use complements to turn subtraction into addition.
- Always check: add back, digit sums, estimate.

## ✎ Part III practice

**115.** Add mentally from left to right: 481 + 634

**116.** 8,678 + 98

**117.** Add: 1,775 + 5,212 + 9,955 + 8,045 + 9,959

**118.** 100 − 65

**119.** 7,279 − 619

**120.** Calculate 7 − (−7) + 2.

**121.** Add mentally from left to right: 966 + 533

**122.** 7,409 + 9

**123.** Add: 5,101 + 3,889 + 8,977 + 6,617

**124.** 100 − 54

**125.** 4,672 − 1,167

**126.** Calculate (−9) − (−15) + (−9).

**127.** Add mentally from left to right: 668 + 377

**128.** 1,540 + 19

**129.** Add: 1,041 + 6,157 + 2,552 + 5,449

**130.** 100 – 16

**131.** 6,176 – 2,567

**132.** Calculate (–2) – 3 + 8.

**133.** Add mentally from left to right: 336 + 893

**134.** 9,559 + 9

**135.** Add: 3,144 + 318 + 6,810 + 6,234 + 8,017 + 3,902

**136.** 100 – 95

**137.** 5,191 – 2,597

**138.** Calculate (–2) – (–20) + (–16).

*Answers: see the answer key (questions 115–138).*

### Five-minute self-test

(1) 587 + 296 (2) 1,000 – 438 (3) 723 – 397 (4) 48 + 37 + 52 + 63 (5) 2,004 – 1,876. *Answers: 883; 562; 326; 200; 128.*

## Part III answer key

**115.** **1,115** — Hundreds 1000, tens 110, units 5: 1000 + 110 + 5 = 1115.

**116.** **8,776** — 98 = 100 – 2. 8,678 + 100 = 8,778, then – 2 = 8,776.

**117.** **34,946** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 8.

**118.** **35** — All from 9, last from 10: 9–6=3, 10–5=5 → 35.

**119.** **6,660** — Subtract 620 instead: 7,279 – 620 = 6,659; add back 1 → 6,660.

**120.** **16** — Subtracting (–7) is the same as adding 7. 7 + 7 + 2 = 16.

**121.** **1,499** — Hundreds 1400, tens 90, units 9: 1400 + 90 + 9 = 1499.

**122.** **7,418** — 9 = 10 – 1. 7,409 + 10 = 7,419, then – 1 = 7,418.

**123.** **24,584** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 5.

**124.** **46** — All from 9, last from 10: 9–5=4, 10–4=6 → 46.

**125.** **3,505** — Subtract 1,170 instead: 4,672 – 1,170 = 3,502; add back 3 → 3,505.

**126.** **–3** — Subtracting (–15) is the same as adding 15. –9 + 15 + –9 = –3.

**127.** **1,045** — Hundreds 900, tens 130, units 15: 900 + 130 + 15 = 1045.

**128.** **1,559** — 19 = 20 – 1. 1,540 + 20 = 1,560, then – 1 = 1,559.

**129.** **15,199** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 7.

**130.** **84** — All from 9, last from 10: 9–1=8, 10–6=4 → 84.

**131.** **3,609** — Subtract 2,570 instead: 6,176 – 2,570 = 3,606; add back 3 → 3,609.

**132.** **3** — Subtracting 3 is the same as adding –3. –2 + –3 + 8 = 3.

**133.** **1,229** — Hundreds 1100, tens 120, units 9: 1100 + 120 + 9 = 1229.

**134.** **9,568** — 9 = 10 – 1. 9,559 + 10 = 9,569, then – 1 = 9,568.

**135.** **28,425** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 3.

**136.** **5** — All from 9, last from 10: 9–9=0, 10–5=5 → 05.

**137.** **2,594** — Subtract 2,600 instead: 5,191 – 2,600 = 2,591; add back 3 → 2,594.

**138.** **2** — Subtracting (–20) is the same as adding 20. –2 + 20 + –16 = 2.

# Part IV

## Multiplication from Basic to Advanced

Multiplication is where Vedic methods shine most. This part moves from tables and patterns, through special multipliers (5, 9, 11, 25, 99, 125 …), to the base methods and the universal vertically-and-crosswise method. Every method shows the **standard** working beside the **shortcut**, the **carry** process, and a box headed **"When this trick does not apply directly"**.

# Chapter 8: Tables, Patterns and Special Multipliers

## 8.1 Multiplication tables and number patterns

| × | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 2 | 22 | 24 | 26 | 28 | 30 | 32 | 34 | 36 | 38 | 40 |
| 3 | 33 | 36 | 39 | 42 | 45 | 48 | 51 | 54 | 57 | 60 |
| 4 | 44 | 48 | 52 | 56 | 60 | 64 | 68 | 72 | 76 | 80 |
| 5 | 55 | 60 | 65 | 70 | 75 | 80 | 85 | 90 | 95 | 100 |
| 6 | 66 | 72 | 78 | 84 | 90 | 96 | 102 | 108 | 114 | 120 |
| 7 | 77 | 84 | 91 | 98 | 105 | 112 | 119 | 126 | 133 | 140 |
| 8 | 88 | 96 | 104 | 112 | 120 | 128 | 136 | 144 | 152 | 160 |
| 9 | 99 | 108 | 117 | 126 | 135 | 144 | 153 | 162 | 171 | 180 |
| 10 | 110 | 120 | 130 | 140 | 150 | 160 | 170 | 180 | 190 | 200 |

### Building the teens tables from what you know

13 × 7 = 10 × 7 + 3 × 7 = 70 + 21 = 91. Any "teen × single digit" is (10 × digit) + (units × digit). With practice the 11–20 tables become as easy as 1–10.

### Patterns worth noticing

- The units digits of the 4-table repeat 4, 8, 2, 6, 0.
- Multiples of 9 have digit sums of 9 (or a multiple of 9).
- $n \times (n + 1)$ is always even; $n^2 - 1 = (n - 1)(n + 1)$.

## 8.2 Multiplying by 10, 100, 1,000 and powers of 10

### Shifting place value

× 10 → every digit moves **one place left** (write one zero, or move the decimal point one place right). × 100 → two places; × 1,000 → three places; × $10^n$ → n places.

### Solved examples

1. 47 × 10 = 470
2. 3.6 × 100 = 360
3. 0.052 × 1,000 = 52
4. 245 × 10,000 = 24,50,000
5. 4.7 × $10^5$ = 4,70,000

### Not "add a zero" for decimals

2.5 × 10 is 25, not 2.50. Think of moving the **digits** (or the point), not of adding zeros.

## 8.3 Multiplying by 5, 25, 50 and 125

### Divide instead of multiply

5 = 10 ÷ 2 · 25 = 100 ÷ 4 · 50 = 100 ÷ 2 · 125 = 1,000 ÷ 8

### Standard method

**68 × 25**

```
  68
× 25
----
 340
1360
----
1700
```

### Shortcut

**68 × 25** 68 × 100 = 6,800 6,800 ÷ 4 = **1,700** (or: 68 ÷ 4 = 17, then × 100)

| # | Problem | Shortcut working | Answer |
|---|---|---|---|
| 1 | 486 × 5 | 4,860 ÷ 2 | 2,430 |
| 2 | 37 × 5 | 370 ÷ 2 | 185 |
| 3 | 84 × 25 | 84 ÷ 4 = 21, × 100 | 2,100 |
| 4 | 37 × 25 | 3,700 ÷ 4 | 925 |
| 5 | 68 × 50 | 6,800 ÷ 2 | 3,400 |
| 6 | 64 × 125 | 64 ÷ 8 = 8, × 1,000 | 8,000 |
| 7 | 36 × 125 | 36,000 ÷ 8 | 4,500 |
| 8 | 4.8 × 25 | 480 ÷ 4 | 120 |
| 9 | 125 × 72 | 72 ÷ 8 = 9, × 1,000 | 9,000 |
| 10 | 25 × 25 | 2,500 ÷ 4 | 625 |

### When this trick does not apply directly

It always gives the right answer, but it is only *faster* when the other number divides easily by 2, 4 or 8. For 37 × 125, 37,000 ÷ 8 = 4,625 needs a careful division — still fine, but not instant.

## 8.4 Multiplying by 9, 99, 999 (Ekanyunena Purvena)

### Same number of digits as the nines

n × 99…9 = (n − 1) | (complement of n). The right part has as many digits as there are nines.

| # | Problem | Left: n − 1 | Right: 99… − left | Answer |
|---|---|---|---|---|
| 11 | 8 × 9 | 7 | 2 | 72 |
| 12 | 57 × 99 | 56 | 43 | 5,643 |
| 13 | 23 × 99 | 22 | 77 | 2,277 |
| 14 | 846 × 999 | 845 | 154 | 8,45,154 |

| # | Problem | Left: n – 1 | Right: 99… – left | Answer |
|---|---|---|---|---|
| 15 | 98 × 99 | 97 | 02 | 9,702 |

### Why it works

n × 99 = n × 100 – n = 100(n – 1) + (100 – n). The left part is n – 1; the right part is 100 – n, which equals 99 – (n – 1).

### When this trick does not apply directly

If the number has **more digits** than the nines (e.g. 345 × 99), the left/right split does not hold. Use **n × 100 – n**: 34,500 – 345 = **34,155**. If it has **fewer** digits (7 × 99), pad with zeros: 07 × 99 → 06 | 93 → **693**.

## 8.5 Multiplying by 11

### The "neighbour sum" rule

Write the first digit, then the sum of each pair of neighbouring digits, then the last digit. Carry when a sum is 10 or more.

### Standard method

**352 × 11**

```
  352
×  11
-----
  352
 3520
-----
 3872
```

### Shortcut

**352 × 11** 3 | 3+5 | 5+2 | 2 3 | 8 | 7 | 2 = **3,872**

**The carry process (789 × 11):** write 7 | 7+8 | 8+9 | 9 = 7 | 15 | 17 | 9. From the right: 9; 17 → write 7, carry 1; 15 + 1 = 16 → write 6, carry 1; 7 + 1 = 8. **789 × 11 = 8,679.**

| # | Problem | Neighbour sums | Answer |
|---|---|---|---|
| 16 | 43 × 11 | 4 \| 7 \| 3 | 473 |
| 17 | 75 × 11 | 7 \| 12 \| 5 → carry | 825 |
| 18 | 99 × 11 | 9 \| 18 \| 9 → carry | 1,089 |
| 19 | 2,314 × 11 | 2 \| 5 \| 4 \| 5 \| 4 | 25,454 |
| 20 | 6,789 × 11 | 6 \| 13 \| 15 \| 17 \| 9 → carries | 74,679 |

### Multiplying by 111 and 22

**× 111:** add digits in groups of three. 32 × 111 → 3 | 3+2 | 3+2 | 2 → 3,552. **× 22:** multiply by 11, then double. 34 × 22 = 374 × 2 = 748.

### When this trick does not apply directly

The rule is for **11** exactly. For 12 or 13, use Paravartya ideas or distribution: 47 × 12 = 470 + 94 = 564.

## 8.6 Multiplying by 101, 1,001 and similar

### Formula

n × 101 = n × 100 + n. For a two-digit n, just write it twice: 47 × 101 = 4,747. n × 1,001 = n × 1,000 + n. For a three-digit n, write it twice: 365 × 1,001 = 3,65,365.

| # | Problem | Working | Answer |
|---|---|---|---|
| 21 | 47 × 101 | 4,700 + 47 | 4,747 |
| 22 | 236 × 101 | 23,600 + 236 | 23,836 |
| 23 | 8 × 101 | 800 + 8 | 808 |
| 24 | 365 × 1,001 | 3,65,000 + 365 | 3,65,365 |

### Writing twice only works when the digits fit

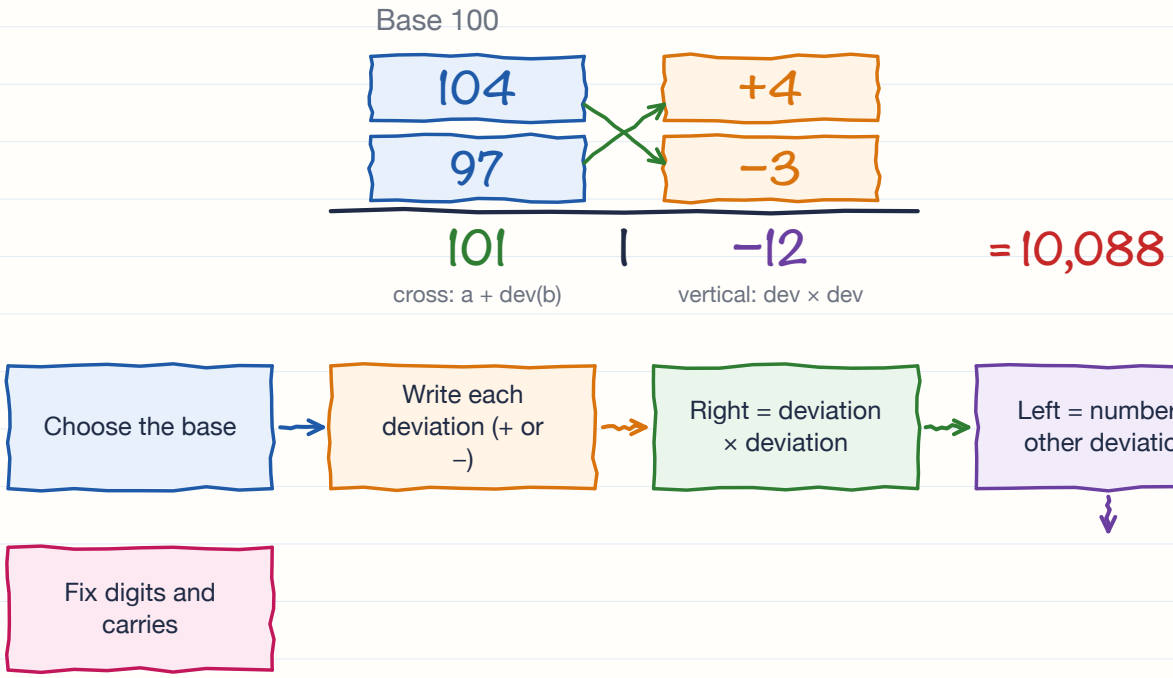
236 × 101 is **not** 236236. A three-digit number times 101 overlaps: 23,600 + 236 = 23,836.

# Chapter 9: Numbers Close to 10, 100 and 1,000 (Nikhilam)

## 9.1 The method

### Learning objective

Multiply two numbers that are both near the same base, in one line.




### Standard method

**98 × 97**

```
  98
× 97
----
 686
882
----
9506
```

### Nikhilam

**98 × 97** (base 100) Deviations −2, −3 Right: (−2)(−3) = 06 Left: 98 − 3 = 95 **9,506**

## 9.2 Both numbers below the base

| # | Problem | Deviations | Left | Right | Answer |
|---|---|---|---|---|---|
| 1 | 9 × 8 | −1, −2 | 7 | 2 | 72 |
| 2 | 96 × 93 | −4, −7 | 89 | 28 | 8,928 |
| 3 | 89 × 91 | −11, −9 | 80 | 99 | 8,099 |
| 4 | 86 × 87 | −14, −13 | 73 | 182 → carry 1 | 7,482 |
| 5 | 993 × 989 | −7, −11 | 982 | 077 | 9,82,077 |

## 9.3 Both numbers above the base

| # | Problem | Deviations | Left | Right | Answer |
|---|---|---|---|---|---|
| 6 | 12 × 13 | +2, +3 | 15 | 6 | 156 |
| 7 | 104 × 108 | +4, +8 | 112 | 32 | 11,232 |
| 8 | 111 × 112 | +11, +12 | 123 | 132 → carry 1 | 12,432 |
| 9 | 1,007 × 1,015 | +7, +15 | 1,022 | 105 | 10,22,105 |

## 9.4 One above, one below

The right part is **negative**. Subtract it from (left × base).

> **104 × 97**
>
> Deviations +4 and −3. Right = −12. Left = 104 − 3 = 101. Answer = 101 × 100 − 12 = 10,100 − 12 = **10,088**.

| # | Problem | Left | Right | Calculation | Answer |
|---|---|---|---|---|---|
| 10 | 108 × 95 | 103 | −40 | 10,300 − 40 | 10,260 |
| 11 | 112 × 93 | 105 | −84 | 10,500 − 84 | 10,416 |
| 12 | 1,004 × 998 | 1,002 | −8 | 10,02,000 − 8 | 10,01,992 |

## 9.5 Numbers near 10 (the finger-friendly version)

For single digits near 10: 7 × 8 → deviations −3, −2 → left 7 − 2 = 5, right 6 → **56**. This is a lovely way for children to check tables above 5 × 5.

## 9.6 Working bases: near 50, 200, 300, 500

> **Working base = multiplier × power of 10**
>
> Find the left part as usual, then **multiply** it by the multiplier (for bases like 200 = 2 × 100) or **divide** it (for 50 = 100 ÷ 2), and attach the right part with the correct number of digits.

| # | Problem | Base | Left | × or ÷ | Right | Answer |
|---|---|---|---|---|---|---|
| 13 | 52 × 54 | 50 = 5 × 10 | 56 | × 5 = 280 | 8 → 280 \| 8 | 2,808 |
| 14 | 46 × 43 | 50 = 5 × 10 | 39 | × 5 = 195 | 28 → carry 2 | 1,978 |

| # | Problem | Base | Left | × or ÷ | Right | Answer |
|---|---|---|---|---|---|---|
| 15 | 212 × 207 | 200 = 2 × 100 | 219 | × 2 = 438 | 84 | 43,884 |
| 16 | 304 × 296 | 300 = 3 × 100 | 300 | × 3 = 900 | −16 | 89,984 |

### Why it works

With base $B = m \times 10^k$: $(B + a)(B + b) = B(B + a + b) + ab = m \times (B + a + b) \times 10^k + ab$. So the left part is multiplied by m and the right part fills k digits.

### Common mistakes with working bases

- Forgetting to multiply (or divide) the left part.
- Using the wrong number of digits on the right: for base 50 = 5 × 10, the right part has **one** digit; for 200 = 2 × 100 it has **two**.

### When this trick does not apply directly

- Numbers on **different bases**: 12 × 97 — no shared base helps. Use vertically and crosswise or distribution (12 × 97 = 1,200 − 36 = 1,164).
- Deviations **too large**: 63 × 78 on base 100 gives right = 37 × 22 = 814, a big carry; and on base 50 left = 91 × 5 = 455, right = 13 × 28 = 364 → still a heavy carry. Use Urdhva.
- A rough guide: Nikhilam is ideal when both deviations are single digits or small two-digit numbers.

## ✎ Chapters 8–9 practice

**139.** 980 × 5

**140.** 306 × 999

**141.** 5,125 × 11

**142.** 18 × 101

**143.** 98 × 98 (Nikhilam, base 100)

**144.** 107 × 108 (base 100)

**145.** 108 × 91

**146.** 494 × 491 (working base 500)

**147.** 825 × 50

**148.** 10 × 99

**149.** 8,021 × 11

**150.** 98 × 101

**151.** 996 × 997 (Nikhilam, base 1,000)

**152.** 115 × 110 (base 100)

**153.** 111 × 98

**154.** 398 × 392 (working base 400)

**155.** 397 × 5

**156.** 9,834 × 9,999

**157.** 4,402 × 11

**158.** 449 × 1,001

**159.** 968 × 970 (Nikhilam, base 1,000)

**160.** 108 × 104 (base 100)

**161.** 103 × 98

**162.** 405 × 393 (working base 400)

*Answers: see the answer key (questions 139–162).*

## Chapters 8–9 answer key

**139.** **4,900** — 5 = 10 / 2 → 980 × 10 ÷ 2 = 4,900.

**140.** **3,05,694** — Left: 306 − 1 = 305 Right: 999 − 305 = 694 Answer = 3,05,694

**141.** **56,375** — Write first digit 5 and last digit 5 In between, add neighbours: 5+1=6, 1+2=3, 2+5=7 Carry where a sum ≥ 10 Answer = 56,375

**142.** **1,818** — 101 = 100 + 1, so write 18 twice (padded): 18 × 101 = 1,818.

**143.** **9,604** — Base = 100 Deviations: 98 → −2, 98 → −2 Left part (cross): 98 − 2 = 96 Right part: (−2) × (−2) = 4 Right part padded to 2 digit(s): 04 Answer = 9,604

**144.** **11,556** — Base = 100 Deviations: 107 → +7, 108 → +8 Left part (cross): 107 + 8 = 115 Right part: (+7) × (+8) = 56 Right part padded to 2 digit(s): 56 Answer = 11,556

**145.** **9,828** — Base = 100 Deviations: 108 → +8, 91 → −9 Left part (cross): 108 − 9 = 99 Right part: (+8) × (−9) = −72 Right part is negative: borrow 1 from left → 99 × 100 − 72 Answer = 9,828

**146.** **2,42,554** — Working base 500 = 5 × 100. Deviations −6, −9. Left: (494 −9) × 5 = 2425. Right: −6 × −9 = 54. 2425 × 100 + 54 = 2,42,554.

**147.** **41,250** — 50 = 100 / 2 → 825 × 100 ÷ 2 = 41,250.

**148.** **990** — Left: 10 − 1 = 9 Right: 99 − 9 = 90 Answer = 990

**149.** **88,231** — Write first digit 8 and last digit 1 In between, add neighbours: 8+0=8, 0+2=2, 2+1=3 Carry where a sum ≥ 10 Answer = 88,231

**150.** **9,898** — 101 = 100 + 1, so write 98 twice (padded): 98 × 101 = 9,898.

**151.** **9,93,012** — Base = 1,000 Deviations: 996 → −4, 997 → −3 Left part (cross): 996 − 3 = 993 Right part: (−4) × (−3) = 12 Right part padded to 3 digit(s): 012 Answer = 9,93,012

**152.** **12,650** — Base = 100 Deviations: 115 → +15, 110 → +10 Left part (cross): 115 + 10 = 125 Right part: (+15) × (+10) = 150 Right part has more than 2 digit(s): carry 1 to the left Answer = 12,650

**153.** **10,878** — Base = 100 Deviations: 111 → +11, 98 → −2 Left part (cross): 111 − 2 = 109 Right part: (+11) × (−2) = −22 Right part is negative: borrow 1 from left → 109 × 100 − 22 Answer = 10,878

**154.** **1,56,016** — Working base 400 = 4 × 100. Deviations −2, −8. Left: (398 −8) × 4 = 1560. Right: −2 × −8 = 16. 1560 × 100 + 16 = 1,56,016.

**155.** **1,985** — 5 = 10 / 2 → 397 × 10 ÷ 2 = 1,985.

**156.** **9,83,30,166** — Left: 9834 − 1 = 9833 Right: 9,999 − 9833 = 0166 Answer = 9,83,30,166

**157.** **48,422** — Write first digit 4 and last digit 2 In between, add neighbours: 4+4=8, 4+0=4, 0+2=2 Carry where a sum ≥ 10 Answer = 48,422

**158.** **4,49,449** — 1001 = 1000 + 1, so write 449 twice (padded): 449 × 1001 = 4,49,449.

**159.** **9,38,960** — Base = 1,000 Deviations: 968 → −32, 970 → −30 Left part (cross): 968 − 30 = 938 Right part: (−32) × (−30) = 960 Right part padded to 3 digit(s): 960 Answer = 9,38,960

**160.** **11,232** — Base = 100 Deviations: 108 → +8, 104 → +4 Left part (cross): 108 + 4 = 112 Right part: (+8) × (+4) = 32 Right part padded to 2 digit(s): 32 Answer = 11,232

**161.** **10,094** — Base = 100 Deviations: 103 → +3, 98 → −2 Left part (cross): 103 − 2 = 101 Right part: (+3) × (−2) = −6 Right part is negative: borrow 1 from left → 101 × 100 − 6 Answer = 10,094

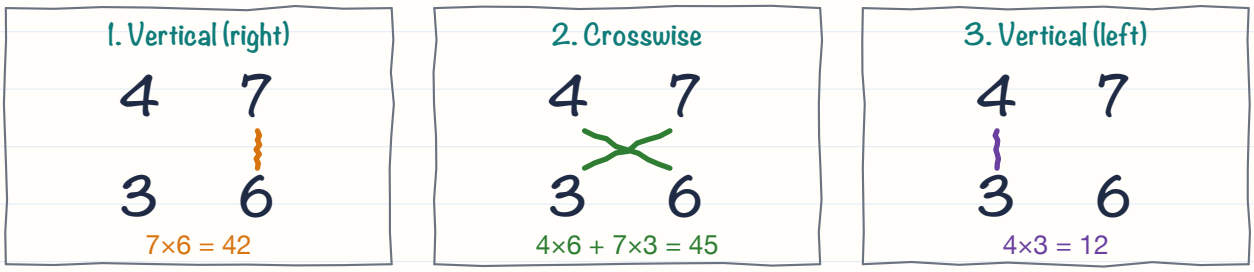
**162.** **1,59,165** — Working base 400 = 4 × 100. Deviations +5, −7. Left: (405 −7) × 4 = 1592. Right: 5 × −7 = −35. 1592 × 100 + −35 = 1,59,165.

# Chapter 10: Vertically and Crosswise for Any Numbers

## 10.1 Two-digit by two-digit

### Learning objective

Multiply any two 2-digit numbers mentally, carrying correctly.




### Standard method

**47 × 36**

```
  47
× 36
----
 282
141
----
1692
```

### Vertically and crosswise

**47 × 36** Units: 7 × 6 = 42 → 2, carry 4 Cross: 4×6 + 7×3 = 45, +4 = 49 → 9, carry 4 Left: 4 × 3 = 12, +4 = 16 **1,692**

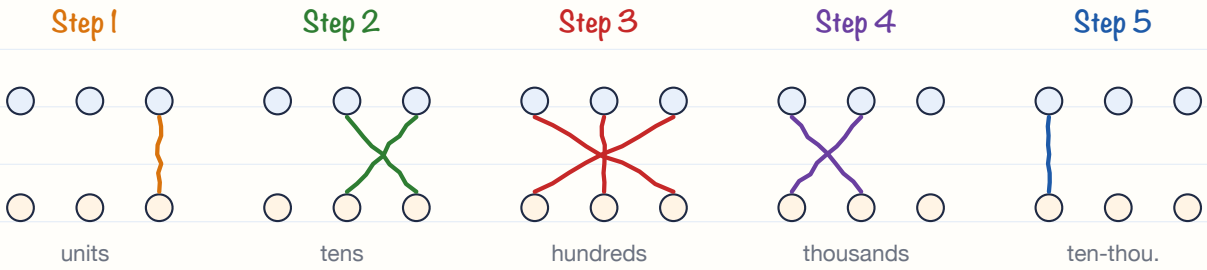
**The carry process.** Work from right to left. Each step produces a number; keep its **units digit** and carry the rest to the next step. The final step is written in full.

| # | Problem | Right (v) | Cross (x) | Left (v) | Answer |
|---|---|---|---|---|---|
| 1 | 21 × 13 | 3 | 1+6 = 7 | 2 | 273 |
| 2 | 32 × 24 | 8 | 12+4 = 16 | 6 | 768 |
| 3 | 54 × 23 | 12 | 15+8 = 23 | 10 | 1,242 |
| 4 | 67 × 89 | 63 | 54+56 = 110 | 48 | 5,963 |
| 5 | 78 × 46 | 48 | 32+42 = 74 | 28 | 3,588 |
| 6 | 95 × 37 | 35 | 63+15 = 78 | 27 | 3,515 |
| 7 | 83 × 83 | 9 | 24+24 = 48 | 64 | 6,889 |
| 8 | 19 × 91 | 9 | 1+81 = 82 | 9 | 1,729 |

### Combine the step values without fuss

A quick way to finish: answer = left × 100 + cross × 10 + right. For 67 × 89: 4,800 + 1,100 + 63 = 5,963.

## 10.2 Three-digit by three-digit




### Five steps for (abc) × (def)

1: c·f 2: b·f + c·e 3: a·f + b·e + c·d 4: a·e + b·d 5: a·d

### 234 × 512

```
Step 1:  4×2                  =  8          → 8, carry 0
Step 2:  3×2 + 4×1            = 10          → 0, carry 1
Step 3:  2×2 + 3×1 + 4×5      = 27, +1 = 28 → 8, carry 2
Step 4:  2×1 + 3×5            = 17, +2 = 19 → 9, carry 1
Step 5:  2×5                  = 10, +1 = 11
Answer: 1,19,808
```

Check by digit sums: 234 → 9 → 0, so the product's digit sum must be 0 (i.e. 9). 1,19,808 → 27 → 9 ✓.

| # | Problem | Step values (5 → 1) | Answer |
|---|---|---|---|
| 9 | 123 × 321 | 3, 8, 14, 8, 3 | 39,483 |
| 10 | 405 × 213 | 8, 4, 22, 5, 15 | 86,265 |
| 11 | 672 × 348 | 18, 45, 82, 64, 16 | 2,33,856 |

### Middle step has three products

The commonest error is forgetting one of a·f, b·e, c·d in step 3. Say them aloud: "outer pair, then middle pair, then… the other outer pair."

### When this trick does not apply directly

It always applies — that is its strength. But it is not always the *fastest*: 998 × 997 is quicker with Nikhilam, and 248 × 125 is quicker as 248 × 1,000 ÷ 8 = 31,000. Look at the numbers before choosing.

## 10.3 Different numbers of digits

Pad the shorter number with leading zeros: 243 × 56 = 243 × 056.

### 243 × 56

Steps: 3×6 = 18 → 8 c1; 4×6 + 3×5 = 39 +1 = 40 → 0 c4; 2×6 + 4×5 + 3×0 = 32 +4 = 36 → 6 c3; 2×5 + 4×0 = 10 +3 = 13 → 3 c1; 2×0 + 1 = 1 → **13,608**.

# Chapter 11: Special Pairs and Squares

## 11.1 Same leading digits, last digits adding to 10

| # | Problem | Left: a(a + 1) | Right: u × v (2 digits) | Answer |
|---|---|---|---|---|
| 1 | 34 × 36 | 3 × 4 = 12 | 24 | 1,224 |
| 2 | 57 × 53 | 5 × 6 = 30 | 21 | 3,021 |
| 3 | 91 × 99 | 9 × 10 = 90 | 09 | 9,009 |
| 4 | 128 × 122 | 12 × 13 = 156 | 16 | 15,616 |

### When this trick does not apply directly

34 × 46 (different tens digits, even though units add to 10) — **does not apply**. Use $40^2 - 6^2$ = 1,600 − 36 = 1,564 (numbers equidistant from 40), or Urdhva. A related pattern **does** work: tens digits adding to 10 and **equal** units, e.g. 37 × 77 → left (3 × 7 + 7) is 28, right (7 × 7) is 49, giving **2,849**.

## 11.2 Squaring numbers ending in 5

(a)$5^2$ = a(a + 1) | 25. See Sutra 1. Extended examples:

| # | Problem | a × (a + 1) | Answer |
|---|---|---|---|
| 5 | $35^2$ | 3 × 4 = 12 | 1,225 |
| 6 | $85^2$ | 8 × 9 = 72 | 7,225 |
| 7 | $105^2$ | 10 × 11 = 110 | 11,025 |
| 8 | $135^2$ | 13 × 14 = 182 | 18,225 |
| 9 | $0.65^2$ | 6 × 7 = 42 → 4,225 → 4 decimal places | 0.4225 |

## 11.3 Squaring numbers near 50

### $(50 + d)^2 = (25 + d) \mid d^2$ (right part: 2 digits)

### Why it works

$(50 + d)^2 = 2{,}500 + 100d + d^2 = 100(25 + d) + d^2$.

| # | Problem | d | 25 + d | $d^2$ | Answer |
|---|---|---|---|---|---|
| 10 | $53^2$ | +3 | 28 | 09 | 2,809 |
| 11 | $58^2$ | +8 | 33 | 64 | 3,364 |
| 12 | $47^2$ | −3 | 22 | 09 | 2,209 |
| 13 | $41^2$ | −9 | 16 | 81 | 1,681 |

## 11.4 Squaring numbers near 100 and 1,000 (Yavadunam)

| # | Problem | n ± d | $d^2$ | Answer |
|---|---|---|---|---|
| 14 | $96^2$ | 92 | 16 | 9,216 |
| 15 | $113^2$ | 126 | 169 → carry 1 | 12,769 |
| 16 | $989^2$ | 978 | 121 → 121 | 9,78,121 |
| 17 | $1{,}016^2$ | 1,032 | 256 | 10,32,256 |

## 11.5 Squaring any number: the duplex method

The **duplex** D of a group of digits is: for one digit, its square; for two digits ab, 2ab; for three digits abc, $2ac + b^2$; for four digits abcd, 2ad + 2bc.

### $347^2$

Duplexes from the left: D(3) = 9, D(34) = 24, D(347) = $2 \cdot 3 \cdot 7 + 4^2 = 58$, D(47) = 56, D(7) = 49.

```
column values:  9 | 24 | 58 | 56 | 49
from the right: 49 → 9 c4;  56+4 = 60 → 0 c6;  58+6 = 64 → 4 c6;
                24+6 = 30 → 0 c3;  9+3 = 12
answer: 1,20,409
```

### When to use the duplex method

It is a general method for squares and is efficient for 3- and 4-digit numbers once practised. For numbers near a base, Yavadunam is faster; for numbers ending in 5, Ekadhikena is fastest.

# Chapter 12: Decomposition, Decimals, Negatives and Checking

## 12.1 Distributive decomposition

| # | Problem | Decomposition | Answer |
|---|---|---|---|
| 1 | 7 × 64 | 7 × 60 + 7 × 4 = 420 + 28 | 448 |
| 2 | 18 × 35 | 18 × 35 = 9 × 70 (halve and double) | 630 |
| 3 | 49 × 16 | 50 × 16 − 16 | 784 |
| 4 | 125 × 36 | 125 × 4 × 9 = 500 × 9 | 4,500 |
| 5 | 302 × 47 | 300 × 47 + 2 × 47 = 14,100 + 94 | 14,194 |
| 6 | 1.5 × 46 | 46 + 23 | 69 |

## 12.2 Multiplying decimals

### Decimal places add

Ignore the points, multiply the whole numbers, then put in as many decimal places as the two numbers have together.

| # | Problem | Whole-number product | Decimal places | Answer |
|---|---|---|---|---|
| 7 | 0.7 × 0.8 | 56 | 2 | 0.56 |
| 8 | 1.2 × 1.3 | 156 | 2 | 1.56 |
| 9 | 9.8 × 9.7 | 9,506 (Nikhilam) | 2 | 95.06 |
| 10 | 0.25 × 0.48 | 1,200 | 4 | 0.12 |
| 11 | 3.5 × 3.5 | 1,225 | 2 | 12.25 |

### Check decimals by estimation

9.8 × 9.7 ≈ 10 × 10 = 100, so 95.06 is sensible; 9.506 or 950.6 would not be.

## 12.3 Multiplication involving negative numbers

### Sign rules

(+)(+) = + · (−)(−) = + · (+)(−) = − · (−)(+) = −. With several factors: an **even** number of negatives gives +, an **odd** number gives −.

| # | Problem | Sign | Size | Answer |
|---|---|---|---|---|
| 12 | (−6) × 7 | − | 42 | −42 |
| 13 | (−12) × (−11) | + | 132 | 132 |
| 14 | (−2) × (−3) × (−5) | − (three negatives) | 30 | −30 |
| 15 | 25 × (−16) | − | 400 | −400 |

The vinculum (bar-digit) idea uses negatives on purpose: 9 = 1 $\bar{1}$ (ten minus one), so 39 × 7 = (40 − 1) × 7 = 280 − 7 = **273**.

## 12.4 Checking multiplication

### Four independent checks

1. **Estimate:** round both numbers. 672 × 348 ≈ 700 × 350 = 2,45,000 (actual 2,33,856 ✓ close).
2. **Last digit:** 672 × 348 must end in 6 because 2 × 8 = 16 ✓.
3. **Digit sum (casting out nines):** 672 → 15 → 6; 348 → 15 → 6; 6 × 6 = 36 → 9. 2,33,856 → 27 → 9 ✓.
4. **Casting out elevens:** alternating digit sums from the right also multiply — this catches swapped digits that the nines-check misses.

### Catching an error with digit sums

Someone claims that 467 × 38 is 17,646. Digit sums: 8 × 2 = 16 → 7. But 17,646 → 24 → 6 ✗. So the answer is wrong. (Correct: 17,746.)

### Part IV — choose your method

| Numbers look like… | Best method |
|---|---|
| × 5, 25, 50, 125 | Multiply by 10/100/1,000 and divide by 2/4/8 |
| × 9, 99, 999 | Ekanyunena: (n − 1) \| complement |
| × 11 | Neighbour sums |
| Both near 10/100/1,000 | Nikhilam |
| Near 50, 200, 300 | Nikhilam with a working base |
| Same tens, units add to 10 | a(a + 1) \| u·v |
| Ends in 5, squared | a(a + 1) \| 25 |
| Anything else | Vertically and crosswise, or decomposition |

### ✎ Chapters 10–12 practice

**163.** 49 × 25 (vertically and crosswise)

**164.** 832 × 286

**165.** 77 × 73

**166.** $53^2$

**167.** $968^2$

**168.** $598^2$ (duplex method)

**169.** 79 × 4

**170.** 9.8 × 1.7

**171.** 20 × (−30)

**172.** 81 × 52 (vertically and crosswise)

**173.** 119 × 370

**174.** 69 × 61

**175.** $25^2$

**176.** $51^2$

**177.** $97^2$

**178.** $296^2$ (duplex method)

**179.** 38 × 9

**180.** 4.1 × 0.86

**181.** (−14) × (−3)

**182.** 52 × 66 (vertically and crosswise)

**183.** 343 × 792

**184.** 93 × 97

**185.** $44^2$

**186.** $967^2$ (duplex method)

**187.** 23 × 3

**188.** 9.3 × 0.98

*Answers: see the answer key (questions 163–188).*

### Part IV mastery checklist

☐ 2-digit × 2-digit in under 15 seconds, mentally ☐ 3-digit × 3-digit on paper in one line ☐ any square from 1 to 100 in under 10 seconds ☐ × 11, × 25, × 99 instantly ☐ check every product with digit sums.

## Chapters 10–12 answer key

**163.** **1,225** — Column 1 (from right): 9×5 = 45 → write 5, carry 4 Column 2 (from right): 9×2 + 4×5 = 38, + carry 4 = 42 → write 2, carry 4 Column 3 (from right): 4×2 = 8, + carry 4 = 12 → write 2, carry 1 Answer = 1,225

**164.** **2,37,952** — Column 1 (from right): 2×6 = 12 → write 2, carry 1 Column 2 (from right): 2×8 + 3×6 = 34, + carry 1 = 35 → write 5, carry 3 Column 3 (from right): 2×2 + 3×8 + 8×6 = 76, + carry 3 = 79 → write 9, carry 7 Column 4 (from right): 3×2 + 8×8 = 70, + carry 7 = 77 → write 7, carry 7 Column 5 (from right): 8×2 = 16, + carry 7 = 23 → write 3, carry 2 Answer = 2,37,952

**165.** **5,621** — Same leading part 7; last digits 7 + 3 = 10 ✓ Left: 7 × 8 = 56 Right: 7 × 3 = 21 (always two digits) Answer = 5,621

**166.** **2,809** — Deviation from 50: +3 Left: 25 + 3 = 28 Right: $3^2$ = 09 Answer = 2,809

**167.** **9,37,024** — Base 1,000, deviation −32 Left: 968 − 32 = 936 Right: $32^2$ = 1024 (needs 3 digit(s), carry 1) Answer = 9,37,024

**168.** **3,57,604** — Duplexes from left: 25, 90, 161, 144, 64 Combine with carries (right to left) → 3,57,604

**169.** **316** — (70 + 9) × 4 = 280 + 36 = 316.

**170.** **16.66** — Multiply without points: 98 × 17 = 1666. Decimal places: 1 + 1 = 2, so the answer is 16.66.

**171.** **−600** — Same signs → positive; different signs → negative. 20 × 30 = 600.

**172.** **4,212** — Column 1 (from right): 1×2 = 2 → write 2, carry 0 Column 2 (from right): 1×5 + 8×2 = 21 → write 1, carry 2 Column 3 (from right): 8×5 = 40, + carry 2 = 42 → write 2, carry 4 Answer = 4,212

**173.** **44,030** — Column 1 (from right): 9×0 = 0 → write 0, carry 0 Column 2 (from right): 9×7 + 1×0 = 63 → write 3, carry 6 Column 3 (from right): 9×3 + 1×7 + 1×0 = 34, + carry 6 = 40 → write 0, carry 4 Column 4 (from right): 1×3 + 1×7 = 10, + carry 4 = 14 → write 4, carry 1 Column 5 (from right): 1×3 = 3, + carry 1 = 4 → write 4, carry 0 Answer = 44,030

**174.** **4,209** — Same leading part 6; last digits 9 + 1 = 10 ✓ Left: 6 × 7 = 42 Right: 9 × 1 = 09 (always two digits) Answer = 4,209

**175.** **625** — Digits before 5: 2 2 × (one more = 3) = 6 Attach 25 → 625

**176.** **2,601** — Deviation from 50: +1 Left: 25 + 1 = 26 Right: $1^2$ = 01 Answer = 2,601

**177.** **9,409** — Base 100, deviation –3 Left: 97 – 3 = 94 Right: $3^2$ = 9 (needs 2 digit(s)) Answer = 9,409

**178.** **87,616** — Duplexes from left: 4, 36, 105, 108, 36 Combine with carries (right to left) → 87,616

**179.** **342** — (30 + 8) × 9 = 270 + 72 = 342.

**180.** **3.526** — Multiply without points: 41 × 86 = 3526. Decimal places: 1 + 2 = 3, so the answer is 3.526.

**181.** **42** — Same signs → positive; different signs → negative. 14 × 3 = 42.

**182.** **3,432** — Column 1 (from right): 2×6 = 12 → write 2, carry 1 Column 2 (from right): 2×6 + 5×6 = 42, + carry 1 = 43 → write 3, carry 4 Column 3 (from right): 5×6 = 30, + carry 4 = 34 → write 4, carry 3 Answer = 3,432

**183.** **2,71,656** — Column 1 (from right): 3×2 = 6 → write 6, carry 0 Column 2 (from right): 3×9 + 4×2 = 35 → write 5, carry 3 Column 3 (from right): 3×7 + 4×9 + 3×2 = 63, + carry 3 = 66 → write 6, carry 6 Column 4 (from right): 4×7 + 3×9 = 55, + carry 6 = 61 → write 1, carry 6 Column 5 (from right): 3×7 = 21, + carry 6 = 27 → write 7, carry 2 Answer = 2,71,656

**184.** **9,021** — Same leading part 9; last digits 3 + 7 = 10 ✓ Left: 9 × 10 = 90 Right: 3 × 7 = 21 (always two digits) Answer = 9,021

**185.** **1,936** — Deviation from 50: –6 Left: 25 – 6 = 19 Right: $6^2$ = 36 Answer = 1,936

**186.** **9,35,089** — Duplexes from left: 81, 108, 162, 84, 49 Combine with carries (right to left) → 9,35,089

**187.** **69** — (20 + 3) × 3 = 60 + 9 = 69.

**188.** **9.114** — Multiply without points: 93 × 98 = 9114. Decimal places: 1 + 2 = 3, so the answer is 9.114.

# Part V

## Division and Remainders

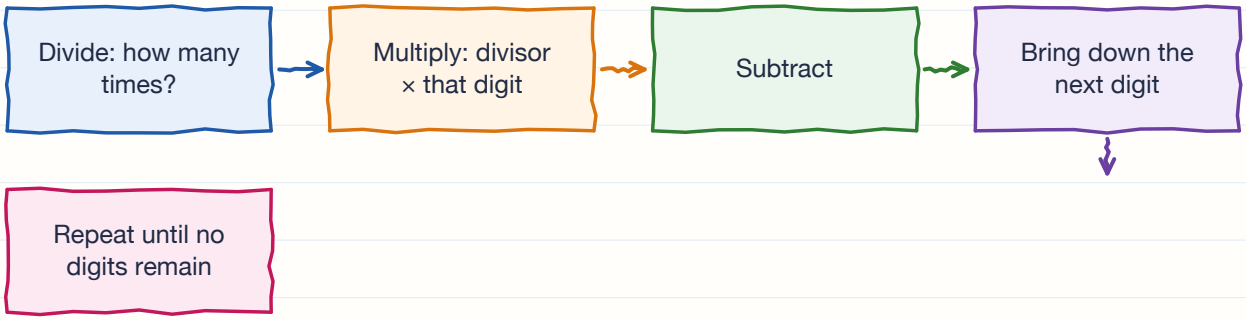
Division is the operation learners fear most, but it is simply multiplication in reverse. This part starts with long division (which always works), then shows the special divisors where shortcuts genuinely help — and clearly says where they do not.

# Chapter 13: Division Fundamentals

## 13.1 Long division, step by step

### Learning objective

Divide any whole number by a one- or two-digit divisor, and state the quotient and remainder correctly.




### 7,385 ÷ 23

```
        3 2 1
     -------
23 ) 7 3 8 5
     6 9          23 × 3 = 69
     ---
       4 8        73 − 69 = 4, bring down 8
       4 6        23 × 2 = 46
       ---
         2 5      48 − 46 = 2, bring down 5
         2 3      23 × 1 = 23
         ---
           2      remainder
```

**Quotient 321, remainder 2.** Check: 23 × 321 + 2 = 7,383 + 2 = 7,385 ✓.

### Division check

Dividend = Divisor × Quotient + Remainder, with 0 ≤ Remainder < Divisor.

### Common long-division mistakes

- **Missing zero in the quotient.** 6,024 ÷ 6 = 1,004, not 14. When the divisor does not go into the current number, write 0 in the quotient before bringing down the next digit.
- **Remainder bigger than the divisor.** If it happens, the quotient digit was too small.

## 13.2 Quotient, remainder and decimals

When a division does not come out exactly, you can either state a remainder or continue into decimals by adding zeros after the decimal point.

### 47 ÷ 8

As quotient and remainder: 47 = 8 × 5 + 7 → **5 remainder 7**. As a decimal: 47 ÷ 8 = 5.875 (because 7/8 = 0.875). As a mixed number: **5 7/8**.

## 13.3 Decimal division

### Make the divisor a whole number

Move the decimal point in the divisor to the end, and move the dividend's point the **same number of places**. 2.56 ÷ 0.16 → 256 ÷ 16 = 16. 4.5 ÷ 0.009 → 4,500 ÷ 9 = 500.

| # | Problem | Rewrite | Answer |
|---|---|---|---|
| 1 | 7.2 ÷ 0.8 | 72 ÷ 8 | 9 |
| 2 | 0.144 ÷ 1.2 | 1.44 ÷ 12 | 0.12 |
| 3 | 15 ÷ 0.25 | 1,500 ÷ 25 | 60 |
| 4 | 3.6 ÷ 0.04 | 360 ÷ 4 | 90 |

# Chapter 14: Shortcuts for Special Divisors

## 14.1 Dividing by 5, 25, 50 and 125

### Multiply instead of divide

÷ 5 = × 2 ÷ 10 · ÷ 25 = × 4 ÷ 100 · ÷ 50 = × 2 ÷ 100 · ÷ 125 = × 8 ÷ 1,000

### Standard method

**1,350 ÷ 25** Long division: 25 into 135 goes 5 times (125), remainder 10; bring down 0 → 100; 25 into 100 goes 4 → **54**.

### Shortcut

**1,350 ÷ 25** 1,350 × 4 = 5,400 5,400 ÷ 100 = **54**

| # | Problem | Shortcut | Answer |
|---|---|---|---|
| 5 | 347 ÷ 5 | 694 ÷ 10 | 69.4 |
| 6 | 2,375 ÷ 25 | 9,500 ÷ 100 | 95 |
| 7 | 735 ÷ 50 | 1,470 ÷ 100 | 14.7 |
| 8 | 3,000 ÷ 125 | 24,000 ÷ 1,000 | 24 |
| 9 | 1,234 ÷ 125 | 9,872 ÷ 1,000 | 9.872 |

## 14.2 Dividing by 9 (Nikhilam)

### Running totals

For a dividend with digits $d_1\ d_2 \ldots d_n$: the quotient digits are the running totals $d_1$, $d_1+d_2$, …; the remainder is the total of **all** digits. Then resolve carries and reduce the remainder below 9.

### 2,341 ÷ 9

```
  9 | 2   3   4 | 1
    |     2   5 | 9        running totals: 2, 2+3 = 5, 5+4 = 9; rema
inder 9+1 = 10
    |-----------|---
      2   5   9 | 10
```

Quotient 259, remainder 10. Since 10 ≥ 9: quotient **260**, remainder **1**. Check: 9 × 260 + 1 = 2,341 ✓.

### The remainder alone

The remainder when dividing by 9 is just the digit sum (reduced). 2,341 → 2 + 3 + 4 + 1 = 10 → 1. This is the fastest remainder trick in all of mathematics.

## 14.3 Dividing by 11

Use Paravartya: 11 = 10 + 1, so the flag is **−1**. Each new column = next digit − previous column value.

### 7,546 ÷ 11

Columns: 7; 5 − 7 = −2; 4 − (−2) = 6; remainder column 6 − 6 = 0. Quotient columns 7, −2, 6 → 700 − 20 + 6 = **686**, remainder **0**. Check: 11 × 686 = 7,546 ✓.

### Remainder on division by 11

Alternate sum of digits from the right: for 7,546 → 6 − 4 + 5 − 7 = 0 → remainder 0. If the result is negative, add 11.

## 14.4 Dividing by numbers just below a power of 10 (8, 89, 98, 997 ...)

Use Nikhilam division with the complement (see Sutra 2D for full detail).

### 10,512 ÷ 97

Complement 03. Split 105 | 12.

```
97 | 1   0   5 |  1   2
03 |     0   3 |              ← column 1 value (1) × 03
   |         0 |  0           ← column 2 value (0) × 03
   |           |  0  24       ← column 3 value (8) × 03
   |-----------|--------
     1   0   8 |  1  26
```

Track the **column totals** from left to right: Column 1 = 1. Column 2 = 0 + 1×0 = 0. Column 3 = 5 + 1×3 + 0×0 = 8. Column 4 = 1 + 0×3 + 8×0 = 1. Column 5 = 2 + 8×3 = 26. Quotient part 1, 0, 8 → **108**; remainder part 1, 26 → 10 + 26 = **36**. Check: 97 × 108 + 36 = 10,476 + 36 = 10,512 ✓.

### Keep track of columns

Each quotient column's value is multiplied by **every** complement digit, which is added to the columns that follow. Writing the column totals in a row (as above) is far less error-prone than writing many small products.

## 14.5 Dividing by 99 and 101

### ÷ 99 for a four-digit number abcd

Quotient = ab; remainder = ab + cd. If the remainder ≥ 99, add 1 to the quotient and subtract 99 from the remainder. 4,567 ÷ 99 → quotient 45, remainder 45 + 67 = 112 → quotient **46**, remainder **13**.

### ÷ 101 for a four-digit number abcd

Quotient = ab; remainder = cd − ab. If negative, subtract 1 from the quotient and add 101. 5,353 ÷ 101 → quotient 53, remainder 53 − 53 = 0 → **53**. 6,120 ÷ 101 → quotient 61, remainder 20 − 61 = −41 → quotient **60**, remainder **60**. Check: 101 × 60 + 60 = 6,120 ✓.

## 14.6 Dividing by numbers just above a power of 10 (12, 13, 112 ...)

See Paravartya (Sutra 4). Example: 1,452 ÷ 12 = 121; 13,789 ÷ 112 = 123 remainder 13.

### When the shortcuts do not apply directly

- Divisors like **17, 23, 47, 73** are not close to a base. Use ordinary long division — it is clearer and just as fast.
- Paravartya with large flags (e.g. ÷ 18 = 10 + 8, flag −8) creates big negative columns. Use long division or Nikhilam with base 20 only if you are experienced.
- **Do not force a shortcut** when ordinary division is clearer.

# Chapter 15: Divisibility, Checking and Exam Tricks

## 15.1 Divisibility rules (complete table)

| Divisor | Rule | Example |
|---|---|---|
| 2 | Last digit even (0, 2, 4, 6, 8) | 3,574 ✓ |
| 3 | Digit sum divisible by 3 | 4,512 → 12 ✓ |
| 4 | Last two digits divisible by 4 | 7,316 → 16 ✓ |
| 5 | Last digit 0 or 5 | 8,435 ✓ |
| 6 | Divisible by 2 and by 3 | 2,154 ✓ |
| 7 | Double the last digit, subtract from the rest; repeat | 203 → 20 − 6 = 14 ✓ |
| 8 | Last three digits divisible by 8 | 51,128 → 128 ✓ |
| 9 | Digit sum divisible by 9 | 6,723 → 18 ✓ |
| 10 | Last digit 0 | 4,560 ✓ |
| 11 | Alternating digit sum divisible by 11 | 9,537 → (9+3) − (5+7) = 0 ✓ |
| 12 | Divisible by 3 and by 4 | 1,524 ✓ |
| 13 | Add 4 × last digit to the rest; repeat | 247 → 24 + 28 = 52 ✓ |
| 25 | Last two digits 00, 25, 50, 75 | 3,475 ✓ |

### Osculation test for 7 (sub-sutra Vestanam)

Is 2,149 divisible by 7? 214 − 2 × 9 = 196 → 19 − 2 × 6 = 7 ✓. So 2,149 = 7 × 307.

### Test for 13 (add 4 × last digit)

Is 1,183 divisible by 13? 118 + 4 × 3 = 130 → 13 + 0 = 13 ✓. So 1,183 = 13 × 91.

## 15.2 Checking division with multiplication

Every division gives you a free check: **divisor × quotient + remainder = dividend.** In an exam, check by estimating: 7,385 ÷ 23 ≈ 7,000 ÷ 20 = 350 — so 321 is reasonable, but 32 or 3,210 are not.

## 15.3 Exam-oriented division tricks

| Divide by | Instead | Example |
|---|---|---|
| 0.5 | multiply by 2 | 37 ÷ 0.5 = 74 |
| 0.25 | multiply by 4 | 18 ÷ 0.25 = 72 |
| 1.25 | multiply by 0.8 | 60 ÷ 1.25 = 48 |
| 12.5 | multiply by 0.08 | 450 ÷ 12.5 = 36 |
| 1.5 | multiply by 2/3 | 27 ÷ 1.5 = 18 |
| 2.5 | multiply by 0.4 | 35 ÷ 2.5 = 14 |
| 33 1/3 | multiply by 3, ÷ 100 | 250 ÷ 33 1/3 = 7.5 |

### Cancelling before dividing

In fraction-style questions like (84 × 45) ÷ (35 × 36), cancel common factors first: 84/35 = 12/5 and 45/36 = 5/4, so the answer is 12/5 × 5/4 = 3. No large multiplication needed.

### Estimating quotients for options

(4,987 ÷ 24.9) ≈ 5,000 ÷ 25 = 200. If the options are 150, 200, 250, 300, choose 200 without dividing exactly. Use this **only** when the options are well separated.

## 15.4 Common errors and how to correct them

| Error | Example | Fix |
|---|---|---|
| Missing zero in quotient | 3,612 ÷ 12 written as 31 | Each "does not go" step gives a 0: answer 301 |
| Moving decimal points unequally | 4.8 ÷ 0.12 treated as 48 ÷ 12 | Move both points 2 places: 480 ÷ 12 = 40 |
| Remainder too large | 50 ÷ 7 = 6 remainder 8 | Remainder must be < 7: 7 remainder 1 |
| Confusing a ÷ b with b ÷ a | "share 5 among 20" | Read carefully: 5 ÷ 20 = 0.25 each |

### Part V summary

- Dividend = divisor × quotient + remainder.
- ÷ 5, 25, 50, 125 → × 2, 4, 2, 8 and shift the point.
- ÷ 9: running digit totals; remainder = digit sum.
- ÷ 11: flag −1; remainder from the alternating digit sum.
- Near-base divisors: Nikhilam (below) or Paravartya (above).
- Otherwise: plain long division — and always check by multiplying back.

### ✎ Part V practice

**189.** 1,215 ÷ 14

**190.** 7,826 ÷ 5 (as a decimal)

**191.** 21,779 ÷ 9 (quotient and remainder)

**192.** 36,962 ÷ 996

**193.** 5,150 ÷ 113

**194.** Is 61,890 divisible by 6? Use the divisibility rule.

**195.** Find the remainder when 31,544 is divided by 13.

**196.** Find the recurring block of 1/49.

**197.** 7,907 ÷ 3

**198.** 8,907 ÷ 5 (as a decimal)

**199.** 39,691 ÷ 9 (quotient and remainder)

**200.** 3,897 ÷ 88

**201.** 5,955 ÷ 113

**202.** Is 69,228 divisible by 4? Use the divisibility rule.

**203.** Find the remainder when 30,515 is divided by 11.

**204.** 6,657 ÷ 22

**205.** 4,339 ÷ 50 (as a decimal)

**206.** 33,462 ÷ 9 (quotient and remainder)

**207.** 426 ÷ 8

**208.** 373 ÷ 13

**209.** Is 76,825 divisible by 4? Use the divisibility rule.

**210.** Find the remainder when 63,708 is divided by 11.

**211.** 3,324 ÷ 14

**212.** 1,247 ÷ 125 (as a decimal)

*Answers: see the answer key (questions 189–212).*

## Part V answer key

**189.** **Q = 86, R = 11** — Check: 14 × 86 + 11 = 1,215.

**190.** **1,565.2** — ÷ 5 = × 2 ÷ 10: 7,826 × 2 = 15,652, ÷ 10 = 1,565.2.

**191.** **Q = 2,419, R = 8** — Running digit sums (quotient columns): 2, 3, 10, 17 Last running sum + last digit = remainder column 26 Adjust carries/remainder ≥ 9 → quotient 2,419, remainder 8

**192.** **Q = 37, R = 110** — Divisor 996 = 1,000 − 4; write complement digits 004 Split dividend: 36 | 962 Column totals after multiplying by the complement: [3, 6] | [9, 18, 26] Resolve carries and remainder → quotient 37, remainder 110

**193.** **Q = 45, R = 65** — Divisor 113 = 100 + 13; transposed flags: −1, −3 Column values after applying flags: [5, -4] | [-6, 12] Resolve negatives/carries → quotient 45, remainder 65

**194.** **Yes** — Rule for 6: divisible by both 2 and 3. 61,890 ÷ 6 leaves remainder 0.

**195.** **6** — 31,544 = 13 × 2,426 + 6.

**196.** **0.(020408163265306122448979…)** — Ekadhika (one more than 4) = 5 Start with 1 on the right; repeatedly multiply by 5, write the unit digit, carry the rest Recurring block (42 digits): 020408163265306122448979…

**197.** **Q = 2,635, R = 2** — Check: 3 × 2,635 + 2 = 7,907.

**198.** **1,781.4** — ÷ 5 = × 2 ÷ 10: 8,907 × 2 = 17,814, ÷ 10 = 1,781.4.

**199.** **Q = 4,410, R = 1** — Running digit sums (quotient columns): 3, 12, 18, 27 Last running sum + last digit = remainder column 28 Adjust carries/remainder ≥ 9 → quotient 4,410, remainder 1

**200.** **Q = 44, R = 25** — Divisor 88 = 100 − 12; write complement digits 12 Split dividend: 38 | 97 Column totals after multiplying by the complement: [3, 11] | [26, 29] Resolve carries and remainder → quotient 44, remainder 25

**201.** **Q = 52, R = 79** — Divisor 113 = 100 + 13; transposed flags: −1, −3 Column values after applying flags: [5, 4] | [-14, -7] Resolve negatives/carries → quotient 52, remainder 79

**202.** **Yes** — Rule for 4: last two digits divisible by 4. 69,228 ÷ 4 leaves remainder 0.

**203.** **1** — 30,515 = 11 × 2,774 + 1.

**204.** **Q = 302, R = 13** — Check: 22 × 302 + 13 = 6,657.

**205.** **86.78** — ÷ 50 = × 2 ÷ 100: 4,339 × 2 = 8,678, ÷ 100 = 86.78.

**206.** **Q = 3,718, R = 0** — Running digit sums (quotient columns): 3, 6, 10, 16 Last running sum + last digit = remainder column 18 Adjust carries/remainder ≥ 9 → quotient 3,718, remainder 0

**207.** **Q = 53, R = 2** — Divisor 8 = 10 − 2; write complement digits 2 Split dividend: 42 | 6 Column totals after multiplying by the complement: [4, 10] | [26] Resolve carries and remainder → quotient 53, remainder 2

**208.** **Q = 28, R = 9** — Divisor 13 = 10 + 3; transposed flags: −3 Column values after applying flags: [3, -2] | [9] Resolve negatives/carries → quotient 28, remainder 9

**209.** **No** — Rule for 4: last two digits divisible by 4. 76,825 ÷ 4 leaves remainder 1.

**210.** **7** — 63,708 = 11 × 5,791 + 7.

**211.** **Q = 237, R = 6** — Check: 14 × 237 + 6 = 3,324.

**212.** **9.976** — ÷ 125 = × 8 ÷ 1000: 1,247 × 8 = 9,976, ÷ 1000 = 9.976.

# Part VI

## Squares, Cubes and Roots

Squares and roots appear in geometry, simplification, compound interest and data interpretation. Knowing the squares up to 50 by heart — and being able to *produce* any square up to 100 in a few seconds — is one of the most valuable skills for competitive exams.

# Chapter 16: Squares

## 16.1 Memorisation chart: squares 1 to 50

| n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 11 | 121 | 21 | 441 | 31 | 961 | 41 | 1,681 |
| 2 | 4 | 12 | 144 | 22 | 484 | 32 | 1,024 | 42 | 1,764 |
| 3 | 9 | 13 | 169 | 23 | 529 | 33 | 1,089 | 43 | 1,849 |
| 4 | 16 | 14 | 196 | 24 | 576 | 34 | 1,156 | 44 | 1,936 |
| 5 | 25 | 15 | 225 | 25 | 625 | 35 | 1,225 | 45 | 2,025 |
| 6 | 36 | 16 | 256 | 26 | 676 | 36 | 1,296 | 46 | 2,116 |
| 7 | 49 | 17 | 289 | 27 | 729 | 37 | 1,369 | 47 | 2,209 |
| 8 | 64 | 18 | 324 | 28 | 784 | 38 | 1,444 | 48 | 2,304 |
| 9 | 81 | 19 | 361 | 29 | 841 | 39 | 1,521 | 49 | 2,401 |
| 10 | 100 | 20 | 400 | 30 | 900 | 40 | 1,600 | 50 | 2,500 |

## 16.2 Important squares from 51 to 100

| n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ |
|---|---|---|---|---|---|---|---|---|---|
| 51 | 2,601 | 61 | 3,721 | 71 | 5,041 | 81 | 6,561 | 91 | 8,281 |
| 52 | 2,704 | 62 | 3,844 | 72 | 5,184 | 82 | 6,724 | 92 | 8,464 |
| 53 | 2,809 | 63 | 3,969 | 73 | 5,329 | 83 | 6,889 | 93 | 8,649 |
| 54 | 2,916 | 64 | 4,096 | 74 | 5,476 | 84 | 7,056 | 94 | 8,836 |
| 55 | 3,025 | 65 | 4,225 | 75 | 5,625 | 85 | 7,225 | 95 | 9,025 |
| 56 | 3,136 | 66 | 4,356 | 76 | 5,776 | 86 | 7,396 | 96 | 9,216 |
| 57 | 3,249 | 67 | 4,489 | 77 | 5,929 | 87 | 7,569 | 97 | 9,409 |
| 58 | 3,364 | 68 | 4,624 | 78 | 6,084 | 88 | 7,744 | 98 | 9,604 |
| 59 | 3,481 | 69 | 4,761 | 79 | 6,241 | 89 | 7,921 | 99 | 9,801 |
| 60 | 3,600 | 70 | 4,900 | 80 | 6,400 | 90 | 8,100 | 100 | 10,000 |

### You do not need to memorise 51–100

Every square from 51 to 100 can be produced in seconds:

- **51–59:** near 50 → (25 + d) | $d^2$. For $57^2$, 25 + 7 is 32 and $7^2$ is 49, giving 3,249.
- **61–89:** split by the nearest ten, or use $(a \pm b)^2$. $68^2 = 70^2 - 2 \times 70 \times 2 + 4 = 4{,}900 - 280 + 4 = 4{,}624$.
- **90–99:** near 100 → (n – d) | $d^2$. For $93^2$, 93 – 7 is 86 and $7^2$ is 49, giving 8,649.
- **Ending in 5:** a(a + 1) | 25.

## 16.3 Square patterns

### Five patterns worth knowing

1. **Consecutive squares differ by the sum of the numbers:** $41^2 = 40^2 + 40 + 41 = 1{,}681$.
2. **Difference of squares:** $a^2 - b^2 = (a + b)(a - b)$. $26^2 - 24^2 = 50 \times 2 = 100$.
3. **Squares of numbers ending in 1:** $71^2 = 70^2 + 2 \times 70 + 1 = 4{,}900 + 141 = 5{,}041$.
4. **Squares of numbers ending in 9:** $79^2 = 80^2 - 2 \times 80 + 1 = 6{,}400 - 159 = 6{,}241$.
5. **Squares of 25 + d and 25 – d end in the same two digits:** $23^2 = 529$ and $27^2 = 729$.

## 16.4 Squaring numbers ending in 5

(a)$5^2$ = a(a + 1) | 25. Practised in Parts II and IV.

## 16.5 Squaring numbers near 100 and 1,000

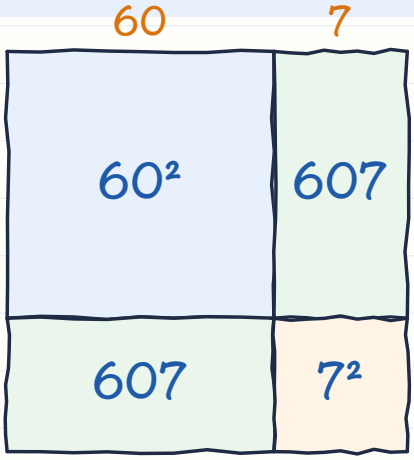
(Yavadunam) $n^2 = (n + d) \mid d^2$ with the right part having 2 digits (base 100) or 3 digits (base 1,000).

| # | Problem | Left n + d | Right $d^2$ | Answer |
|---|---|---|---|---|
| 1 | $92^2$ | 84 | 64 | 8,464 |
| 2 | $107^2$ | 114 | 49 | 11,449 |
| 3 | $985^2$ | 970 | 225 | 9,70,225 |
| 4 | $1{,}032^2$ | 1,064 | 1,024 → carry 1 | 10,65,024 |

## 16.6 Squaring two-digit and three-digit numbers

### The $(a + b)^2$ split

$(a + b)^2 = a^2 + 2ab + b^2$. Choose a as a round number.




$(60 + 7)^2 = 60^2 + 2607 + 7^2$

### $67^2$ by the split

$(60 + 7)^2 = 3{,}600 + 2 \times 60 \times 7 + 49 = 3{,}600 + 840 + 49 =$ **4,489**.

### 3-digit: $304^2$ and $347^2$

$304^2 = (300 + 4)^2 = 90{,}000 + 2{,}400 + 16 =$ **92,416**. $347^2$ by the duplex method (Part IV): column values 9 | 24 | 58 | 56 | 49 → **1,20,409**.

# Chapter 17: Cubes

## 17.1 Cubes 1 to 30

| n | $n^3$ | n | $n^3$ | n | $n^3$ |
|---|---|---|---|---|---|
| 1 | 1 | 11 | 1,331 | 21 | 9,261 |
| 2 | 8 | 12 | 1,728 | 22 | 10,648 |
| 3 | 27 | 13 | 2,197 | 23 | 12,167 |
| 4 | 64 | 14 | 2,744 | 24 | 13,824 |
| 5 | 125 | 15 | 3,375 | 25 | 15,625 |
| 6 | 216 | 16 | 4,096 | 26 | 17,576 |
| 7 | 343 | 17 | 4,913 | 27 | 19,683 |
| 8 | 512 | 18 | 5,832 | 28 | 21,952 |
| 9 | 729 | 19 | 6,859 | 29 | 24,389 |
| 10 | 1,000 | 20 | 8,000 | 30 | 27,000 |

## 17.2 Useful cube patterns

| Number ends in | Square ends in | Cube ends in |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 1 | 1 |
| 2 | 4 | 8 |
| 3 | 9 | 7 |
| 4 | 6 | 4 |
| 5 | 5 | 5 |
| 6 | 6 | 6 |
| 7 | 9 | 3 |
| 8 | 4 | 2 |
| 9 | 1 | 9 |

### Cubes near a base (Yavadunam for cubes)

$(B + d)^3 = (n + 2d) \mid 3d^2 \mid d^3$, where each right-hand part has as many digits as the base has zeros. $103^3$: 103 + 6 = 109 | 3 × 9 = 27 | 27 → **10,92,727**. $98^3$: 98 − 4 = 94 | 3 × 4 = 12 | −8 → 9,41,200 − 8 = **9,41,192**.

### Why it works

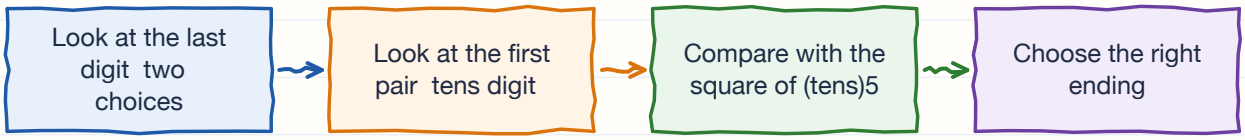
$(B + d)^3 = B^3 + 3B^2d + 3Bd^2 + d^3 = B^2(B + 3d) + B(3d^2) + d^3$. The left part B + 3d is n + 2d.

### Algebraic cube identities

$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ · $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ · $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

# Chapter 18: Square Roots and Cube Roots

## 18.1 Square roots of perfect squares (up to 4 digits) by observation




### √7,744

- Last digit 4 → root ends in **2 or 8**.
- First pair 77 lies between $8^2 = 64$ and $9^2 = 81$ → tens digit **8**.
- $85^2 = 7{,}225 < 7{,}744$, so the root is above 85 → **88**.

### √5,329

Last digit 9 → 3 or 7. 53 lies between 49 and 64 → 7_. $75^2 = 5{,}625 > 5{,}329$ → below 75 → **73**.

### √11,449 (5 digits)

Last digit 9 → 3 or 7. Leading part 114 lies between $10^2 = 100$ and $11^2 = 121$ → 10_. $105^2 = 11{,}025 < 11{,}449$ → **107**.

### This only works for perfect squares

For 7,750 the method gives nonsense. If the question does not promise a perfect square, use estimation or long division.

## 18.2 The long-division method for square roots

### √53,824

Pair the digits from the right: 5 | 38 | 24.

```
          2    3    2
       ------------
 2  )   5 | 38 | 24
         4                  2² = 4
       ---
43  )   1    38            double 2 → 4_; 43 × 3 = 129
         1    29
       --------
462 )          9    24     double 23 → 46_; 462 × 2 = 924
               9    24
              -------
                   0
```

**√53,824 = 232.** Check: $232^2 = 53{,}824$ ✓.

### Algorithm

1. Pair digits from the decimal point outwards.
2. Find the largest square ≤ the first pair; write its root as the first digit.
3. Subtract, bring down the next pair.
4. Double the root found so far, write it with a blank: (2r)_. Find the largest digit x with (2r)x × x ≤ current number.
5. Repeat. For decimals, bring down pairs of zeros.

### √2 to three decimal places

Pairs 2 | 00 | 00 | 00. Digits found: 1, then 24 × 4 = 96 ≤ 100, then 281 × 1 = 281 ≤ 400, then 2824 × 4 = 11,296 ≤ 11,900. So √2 ≈ **1.414** (more precisely 1.41421…).

## 18.3 Estimating square roots

### The tangent-line estimate

$\sqrt{n} \approx a + (n - a^2) \div (2a)$, where $a^2$ is the nearest perfect square. √300: a = 17 (289). √300 ≈ 17 + 11/34 ≈ 17.32 (true 17.3205…). √70: a = 8 (64). √70 ≈ 8 + 6/16 = 8.375 (true 8.3666…). The estimate is always slightly **too big**.

### Useful values to remember

√2 ≈ 1.414 · √3 ≈ 1.732 · √5 ≈ 2.236 · √6 ≈ 2.449 · √7 ≈ 2.646 · √10 ≈ 3.162

## 18.4 Cube roots of perfect cubes

### Two-step method for perfect cubes up to 6 digits

1. **Last digit** of the cube → last digit of the root (from the cube-endings table: 2↔8, 3↔7, others unchanged).
2. **Thousands part** (drop the last three digits) → find the largest cube ≤ it → that is the tens digit.

| # | Problem | Last digit → | Thousands part → | Answer |
|---|---|---|---|---|
| 1 | ∛2,74,625 | 5 → 5 | 274: $6^3 = 216 \le 274 < 343$ | 65 |
| 2 | ∛4,93,039 | 9 → 9 | 493: $7^3 = 343 \le 493 < 512$ | 79 |
| 3 | ∛1,03,823 | 3 → 7 | 103: $4^3 = 64 \le 103 < 125$ | 47 |
| 4 | ∛6,81,472 | 2 → 8 | 681: $8^3 = 512 \le 681 < 729$ | 88 |
| 5 | ∛12,167 | 7 → 3 | 12: $2^3 = 8 \le 12 < 27$ | 23 |

### Only for perfect cubes

The question must guarantee a perfect cube. Otherwise, estimate: ∛100 ≈ 4.64 because $4.6^3 \approx 97$ and $4.7^3 \approx 104$.

## 18.5 Approximation and verification

Always verify a root by squaring or cubing back, at least roughly: $88^2$ ends in 4 ✓ and is close to $90^2 = 8{,}100$ ✓.

# Chapter 19: Daily Drills for Squares and Roots

### A 10-minute daily routine

- **Minutes 1–3:** recite squares 11–30 aloud (forward, then backward).
- **Minutes 4–6:** produce five random squares 31–99 using the near-50 / near-100 / ending-in-5 methods.
- **Minutes 7–8:** three perfect-square roots and two perfect-cube roots.
- **Minutes 9–10:** two estimates of non-perfect square roots, checked with a calculator afterwards.

| Day | Squares to recite | Roots to find |
|---|---|---|
| Monday | 11–20 | $\sqrt{1,849}$, $\sqrt{3,481}$, $\sqrt[3]{9,261}$ |
| Tuesday | 21–30 | $\sqrt{6,241}$, $\sqrt{2,209}$, $\sqrt[3]{32,768}$ |
| Wednesday | 31–40 | $\sqrt{9,409}$, $\sqrt{1,369}$, $\sqrt[3]{1,57,464}$ |
| Thursday | 41–50 | $\sqrt{5,776}$, $\sqrt{8,281}$, $\sqrt[3]{2,16,000}$ |
| Friday | 51–60 (near 50) | $\sqrt{7,056}$, $\sqrt{4,356}$, $\sqrt[3]{3,89,017}$ |
| Saturday | 90–100 (near 100) | $\sqrt{9,801}$, $\sqrt{9,025}$, $\sqrt[3]{7,04,969}$ |
| Sunday | mixed revision | any five |

*Answers to the drill roots:* Mon 43, 59, 21 · Tue 79, 47, 32 · Wed 97, 37, 54 · Thu 76, 91, 60 · Fri 84, 66, 73 · Sat 99, 95, 89.

### Part VI summary

- Know squares 1–50; produce 51–100 with near-50, near-100, ending-in-5 or $(a \pm b)^2$.
- Cubes near a base: $(n + 2d) \mid 3d^2 \mid d^3$.
- Perfect square roots: last digit → two choices; first pair → tens; compare with $(\text{tens})5^2$.
- Perfect cube roots: last digit ($2 \leftrightarrow 8$, $3 \leftrightarrow 7$) and the thousands part.
- Non-perfect roots: long division (exact digits) or $\sqrt{n} \approx a + (n - a^2)/2a$ (approximate).

## ✎ Part VI practice

**213.** $31^2$

**214.** $58^2$

**215.** $993^2$

**216.** Use $(a \pm b)^2$ to find $82^2$

**217.** $953^2$ (duplex method)

**218.** $21^3$

**219.** $\sqrt{7,921}$

**220.** $\sqrt[3]{9,12,673}$

**221.** Estimate $\sqrt{234}$ to 2 decimal places (approximate). *(approximate)*

**222.** $40^2$

**223.** $49^2$

**224.** $108^2$

**225.** Use $(a \pm b)^2$ to find $298^2$

**226.** $229^2$ (duplex method)

**227.** $\sqrt{1,024}$

**228.** $\sqrt[3]{7,78,688}$

**229.** Estimate $\sqrt{127}$ to 2 decimal places (approximate). *(approximate)*

**230.** $21^2$

**231.** $41^2$

**232.** $107^2$

**233.** Use $(a \pm b)^2$ to find $69^2$

**234.** $332^2$ (duplex method)

**235.** $\sqrt{1,296}$

**236.** $\sqrt[3]{5,92,704}$

*Answers: see the answer key (questions 213–236).*

## Part VI answer key

**213.** **961** — $31 \times 31 = 961$.

**214.** **3,364** — Deviation from 50: +8 Left: 25 + 8 = 33 Right: $8^2 = 64$ Answer = 3,364

**215.** **9,86,049** — Base 1,000, deviation −7 Left: 993 − 7 = 986 Right: $7^2 = 49$ (needs 3 digit(s)) Answer = 9,86,049

**216.** **6,724** — $(80 + 2)^2 = 6400 + 2 \times 80 \times 2 + 4 = 6,724$.

**217.** **9,08,209** — Duplexes from left: 81, 90, 79, 30, 9 Combine with carries (right to left) → 9,08,209

**218.** **9,261** — $21 \times 21 \times 21 = 9,261$.

**219.** **89** — Last digit 1 → root ends in 1 or 9. Leading part 79 lies between $8^2$ and $9^2$, so tens digit is 8. Check $85^2 = 7225$ to choose → 89.

**220.** **97** — Last digit 3 → cube root ends in 7 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 912 lies between $9^3$ and $10^3$ → tens digit 9.

**221.** **≈ 15.30 (true value 15.297)** — Nearest perfect square 225 = $15^2$. $\sqrt{n} \approx 15 + (234 - 225) \div (2 \times 15) = 15.300$.

**222.** **1,600** — $40 \times 40 = 1600$.

**223.** **2,401** — Deviation from 50: −1 Left: 25 − 1 = 24 Right: $1^2 = 01$ Answer = 2,401

**224.** **11,664** — Base 100, deviation +8 Left: 108 + 8 = 116 Right: $8^2 = 64$ (needs 2 digit(s)) Answer = 11,664

**225.** **88,804** — $(300 - 2)^2 = 90000 - 2 \times 300 \times 2 + 4 = 88,804$.

**226.** **52,441** — Duplexes from left: 4, 8, 40, 36, 81 Combine with carries (right to left) → 52,441

**227.** **32** — Last digit 4 → root ends in 2 or 8. Leading part 10 lies between $3^2$ and $4^2$, so tens digit is 3. Check $35^2 = 1225$ to choose → 32.

**228.** **92** — Last digit 8 → cube root ends in 2 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 778 lies between $9^3$ and $10^3$ → tens digit 9.

**229.** **≈ 11.27 (true value 11.269)** — Nearest perfect square 121 = $11^2$. $\sqrt{n} \approx 11 + (127 - 121) \div (2 \times 11) = 11.273$.

**230.** **441** — $21 \times 21 = 441$.

**231.** **1,681** — Deviation from 50: −9 Left: 25 − 9 = 16 Right: $9^2 = 81$ Answer = 1,681

**232.** **11,449** — Base 100, deviation +7 Left: 107 + 7 = 114 Right: $7^2 = 49$ (needs 2 digit(s)) Answer = 11,449

**233.** **4,761** — $(70 - 1)^2 = 4900 - 2 \times 70 \times 1 + 1 = 4,761$.

**234.** **1,10,224** — Duplexes from left: 9, 18, 21, 12, 4 Combine with carries (right to left) → 1,10,224

**235.** **36** — Last digit 6 → root ends in 4 or 6. Leading part 12 lies between $3^2$ and $4^2$, so tens digit is 3. Check $35^2 = 1225$ to choose → 36.

**236.** **84** — Last digit 4 → cube root ends in 4 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 592 lies between $8^3$ and $9^3$ → tens digit 8.

# Part VII

## Factorisation and Algebra

Algebra is arithmetic with letters. Every Vedic multiplication method you have learned works for algebraic expressions too — in fact, the algebra is *why* the methods work. This part builds algebra from the beginning and connects each idea back to mental arithmetic.

# Chapter 20: Expressions, Variables and Equations

## 20.1 Variables, constants and expressions

- A **variable** is a letter that stands for a number that can change (x, y, n).
- A **constant** is a fixed number (5, –3, ½).
- A **term** is a number, a variable, or their product: $7x$, $-3y^2$, $12$.
- An **expression** is terms joined by + or –: $3x^2 - 5x + 2$.
- **Like terms** have the same variables with the same powers: $4x$ and $-9x$ are like terms; $4x$ and $4x^2$ are not.

### Simplifying expressions

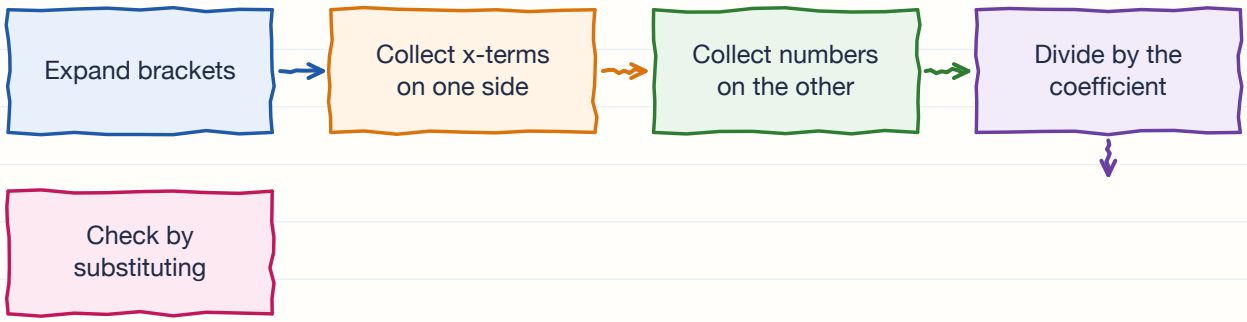
- $5x + 3y - 2x + 7y = (5 - 2)x + (3 + 7)y =$ **$3x + 10y$**
- $4(2a - 3) - (a + 5) = 8a - 12 - a - 5 =$ **$7a - 17$**
- Evaluate $3x^2 - 5x + 2$ at $x = 4$: $3(16) - 20 + 2 =$ **$30$**

### A minus sign outside brackets

$-(a + 5) = -a - 5$. The minus applies to **every** term inside. Forgetting this is the most common algebra error.

## 20.2 Linear equations in one variable

An **equation** says two expressions are equal. To solve, do the same operation to both sides — or, equivalently, **transpose** (Paravartya): move a term across the equals sign and change its sign.




### Five linear equations

1. $4x - 7 = 21 \rightarrow 4x = 28 \rightarrow$ **$x = 7$**
2. $3(x + 2) = 2x + 11 \rightarrow 3x + 6 = 2x + 11 \rightarrow$ **$x = 5$**
3. $5x - 3 = 2x + 15 \rightarrow 3x = 18 \rightarrow$ **$x = 6$**
4. $x/3 + x/4 = 14 \rightarrow (4x + 3x)/12 = 14 \rightarrow 7x = 168 \rightarrow$ **$x = 24$**
5. $0.6x - 1.5 = 0.2x + 0.5 \rightarrow 0.4x = 2 \rightarrow$ **$x = 5$**

### Paravartya formula

$ax + b = cx + d \Rightarrow$ **$x = (d - b)/(a - c)$**. Example 3: $x = (15 - (-3))/(5 - 2) = 18/3 = 6$.

## 20.3 Word problems into equations

### "A number doubled and increased by 7 gives 31."

$2x + 7 = 31 \rightarrow 2x = 24 \rightarrow x =$ **12**.

### Ages

A mother is 26 years older than her daughter. In 4 years she will be three times as old as the daughter. Let the daughter's age be d. $d + 26 + 4 = 3(d + 4) \rightarrow d + 30 = 3d + 12 \rightarrow 18 = 2d \rightarrow$ **d = 9**, mother **35**. Check: in 4 years, $39 = 3 \times 13$ ✓.

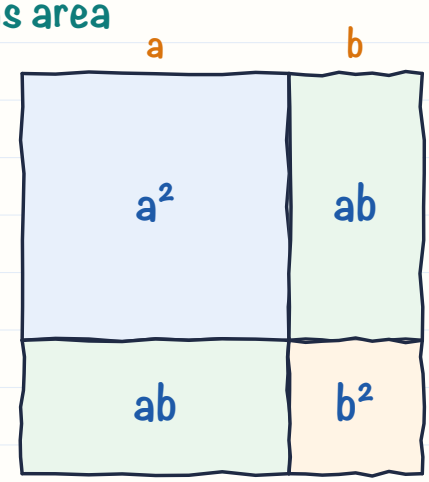
# Chapter 21: Algebraic Identities

An **identity** is true for **every** value of the variables. These five are used constantly.

### The essential identities

1. $(a + b)^2 = a^2 + 2ab + b^2$
2. $(a - b)^2 = a^2 - 2ab + b^2$
3. $(a + b)(a - b) = a^2 - b^2$
4. $(x + a)(x + b) = x^2 + (a + b)x + ab$
5. $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

## 21.1 $(a + b)^2$ — seen as area




$$(a + b)^2 = a^2 + 2ab + b^2$$

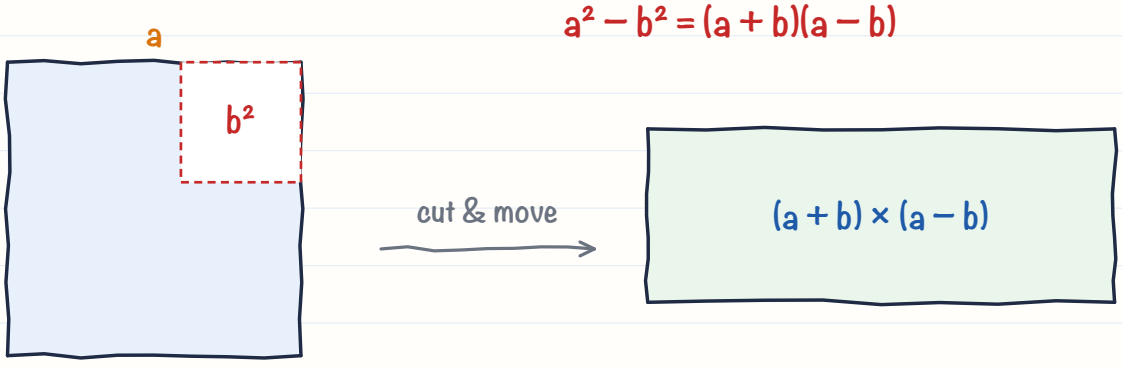
A square of side $(a + b)$ splits into a square $a^2$, a square $b^2$, and **two** rectangles $ab$. That is why there is a **2ab** term — the most commonly forgotten part.

### $(a + b)^2$ is not $a^2 + b^2$

$(3 + 4)^2 = 49$, but $3^2 + 4^2 = 25$. The missing 24 is $2 \times 3 \times 4$.

## 21.2 Difference of two squares — seen as cut-and-move

$$a^2 - b^2 = (a + b)(a - b)$$




### Using $a^2 - b^2$ for arithmetic

- $53 \times 47 = (50 + 3)(50 - 3) = 2{,}500 - 9 =$ **2,491**

- $102 \times 98 = 10{,}000 - 4 =$ **9,996**
- $87^2 - 13^2 = (87 + 13)(87 - 13) = 100 \times 74 =$ **7,400**
- $1{,}005 \times 995 = 10{,}00{,}000 - 25 =$ **9,99,975**
- $6.2 \times 5.8 = 36 - 0.04 =$ **35.96**

## 21.3 Perfect-square identities for mental squaring

### Worked example

- $103^2 = 10{,}000 + 600 + 9 =$ **10,609**
- $97^2 = 10{,}000 - 600 + 9 =$ **9,409**
- $4.1^2 = 16 + 0.8 + 0.01 =$ **16.81**
- $(x + 5)^2 = x^2 + 10x + 25$
- $(2y - 3)^2 = 4y^2 - 12y + 9$

## 21.4 Using identities to evaluate expressions

### If $a + b = 10$ and $ab = 21$, find $a^2 + b^2$.

$a^2 + b^2 = (a + b)^2 - 2ab = 100 - 42 =$ **58**. (Check: a = 7, b = 3 gives 49 + 9 = 58 ✓.)

### If $x + 1/x = 5$, find $x^2 + 1/x^2$.

Square both sides: $x^2 + 2 + 1/x^2 = 25 \rightarrow x^2 + 1/x^2 =$ **23**. A favourite exam pattern.

# Chapter 22: Multiplying Polynomials (Vertically and Crosswise)

The Urdhva pattern multiplies polynomials just as it multiplies numbers — the "digits" are the coefficients, and there is **no carrying**.

### Standard method (FOIL)

**(3x + 2)(4x – 5)** First: $12x^2$ Outer: $-15x$ Inner: $8x$ Last: $-10$ Sum: $12x^2 - 7x - 10$

### Vertically and crosswise

**(3x + 2)(4x – 5)** Vertical (left): $3 \times 4 = \mathbf{12x^2}$ Crosswise: $3(-5) + 2(4) = \mathbf{-7x}$ Vertical (right): $2(-5) = \mathbf{-10}$

### Three-term by three-term: $(x^2 + 2x + 3)(2x^2 - x + 4)$

Coefficients (1, 2, 3) and (2, –1, 4): $x^4$: $1 \cdot 2 = 2$ · $x^3$: $1(-1) + 2 \cdot 2 = 3$ · $x^2$: $1 \cdot 4 + 2(-1) + 3 \cdot 2 = 8$ · x: $2 \cdot 4 + 3(-1) = 5$ · constant: $3 \cdot 4 = 12$ **$= 2x^4 + 3x^3 + 8x^2 + 5x + 12$.** Check (Gunitasamuccaya, x = 1): $(1 + 2 + 3)(2 - 1 + 4) = 6 \times 5 = 30$, and $2 + 3 + 8 + 5 + 12 = 30$ ✓.

### The x = 1 check works for every product

Put x = 1 in each factor and in your answer; the numbers must agree. Putting x = –1 gives a second, independent check.

# Chapter 23: Factorisation

**Factorising** is multiplying in reverse: writing an expression as a product.

## 23.1 Common factors

$6x^2 + 9x = 3x(2x + 3)$. Always take out the highest common factor first.

## 23.2 Quadratics $x^2 + bx + c$

### Find two numbers

$x^2 + bx + c = (x + m)(x + n)$ where **$m + n = b$** and **$m \times n = c$**.

| # | Expression | m, n (sum b, product c) | Factorised |
|---|---|---|---|
| 1 | $x^2 + 7x + 12$ | 3, 4 | $(x + 3)(x + 4)$ |
| 2 | $x^2 - 9x + 20$ | −4, −5 | $(x - 4)(x - 5)$ |
| 3 | $x^2 + 2x - 15$ | 5, −3 | $(x + 5)(x - 3)$ |
| 4 | $x^2 - 4x - 21$ | −7, 3 | $(x - 7)(x + 3)$ |
| 5 | $x^2 - 49$ | 7, −7 | $(x + 7)(x - 7)$ |

## 23.3 Quadratics $ax^2 + bx + c$ (Adyamadyena method)

*Adyamadyenantyamantyena* — "the first by the first and the last by the last". Split the middle term so that the **first** coefficient splits as the product of the first terms and the **last** as the product of the last terms.

### $6x^2 + 13x + 6$

Find two numbers with product $a \times c = 36$ and sum 13: **4 and 9**. $6x^2 + 4x + 9x + 6 = 2x(3x + 2) + 3(3x + 2) =$ **$(2x + 3)(3x + 2)$**. Check crosswise: $2 \cdot 2 + 3 \cdot 3 = 13$ ✓.

### $3x^2 - 10x - 8$

$a \times c = -24$, sum −10: **−12 and 2**. $3x^2 - 12x + 2x - 8 = 3x(x - 4) + 2(x - 4) =$ **$(3x + 2)(x - 4)$**.

## 23.4 Difference of squares and perfect squares

- $4x^2 - 25 = (2x + 5)(2x - 5)$
- $9y^2 - 12y + 4 = (3y - 2)^2$
- $x^4 - 16 = (x^2 + 4)(x^2 - 4) = (x^2 + 4)(x + 2)(x - 2)$

# Chapter 24: Simultaneous Linear Equations

## 24.1 Elimination and substitution

### $2x + 3y = 13$ and $5x - 2y = 4$

Multiply the first by 2 and the second by 3: $4x + 6y = 26$ and $15x - 6y = 12$. Add: $19x = 38 \rightarrow$ **$x = 2$**. Then $3y = 13 - 4 = 9 \rightarrow$ **$y = 3$**.

## 24.2 Cross-multiplication (a Vedic-style formula)

### For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$

$x = (c_1b_2 - c_2b_1) \div (a_1b_2 - a_2b_1)$ $y = (a_1c_2 - a_2c_1) \div (a_1b_2 - a_2b_1)$ Same example: $x = (13\cdot(-2) - 4\cdot3)/(2\cdot(-2) - 5\cdot3) = (-26 - 12)/(-4 - 15) = -38/-19 = 2$; $y = (2\cdot4 - 5\cdot13)/(-19) = (8 - 65)/(-19) = 3$.

This is the method Tirthaji presents using Paravartya ideas; it is identical to Cramer's rule taught in higher classes.

## 24.3 Special patterns (from Part II)

| Pattern | Sutra | Example | Solution |
|---|---|---|---|
| Coefficients swapped | Sankalana-Vyavakalanabhyam | $37x + 23y = 97$, $23x + 37y = 83$ | add: $x + y = 3$; subtract: $x - y = 1 \rightarrow x = 2, y = 1$ |
| One pair in ratio with constants | Anurupye Shunyamanyat | $6x + 5y = 10$, $12x + 9y = 18$ | x-coeff 6:12 = constants 10:18? No (1:2 vs 5:9) — pattern fails; solve normally |
| y-coeffs and constants in ratio | Anurupye Shunyamanyat | $2x + 3y = 6$, $7x + 9y = 18$ | $3:9 = 6:18 \rightarrow x = 0, y = 2$ |

### Check the pattern first

The second row shows that a pattern **can fail**. Always check the exact condition before applying a special sutra.

## 24.4 Conventional algebra versus shortcuts

### When to use / when not to use

Shortcut sutras are fast for the patterns they fit. Elimination and cross-multiplication **always** work. In exams: spend two seconds looking for a pattern; if none jumps out, solve conventionally.

### Part VII summary

- Collect like terms; a minus outside a bracket changes every sign inside.
- $ax + b = cx + d \Rightarrow x = (d - b)/(a - c)$.
- $(a \pm b)^2 = a^2 \pm 2ab + b^2$; $a^2 - b^2 = (a + b)(a - b)$.
- Multiply polynomials vertically and crosswise; check with $x = 1$.
- $x^2 + bx + c$: find m, n with sum b and product c. $ax^2 + bx + c$: split using $a \times c$.

- Simultaneous equations: elimination, or cross-multiplication formula; look for swapped coefficients.

## ✎ Part VII practice

**237.** Solve $2x - 10 = 2$

**238.** Solve $6x - 12 = 9x - 42$

**239.** Expand $(3x + 5)(2x - 4)$

**240.** Factorise $x^2 - 7x + 10$

**241.** $79^2 - 5^2$

**242.** Use $(a \pm b)^2$ to find $57^2$

**243.** Solve: $7x + 2y = 32$, $2x + 7y = -23$

**244.** Solve: $19x + 40y = -143$, $40x + 19y = 25$

**245.** Solve $(x + 11)(x + 11) = (x + 6)(x + 1)$

**246.** Solve $x^2 + 0x - 49 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**247.** Solve $8x + 1 = 105$

**248.** Solve $10x + 6 = 4x - 30$

**249.** Expand $(2x + 2)(2x - 9)$

**250.** Factorise $x^2 + 3x + 2$

**251.** $93^2 - 14^2$

**252.** Use $(a \pm b)^2$ to find $32^2$

**253.** Solve: $5x + 4y = 33$, $4x + 9y = 38$

**254.** Solve: $20x + 13y = 144$, $13x + 20y = 186$

**255.** Solve $(x + 12)(x + 5) = (x + 8)(x + 6)$

**256.** Solve $x^2 - 12x + 35 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**257.** Solve $8x - 17 = 23$

**258.** Solve $8x - 10 = 2x - 28$

**259.** Expand $(x + 6)(3x - 7)$

**260.** Factorise $x^2 - 10x + 24$

**261.** $47^2 - 9^2$

**262.** Use $(a \pm b)^2$ to find $98^2$

*Answers: see the answer key (questions 237–262).*

## Part VII answer key

**237.** **x = 6** — Transpose −10 (change sign): $2x = 2 + 10 = 12$; divide by 2: $x = 6$.

**238.** **x = 10** — Paravartya formula for $ax + b = cx + d$: $x = (d - b)/(a - c) = (-42 - -12)/(6 - 9) = -30/-3 = 10$.

**239.** **$6x^2 - 2x - 20$** — Vertical (left): $3 \times 2 = 6$. Crosswise: $3 \times (-4) + 2 \times (5) = -2$. Vertical (right): $(5) \times (-4) = -20$.

**240.** **$(x - 2)(x - 5)$** — Find two numbers with product 10 and sum −7: −2 and −5. Check with Gunita-samuccaya (put $x = 1$): $(1 - 2)(1 - 5) = 4$ and $1 - 7 + 10 = 4$.

**241.** **6,216** — $a^2 - b^2 = (a + b)(a - b) = 84 \times 74 = 6{,}216$.

**242.** **3,249** — $(60 - 3)^2 = 3600 - 2{\times}60{\times}3 + 9 = 3{,}249$.

**243.** **$x = 6, y = -5$** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (224 - -46)/45 = 6$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (-161 - 64)/45 = -5$.

**244.** **x = 3, y = –5** — Add: 59(x + y) = –118 → x + y = –2. Subtract: -21(x – y) = –168 → x – y = 8. Solve: x = 3, y = –5.

**245.** **x = –7 2/3** — x = (cd – ab)/(a + b – c – d) = (6 – 121)/(22 – 7).

**246.** **x = 7 or –7** — Differential: 2x + 0. Discriminant = 0 – -196 = 196, √ = 14. 2x + 0 = ±14.

**247.** **x = 13** — Transpose 1 (change sign): 8x = 105 – 1 = 104; divide by 8: x = 13.

**248.** **x = –6** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (–30 – 6)/(10 – 4) = –36/6 = –6.

**249.** **4x² – 14x – 18** — Vertical (left): 2 × 2 = 4. Crosswise: 2 × (–9) + 2 × (2) = –14. Vertical (right): (2) × (–9) = –18.

**250.** **(x + 1)(x + 2)** — Find two numbers with product 2 and sum 3: 1 and 2. Check with Gunita-samuccaya (put x = 1): (1 + 1)(1 + 2) = 6 and 1 + 3 + 2 = 6.

**251.** **8,453** — $a^2 - b^2 = (a + b)(a - b) = 107 \times 79 = 8{,}453$.

**252.** **1,024** — $(30 + 2)^2 = 900 + 2{\times}30{\times}2 + 4 = 1{,}024$.

**253.** **x = 5, y = 2** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (297 - 152)/29 = 5$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (190 - 132)/29 = 2$.

**254.** **x = 2, y = 8** — Add: 33(x + y) = 330 → x + y = 10. Subtract: 7(x – y) = –42 → x – y = –6. Solve: x = 2, y = 8.

**255.** **x = –4** — x = (cd – ab)/(a + b – c – d) = (48 – 60)/(17 – 14).

**256.** **x = 7 or 5** — Differential: 2x – 12. Discriminant = 144 – 140 = 4, √ = 2. 2x – 12 = ±2.

**257.** **x = 5** — Transpose –17 (change sign): 8x = 23 + 17 = 40; divide by 8: x = 5.

**258.** **x = –3** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (–28 – –10)/(8 – 2) = –18/6 = –3.

**259.** **3x² + 11x – 42** — Vertical (left): 1 × 3 = 3. Crosswise: 1 × (–7) + 3 × (6) = 11. Vertical (right): (6) × (–7) = –42.

**260.** **(x – 6)(x – 4)** — Find two numbers with product 24 and sum –10: –6 and –4. Check with Gunita-samuccaya (put x = 1): (1 – 6)(1 – 4) = 15 and 1 – 10 + 24 = 15.

**261.** **2,128** — $a^2 - b^2 = (a + b)(a - b) = 56 \times 38 = 2{,}128$.

**262.** **9,604** — $(100 - 2)^2 = 10000 - 2{\times}100{\times}2 + 4 = 9{,}604$.

# Part VIII

## Advanced Arithmetic

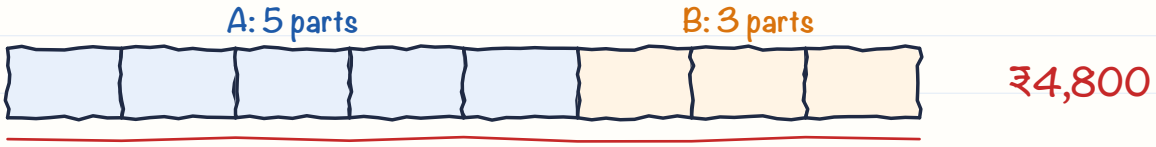
This part covers the arithmetic topics that make up most of the quantitative-aptitude syllabus in SSC, Banking and Railway exams. For every topic the **standard concept** comes first, then the **mental-math shortcuts** that genuinely help.

> **An honest note on shortcuts**
> Most aptitude topics do **not** have a special Vedic sutra. What helps is: clear formulas, the fraction–percentage chart, LCM thinking for work problems, ratio thinking, and the fast multiplication/division methods from Parts III–VI. We do not attach Vedic names where they do not belong.

# Chapter 25: Ratio, Proportion and Averages

## 25.1 Ratio

A **ratio** a : b compares two quantities of the same kind. It can be simplified like a fraction: 18 : 24 = 3 : 4.




8 equal parts in all

### Dividing in a ratio

**Divide ₹7,200 among A, B, C in the ratio 5 : 3 : 4.** Total parts = 12; one part = 7,200 ÷ 12 = 600. A = ₹**3,000**, B = ₹**1,800**, C = ₹**2,400**.

### Combining ratios

If A : B = 2 : 3 and B : C = 4 : 5, make B the same: A : B = 8 : 12, B : C = 12 : 15 → **A : B : C = 8 : 12 : 15**.

## 25.2 Direct and inverse proportion

### Two kinds of proportion

**Direct:** both increase together; the ratio stays constant. 6 pens cost ₹90 → 15 pens cost 90 × 15/6 = ₹**225**. **Inverse:** one increases as the other decreases; the product stays constant. 12 workers take 15 days → 18 workers take 12 × 15/18 = **10 days**.

### The "men-days-work" formula

$M_1 \times D_1 \times H_1 / W_1 = M_2 \times D_2 \times H_2 / W_2$ (men, days, hours per day, work). One formula handles most "chain rule" questions.

## 25.3 Averages

Average = sum of values ÷ number of values.

### Assumed-mean (deviation) method

Choose a convenient value near the data, average the **deviations**, and add. Average of 47, 52, 49, 55, 57: assume 50; deviations –3, 2, –1, 5, 7 → total 10 → 10 ÷ 5 = 2. Average = **52**.

### A number removed

The average of 5 numbers is 42. When one number is removed, the average of the rest is 40. Removed number = 5 × 42 – 4 × 40 = 210 – 160 = **50**.

## 25.4 Weighted averages

### Formula

Weighted average = $(n_1a_1 + n_2a_2 + \ldots) \div (n_1 + n_2 + \ldots)$

### Two sections

30 students average 60; 20 students average 75. Combined average = (1,800 + 1,500) ÷ 50 = **66**. **Shortcut (alligation view):** the combined average lies between 60 and 75, closer to the bigger group. Distances are in the inverse ratio of sizes 30 : 20 = 3 : 2, so the gap 15 splits as 6 : 9 → 60 + 6 = 66.

# Chapter 26: Percentages in Depth

## 26.1 Percentage as a fraction (the master chart)

The fraction chart of Chapter 3 is your best tool. For example, 37.5% is 3/8, so 37.5% of 1,624 is 3 × (1,624 ÷ 8) = 3 × 203 = **609**.

## 26.2 "A is r% more than B" — how much less is B than A?

### Formula

If A is r% more than B, then B is **r/(100 + r) × 100 %** less than A. If A is r% less than B, then B is **r/(100 – r) × 100 %** more than A.

### Salaries

A's salary is 25% more than B's. B's salary is 25/125 × 100 = **20%** less than A's. Price rises by 25%. To keep spending the same, consumption must fall by **20%** (same formula).

## 26.3 Population and depreciation

### Formula

After n years at r% growth: $P \times (1 + r/100)^n$. At r% depreciation: $P \times (1 – r/100)^n$.

### Worked example

A town of 50,000 grows by 10% a year. After 2 years: 50,000 × 1.1 × 1.1 = **60,500**. A machine worth ₹80,000 depreciates 10% a year. After 2 years: 80,000 × 0.9 × 0.9 = **₹64,800**.

## 26.4 Successive percentage changes

### Net change of a% then b%

a + b + ab/100 (use negative numbers for decreases). Discounts of 20% and 10%: –20 – 10 + 200/100 = –28 → a single discount of **28%**.

# Chapter 27: Profit, Loss and Discount

## 27.1 Core terms

- **Cost price (CP):** what the seller paid.
- **Selling price (SP):** what the buyer pays.
- **Profit** = SP − CP (if SP > CP); **Loss** = CP − SP.
- **Profit % and loss % are always on CP.**
- **Marked price (MP):** the price label. **Discount** is always on MP.

### Multiplier forms

SP = CP × (100 + profit%)/100 · SP = CP × (100 − loss%)/100 · SP = MP × (100 − discount%)/100

### Five profit-and-loss examples

1. CP ₹800, profit 15% → SP = 800 × 1.15 = **₹920**.
2. SP ₹1,170 after a 10% loss → CP = 1,170 ÷ 0.9 = **₹1,300**.
3. MP ₹2,000, discount 15% → SP = **₹1,700**. If CP = ₹1,360, profit = 340/1,360 = **25%**.
4. Bought 12 pens for ₹100 and sold 10 pens for ₹100. CP per pen 100/12, SP per pen 100/10 → profit = (1/10 − 1/12)/(1/12) = 20%. **Shortcut:** profit % = (12 − 10)/10 × 100 = **20%**.
5. A dishonest trader sells at cost price but uses a 900 g weight for 1 kg. Profit = 100/900 × 100 = **11 1/9 %**.

### Profit % is on cost price

If CP = 400 and SP = 500, profit is 100, which is 25% of CP — **not** 20% (that would be on SP).

### Same SP, one profit and one loss

Two items sold at the same price, one at x% profit and one at x% loss, always give an **overall loss of $x^2/100$ %**. For x = 20: loss 4%.

# Chapter 28: Simple and Compound Interest

## 28.1 Simple interest

### Formula

SI = P × R × T / 100. Amount A = P + SI.

### Worked example

₹12,000 at 8% per annum for 3 years: SI = 12,000 × 8 × 3 / 100 = **₹2,880**. At what rate does money double in 8 years (SI)? SI = P, so R = 100/T = 100/8 = **12.5%**.

## 28.2 Compound interest

### Formula

$A = P(1 + R/100)^T$. CI = A − P. Half-yearly: use R/2 and 2T. Quarterly: R/4 and 4T.

### Compound interest

1. ₹10,000 at 10% for 2 years: A = 10,000 × 1.21 = 12,100 → **CI ₹2,100**.
2. ₹8,000 at 10% per annum compounded half-yearly for 1 year: $A = 8,000 \times 1.05^2 = 8,820$ → **CI ₹820**.
3. ₹15,625 at 4% for 3 years: $A = 15,625 \times (26/25)^3 = 15,625 \times 17,576/15,625$ = **₹17,576**. (Choosing 4% = 1/25 makes the arithmetic exact.)

### CI − SI shortcut

For 2 years: $CI - SI = P(R/100)^2$. For 3 years: $CI - SI = P(R/100)^2(3 + R/100)$. ₹10,000 at 10%, 2 years: 10,000 × 0.01 = **₹100** (CI 2,100 vs SI 2,000 ✓).

### Year-by-year layout (the "percentage tree")

10% CI on ₹10,000: year 1 interest 1,000; year 2 interest is 1,000 plus 10% of that 1,000, i.e. 1,100. Total 2,100. For 3 years add 1,000 + 200 + 10 = 1,210 → total 3,310. Each year's interest = previous interest + R% of it.

# Chapter 29: Time and Work, Pipes and Cisterns

## 29.1 The LCM method

### Total work = LCM of the days

Let the total work be the LCM of the individual times. Each person's daily work = total ÷ their days. Combined time = total ÷ combined daily work.

### A in 12 days, B in 18 days

Total = LCM(12, 18) = 36 units. A does 3/day, B does 2/day; together 5/day. Time = 36/5 = **7 1/5 days**.

### Finding one person's time

A and B together take 10 days; A alone takes 15. Total = 30 units: together 3/day, A 2/day → B 1/day → B alone **30 days**.

### Efficiency

A is twice as efficient as B, and together they finish in 14 days. Efficiencies 2 : 1, total = 3 × 14 = 42 units. A alone: 42/2 = **21 days**; B alone: **42 days**.

## 29.2 Pipes and cisterns

Filling pipes do positive work; outlets (leaks) do negative work.

### Worked example

Pipe A fills a tank in 20 min, B in 30 min, and C empties it in 15 min. Total = LCM(20, 30, 15) = 60 units. Per minute: +3, +2, −4 → net +1. Time = **60 minutes**.

### Check whether a pipe empties

If the outlet is faster than all inlets combined, the tank never fills. Read the question carefully.

### Chapters 25–29

- Ratios: total parts → one part → each share.
- Direct: ratio constant. Inverse: product constant.
- Averages: use deviations from an assumed mean.
- r% more ⇒ r/(100 + r) less. Successive: a + b + ab/100.
- Profit % on CP; discount % on MP.
- SI = PRT/100; A = $P(1 + R/100)^T$; 2-year CI − SI = $P(R/100)^2$.
- Work and pipes: total work = LCM; outlets are negative.

## ✎ Chapters 25–29 practice

**263.** Divide ₹2,556 between A and B in the ratio 5 : 4.

**264.** Find the average of 43, 34, 40, 39, 42, 50, 46.

**265.** Section A has 28 students with average 55; section B has 11 students with average 81. Find the combined average.

**266.** A price changes from ₹800 to ₹600. Find the percentage decrease.

**267.** Find 9 1/11% of 407 using a fraction.

**268.** An item costing ₹1,500 is sold for ₹1,125. Find the loss percentage.

**269.** Find the single discount equal to successive discounts of 30% and 20%.

**270.** Find the simple interest on ₹12,000 at 12% per annum for 4 years.

**271.** Find the compound interest on ₹80,000 at 10% per annum, compounded yearly, for 2 years.

**272.** A can finish a job in 6 days and B in 12 days. How long will they take together?

**273.** Pipe A fills a tank in 5 hours; pipe B empties it in 20 hours. If both are open, how long to fill the empty tank?

**274.** Divide ₹5,434 between A and B in the ratio 6 : 5.

**275.** Find the average of 65, 58, 65, 62, 50, 60.

**276.** Section A has 19 students with average 64; section B has 35 students with average 80. Find the combined average.

**277.** Find 12.5% of 1,136 using a fraction.

**278.** An item costing ₹250 is sold for ₹280. Find the profit percentage.

**279.** Find the single discount equal to successive discounts of 10% and 20%.

**280.** Find the simple interest on ₹10,000 at 12% per annum for 5 years.

**281.** Find the compound interest on ₹10,000 at 10% per annum, compounded yearly, for 2 years.

**282.** A can finish a job in 20 days and B in 5 days. How long will they take together?

**283.** Pipe A fills a tank in 8 hours; pipe B empties it in 24 hours. If both are open, how long to fill the empty tank?

**284.** Divide ₹222 between A and B in the ratio 1 : 1.

**285.** Find the average of 55, 47, 55, 45, 48.

**286.** Section A has 18 students with average 65; section B has 19 students with average 61. Find the combined average.

**287.** A price changes from ₹4,000 to ₹5,600. Find the percentage increase.

**288.** Find 33 1/3% of 96 using a fraction.

*Answers: see the answer key (questions 263–288).*

# Chapters 25–29 answer key

**263.** **A ₹1,420, B ₹1,136** — Total parts 9; one part = 2,556 ÷ 9 = 284.

**264.** **42** — Assume 42. Deviations: +1, −8, −2, −3, +0, +8, +4; total +0; ÷ 7 = 0. Average = 42.

**265.** **62.33** — (28×55 + 11×81) ÷ 39 = 2431 ÷ 39.

**266.** **25%** — Change = 200. % change = change ÷ original × 100 = 200/800 × 100 = 25%.

**267.** **37** — 9 1/11% = 1/11. 1/11 × 407 = 37.

**268.** **25% loss** — Loss = 375. % = 375/1500 × 100 = 25%.

**269.** **44%** — Net = a + b − ab/100 = 30 + 20 − 600/100 = 44%.

**270.** **₹5,760** — SI = PRT/100 = 12,000 × 12 × 4 / 100 = 5,760.

**271.** **₹16,800** — $A = P(1 + R/100)^n$ = 80,000 × $(110/100)^2$ = ₹96,800. CI = A − P = ₹16,800.

**272.** **4 days** — Total work = LCM(6,12) = 12 units. A does 2/day, B does 1/day; together 3/day. Time = 12/3 = 4 days.

**273.** **6 2/3 hours** — Net rate = 1/5 − 1/20 = 3/20 tank/hour, so time = 6 2/3 hours.

**274.** **A ₹2,964, B ₹2,470** — Total parts 11; one part = 5,434 ÷ 11 = 494.

**275.** **60** — Assume 59. Deviations: +6, −1, +6, +3, −9, +1; total +6; ÷ 6 = 1. Average = 60.

**276.** **74.37** — (19×64 + 35×80) ÷ 54 = 4016 ÷ 54.

**277.** **142** — 12.5% = 1/8. 1/8 × 1,136 = 142.

**278.** **12% profit** — Profit = 30. % = 30/250 × 100 = 12%.

**279.** **28%** — Net = a + b − ab/100 = 10 + 20 − 200/100 = 28%.

**280.** **₹6,000** — SI = PRT/100 = 10,000 × 12 × 5 / 100 = 6,000.

**281.** **₹2,100** — $A = P(1 + R/100)^n$ = 10,000 × $(110/100)^2$ = ₹12,100. CI = A − P = ₹2,100.

**282.** **4 days** — Total work = LCM(20,5) = 20 units. A does 1/day, B does 4/day; together 5/day. Time = 20/5 = 4 days.

**283.** **12 hours** — Net rate = 1/8 − 1/24 = 1/12 tank/hour, so time = 12 hours.

**284.** **A ₹111, B ₹111** — Total parts 2; one part = 222 ÷ 2 = 111.

**285.** **50** — Assume 51. Deviations: +4, −4, +4, −6, −3; total −5; ÷ 5 = −1. Average = 50.

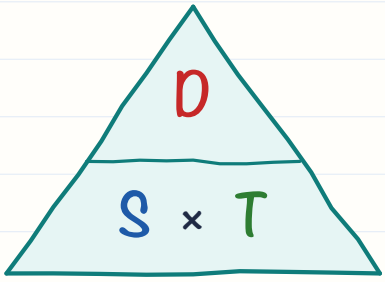
**286.** **62.95** — (18×65 + 19×61) ÷ 37 = 2329 ÷ 37.

**287.** **40%** — Change = 1,600. % change = change ÷ original × 100 = 1600/4000 × 100 = 40%.

**288.** **32** — 33 1/3% = 1/3. 1/3 × 96 = 32.

# Chapter 30: Speed, Distance and Time

## 30.1 The basic relation




D = S × T

S = D ÷ T

T = D ÷ S

km/h × 5/18 = m/s

### Formula

Distance = Speed × Time. To convert km/h to m/s multiply by **5/18**; m/s to km/h multiply by **18/5**. 72 km/h = 72 × 5/18 = **20 m/s**. 15 m/s = 15 × 18/5 = **54 km/h**.

## 30.2 Average speed

### Equal distances at speeds a and b

Average speed = **2ab/(a + b)** — not (a + b)/2. Going at 40 km/h and returning at 60 km/h: 2 × 40 × 60/100 = **48 km/h**.

### Average speed is total distance ÷ total time

Never average the speeds directly unless the **times** are equal.

## 30.3 Relative speed

Same direction: subtract the speeds. Opposite directions: add them.

## 30.4 Trains

### Formula

A train crossing a pole or a person: distance = length of the train. Crossing a platform, bridge or another train: distance = sum of the lengths.

### Trains

1. A 240 m train at 54 km/h (15 m/s) passes a pole in 240 ÷ 15 = **16 s**.
2. The same train crosses a 360 m platform in (240 + 360) ÷ 15 = **40 s**.
3. Trains of 150 m and 100 m run in opposite directions at 50 and 40 km/h. Relative speed 90 km/h = 25 m/s; time = 250 ÷ 25 = **10 s**.
4. Same trains in the same direction: relative speed 10 km/h = 25/9 m/s; time = 250 ÷ (25/9) = **90 s**.

## 30.5 Boats and streams

### Formula

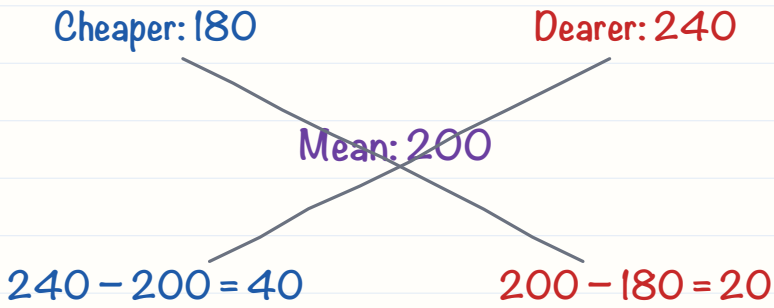
Downstream speed = boat + stream. Upstream speed = boat − stream. Boat = (down + up)/2. Stream = (down − up)/2.

### Worked example

Downstream 18 km/h, upstream 12 km/h → boat **15 km/h**, stream **3 km/h**. Time to go 30 km downstream and come back: 30/18 + 30/12 = 5/3 + 5/2 = 25/6 hours = **4 h 10 min**.

# Chapter 31: Mixtures, Alligation and Partnership

## 31.1 Alligation




Ratio cheaper : dearer = 2 : 1

### Rule of alligation

Cheaper : Dearer = (Dearer − Mean) : (Mean − Cheaper) Tea at ₹180/kg and ₹240/kg mixed to give ₹200/kg: (240 − 200) : (200 − 180) = 40 : 20 = **2 : 1**.

### Alligation works for any "average of two groups"

Speeds, marks, prices, percentages of concentration — whenever two groups combine into one average, the cross diagram gives the ratio of the groups.

## 31.2 Replacement

### Repeated replacement

A container holds x litres of liquid. Each time y litres are removed and replaced with water. After n operations, liquid left = $x(1 - y/x)^n$. 40 L milk, 8 L replaced twice: $40 \times (32/40)^2 = 40 \times 0.64 =$ **25.6 L** of milk.

## 31.3 Partnership

### Formula

Profit is shared in the ratio of (capital × time).

### Worked example

A invests ₹50,000 for 12 months; B invests ₹75,000 for 8 months. Ratio 6,00,000 : 6,00,000 = 1 : 1. A profit of ₹24,000 is shared **₹12,000 each**.

# Chapter 32: Problems on Ages

### Let the present ages be the unknowns

Write "n years ago" as (age – n) and "n years hence" as (age + n). The **difference** of two people's ages never changes.

### Worked example

The sum of a father's and son's ages is 50. Five years ago the father was 7 times as old as the son. Let the son be s: father = 50 – s. 50 – s – 5 = 7(s – 5) → 45 – s = 7s – 35 → 8s = 80 → **s = 10**, father **40**. Check: 5 years ago 35 = 7 × 5 ✓.

# Chapter 33: Number Systems, Remainders and Cyclicity

## 33.1 Types of numbers (review)

Natural ⊂ whole ⊂ integers ⊂ rational ⊂ real. Rational numbers can be written p/q; irrational numbers ($\sqrt{2}$, π) cannot. Every recurring decimal is rational: 0.333… = 1/3.

## 33.2 Remainders

### Remainder theorem for numbers

Remainder of (a × b) = remainder of (remainder of a × remainder of b). Same for sums and powers. (17 × 23 × 29) ÷ 5 → remainders 2, 3, 4 → 2 × 3 × 4 = 24 → remainder **4**.

### Remainder of a power

$2^{100} \div 7$: since $2^3 = 8$ leaves remainder 1, $2^{99} = (2^3)^{33}$ leaves remainder 1, so $2^{100}$ leaves remainder **2**.

## 33.3 Cyclicity of unit digits

| Unit digit | Last digits of powers 1, 2, 3, 4 … | Cycle length |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 1 |
| 2 | 2, 4, 8, 6 | 4 |
| 3 | 3, 9, 7, 1 | 4 |
| 4 | 4, 6 | 2 |
| 5 | 5 | 1 |
| 6 | 6 | 1 |
| 7 | 7, 9, 3, 1 | 4 |
| 8 | 8, 4, 2, 6 | 4 |
| 9 | 9, 1 | 2 |

### Unit digit of $7^{95}$

Powers of 7 end in 7, 9, 3, 1 (cycle of 4). 95 = 4 × 23 + 3 → third in the cycle → **3**.

### Trailing zeros of 100!

Count factors of 5: 100/5 = 20, 100/25 = 4 → **24 zeros**.

## 33.4 Number of factors, sum of factors

For $n = 2^3 \times 3^2 \times 5$: number of factors = $4 \times 3 \times 2 = 24$. Sum of factors = $(1 + 2 + 4 + 8)(1 + 3 + 9)(1 + 5) = 15 \times 13 \times 6 =$ **1,170**.

# Chapter 34: Permutations, Combinations and Probability

## 34.1 Counting principles

### Formula

**Multiplication principle:** m ways then n ways → m × n ways. **Permutations** (order matters): ${}^{n}P_{r} = n!/(n - r)!$ **Combinations** (order does not matter): ${}^{n}C_{r} = n!/(r!(n - r)!)$

### Counting

1. Arrangements of the letters of INDIA (I repeats twice): 5!/2! = **60**.
2. Committee of 3 men from 6 and 2 women from 5: ${}^{6}C_{3} \times {}^{5}C_{2} = 20 \times 10 =$ **200**.
3. Handshakes among 10 people: ${}^{10}C_{2} =$ **45**.
4. 4-digit PINs with no repeated digit: 10 × 9 × 8 × 7 = **5,040**.

## 34.2 Basic probability

### Formula

P(event) = favourable outcomes ÷ total equally likely outcomes. $0 \le P \le 1$. P(not A) = 1 − P(A).

### Probability

1. Two dice, sum 7: favourable (1,6) (2,5) (3,4) (4,3) (5,2) (6,1) = 6 of 36 → **1/6**.
2. At least one head in 3 tosses: 1 − P(no heads) = 1 − 1/8 = **7/8**.
3. A king or a heart from 52 cards: (4 + 13 − 1)/52 = 16/52 = **4/13** (the king of hearts is counted once).

### "Or" means add, but subtract the overlap

P(A or B) = P(A) + P(B) − P(A and B).

# Chapter 35: Geometry and Mensuration

## 35.1 Formula sheet

| Shape | Area / Volume | Perimeter / Surface area |
|---|---|---|
| Rectangle l × b | lb | 2(l + b) |
| Square side a | $a^2$ | 4a; diagonal $a\sqrt{2}$ |
| Triangle | ½ × base × height; Heron: $\sqrt{s(s-a)(s-b)(s-c)}$ | a + b + c |
| Circle radius r | $\pi r^2$ | $2\pi r$ |
| Cuboid l × b × h | lbh | 2(lb + bh + hl) |
| Cube side a | $a^3$ | $6a^2$ |
| Cylinder r, h | $\pi r^2 h$ | curved $2\pi rh$; total $2\pi r(r + h)$ |
| Cone r, h, slant l | $\frac{1}{3}\pi r^2 h$ | curved $\pi rl$; $l = \sqrt{r^2 + h^2}$ |
| Sphere r | $4/3\ \pi r^3$ | $4\pi r^2$ |

## 35.2 Worked examples (π = 22/7 where needed)

### Mensuration

1. Rectangle 24 m × 15 m: area **360 m²**, perimeter **78 m**.
2. Circle of radius 14 cm: area 22/7 × 196 = **616 cm²**; circumference 2 × 22/7 × 14 = **88 cm**.
3. Triangle with sides 13, 14, 15: s = 21; area = $\sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{7{,}056} =$ **84**.
4. Cylinder r = 7, h = 10: volume 22/7 × 49 × 10 = **1,540 cm³**; curved surface 2 × 22/7 × 7 × 10 = **440 cm²**.
5. Sphere r = 21: volume 4/3 × 22/7 × 9,261 = **38,808 cm³**.
6. Right triangle with legs 9 and 12: hypotenuse $\sqrt{(81 + 144)} = \sqrt{225} =$ **15**.

### Pythagorean triples to memorise

(3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25), (20, 21, 29) — and their multiples, such as (9, 12, 15) and (6, 8, 10).

### Choose π = 22/7 when the radius is a multiple of 7

The 7 cancels, and the arithmetic stays exact.

### Chapters 30–35

- D = S × T; km/h × 5/18 = m/s; average speed for equal distances = 2ab/(a + b).
- Trains: add lengths when crossing a long object; relative speed adds (opposite) or subtracts (same).
- Alligation: (dearer − mean) : (mean − cheaper).
- Partnership: capital × time.
- Remainders multiply; unit digits cycle with period 1, 2 or 4.

- ${}^nP_r$ for arrangements, ${}^nC_r$ for selections; P(not A) = 1 − P(A).

## ✎ Chapters 30–35 practice

**289.** A car travels at 72 km/h for 3 hours. Find the distance, and the speed in m/s.

**290.** Two trains 150 m and 130 m long run at 54 km/h and 54 km/h in opposite directions. How long do they take to cross each other?

**291.** A boat goes downstream at 13 km/h and upstream at 7 km/h. Find the speed of the boat in still water and of the stream.

**292.** In what ratio should rice at ₹50/kg be mixed with rice at ₹90/kg to get a mixture worth ₹62/kg?

**293.** A invests ₹30,000 for 12 months and B invests ₹60,000 for 9 months. Profit is ₹7,330. Find A's share.

**294.** A father is 4 times as old as his son. In 3 years the sum of their ages will be 36. Find their present ages.

**295.** Find the last digit of $17^{313}$.

**296.** In how many ways can a committee of 3 be chosen from 12 people?

**297.** A bag has 5 red and 6 blue balls. Two are drawn at random. Probability both are red?

**298.** A rectangular field is 44 m by 34 m. Find its area and perimeter.

**299.** A car travels at 60 km/h for 3 hours. Find the distance, and the speed in m/s.

**300.** Two trains 120 m and 160 m long run at 90 km/h and 45 km/h in the same direction. How long do they take to cross each other?

**301.** A boat goes downstream at 17 km/h and upstream at 15 km/h. Find the speed of the boat in still water and of the stream.

**302.** In what ratio should rice at ₹20/kg be mixed with rice at ₹90/kg to get a mixture worth ₹29/kg?

**303.** A invests ₹50,000 for 12 months and B invests ₹45,000 for 10 months. Profit is ₹19,600. Find A's share.

**304.** A father is 2 times as old as his son. In 4 years the sum of their ages will be 50. Find their present ages.

**305.** Find the last digit of $22^{608}$.

**306.** In how many ways can a committee of 4 be chosen from 7 people?

**307.** A bag has 5 red and 4 blue balls. Two are drawn at random. Probability both are red?

**308.** Volume of a cylinder with radius 21 cm and height 12 cm (π = 22/7)?

**309.** A car travels at 60 km/h for 2 hours. Find the distance, and the speed in m/s.

**310.** Two trains 240 m and 160 m long run at 90 km/h and 36 km/h in opposite directions. How long do they take to cross each other?

**311.** A boat goes downstream at 22 km/h and upstream at 16 km/h. Find the speed of the boat in still water and of the stream.

**312.** In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹46/kg?

*Answers: see the answer key (questions 289–312).*

# Chapters 30–35 answer key

**289.** **216 km; 20 m/s** — D = S × T = 72 × 3 = 216 km. km/h → m/s: × 5/18 → 20 m/s.

**290.** **9 1/3 s** — Relative speed = 108 km/h = 30 m/s. Distance = 150 + 130 = 280 m. Time = 9 1/3 s.

**291.** **Boat 10 km/h, stream 3 km/h** — Boat = (down + up)/2 = (13 + 7)/2 = 10. Stream = (down − up)/2 = 3.

**292.** **7 : 3** — Alligation cross: (90 − 62) : (62 − 50) = 28 : 12 = 7 : 3.

**293.** **₹2,932** — Ratio = 30,000×12 : 60,000×9 = 2 : 3. A's share = 2/5 × 7,330 = 2,932.

**294.** **Son 6, father 24** — Let son = x, father = 4x. (4x + 3) + (x + 3) = 36 → 5x = 30 → x = 6.

**295.** **7** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 313 mod 4 = 1 → 7.

**296.** **220** — Order does not matter: ${}^{n}C_{r}$ = 12!/(3!·9!) = 220.

**297.** **2/11** — C(red, 2)/C(total, 2) = 10/55 = 2/11.

**298.** **Area 1,496 m², perimeter 156 m** — A = l × w; P = 2(l + w).

**299.** **180 km; 16 2/3 m/s** — D = S × T = 60 × 3 = 180 km. km/h → m/s: × 5/18 → 16 2/3 m/s.

**300.** **22 2/5 s** — Relative speed = 45 km/h = 12 1/2 m/s. Distance = 120 + 160 = 280 m. Time = 22 2/5 s.

**301.** **Boat 16 km/h, stream 1 km/h** — Boat = (down + up)/2 = (17 + 15)/2 = 16. Stream = (down − up)/2 = 1.

**302.** **61 : 9** — Alligation cross: (90 − 29) : (29 − 20) = 61 : 9 = 61 : 9.

**303.** **₹11,200** — Ratio = 50,000×12 : 45,000×10 = 4 : 3. A's share = 4/7 × 19,600 = 11,200.

**304.** **Son 14, father 28** — Let son = x, father = 2x. (2x + 4) + (x + 4) = 50 → 3x = 42 → x = 14.

**305.** **6** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 608 mod 4 = 4 → 6.

**306.** **35** — Order does not matter: ${}^{n}C_{r}$ = 7!/(4!·3!) = 35.

**307.** **5/18** — C(red, 2)/C(total, 2) = 10/36 = 5/18.

**308.** **16,632 cm³** — V = πr²h = 22/7 × 441 × 12 = 16,632.

**309.** **120 km; 16 2/3 m/s** — D = S × T = 60 × 2 = 120 km. km/h → m/s: × 5/18 → 16 2/3 m/s.

**310.** **11 3/7 s** — Relative speed = 126 km/h = 35 m/s. Distance = 240 + 160 = 400 m. Time = 11 3/7 s.

**311.** **Boat 19 km/h, stream 3 km/h** — Boat = (down + up)/2 = (22 + 16)/2 = 19. Stream = (down − up)/2 = 3.

**312.** **7 : 8** — Alligation cross: (60 − 46) : (46 − 30) = 14 : 16 = 7 : 8.

# Part IX

## Mental Maths Tricks

This part collects everyday mental tricks. Each trick is labelled either **Exact**, meaning it always gives the true answer, or **Approximate**, meaning it gives a quick estimate that you must not present as exact.

> **Exact vs approximate**
> Mixing these up is the biggest danger with "tricks". Exam options are often designed so that a careless estimate falls on a wrong option. Estimate to **eliminate** options, then compute exactly if two options are close.

# Chapter 36: Estimation and Approximation

## 36.1 Rounding to friendly numbers — Approximate

### Rounding products

4,987 × 31 ≈ 5,000 × 31 = 1,55,000 (exact: 1,54,597; error about 0.3%). 39.98 × 25.03 ≈ 40 × 25 = 1,000 (exact 1,000.6994).

### Compensate when you round both numbers

If you round one number **up** and the other **down**, errors partly cancel. Rounding both in the same direction doubles the error.

## 36.2 Estimating quotients and roots — Approximate

- 7,215 ÷ 59 ≈ 7,200 ÷ 60 = 120 (exact 122.28…).
- √500 lies between 22 (484) and 23 (529), much closer to 22 → about 22.4 (true value ≈ 22.36).
- ∛1,000 is exactly 10, so ∛1,100 is a little more than 10 → about 10.3.

## 36.3 Order-of-magnitude check — Approximate

Before writing any answer, ask: "Roughly how big should this be?" 312 × 48 must be near 300 × 50 = 15,000, so 1,497.6 or 1,49,760 are clearly wrong. (The exact answer is 14,976.)

# Chapter 37: Doubling, Halving and Re-grouping

## 37.1 Double and halve — Exact

**Formula**

a × b = (a ÷ 2) × (b × 2). Halve one factor and double the other until one becomes easy.

**Worked example**

- 16 × 25 = 8 × 50 = 4 × 100 = **400**
- 35 × 14 = 70 × 7 = **490**
- 48 × 125 = 24 × 250 = 12 × 500 = 6 × 1,000 = **6,000**
- 4.5 × 18 = 9 × 9 = **81**

## 37.2 Factor re-grouping — Exact

- 25 × 36 × 4 = (25 × 4) × 36 = **3,600**
- 15 × 28 = 15 × 2 × 14 = 30 × 14 = **420**
- 125 × 64 = 125 × 8 × 8 = 1,000 × 8 = **8,000**

## 37.3 Multiplying by 4, 8, 16 — Exact

Double repeatedly: 37 × 8 → 74 → 148 → **296**.

# Chapter 38: Percentage Tricks

## 38.1 The swap rule — Exact

> **Formula**
> **x% of y = y% of x.** 8% of 25 = 25% of 8 = **2**. 16% of 50 = 50% of 16 = **8**.

## 38.2 Build from 10%, 5% and 1% — Exact

18% of 250 = 10% (25) + 5% (12.5) + 3 × 1% (7.5) = **45**.

## 38.3 Fraction equivalents — Exact

| Percentage | Fraction | Example |
|---|---|---|
| 12.5% | 1/8 | 12.5% of 640 = 80 |
| 16.67% | 1/6 (approx. %) | 1/6 of 360 = 60 |
| 33.33% | 1/3 (approx. %) | 1/3 of 2,700 = 900 |
| 37.5% | 3/8 | 3/8 of 960 = 360 |
| 62.5% | 5/8 | 5/8 of 480 = 300 |
| 87.5% | 7/8 | 7/8 of 800 = 700 |

> **About 16.67% and 33.33%**
> 16.67% is a rounded form of 16 2/3 %. Using the fraction 1/6 is **exact**; multiplying by 0.1667 is **approximate**.

## 38.4 Approximate percentages — Approximate

23.8% of 4,012 ≈ 24% of 4,000 = 960 (exact ≈ 954.9). Use only for elimination.

# Chapter 39: Calendar Calculations — Exact

## 39.1 Odd days

An **odd day** is a day left over after complete weeks. Day numbers: Sunday 0, Monday 1, … Saturday 6.

### Odd-day facts

- Ordinary year = 365 days = 52 weeks + **1** odd day; leap year = **2** odd days.
- 100 years = **5** odd days; 200 years = **3**; 300 years = **1**; 400 years = **0**.
- Month odd days: Jan 3, Feb 0 (1 in a leap year), Mar 3, Apr 2, May 3, Jun 2, Jul 3, Aug 3, Sep 2, Oct 3, Nov 2, Dec 3.
- Leap year: divisible by 4, but century years only if divisible by 400 (1900 no, 2000 yes).

### What day was 15 August 1947?

- 1600 years: 0 odd days.
- Years 1601–1900 (300 years): 1.
- Years 1901–1946 (46 years, 11 leap years): 46 + 11 = 57, and 57 leaves remainder 1 after dividing by 7. So 1.
- 1947 up to 15 August: 3 + 0 + 3 + 2 + 3 + 2 + 3 + 15, which is 31, leaving remainder 3. So 3.
- Total 0 + 1 + 1 + 3 = 5 → **Friday**.

### Same date, later year

26 January 2025 was a Sunday. 26 January 2026 is 365 days later → 1 odd day → **Monday**. (2026 is not a leap year, and no 29 February lies between them.)

# Chapter 40: Divisibility and Last-Digit Tricks

## 40.1 Divisibility by 7 and 13 with an osculator — Exact

### Ekadhika osculator

Find a multiple of the divisor ending in 9: $7 \times 7 = 49$ and $13 \times 3 = 39$. "One more than" the digits before the 9 gives the **osculator**: 5 for 7, and 4 for 13. Repeatedly: (number without last digit) + (last digit × osculator). If the result is divisible, so is the original number.

### Worked example

- Is 2,401 divisible by 7? $240 + 1 \times 5 = 245 \rightarrow 24 + 5 \times 5 = 49 \rightarrow$ yes. ($2{,}401 = 7^4$.)
- Is 1,001 divisible by 13? $100 + 1 \times 4 = 104 \rightarrow 10 + 4 \times 4 = 26 \rightarrow$ yes.

## 40.2 Last digit of a power — Exact

### Last digit of $3^{47}$

Powers of 3 end in 3, 9, 7, 1 (cycle 4). 47 leaves remainder 3 after dividing by 4 → third in cycle → **7**.

### Digits that never change

Any power of a number ending in 0, 1, 5 or 6 ends in the same digit. Numbers ending in 76 keep ending in 76 when raised to any power.

## 40.3 Perfect-square screening — Exact (as a rejection test)

A perfect square never ends in 2, 3, 7 or 8, and its digital root is always 1, 4, 7 or 9. So 3,528 cannot be a perfect square. **Passing** the test does not prove a number is a square.

# Chapter 41: Comparing Fractions and Quick Checks

## 41.1 Cross-multiplication comparison — Exact

To compare 7/9 and 11/14: compare $7 \times 14 = 98$ with $9 \times 11 = 99$. $98 < 99$, so **11/14 is larger**.

## 41.2 Digit-sum check — Exact as a rejection test

If the digit-sum check fails, the answer is definitely wrong. If it passes, the answer is *probably* right (swapped digits are not detected). See Chapter 12.

## 41.3 Units-digit check — Exact as a rejection test

The units digit of $47 \times 83$ must be the units digit of $7 \times 3 = 21 \rightarrow 1$. An option ending in 3 is wrong.

### Part IX summary

- Label every trick Exact or Approximate.
- Approximate: rounding, estimates for elimination only.
- Exact: double and halve, re-grouping, x% of y = y% of x, fraction equivalents, odd-days calendar, osculator divisibility, power cycles.
- Rejection tests (digit sums, units digits, square endings) can prove an answer wrong but not right.

### ✎ Part IX practice

**313.** Estimate $89 \times 326$ by rounding (approximate answer). *(approximate)*

**314.** Find 120% of 7,200.

**315.** Write 1/12 as a percentage.

**316.** What day of the week was/is 24 February 1963?

**317.** Find the last digit of $27^{771}$.

**318.** Is 60,744 divisible by 3? Use the divisibility rule.

**319.** Approximate: $982.98 \times 39.98 \div 10 + 812.9$ *(approximate)*

**320.** $13 \times 3$

**321.** Estimate $68 \times 775$ by rounding (approximate answer). *(approximate)*

**322.** Find 45% of 640.

**323.** Write 1/25 as a percentage.

**324.** What day of the week was/is 21 December 1995?

**325.** Find the last digit of $38^{354}$.

**326.** Is 96,399 divisible by 3? Use the divisibility rule.

**327.** Approximate: $448.03 \times 13.98 \div 10 + 632.9$ *(approximate)*

**328.** $16 \times 3$

**329.** Estimate $29 \times 370$ by rounding (approximate answer). *(approximate)*

**330.** Find 20% of 3,600.

**331.** Write 2/5 as a percentage.

**332.** What day of the week was/is 7 July 1984?

*Answers: see the answer key (questions 313–332).*

## Part IX answer key

**313.** **≈ 29,700 (exact 29,014)** — 89 ≈ 90, 326 ≈ 330; 90 × 330 = 29,700. This is an estimate, not the exact product.

**314.** **8,640** — Split 120% into easy parts (10%, 5%, 1% …) or use fractions: 120% = 6/5. Result 8,640.

**315.** **8 1/3%** — 1/12 × 100 = 8 1/3%.

**316.** **Sunday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**317.** **3** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 771 mod 4 = 3 → 3.

**318.** **Yes** — Rule for 3: digit sum divisible by 3. 60,744 ÷ 3 leaves remainder 0.

**319.** **≈ 4745 (exact 4742.85)** — Round each number: 983 × 40 ÷ 10 + 813 ≈ 4745.0.

**320.** **39** — (10 + 3) × 3 = 30 + 9 = 39.

**321.** **≈ 56,000 (exact 52,700)** — 68 ≈ 70, 775 ≈ 800; 70 × 800 = 56,000. This is an estimate, not the exact product.

**322.** **288** — Split 45% into easy parts (10%, 5%, 1% …) or use fractions: 45% = 9/20. Result 288.

**323.** **4%** — 1/25 × 100 = 4%.

**324.** **Thursday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**325.** **4** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 354 mod 4 = 2 → 4.

**326.** **Yes** — Rule for 3: digit sum divisible by 3. 96,399 ÷ 3 leaves remainder 0.

**327.** **≈ 1260 (exact 1259.25)** — Round each number: 448 × 14 ÷ 10 + 633 ≈ 1260.2.

**328.** **48** — (10 + 6) × 3 = 30 + 18 = 48.

**329.** **≈ 11,100 (exact 10,730)** — 29 ≈ 30, 370 ≈ 370; 30 × 370 = 11,100. This is an estimate, not the exact product.

**330.** **720** — Split 20% into easy parts (10%, 5%, 1% …) or use fractions: 20% = 1/5. Result 720.

**331.** **40%** — 2/5 × 100 = 40%.

**332.** **Saturday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

# Part X

## Mental Maths for Children (Ages 5–12)

This part is written for **parents and teachers**, with child-facing pages they can read together. It builds number sense first and shortcuts later. A child who understands *why* 9 + 6 = 15 will learn the shortcuts easily; a child who only memorises tricks will forget them.

### Five rules for parents

1. **Short and daily beats long and weekly.** 10–15 minutes a day is ideal.
2. **Use objects before symbols.** Beads, coins, blocks and fingers come before written sums.
3. **Praise the method, not just the answer.** "I like how you made a ten first!"
4. **Never rush or time young children.** Speed comes from understanding, not pressure.
5. **Stop while it is still fun.** End every session with a success.

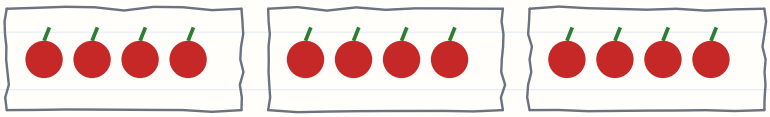
### Avoid comparison and anxiety

Do not compare your child with siblings or classmates. Maths anxiety, once it starts, is hard to undo. Mistakes are a normal part of learning. Say "Let's check it together", not "Wrong!"

# Chapter 42: Level 1 — Ages 5 to 7 (Number Sense)

## 42.1 Counting and number recognition

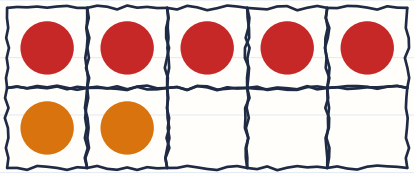
- Count real objects, touching each one once.
- Count forwards and backwards from any number (from 7 back to 0).
- Recognise small groups without counting (dice patterns, fingers). This skill is called **subitising**.



3 × 4 = 12

"*How many apples are in each basket? How many baskets? How many altogether?*" (3 groups of 4 make 12.)

## 42.2 Ten frames and number bonds

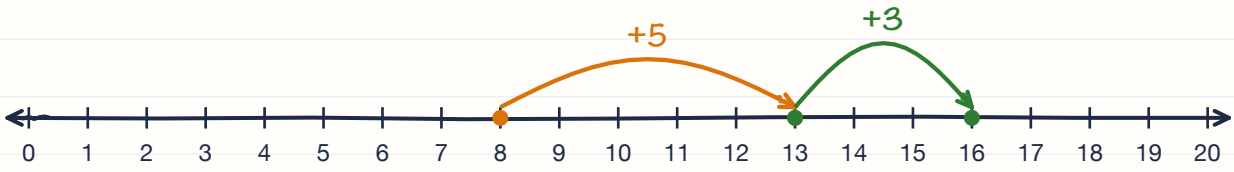


7 + 3 = 10

A ten frame shows how far a number is from ten. 7 dots leave **3 empty spaces**, so 7 and 3 make 10. This is the beginning of *Nikhilam* thinking: "how far from 10?"

> **Number bonds of 10 — the most important facts at this level**
> 1 + 9 · 2 + 8 · 3 + 7 · 4 + 6 · 5 + 5. Sing them, clap them, play "snap" with them.

## 42.3 Adding and subtracting within 20




> **Make a ten (child version)**
> 8 + 5: "8 needs 2 to make 10. Take 2 from the 5. Now 10 + 3 = 13."

> **Child-friendly practice**
> - 9 + 4: 9 needs 1 → 10 + 3 = **13**
> - 7 + 6: 7 needs 3 → 10 + 3 = **13**
> - 15 − 7: 15 − 5 = 10, then 2 more → **8**

- 12 – 9: from 9 up to 12 is **3** (count up!)

## 42.4 Doubles and halves

Doubles: 1 + 1, 2 + 2, … 10 + 10. **Near doubles**: 6 + 7 is double 6 plus 1, which gives **13**.

## 42.5 Patterns and shapes

- Continue the pattern: 2, 4, 6, 8, __ (10).
- Spot shapes around the house: circles, squares, triangles, rectangles.

### Level I games

- **Bond snap:** say a number; the child says its partner to 10.
- **Shop:** price toys at ₹1–₹10 and pay with real coins.
- **Stairs counting:** count up and down stairs in 1s and 2s.
- **Dice race:** roll two dice and add; the first to 50 wins.

# Chapter 43: Level 2 — Ages 8 to 10 (Fluency)

## 43.1 Place value and bigger numbers

Use bundles of ten sticks and a place-value chart (Chapter 1). Read 4,07,315 aloud as "four lakh seven thousand three hundred fifteen".

## 43.2 Mental addition and subtraction to 1,000

- Add left to right: 346 + 237 = 500 + 70 + 13 = **583**.
- Round and adjust: 198 + 456 = 200 + 456 − 2 = **654**.
- Subtract from 100 and 1,000 with **"all from 9 and the last from 10"**: 1,000 − 368 = **632**.

## 43.3 Times tables — understanding before memorising




An array shows that 4 × 6 and 6 × 4 are the same rectangle turned around, so the child learns **half** as many facts.

> **Learn tables in this order**
> ×2, ×10, ×5 (half of ×10), ×4 (double ×2), ×9 (×10 minus one group), ×3, ×6 (double ×3), ×8 (double ×4), ×7 last. By then, only 7 × 7 is genuinely new.

> **The 9 times table finger trick**
> Hold up 10 fingers. For 9 × 4, fold down the 4th finger: 3 fingers on the left, 6 on the right → **36**. Works for 9 × 1 to 9 × 10.

## 43.4 First Vedic shortcuts

- **×11** for 2-digit numbers: for 34 × 11, write 3, then 3 + 4 = 7 in the middle, then 4 → **374**.
- **Squares ending in 5**: $35^2$ = (3 × 4) | 25 = **1,225**.
- **×5**: halve and multiply by 10: 46 × 5 = **230**.

## 43.5 Fractions with pictures




3 of 4 equal parts are shaded

"Three of the four equal parts are coloured." Always draw before writing fractions. Pizza, roti and chocolate bars are excellent models.

## 43.6 Money and time

### Money and time

- A pen costs ₹18 and a notebook ₹35. Total ₹**53**. Change from ₹100: ₹**47**.
- School starts at 8:45 and ends at 2:15. That is **5 hours 30 minutes**.

### Level 2 games

- **Bus stop maths:** add the digits of number plates.
- **Price guess:** at the shop, estimate the bill before paying.
- **Table bingo:** call out products; the child marks the matching multiplication.
- **Magic squares:** see Chapter 44.

# Chapter 44: Level 3 — Ages 11 to 12 (Confidence and Problem Solving)

## 44.1 Two-digit multiplication

Teach both the standard column method and **vertically and crosswise** (Chapter 10). Let the child choose the one they prefer, and check with the other.

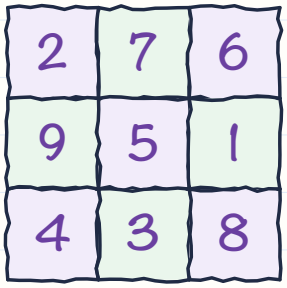
## 44.2 Base methods near 10 and 100

97 × 96: both are below 100 by 3 and 4. 97 − 4 = 93; 3 × 4 = 12 → **9,312**.

## 44.3 Fractions, decimals and percentages

Memorise the halves, quarters, fifths and eighths chart (Chapter 3). 25% of 60 is a quarter of 60, which is **15**.

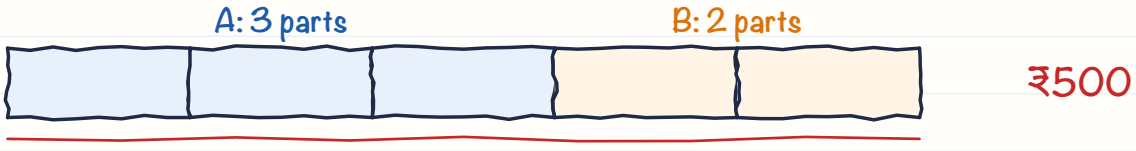
## 44.4 Puzzles and logic




Every row, column and diagonal adds to 15

Every row, column and diagonal adds to **15**. Challenge: make a magic square with 2 to 10 (each line will add to 18).

## 44.5 Bar models for word problems




5 equal parts in all

"*Riya and Arjun share ₹500 in the ratio 3 : 2.*" Five equal parts → one part ₹100 → Riya **₹300**, Arjun **₹200**.

### What each level should achieve

- **Level 1:** count confidently, bonds of 10, add and subtract within 20.
- **Level 2:** place value to lakhs, mental add/subtract to 1,000, all tables to 10 × 10, first shortcuts.
- **Level 3:** 2-digit multiplication two ways, base methods, fraction–percentage chart, bar models.

# Chapter 45: Daily Plans, Rewards and Progress

## 45.1 A 15-minute daily routine

| Minutes | Activity |
|---|---|
| 0–3 | Warm-up game (bonds, doubles, or tables) |
| 3–8 | One new idea with objects or a picture |
| 8–12 | Practice (5–8 questions) |
| 12–15 | A puzzle or game; finish on a success |

## 45.2 A 4-week starter plan (Level 2)

| Week | Focus | Goal by the end of the week |
|---|---|---|
| 1 | Bonds of 10 and 100; make-a-ten | 20 bond questions in 2 minutes, no stress |
| 2 | Left-to-right addition; round and adjust | add two 3-digit numbers mentally |
| 3 | Tables ×2, ×5, ×10, ×4, ×9 | answer any of these in 3 seconds |
| 4 | ×11 and squares ending in 5 | explain *why* the trick works using an array |

## 45.3 Rewards that work

- Sticker charts for **effort** (practised today), not only for speed.
- A "maths champion" certificate after every completed week.
- Let the child **teach** a trick to a family member. Teaching is the best reward and the best revision.

## 45.4 Progress tracker

| Skill | Not yet | Getting there | Confident |
|---|---|---|---|
| Bonds of 10 | | | |
| Add within 20 | | | |
| Tables to 10 × 10 | | | |
| Mental add/subtract to 1,000 | | | |
| ×11 and squares ending in 5 | | | |
| Fractions with pictures | | | |
| Word problems with bar models | | | |

## 45.5 Sample worksheets

The separate **Children's Worksheets** PDF contains printable pages for all three levels, with writing space and answer keys. A sample follows.

## ✎ Children's sample worksheet

**333.** 8 + □ = 10

**334.** Riya has 3 toy cars. Her friend gives her 6 more. How many toy cars does she have now?

**335.** There are 21 birds on a tree. 8 are taken away. How many are left?

**336.** Find the missing number: 1, □, 9, 13, 17, 21

**337.** Half of 44

**338.** 6 × 12

**339.** A toy costs ₹65. You pay with a ₹100 note. How much change do you get?

**340.** 24 sweets are shared equally among 6 children. How many does each child get, and how many are left over?

**341.** 94 + □ = 100

**342.** Riya has 6 mangoes. Her friend gives her 9 more. How many mangoes does she have now?

**343.** There are 20 apples in a basket. 16 are taken away. How many are left?

**344.** Find the missing number: 13, 16, 19, □, 25, 28

**345.** Double 13

**346.** 10 × 7

**347.** A toy costs ₹42. You pay with a ₹100 note. How much change do you get?

**348.** 49 sweets are shared equally among 5 children. How many does each child get, and how many are left over?

**349.** 3 + □ = 10

**350.** Riya has 7 marbles. Her friend gives her 10 more. How many marbles does she have now?

**351.** There are 18 birds on a tree. 7 are taken away. How many are left?

**352.** Find the missing number: 9, □, 15, 18, 21, 24

**353.** Half of 104

**354.** 6 × 2

**355.** 27 sweets are shared equally among 4 children. How many does each child get, and how many are left over?

**356.** 6 + □ = 10

*Answers: see the answer key (questions 333–356).*

## Children's worksheet answers

**333.** **2** — Count on from 8 to 10, or think: which number makes 10 with 8?

**334.** **9** — Start at 3 and count on 6: 3 + 6 = 9.

**335.** **13** — On a number line, start at 21 and jump back 8: 21 − 8 = 13.

**336.** **5** — The numbers go up by 4 each time.

**337.** **22** — Split 44 into two equal groups.

**338.** **72** — 6 groups of 12 = 72.

**339.** **₹35** — Count up from 65 to 100: 35.

**340.** **4 each, 0 left over** — 6 × 4 = 24; 24 − 24 = 0.

**341.** **6** — Count on from 94 to 100, or think: which number makes 100 with 94?

**342.** **15** — Start at 6 and count on 9: 6 + 9 = 15.

**343.** **4** — On a number line, start at 20 and jump back 16: 20 − 16 = 4.

**344.** **22** — The numbers go up by 3 each time.

**345.** **26** — Double = add the number to itself: 13 + 13 = 26.

**346.** **70** — 10 groups of 7 = 70.

**347.** **₹58** — Count up from 42 to 100: 58.

**348.** **9 each, 4 left over** — 5 × 9 = 45; 49 − 45 = 4.

**349.** **7** — Count on from 3 to 10, or think: which number makes 10 with 3?

**350.** **17** — Start at 7 and count on 10: 7 + 10 = 17.

**351.** **11** — On a number line, start at 18 and jump back 7: 18 − 7 = 11.

**352.** **12** — The numbers go up by 3 each time.

**353.** **52** — Split 104 into two equal groups.

**354.** **12** — 6 groups of 2 = 12.

**355.** **6 each, 3 left over** — 4 × 6 = 24; 27 − 24 = 3.

**356.** **4** — Count on from 6 to 10, or think: which number makes 10 with 6?

# Part XI

## Vedic Maths for Government Exams

This part shows how to apply mental maths to the quantitative sections of Indian competitive exams, such as SSC (CGL, CHSL, MTS, CPO), Banking (IBPS and SBI PO and Clerk, RRB), Railways (RRB NTPC and Group D), state PSCs, police recruitment, and defence entrance tests.

### About exam patterns and questions

- Exam patterns, syllabi, marks and negative marking **change from year to year**. Always read the **official notification** for your exam.
- Every question in this book is an **original question written in the style of** these exams. None is presented as a previous-year question.
- No method can guarantee selection. Speed comes from understanding plus regular, timed practice.

# Chapter 46: How Quantitative Sections Work

## 46.1 Common question families

| Family | Typical in | Most useful skills |
|---|---|---|
| Simplification and approximation | Banking prelims, SSC, Railways | BODMAS, fraction chart, squares, rounding |
| Number series | Banking | differences, squares/cubes, ×n ± k patterns |
| Quadratic comparison | Banking | factorising quadratics quickly |
| Arithmetic word problems | All | percentages, ratio, profit, interest, work, speed |
| Data interpretation (DI) | Banking mains, SSC CGL | percentages, ratios, averages, estimation |
| Number system, algebra, geometry, trigonometry | SSC CGL, CPO | identities, remainders, mensuration formulas |

## 46.2 Where Vedic methods genuinely help

- **Calculation-heavy steps**: multiplying 2–3-digit numbers, squaring, dividing by numbers near a base.
- **Simplification**: × 11, × 5/25/125, near-base products, squares of numbers near 50/100.
- **Checking**: digit-sum and units-digit checks to eliminate options.
- **DI**: percentage ↔ fraction conversions and quick estimation.

## 46.3 Where they do not help much

- Understanding the word problem (that needs reading and modelling).
- Geometry and trigonometry (mostly formulas and properties).
- Reasoning sections.

> ### Choosing method versus speed
> Use a shortcut only if you can recognise it **within 2–3 seconds**. Otherwise do the standard method you trust. A wrong fast answer costs marks if there is negative marking.

# Chapter 47: Simplification and Approximation

## 47.1 Simplification (exact)

### Worked example

**25% of 480 + 3/4 of 160 − √196 = ?** 25% of 480 is 120; 3/4 of 160 is 120; √196 is 14. Answer: 120 + 120 − 14 = **226**.

### Worked example

**(48 × 52) ÷ 13 + 15² = ?** 48 × 52 = 2,500 − 4 = 2,496 (difference of squares). 2,496 ÷ 13 = 192. $15^2$ = 225. Answer: **417**.

## 47.2 Approximation questions

Answer options are usually far apart. Round **each** number to the nearest convenient value.

### Approximate method

**299.8 ÷ 5.02 + 11.97² ≈ ?** Use 300 ÷ 5 = 60 and $12^2$ = 144 → **≈ 204**.

### Do not over-round

Rounding 11.97 to 10 instead of 12 changes 144 into 100, enough to pick the wrong option. Round to the **nearest** easy number, not the nearest ten.

# Chapter 48: Number Series and Quadratic Comparison

## 48.1 Number series: a checklist

1. Find the differences. Are they constant, increasing steadily, or doubling?
2. Try ×2, ×3 with ± a constant.
3. Look for squares/cubes ($n^2$, $n^2 \pm 1$, $n^3$).
4. Look for two interleaved series (odd and even positions).

### Number series

- 3, 7, 15, 31, 63, ? → each term × 2 + 1 → **127**.
- 2, 6, 12, 20, 30, ? → n(n + 1): 1·2, 2·3, … → **42**.
- 5, 11, 23, 47, ? → × 2 + 1 → **95**.
- 1, 8, 27, 64, ? → cubes → **125**.
- 120, 60, 20, 5, ? → ÷ 2, ÷ 3, ÷ 4, ÷ 5 → **1**.

## 48.2 Quadratic comparison

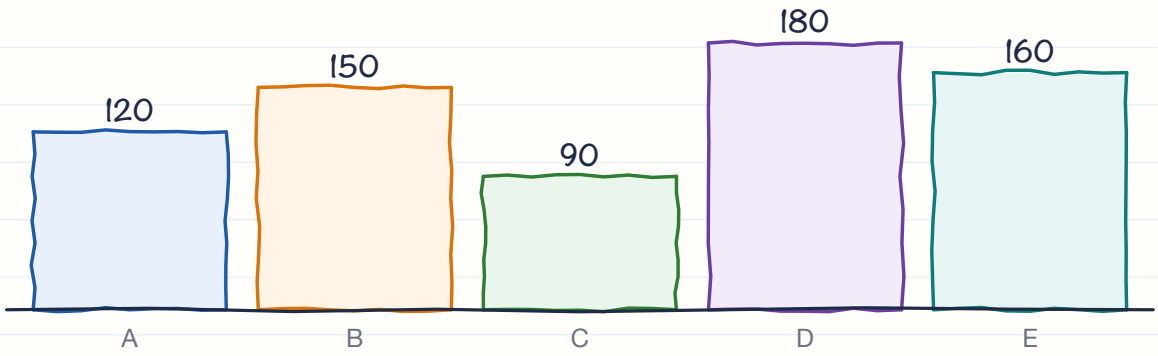
### Worked example

I: $x^2 - 7x + 12 = 0 \rightarrow x = 3$ or 4. II: $y^2 - 9y + 20 = 0 \rightarrow y = 4$ or 5. Compare every pair: $3 < 4$, $3 < 5$, $4 = 4$, $4 < 5$ → **x ≤ y**.

### Sign shortcut for quadratic roots

For $x^2 + bx + c = 0$: if b and c are both positive, both roots are negative. If b is negative and c positive, both roots are positive. If c is negative, the roots have opposite signs.

# Chapter 49: Data Interpretation




### Sales (₹ lakh) of five companies, shown above

1. Average sales = (120 + 150 + 90 + 180 + 160) ÷ 5 = 700 ÷ 5 = **140**.
2. By what percentage are D's sales more than C's? (180 − 90)/90 × 100 = **100%**.
3. Ratio of B to E = 150 : 160 = **15 : 16**.
4. C's share of the total = 90/700 ≈ **12.9%** (≈ 1/8, since 1/8 = 12.5%).

## 49.1 A table-based set

Production of a factory (thousand units):

| Year | Cars | Bikes |
|---|---|---|
| 2021 | 40 | 120 |
| 2022 | 45 | 150 |
| 2023 | 54 | 135 |
| 2024 | 60 | 162 |

### Questions on the table

1. Percentage increase in car production from 2021 to 2024: (60 − 40)/40 × 100 = **50%**.
2. Percentage change in bike production from 2022 to 2023: (135 − 150)/150 × 100 = **−10%** (a 10% fall).
3. Average bike production: 567 ÷ 4 = **141.75** thousand.
4. Ratio of total production in 2023 to 2021: 189 : 160.
5. In which year did car production grow fastest? Growth was 5/40 = 12.5% in 2022, 9/45 = **20% in 2023**, and 6/54 ≈ 11.1% in 2024. **Answer: 2023.**

### Largest increase vs largest percentage increase

The biggest jump in *units* and the biggest jump in *percentage* can happen in different years. Read which one the question asks for.

### DI routine

Read the title, units and scale **first**. Convert percentages into fractions (Chapter 3). Estimate before you calculate, and calculate exactly only when two options are close.

# Chapter 50: Speed Strategy, Traps and Time Management

## 50.1 The two-pass method

1. **Pass 1:** answer every question you can do in under 45 seconds. Mark the rest.
2. **Pass 2:** return to marked questions in order of confidence.
3. **Never** spend more than about 2 minutes on one question in a speed test.

## 50.2 Option elimination

- **Units digit:** 47 × 83 must end in 1.
- **Digital root:** the digital root of the answer must match the digital root of the calculation.
- **Size:** an estimate removes options of the wrong magnitude.
- **Parity and divisibility:** an answer to "how many students" must be a whole number.

## 50.3 Ten common traps

| Trap | How to avoid it |
|---|---|
| Profit % taken on SP instead of CP | Profit % is on CP unless stated |
| Averaging speeds directly | Use total distance ÷ total time |
| Forgetting 2ab in $(a + b)^2$ | Draw the area square mentally |
| Successive % treated as additive | Use a + b + ab/100 |
| Wrong units (km/h vs m/s) | Convert first, × 5/18 |
| Ignoring "at least" | Use 1 − P(none) |
| Counting overlap twice | Subtract P(A and B) |
| Reading "less than" backwards | Write the equation in words first |
| Leap-year error in calendars | Century years must be divisible by 400 |
| Over-rounding in approximation | Round to the nearest easy number |

## 50.4 Accuracy versus speed

> **Why accuracy comes first**
> With negative marking, a 90%-accurate candidate who attempts 40 questions often scores more than a 60%-accurate candidate who attempts 60. Build accuracy first, then speed.

## 50.5 Mock-test analysis

After every mock test, fill in a table like the one below (the Excel tracker included with this book automates it).

| Question | Topic | Right / Wrong / Skipped | Time (s) | Reason for error | Fix |
|---|---|---|---|---|---|
| 7 | Profit and loss | Wrong | 95 | took profit on SP | re-read Chapter 27 |
| | | | | | |
| | | | | | |

### Part XI summary

- Check the official notification; patterns change.
- Shortcuts help with calculation and checking, not with understanding the problem.
- Simplify exactly; approximate carefully.
- Series: differences, ×n ± k, powers, interleaving.
- DI: read the units, convert % to fractions, estimate.
- Two-pass method, eliminate options, analyse every mock test.

### ✎ Exam-style practice (original questions)

**357.** Simplify: 4 + 14 ÷ 7 × 2 − 28

**358.** Approximate: 375.97 × 30.01 ÷ 10 + 528.9 *(approximate)*

**359.** A price changes from ₹2,500 to ₹1,875. Find the percentage decrease.

**360.** An item costing ₹500 is sold for ₹550. Find the profit percentage.

**361.** Find the simple interest on ₹10,000 at 4% per annum for 5 years.

**362.** Find the compound interest on ₹20,000 at 20% per annum, compounded yearly, for 2 years.

**363.** A can finish a job in 10 days and B in 15 days. How long will they take together?

**364.** A car travels at 36 km/h for 2 hours. Find the distance, and the speed in m/s.

**365.** Two trains 200 m and 160 m long run at 54 km/h and 54 km/h in opposite directions. How long do they take to cross each other?

**366.** Divide ₹425 between A and B in the ratio 2 : 3.

**367.** Volume of a sphere of radius 7 cm (π = 22/7)?

**368.** A bag has 7 red and 5 blue balls. Two are drawn at random. Probability both are red?

**369.** Simplify: 18 + 12 ÷ 4 × 3 − 10

**370.** Approximate: 673.02 × 28.98 ÷ 10 + 533.9 *(approximate)*

**371.** A price changes from ₹1,600 to ₹1,120. Find the percentage decrease.

**372.** An item costing ₹1,200 is sold for ₹960. Find the loss percentage.

**373.** Find the simple interest on ₹40,000 at 8% per annum for 2 years.

**374.** Find the compound interest on ₹80,000 at 5% per annum, compounded yearly, for 2 years.

**375.** A can finish a job in 8 days and B in 24 days. How long will they take together?

**376.** A car travels at 90 km/h for 3/4 hours. Find the distance, and the speed in m/s.

**377.** Two trains 150 m and 100 m long run at 90 km/h and 54 km/h in the same direction. How long do they take to cross each other?

**378.** Divide ₹1,833 between A and B in the ratio 4 : 9.

**379.** Volume of a cylinder with radius 7 cm and height 11 cm (π = 22/7)?

**380.** A bag has 4 red and 8 blue balls. Two are drawn at random. Probability both are red?

*Answers: see the answer key (questions 357–380).*

## Exam-style practice answers

**357.** **–20** — Division and multiplication left to right first: 14 ÷ 7 = 2; × 2 = 4. Then 4 + 4 – 28 = –20.

**358.** **≈ 1657 (exact 1657.19)** — Round each number: 376 × 30 ÷ 10 + 529 ≈ 1657.0.

**359.** **25%** — Change = 625. % change = change ÷ original × 100 = 625/2500 × 100 = 25%.

**360.** **10% profit** — Profit = 50. % = 50/500 × 100 = 10%.

**361.** **₹2,000** — SI = PRT/100 = 10,000 × 4 × 5 / 100 = 2,000.

**362.** **₹8,800** — $A = P(1 + R/100)^n$ = 20,000 × $(120/100)^2$ = ₹28,800. CI = A – P = ₹8,800.

**363.** **6 days** — Total work = LCM(10,15) = 30 units. A does 3/day, B does 2/day; together 5/day. Time = 30/5 = 6 days.

**364.** **72 km; 10 m/s** — D = S × T = 36 × 2 = 72 km. km/h → m/s: × 5/18 → 10 m/s.

**365.** **12 s** — Relative speed = 108 km/h = 30 m/s. Distance = 200 + 160 = 360 m. Time = 12 s.

**366.** **A ₹170, B ₹255** — Total parts 5; one part = 425 ÷ 5 = 85.

**367.** **1437 1/3 cm³** — $V = 4/3\ \pi r^3$ = 1437 1/3.

**368.** **7/22** — C(red, 2)/C(total, 2) = 21/66 = 7/22.

**369.** **17** — Division and multiplication left to right first: 12 ÷ 4 = 3; × 3 = 9. Then 18 + 9 – 10 = 17.

**370.** **≈ 2486 (exact 2484.31)** — Round each number: 673 × 29 ÷ 10 + 534 ≈ 2485.7.

**371.** **30%** — Change = 480. % change = change ÷ original × 100 = 480/1600 × 100 = 30%.

**372.** **20% loss** — Loss = 240. % = 240/1200 × 100 = 20%.

**373.** **₹6,400** — SI = PRT/100 = 40,000 × 8 × 2 / 100 = 6,400.

**374.** **₹8,200** — $A = P(1 + R/100)^n$ = 80,000 × $(105/100)^2$ = ₹88,200. CI = A – P = ₹8,200.

**375.** **6 days** — Total work = LCM(8,24) = 24 units. A does 3/day, B does 1/day; together 4/day. Time = 24/4 = 6 days.

**376.** **67 1/2 km; 25 m/s** — D = S × T = 90 × 3/4 = 67 1/2 km. km/h → m/s: × 5/18 → 25 m/s.

**377.** **25 s** — Relative speed = 36 km/h = 10 m/s. Distance = 150 + 100 = 250 m. Time = 25 s.

**378.** **A ₹564, B ₹1,269** — Total parts 13; one part = 1,833 ÷ 13 = 141.

**379.** **1,694 cm³** — $V = \pi r^2 h$ = 22/7 × 49 × 11 = 1,694.

**380.** **1/11** — C(red, 2)/C(total, 2) = 6/66 = 1/11.

# Part XII

## Exercise Bank

Over 400 original questions, arranged by topic and difficulty, followed by mixed drills, timed tests and challenge questions. Complete answers with worked solutions are at the end of this part.

# Chapter 51: Topic-wise Exercises, Drills and Tests

### How to use this bank

- **Topic sections A–H** each have Beginner, Intermediate and Advanced blocks. Finish the Beginner block before moving on.
- **Mixed drills (I)** combine topics, which is how real exams feel.
- **Timed tests (J):** 20 questions in 8 minutes. Write only the answer. Record your score and time in the tracker.
- **Challenge questions (K)** stretch the strongest methods in the book.
- Every answer has a worked solution in the answer key. Read the solution even when you were right; you may find a faster route.

### Marking yourself

Tick correct answers, circle mistakes, and write the **reason** for each mistake (concept, careless, slow). Repeat only the circled questions a week later.

## A. Numbers and foundations

### ✎ Numbers and foundations — Beginner

**381.** In 9,36,54,607, what is the place value of the digit in the thousands place?

**382.** Rewrite 97,48,49,451 using international commas (millions).

**383.** Is 261 prime or composite?

**384.** Is 3,101 divisible by 9? Use the divisibility rule.

**385.** Calculate 10 – 0 + 18.

**386.** How many factors does 96 have? List them.

**387.** In 4,24,70,598, what is the place value of the digit in the units place?

**388.** Rewrite 38,65,74,935 using international commas (millions).

*Answers: see the answer key (questions 381–388).*

### ✎ Numbers and foundations — Intermediate

**389.** In 3,95,86,096, what is the place value of the digit in the ten lakhs place?

**390.** Rewrite 30,32,88,428 using international commas (millions).

**391.** Is 74,790 divisible by 3? Use the divisibility rule.

**392.** Calculate 17 – 28 + 21.

**393.** In 9,14,49,917, what is the place value of the digit in the lakhs place?

**394.** Rewrite 91,05,578 using international commas (millions).

**395.** Is 337 prime or composite?

**396.** Is 87,634 divisible by 8? Use the divisibility rule.

*Answers: see the answer key (questions 389–396).*

### ✎ Numbers and foundations — Advanced

**397.** In 5,23,85,349, what is the place value of the digit in the hundreds place?

**398.** Rewrite 16,85,639 using international commas (millions).

**399.** Is 117 prime or composite?

**400.** Is 34,670 divisible by 3? Use the divisibility rule.

*Answers: see the answer key (questions 397–400).*

## B. Mental addition and subtraction

### ✎ Mental addition and subtraction — Beginner

**401.** Add mentally from left to right: 453 + 173

**402.** 5,785 + 29

**403.** 1,00,000 − 25,327

**404.** 2,172 − 2,088

**405.** Add mentally from left to right: 821 + 511

**406.** 6,708 + 99

**407.** 1,00,000 − 73,792

**408.** 1,818 − 1,419

**409.** Add mentally from left to right: 780 + 609

**410.** 3,260 + 999

*Answers: see the answer key (questions 401–410).*

### ✎ Mental addition and subtraction — Intermediate

**411.** Add: 6,869 + 8,176 + 3,186 + 232 + 7,124

**412.** Estimate 21 × 516 by rounding (approximate answer). *(approximate)*

**413.** Add: 8,857 + 7,818 + 3,558 + 8,719 + 1,587

**414.** Estimate 72 × 550 by rounding (approximate answer). *(approximate)*

**415.** Add: 3,746 + 8,096 + 7,770 + 4,627 + 5,372

**416.** Estimate 30 × 861 by rounding (approximate answer). *(approximate)*

**417.** Add: 1,763 + 142 + 536 + 3,716 + 6,248 + 7,825

**418.** Estimate 90 × 275 by rounding (approximate answer). *(approximate)*

*Answers: see the answer key (questions 411–418).*

### ✎ Mental addition and subtraction — Advanced

**419.** Add: 9,387 + 651 + 3,312 + 8,261

**420.** Estimate 82 × 955 by rounding (approximate answer). *(approximate)*

**421.** Add: 6,783 + 9,446 + 8,064 + 7,958 + 5,149

**422.** Estimate 30 × 927 by rounding (approximate answer). *(approximate)*

*Answers: see the answer key (questions 419–422).*

## C. Fractions, decimals and percentages

### ✎ Fractions, decimals and percentages — Beginner

**423.** Simplify 12/90.

**424.** Write 1/9 as a percentage.

**425.** Find 5% of 320.

**426.** Find 25% of 324 using a fraction.

**427.** Simplify 72/132.

**428.** Write 1/5 as a percentage.

**429.** Find 5% of 160.

**430.** Find 11 1/9% of 873 using a fraction.

**431.** Simplify 18/27.

**432.** Write 1/20 as a percentage.

*Answers: see the answer key (questions 423–432).*

### ✎ Fractions, decimals and percentages — Intermediate

**433.** 4/3 − 3/5

**434.** A price changes from ₹250 to ₹175. Find the percentage decrease.

**435.** 9.2 × 9.5

**436.** 1 + 1/5

**437.** A price changes from ₹500 to ₹525. Find the percentage increase.

**438.** 6.1 × 7.1

**439.** 2/3 + 9/4

**440.** A price changes from ₹400 to ₹420. Find the percentage increase.

**441.** 7.2 × 3.8

**442.** 3/5 − 2

*Answers: see the answer key (questions 433–442).*

### ✎ Fractions, decimals and percentages — Advanced

**443.** 1/4 + 3/11

**444.** A price changes from ₹2,000 to ₹1,600. Find the percentage decrease.

**445.** 2.1 × 0.92

**446.** 3/10 + 1

*Answers: see the answer key (questions 443–446).*

## D. Powers, roots and number properties

### ✎ Powers, roots and number properties — Beginner

**447.** $18^2$

**448.** $57^2$

**449.** $54^2$

**450.** $13^2$

**451.** $34^2$

**452.** $50^2$

**453.** $23^2$

**454.** $19^2$

**455.** $28^2$

**456.** $30^2$

*Answers: see the answer key (questions 447–456).*

### ✎ Powers, roots and number properties — Intermediate

**457.** $24^3$

**458.** $\sqrt{1,764}$

**459.** $\sqrt[3]{7,53,571}$

**460.** Find the HCF and LCM of 27 and 81.

**461.** Write 420 as a product of primes.

**462.** $(2^2)^2$

**463.** Find the remainder when 87,433 is divided by 9.

**464.** $14^3$

**465.** $\sqrt[3]{5,12,000}$

**466.** Find the HCF and LCM of 24 and 36.

**467.** $2^5 \times 2^3$ (give the value)

**468.** Find the remainder when 63,530 is divided by 9.

*Answers: see the answer key (questions 457–468).*

### ✎ Powers, roots and number properties — Advanced

**469.** Estimate $\sqrt{339}$ to 2 decimal places (approximate). *(approximate)*

**470.** Estimate $\sqrt{71}$ to 2 decimal places (approximate). *(approximate)*

**471.** Estimate $\sqrt{393}$ to 2 decimal places (approximate). *(approximate)*

**472.** Estimate $\sqrt{288}$ to 2 decimal places (approximate). *(approximate)*

**473.** Estimate $\sqrt{111}$ to 2 decimal places (approximate). *(approximate)*

**474.** Estimate $\sqrt{74}$ to 2 decimal places (approximate). *(approximate)*

*Answers: see the answer key (questions 469–474).*

## E. Multiplication

## ✎ Multiplication — Beginner

**475.** 5,158 × 11

**476.** 671 × 125

**477.** 29 × 99

**478.** 85 × 9

**479.** $125^2$

**480.** 5,260 × 11

**481.** 191 × 125

**482.** 2,109 × 9,999

**483.** 65 × 8

**484.** $135^2$

**485.** 2,213 × 11

**486.** 754 × 5

**487.** 95 × 99

**488.** 72 × 4

**489.** 2,367 × 11

**490.** 19 × 25

*Answers: see the answer key (questions 475–490).*

## ✎ Multiplication — Intermediate

**491.** 407 × 1,001

**492.** 94 × 95 (Nikhilam, base 100)

**493.** 107 × 115 (base 100)

**494.** 27 × 23 (vertically and crosswise)

**495.** 119 × 111

**496.** (−22) × (−12)

**497.** 768 × 1,001

**498.** 93 × 96 (Nikhilam, base 100)

**499.** 1,030 × 1,016 (base 1,000)

**500.** 70 × 76 (vertically and crosswise)

**501.** 154 × 156

**502.** (−7) × (−29)

**503.** $96^2$

**504.** 437 × 1,001

**505.** 88 × 95 (Nikhilam, base 100)

**506.** 104 × 112 (base 100)

**507.** 83 × 42 (vertically and crosswise)

**508.** 12 × 18

**509.** (−18) × (−19)

**510.** $1{,}011^2$

**511.** $52^2$

**512.** $70 \times 101$

*Answers: see the answer key (questions 491–512).*

### ✎ Multiplication — Advanced

**513.** $105 \times 95$

**514.** $398 \times 403$ (working base 400)

**515.** $273 \times 732$

**516.** $707^2$ (duplex method)

**517.** $111 \times 90$

**518.** $306 \times 309$ (working base 300)

**519.** $883 \times 352$

**520.** $95^2$ (duplex method)

**521.** $112 \times 96$

**522.** $504 \times 496$ (working base 500)

**523.** $945 \times 558$

**524.** $437^2$ (duplex method)

**525.** $105 \times 88$

**526.** $201 \times 191$ (working base 200)

*Answers: see the answer key (questions 513–526).*

## F. Division and remainders

### ✎ Division and remainders — Beginner

**527.** $9{,}330 \div 8$

**528.** $3{,}974 \div 9$ (quotient and remainder)

**529.** $5{,}924 \div 50$ (as a decimal)

**530.** $767 \div 29$

**531.** $46{,}744 \div 9$ (quotient and remainder)

**532.** $2{,}020 \div 5$ (as a decimal)

**533.** $9{,}734 \div 6$

**534.** $30{,}370 \div 9$ (quotient and remainder)

**535.** $4{,}291 \div 50$ (as a decimal)

**536.** $3{,}530 \div 9$

*Answers: see the answer key (questions 527–536).*

### ✎ Division and remainders — Intermediate

**537.** 7,351 ÷ 10

**538.** 38,731 ÷ 9 (quotient and remainder)

**539.** 7,235 ÷ 25 (as a decimal)

**540.** 2,129 ÷ 89

**541.** 5,000 ÷ 112

**542.** 2,482 ÷ 11

*Answers: see the answer key (questions 537–542).*

### ✎ Division and remainders — Advanced

**543.** 46,132 ÷ 998

**544.** 4,292 ÷ 111

**545.** 4,885 ÷ 98

**546.** 822 ÷ 13

**547.** 3,630 ÷ 89

**548.** 1,068 ÷ 12

**549.** 1,678 ÷ 88

**550.** 593 ÷ 11

**551.** Find the recurring block of 1/19.

**552.** 4,023 ÷ 97

*Answers: see the answer key (questions 543–552).*

## G. Algebra

### ✎ Algebra — Beginner

**553.** Solve $5x + 19 = 84$

**554.** Expand $(x - 8)(x + 3)$

**555.** Solve $2x - 20 = -22$

**556.** Expand $(3x + 3)(x - 8)$

**557.** Solve $7x - 15 = -29$

**558.** Expand $(2x + 9)(x - 8)$

**559.** Solve $3x + 9 = -3$

**560.** Expand $(3x - 3)(2x + 1)$

*Answers: see the answer key (questions 553–560).*

### ✎ Algebra — Intermediate

**561.** Solve $12x + 1 = 9x - 17$

**562.** Solve: $5x + 1y = -8$, $3x + 7y = 40$

**563.** Factorise $x^2 - x - 30$

**564.** $63^2 - 19^2$

**565.** Use $(a \pm b)^2$ to find $27^2$

**566.** Solve $11x + 5 = 8x + 5$

**567.** Solve: $4x + 9y = 18$, $4x + 2y = 4$

**568.** Factorise $x^2 + 11x + 18$

**569.** $33^2 - 14^2$

**570.** Use $(a \pm b)^2$ to find $71^2$

**571.** Solve $8x - 10 = 7x - 12$

**572.** Solve: $4x + 7y = -16$, $1x + 1y = -1$

**573.** Factorise $x^2 + 10x + 25$

**574.** $75^2 - 15^2$

**575.** Use $(a \pm b)^2$ to find $83^2$

**576.** Solve $3x + 15 = 4x + 22$

*Answers: see the answer key (questions 561–576).*

### ✎ Algebra — Advanced

**577.** Solve: $36x + 29y = -79$, $29x + 36y = -51$

**578.** Solve $(x + 7)(x + 12) = (x + 8)(x + 3)$

**579.** Solve $x^3 + 12x^2 + 47x + 60 = 0$ by completing the cube

**580.** Solve $x^2 - 13x + 40 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**581.** Solve: $53x + 36y = 193$, $36x + 53y = 74$

**582.** Solve $(x + 5)(x + 9) = (x + 12)(x + 1)$

**583.** Solve $x^2 - 9x + 18 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**584.** Solve: $41x + 22y = 186$, $22x + 41y = 129$

**585.** Solve $(x + 12)(x + 4) = (x + 3)(x + 11)$

**586.** Solve $x^2 - 1x - 30 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**587.** Solve: $19x + 27y = 349$, $27x + 19y = 341$

**588.** Solve $(x + 4)(x + 5) = (x + 3)(x + 5)$

*Answers: see the answer key (questions 577–588).*

## H. Arithmetic aptitude

## ✎ Arithmetic aptitude — Beginner

**589.** Divide ₹4,102 between A and B in the ratio 3 : 4.

**590.** Find the average of 48, 49, 62, 47, 58, 54.

**591.** Find the simple interest on ₹25,000 at 6% per annum for 6 years.

**592.** A car travels at 36 km/h for 2 1/2 hours. Find the distance, and the speed in m/s.

**593.** Divide ₹2,483 between A and B in the ratio 4 : 9.

**594.** Find the average of 61, 58, 55, 49, 60, 53, 42.

**595.** Find the simple interest on ₹10,000 at 4% per annum for 3 years.

**596.** A car travels at 60 km/h for 1/2 hours. Find the distance, and the speed in m/s.

**597.** Divide ₹1,064 between A and B in the ratio 1 : 6.

**598.** Find the average of 50, 65, 59, 61, 50.

**599.** Find the simple interest on ₹15,000 at 10% per annum for 5 years.

**600.** A car travels at 36 km/h for 1/2 hours. Find the distance, and the speed in m/s.

*Answers: see the answer key (questions 589–600).*

## ✎ Arithmetic aptitude — Intermediate

**601.** Section A has 11 students with average 42; section B has 38 students with average 65. Find the combined average.

**602.** An item costing ₹400 is sold for ₹448. Find the profit percentage.

**603.** Find the single discount equal to successive discounts of 25% and 20%.

**604.** Find the compound interest on ₹20,000 at 5% per annum, compounded yearly, for 3 years.

**605.** Pipe A fills a tank in 12 hours; pipe B empties it in 20 hours. If both are open, how long to fill the empty tank?

**606.** Two trains 240 m and 200 m long run at 90 km/h and 45 km/h in opposite directions. How long do they take to cross each other?

**607.** A boat goes downstream at 17 km/h and upstream at 9 km/h. Find the speed of the boat in still water and of the stream.

**608.** A invests ₹30,000 for 12 months and B invests ₹25,000 for 12 months. Profit is ₹24,695. Find A's share.

**609.** A father is 4 times as old as his son. In 8 years the sum of their ages will be 61. Find their present ages.

**610.** Find the last digit of $4^{290}$.

**611.** In how many ways can a committee of 3 be chosen from 8 people?

**612.** 3 fair coins are tossed. Probability of exactly 3 heads?

**613.** Volume of a sphere of radius 28 cm ($\pi = 22/7$)?

**614.** Section A has 10 students with average 44; section B has 17 students with average 85. Find the combined average.

**615.** An item costing ₹1,200 is sold for ₹1,080. Find the loss percentage.

**616.** Find the single discount equal to successive discounts of 20% and 10%.

**617.** Find the compound interest on ₹10,000 at 20% per annum, compounded yearly, for 2 years.

**618.** A can finish a job in 12 days and B in 36 days. How long will they take together?

**619.** Pipe A fills a tank in 6 hours; pipe B empties it in 12 hours. If both are open, how long to fill the empty tank?

**620.** Two trains 120 m and 200 m long run at 63 km/h and 54 km/h in the same direction. How long do they take to cross each other?

**621.** A boat goes downstream at 21 km/h and upstream at 9 km/h. Find the speed of the boat in still water and of the stream.

**622.** A invests ₹30,000 for 12 months and B invests ₹45,000 for 9 months. Profit is ₹44,744. Find A's share.

**623.** A father is 4 times as old as his son. In 7 years the sum of their ages will be 89. Find their present ages.

**624.** Find the last digit of $27^{274}$.

**625.** In how many ways can 2 prizes (1st, 2nd, …) be given to 10 students, one prize each?

**626.** 3 fair coins are tossed. Probability of exactly 1 heads?

**627.** Find the area and circumference of a circle of radius 28 cm ($\pi = 22/7$).

**628.** Section A has 21 students with average 45; section B has 23 students with average 90. Find the combined average.

**629.** An item costing ₹400 is sold for ₹432. Find the profit percentage.

**630.** Find the single discount equal to successive discounts of 30% and 10%.

**631.** A can finish a job in 12 days and B in 24 days. How long will they take together?

**632.** Pipe A fills a tank in 10 hours; pipe B empties it in 15 hours. If both are open, how long to fill the empty tank?

**633.** Two trains 240 m and 100 m long run at 54 km/h and 36 km/h in opposite directions. How long do they take to cross each other?

**634.** A boat goes downstream at 21 km/h and upstream at 19 km/h. Find the speed of the boat in still water and of the stream.

*Answers: see the answer key (questions 601–634).*

## ✎ Arithmetic aptitude — Advanced

**635.** In what ratio should rice at ₹30/kg be mixed with rice at ₹80/kg to get a mixture worth ₹74/kg?

**636.** What day of the week was/is 8 October 1998?

**637.** In what ratio should rice at ₹20/kg be mixed with rice at ₹90/kg to get a mixture worth ₹25/kg?

**638.** What day of the week was/is 18 September 2001?

**639.** In what ratio should rice at ₹50/kg be mixed with rice at ₹90/kg to get a mixture worth ₹58/kg?

**640.** What day of the week was/is 16 September 1971?

**641.** In what ratio should rice at ₹20/kg be mixed with rice at ₹90/kg to get a mixture worth ₹39/kg?

**642.** What day of the week was/is 12 December 1982?

**643.** In what ratio should rice at ₹20/kg be mixed with rice at ₹80/kg to get a mixture worth ₹47/kg?

**644.** What day of the week was/is 18 June 1952?

*Answers: see the answer key (questions 635–644).*

# 1. Mixed-topic drills

## ✎ Mixed drill 1

**645.** In 3,92,13,747, what is the place value of the digit in the lakhs place?

**646.** Calculate (–28) – (–2) + (–7).

**647.** 100 – 51

**648.** Simplify 6/9.

**649.** $23^3$

**650.** Write 924 as a product of primes.

**651.** 6,200 × 11

**652.** 11 × 101

**653.** 26 × 24

**654.** 391 × 394 (working base 400)

**655.** $461^2$ (duplex method)

**656.** 3,218 ÷ 89

*Answers: see the answer key (questions 645–656).*

## ✎ Mixed drill 2

**657.** Rewrite 31,84,30,128 using international commas (millions).

**658.** How many factors does 100 have? List them.

**659.** 7,480 – 1,157

**660.** Write 1/3 as a percentage.

**661.** 2.9 × 0.58

**662.** $\sqrt{625}$

**663.** $2^{-2}$ as a fraction

**664.** 42 × 25

**665.** 99 × 89 (Nikhilam, base 100)

**666.** $105^2$

**667.** 177 × 240

**668.** 853 ÷ 15

*Answers: see the answer key (questions 657–668).*

## ✎ Mixed drill 3

**669.** Is 251 prime or composite?

**670.** Add mentally from left to right: 572 + 178

**671.** Add: 1,902 + 4,969 + 4,333 + 6,567 + 6,955

**672.** Find 12.5% of 4,800.

**673.** Find 37 1/2% of 520 using a fraction.

**674.** Find the remainder when 54,553 is divided by 7.

**675.** 435 × 999

**676.** (–27) × 18

**677.** $962^2$

**678.** 8,384 ÷ 9 (quotient and remainder)

**679.** Solve: 2x + 3y = 39, 2x + 4y = 48

**680.** Solve: 58x + 15y = 348, 15x + 58y = 90

*Answers: see the answer key (questions 669–680).*

### ✎ Mixed drill 4

**681.** Is 54,086 divisible by 4? Use the divisibility rule.

**682.** 2,339 + 98

**683.** Estimate 95 × 923 by rounding (approximate answer). *(approximate)*

**684.** 8/7 + 5/4

**685.** Find the HCF and LCM of 33 and 12.

**686.** Estimate √212 to 2 decimal places (approximate). *(approximate)*

**687.** 39 × 5

**688.** 89 × 74 (vertically and crosswise)

**689.** 105 × 89

**690.** 5,773 ÷ 25 (as a decimal)

**691.** Solve 4x − 15 = 9

**692.** Factorise $x^2 + 5x + 4$

*Answers: see the answer key (questions 681–692).*

## J. Timed mental-math tests (target: 20 questions in 8 minutes; write only the answer)

### ✎ Timed test 1

**693.** 425 × 11

**694.** 100 − 35

**695.** 4,027 + 199

**696.** 318 × 5

**697.** 155 × 155

**698.** 998 × 995 (Nikhilam, base 1,000)

**699.** Find 5% of 2,400.

**700.** 4 × 10

**701.** 6,944 ÷ 50 (as a decimal)

**702.** 73 × 99

**703.** Find 11 1/9% of 1,314 using a fraction.

**704.** Add mentally from left to right: 590 + 470

**705.** $74^2 - 16^2$

**706.** 2,295 × 11

**707.** $15^2$

**708.** 1,00,000 − 45,137

**709.** 391 + 39

**710.** 735 × 125

**711.** 129 × 121

**712.** 962 × 987 (Nikhilam, base 1,000)

*Answers: see the answer key (questions 693–712).*

## ✎ Timed test 2

**713.** 1,000 − 592

**714.** 2,568 + 999

**715.** 137 × 25

**716.** 143 × 147

**717.** 93 × 91 (Nikhilam, base 100)

**718.** Find 30% of 2,400.

**719.** 8 × 11

**720.** 1,049 ÷ 5 (as a decimal)

**721.** $22^2$

**722.** 83 × 99

**723.** Find 66 2/3% of 93 using a fraction.

**724.** Add mentally from left to right: 536 + 808

**725.** $33^2 - 8^2$

**726.** 3,165 × 11

**727.** $55^2$

**728.** 100 − 91

**729.** 690 + 197

**730.** 89 × 25

**731.** 56 × 54

**732.** 98 × 90 (Nikhilam, base 100)

*Answers: see the answer key (questions 713–732).*

## ✎ Timed test 3

**733.** 10,000 − 3,616

**734.** 9,483 + 39

**735.** 415 × 50

**736.** 137 × 133

**737.** 91 × 90 (Nikhilam, base 100)

**738.** Find 35% of 240.

**739.** 10 × 12

**740.** 220 ÷ 50 (as a decimal)

**741.** 66 × 99

**742.** Find 75% of 112 using a fraction.

**743.** Add mentally from left to right: 304 + 362

**744.** $86^2 - 10^2$

**745.** $71 \times 11$

**746.** $75^2$

**747.** 10,000 − 5,695

**748.** 1,215 + 299

**749.** $335 \times 50$

**750.** $97 \times 97$ (Nikhilam, base 100)

**751.** Find 5% of 7,200.

**752.** $4 \times 3$

*Answers: see the answer key (questions 733–752).*

### ✎ Timed test 4

**753.** 7,249 + 39

**754.** $349 \times 125$

**755.** $135 \times 135$

**756.** $99 \times 93$ (Nikhilam, base 100)

**757.** Find 30% of 7,200.

**758.** $9 \times 11$

**759.** $709 \div 25$ (as a decimal)

**760.** $570 \times 999$

**761.** Find 25% of 344 using a fraction.

**762.** Add mentally from left to right: 399 + 387

**763.** $44^2 - 10^2$

**764.** $3{,}853 \times 11$

**765.** $155^2$

**766.** 1,000 − 802

**767.** 618 + 9

**768.** $902 \times 125$

**769.** $104 \times 106$

**770.** $985 \times 973$ (Nikhilam, base 1,000)

**771.** Find 30% of 800.

**772.** $8 \times 7$

*Answers: see the answer key (questions 753–772).*

## K. Challenge questions

## ✎ Challenge set

**773.** Estimate √330 to 2 decimal places (approximate). *(approximate)*

**774.** 106 × 89

**775.** 56 × 42 (working base 50)

**776.** 533 × 410

**777.** $974^2$ (duplex method)

**778.** 586 ÷ 8

**779.** 587 ÷ 13

**780.** Solve: 49x + 58y = 89, 58x + 49y = 125

**781.** Solve (x + 6)(x + 10) = (x + 11)(x + 4)

**782.** Solve $x^3 + 3x^2 + 2x + 0 = 0$ by completing the cube

**783.** Solve $x^2 - 3x - 18 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**784.** In what ratio should rice at ₹40/kg be mixed with rice at ₹90/kg to get a mixture worth ₹54/kg?

**785.** What day of the week was/is 3 June 2001?

**786.** Estimate √348 to 2 decimal places (approximate). *(approximate)*

**787.** 111 × 91

**788.** 57 × 55 (working base 50)

**789.** 120 × 635

**790.** $327^2$ (duplex method)

**791.** 1,609 ÷ 89

**792.** 8,864 ÷ 113

*Answers: see the answer key (questions 773–792).*

## Part XII answer key with worked solutions

**381.** **4,000** — The digit is 4; thousands = 1,000; 4 × 1,000 = 4,000.

**382.** **974,849,451** — International grouping uses threes from the right: thousand, million, billion.

**383.** **Composite** — 261 = 3 × 87, so it is composite.

**384.** **No** — Rule for 9: digit sum divisible by 9. 3,101 ÷ 9 leaves remainder 5.

**385.** **28** — Subtracting 0 is the same as adding 0. 10 + 0 + 18 = 28.

**386.** **12: 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96** — Pair factors from the outside in: 1×96, 2×48, 3×32, 4×24, 6×16, 8×12.

**387.** **8** — The digit is 8; units = 1; 8 × 1 = 8.

**388.** **386,574,935** — International grouping uses threes from the right: thousand, million, billion.

**389.** **90,00,000** — The digit is 9; ten lakhs = 10,00,000; 9 × 10,00,000 = 90,00,000.

**390.** **303,288,428** — International grouping uses threes from the right: thousand, million, billion.

**391.** **Yes** — Rule for 3: digit sum divisible by 3. 74,790 ÷ 3 leaves remainder 0.

**392.** **10** — Subtracting 28 is the same as adding –28. 17 + –28 + 21 = 10.

**393.** **4,00,000** — The digit is 4; lakhs = 1,00,000; 4 × 1,00,000 = 4,00,000.

**394.** **9,105,578** — International grouping uses threes from the right: thousand, million, billion.

**395.** **Prime** — √337 ≈ 18. No prime up to 18 divides 337, so it is prime.

**396.** **No** — Rule for 8: last three digits divisible by 8. 87,634 ÷ 8 leaves remainder 2.

**397.** **300** — The digit is 3; hundreds = 100; 3 × 100 = 300.

**398.** **1,685,639** — International grouping uses threes from the right: thousand, million, billion.

**399.** **Composite** — 117 = 3 × 39, so it is composite.

**400.** **No** — Rule for 3: digit sum divisible by 3. 34,670 ÷ 3 leaves remainder 2.

**401.** **626** — Hundreds 500, tens 120, units 6: 500 + 120 + 6 = 626.

**402.** **5,814** — 29 = 30 – 1. 5,785 + 30 = 5,815, then – 1 = 5,814.

**403.** **74,673** — All from 9, last from 10: 9–2=7 9–5=4 9–3=6 9–2=7, 10–7=3 → 74673.

**404.** **84** — Subtract 2,090 instead: 2,172 – 2,090 = 82; add back 2 → 84.

**405.** **1,332** — Hundreds 1300, tens 30, units 2: 1300 + 30 + 2 = 1332.

**406.** **6,807** — 99 = 100 – 1. 6,708 + 100 = 6,808, then – 1 = 6,807.

**407.** **26,208** — All from 9, last from 10: 9–7=2 9–3=6 9–7=2 9–9=0, 10–2=8 → 26208.

**408.** **399** — Subtract 1,420 instead: 1,818 – 1,420 = 398; add back 1 → 399.

**409.** **1,389** — Hundreds 1300, tens 80, units 9: 1300 + 80 + 9 = 1389.

**410.** **4,259** — 999 = 1000 – 1. 3,260 + 1000 = 4,260, then – 1 = 4,259.

**411.** **25,587** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 9.

**412.** **≈ 10,000 (exact 10,836)** — 21 ≈ 20, 516 ≈ 500; 20 × 500 = 10,000. This is an estimate, not the exact product.

**413.** **30,539** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 2.

**414.** **≈ 42,000 (exact 39,600)** — 72 ≈ 70, 550 ≈ 600; 70 × 600 = 42,000. This is an estimate, not the exact product.

**415.** **29,611** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 1.

**416.** **≈ 27,000 (exact 25,830)** — 30 ≈ 30, 861 ≈ 900; 30 × 900 = 27,000. This is an estimate, not the exact product.

**417.** **20,230** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 7.

**418.** **≈ 25,200 (exact 24,750)** — 90 ≈ 90, 275 ≈ 280; 90 × 280 = 25,200. This is an estimate, not the exact product.

**419.** **21,611** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 2.

**420.** **≈ 80,000 (exact 78,310)** — 82 ≈ 80, 955 ≈ 1000; 80 × 1000 = 80,000. This is an estimate, not the exact product.

**421.** **37,400** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 5.

**422.** **≈ 27,000 (exact 27,810)** — 30 ≈ 30, 927 ≈ 900; 30 × 900 = 27,000. This is an estimate, not the exact product.

**423.** **2/15** — HCF of 12 and 90 is 6. Divide both by 6.

**424.** **11 1/9%** — 1/9 × 100 = 11 1/9%.

**425.** **16** — Split 5% into easy parts (10%, 5%, 1% …) or use fractions: 5% = 1/20. Result 16.

**426.** **81** — 25% = 1/4. 1/4 × 324 = 81.

**427.** **6/11** — HCF of 72 and 132 is 12. Divide both by 12.

**428.** **20%** — 1/5 × 100 = 20%.

**429.** **8** — Split 5% into easy parts (10%, 5%, 1% …) or use fractions: 5% = 1/20. Result 8.

**430.** **97** — 11 1/9% = 1/9. 1/9 × 873 = 97.

**431.** **2/3** — HCF of 18 and 27 is 9. Divide both by 9.

**432.** **5%** — 1/20 × 100 = 5%.

**433.** **11/15** — LCM of denominators = 15. Convert, then subtract numerators and simplify: 11/15.

**434.** **30%** — Change = 75. % change = change ÷ original × 100 = 75/250 × 100 = 30%.

**435.** **87.4** — Multiply without points: 92 × 95 = 8740. Decimal places: 1 + 1 = 2, so the answer is 87.4.

**436.** **1 1/5** — LCM of denominators = 5. Convert, then add numerators and simplify: 6/5.

**437.** **5%** — Change = 25. % change = change ÷ original × 100 = 25/500 × 100 = 5%.

**438.** **43.31** — Multiply without points: 61 × 71 = 4331. Decimal places: 1 + 1 = 2, so the answer is 43.31.

**439.** **2 11/12** — LCM of denominators = 12. Convert, then add numerators and simplify: 35/12.

**440.** **5%** — Change = 20. % change = change ÷ original × 100 = 20/400 × 100 = 5%.

**441.** **27.36** — Multiply without points: 72 × 38 = 2736. Decimal places: 1 + 1 = 2, so the answer is 27.36.

**442.** **–1 2/5** — LCM of denominators = 5. Convert, then subtract numerators and simplify: –7/5.

**443.** **23/44** — LCM of denominators = 44. Convert, then add numerators and simplify: 23/44.

**444.** **20%** — Change = 400. % change = change ÷ original × 100 = 400/2000 × 100 = 20%.

**445.** **1.932** — Multiply without points: 21 × 92 = 1932. Decimal places: 1 + 2 = 3, so the answer is 1.932.

**446.** **1 3/10** — LCM of denominators = 10. Convert, then add numerators and simplify: 13/10.

**447.** **324** — 18 × 18 = 324.

**448.** **3,249** — 57 × 57 = 3249.

**449.** **2,916** — 54 × 54 = 2916.

**450.** **169** — 13 × 13 = 169.

**451.** **1,156** — 34 × 34 = 1156.

**452.** **2,500** — 50 × 50 = 2500.

**453.** **529** — 23 × 23 = 529.

**454.** **361** — 19 × 19 = 361.

**455.** **784** — 28 × 28 = 784.

**456.** **900** — 30 × 30 = 900.

**457.** **13,824** — 24 × 24 × 24 = 13,824.

**458.** **42** — Last digit 4 → root ends in 2 or 8. Leading part 17 lies between $4^2$ and $5^2$, so tens digit is 4. Check $45^2 = 2025$ to choose → 42.

**459.** **91** — Last digit 1 → cube root ends in 1 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 753 lies between $9^3$ and $10^3$ → tens digit 9.

**460.** **HCF 27, LCM 81** — HCF × LCM = product: 27 × 81 = 2187 = 27 × 81.

**461.** **$2^2 \times 3 \times 5 \times 7$** — Divide repeatedly by the smallest prime that works (factor tree).

**462.** **16** — Multiply exponents: $2^4$.

**463.** **7** — 87,433 = 9 × 9,714 + 7. Shortcut: digit sum mod 9.

**464.** **2,744** — 14 × 14 × 14 = 2,744.

**465.** **80** — Last digit 0 → cube root ends in 0 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 512 lies between $8^3$ and $9^3$ → tens digit 8.

**466.** **HCF 12, LCM 72** — HCF × LCM = product: 12 × 72 = 864 = 24 × 36.

**467.** **256** — Add exponents: $2^8 = 256$.

**468.** **8** — 63,530 = 9 × 7,058 + 8. Shortcut: digit sum mod 9.

**469.** **≈ 18.42 (true value 18.412)** — Nearest perfect square 324 = $18^2$. $\sqrt{n} \approx 18 + (339 - 324) \div (2 \times 18) = 18.417$.

**470.** **≈ 8.44 (true value 8.426)** — Nearest perfect square 64 = $8^2$. $\sqrt{n} \approx 8 + (71 - 64) \div (2 \times 8) = 8.438$.

**471.** **≈ 19.82 (true value 19.824)** — Nearest perfect square 400 = $20^2$. $\sqrt{n} \approx 20 + (393 - 400) \div (2 \times 20) = 19.825$.

**472.** **≈ 16.97 (true value 16.971)** — Nearest perfect square 289 = $17^2$. $\sqrt{n} \approx 17 + (288 - 289) \div (2 \times 17) = 16.971$.

**473.** **≈ 10.55 (true value 10.536)** — Nearest perfect square 121 = $11^2$. $\sqrt{n} \approx 11 + (111 - 121) \div (2 \times 11) = 10.545$.

**474.** **≈ 8.61 (true value 8.602)** — Nearest perfect square 81 = $9^2$. $\sqrt{n} \approx 9 + (74 - 81) \div (2 \times 9) = 8.611$.

**475.** **56,738** — Write first digit 5 and last digit 8 In between, add neighbours: 5+1=6, 1+5=6, 5+8=13 Carry where a sum ≥ 10 Answer = 56,738

**476.** **83,875** — 125 = 1000 / 8 → 671 × 1000 ÷ 8 = 83,875.

**477.** **2,871** — Left: 29 − 1 = 28 Right: 99 − 28 = 71 Answer = 2,871

**478.** **765** — (80 + 5) × 9 = 720 + 45 = 765.

**479.** **15,625** — Digits before 5: 12 12 × (one more = 13) = 156 Attach 25 → 15,625

**480.** **57,860** — Write first digit 5 and last digit 0 In between, add neighbours: 5+2=7, 2+6=8, 6+0=6 Carry where a sum ≥ 10 Answer = 57,860

**481.** **23,875** — 125 = 1000 / 8 → 191 × 1000 ÷ 8 = 23,875.

**482.** **2,10,87,891** — Left: 2109 − 1 = 2108 Right: 9,999 − 2108 = 7891 Answer = 2,10,87,891

**483.** **520** — (60 + 5) × 8 = 480 + 40 = 520.

**484.** **18,225** — Digits before 5: 13 13 × (one more = 14) = 182 Attach 25 → 18,225

**485.** **24,343** — Write first digit 2 and last digit 3 In between, add neighbours: 2+2=4, 2+1=3, 1+3=4 Carry where a sum ≥ 10 Answer = 24,343

**486.** **3,770** — 5 = 10 / 2 → 754 × 10 ÷ 2 = 3,770.

**487.** **9,405** — Left: 95 − 1 = 94 Right: 99 − 94 = 05 Answer = 9,405

**488.** **288** — (70 + 2) × 4 = 280 + 8 = 288.

**489.** **26,037** — Write first digit 2 and last digit 7 In between, add neighbours: 2+3=5, 3+6=9, 6+7=13 Carry where a sum ≥ 10 Answer = 26,037

**490.** **475** — 25 = 100 / 4 → 19 × 100 ÷ 4 = 475.

**491.** **4,07,407** — 1001 = 1000 + 1, so write 407 twice (padded): 407 × 1001 = 4,07,407.

**492.** **8,930** — Base = 100 Deviations: 94 → −6, 95 → −5 Left part (cross): 94 − 5 = 89 Right part: (−6) × (−5) = 30 Right part padded to 2 digit(s): 30 Answer = 8,930

**493.** **12,305** — Base = 100 Deviations: 107 → +7, 115 → +15 Left part (cross): 107 + 15 = 122 Right part: (+7) × (+15) = 105 Right part has more than 2 digit(s): carry 1 to the left Answer = 12,305

**494.** **621** — Column 1 (from right): 7×3 = 21 → write 1, carry 2 Column 2 (from right): 7×2 + 2×3 = 20, + carry 2 = 22 → write 2, carry 2 Column 3 (from right): 2×2 = 4, + carry 2 = 6 → write 6, carry 0 Answer = 621

**495.** **13,209** — Same leading part 11; last digits 9 + 1 = 10 ✓ Left: 11 × 12 = 132 Right: 9 × 1 = 09 (always two digits) Answer = 13,209

**496.** **264** — Same signs → positive; different signs → negative. 22 × 12 = 264.

**497.** **7,68,768** — 1001 = 1000 + 1, so write 768 twice (padded): 768 × 1001 = 7,68,768.

**498.** **8,928** — Base = 100 Deviations: 93 → −7, 96 → −4 Left part (cross): 93 − 4 = 89 Right part: (−7) × (−4) = 28 Right part padded to 2 digit(s): 28 Answer = 8,928

**499.** **10,46,480** — Base = 1,000 Deviations: 1030 → +30, 1016 → +16 Left part (cross): 1030 + 16 = 1046 Right part: (+30) × (+16) = 480 Right part padded to 3 digit(s): 480 Answer = 10,46,480

**500.** **5,320** — Column 1 (from right): 0×6 = 0 → write 0, carry 0 Column 2 (from right): 0×7 + 7×6 = 42 → write 2, carry 4 Column 3 (from right): 7×7 = 49, + carry 4 = 53 → write 3, carry 5 Answer = 5,320

**501.** **24,024** — Same leading part 15; last digits 4 + 6 = 10 ✓ Left: 15 × 16 = 240 Right: 4 × 6 = 24 (always two digits) Answer = 24,024

**502.** **203** — Same signs → positive; different signs → negative. 7 × 29 = 203.

**503.** **9,216** — Base 100, deviation −4 Left: 96 − 4 = 92 Right: $4^2$ = 16 (needs 2 digit(s)) Answer = 9,216

**504.** **4,37,437** — 1001 = 1000 + 1, so write 437 twice (padded): 437 × 1001 = 4,37,437.

**505.** **8,360** — Base = 100 Deviations: 88 → −12, 95 → −5 Left part (cross): 88 − 5 = 83 Right part: (−12) × (−5) = 60 Right part padded to 2 digit(s): 60 Answer = 8,360

**506.** **11,648** — Base = 100 Deviations: 104 → +4, 112 → +12 Left part (cross): 104 + 12 = 116 Right part: (+4) × (+12) = 48 Right part padded to 2 digit(s): 48 Answer = 11,648

**507.** **3,486** — Column 1 (from right): 3×2 = 6 → write 6, carry 0 Column 2 (from right): 3×4 + 8×2 = 28 → write 8, carry 2 Column 3 (from right): 8×4 = 32, + carry 2 = 34 → write 4, carry 3 Answer = 3,486

**508.** **216** — Same leading part 1; last digits 2 + 8 = 10 ✓ Left: 1 × 2 = 2 Right: 2 × 8 = 16 (always two digits) Answer = 216

**509.** **342** — Same signs → positive; different signs → negative. 18 × 19 = 342.

**510.** **10,22,121** — Base 1,000, deviation +11 Left: 1011 + 11 = 1022 Right: $11^2$ = 121 (needs 3 digit(s)) Answer = 10,22,121

**511.** **2,704** — Deviation from 50: +2 Left: 25 + 2 = 27 Right: $2^2$ = 04 Answer = 2,704

**512.** **7,070** — 101 = 100 + 1, so write 70 twice (padded): 70 × 101 = 7,070.

**513.** **9,975** — Base = 100 Deviations: 105 → +5, 95 → −5 Left part (cross): 105 − 5 = 100 Right part: (+5) × (−5) = −25 Right part is negative: borrow 1 from left → 100 × 100 − 25 Answer = 9,975

**514.** **1,60,394** — Working base 400 = 4 × 100. Deviations −2, +3. Left: (398 +3) × 4 = 1604. Right: −2 × 3 = −6. 1604 × 100 + −6 = 1,60,394.

**515.** **1,99,836** — Column 1 (from right): 3×2 = 6 → write 6, carry 0 Column 2 (from right): 3×3 + 7×2 = 23 → write 3, carry 2 Column 3 (from right): 3×7 + 7×3 + 2×2 = 46, + carry 2 = 48 → write 8, carry 4 Column 4 (from right): 7×7 + 2×3 = 55, + carry 4 = 59 → write 9, carry 5 Column 5 (from right): 2×7 = 14, + carry 5 = 19 → write 9, carry 1 Answer = 1,99,836

**516.** **4,99,849** — Duplexes from left: 49, 0, 98, 0, 49 Combine with carries (right to left) → 4,99,849

**517.** **9,990** — Base = 100 Deviations: 111 → +11, 90 → −10 Left part (cross): 111 − 10 = 101 Right part: (+11) × (−10) = −110 Right part is negative: borrow 1 from left → 101 × 100 − 110 Answer = 9,990

**518.** **94,554** — Working base 300 = 3 × 100. Deviations +6, +9. Left: (306 +9) × 3 = 945. Right: 6 × 9 = 54. 945 × 100 + 54 = 94,554.

**519.** **3,10,816** — Column 1 (from right): 3×2 = 6 → write 6, carry 0 Column 2 (from right): 3×5 + 8×2 = 31 → write 1, carry 3 Column 3 (from right): 3×3 + 8×5 + 8×2 = 65, + carry 3 = 68 → write 8, carry 6 Column 4 (from right): 8×3 + 8×5 = 64, + carry 6 = 70 → write 0, carry 7 Column 5 (from right): 8×3 = 24, + carry 7 = 31 → write 1, carry 3 Answer = 3,10,816

**520.** **9,025** — Duplexes from left: 81, 90, 25 Combine with carries (right to left) → 9,025

**521.** **10,752** — Base = 100 Deviations: 112 → +12, 96 → −4 Left part (cross): 112 − 4 = 108 Right part: (+12) × (−4) = −48 Right part is negative: borrow 1 from left → 108 × 100 − 48 Answer = 10,752

**522.** **2,49,984** — Working base 500 = 5 × 100. Deviations +4, −4. Left: (504 −4) × 5 = 2500. Right: 4 × −4 = −16. 2500 × 100 + −16 = 2,49,984.

**523.** **5,27,310** — Column 1 (from right): 5×8 = 40 → write 0, carry 4 Column 2 (from right): 5×5 + 4×8 = 57, + carry 4 = 61 → write 1, carry 6 Column 3 (from right): 5×5 + 4×5 + 9×8 = 117, + carry 6 = 123 → write 3, carry 12 Column 4 (from right): 4×5 + 9×5 = 65, + carry 12 = 77 → write 7, carry 7 Column 5 (from right): 9×5 = 45, + carry 7 = 52 → write 2, carry 5 Answer = 5,27,310

**524.** **1,90,969** — Duplexes from left: 16, 24, 65, 42, 49 Combine with carries (right to left) → 1,90,969

**525.** **9,240** — Base = 100 Deviations: 105 → +5, 88 → −12 Left part (cross): 105 − 12 = 93 Right part: (+5) × (−12) = −60 Right part is negative: borrow 1 from left → 93 × 100 − 60 Answer = 9,240

**526.** **38,391** — Working base 200 = 2 × 100. Deviations +1, −9. Left: (201 −9) × 2 = 384. Right: 1 × −9 = −9. 384 × 100 + −9 = 38,391.

**527.** **Q = 1,166, R = 2** — Check: 8 × 1,166 + 2 = 9,330.

**528.** **Q = 441, R = 5** — Running digit sums (quotient columns): 3, 12, 19 Last running sum + last digit = remainder column 23 Adjust carries/remainder ≥ 9 → quotient 441, remainder 5

**529.** **118.48** — ÷ 50 = × 2 ÷ 100: 5,924 × 2 = 11,848, ÷ 100 = 118.48.

**530.** **Q = 26, R = 13** — Check: 29 × 26 + 13 = 767.

**531.** **Q = 5,193, R = 7** — Running digit sums (quotient columns): 4, 10, 17, 21 Last running sum + last digit = remainder column 25 Adjust carries/remainder ≥ 9 → quotient 5,193, remainder 7

**532.** **404** — ÷ 5 = × 2 ÷ 10: 2,020 × 2 = 4,040, ÷ 10 = 404.

**533.** **Q = 1,622, R = 2** — Check: 6 × 1,622 + 2 = 9,734.

**534.** **Q = 3,374, R = 4** — Running digit sums (quotient columns): 3, 3, 6, 13 Last running sum + last digit = remainder column 13 Adjust carries/remainder ≥ 9 → quotient 3,374, remainder 4

**535.** **85.82** — ÷ 50 = × 2 ÷ 100: 4,291 × 2 = 8,582, ÷ 100 = 85.82.

536. **Q = 392, R = 2** — Check: 9 × 392 + 2 = 3,530.

537. **Q = 735, R = 1** — Check: 10 × 735 + 1 = 7,351.

538. **Q = 4,303, R = 4** — Running digit sums (quotient columns): 3, 11, 18, 21 Last running sum + last digit = remainder column 22 Adjust carries/remainder ≥ 9 → quotient 4,303, remainder 4

539. **289.4** — ÷ 25 = × 4 ÷ 100: 7,235 × 4 = 28,940, ÷ 100 = 289.4.

540. **Q = 23, R = 82** — Divisor 89 = 100 − 11; write complement digits 11 Split dividend: 21 | 29 Column totals after multiplying by the complement: [2, 3] | [7, 12] Resolve carries and remainder → quotient 23, remainder 82

541. **Q = 44, R = 72** — Divisor 112 = 100 + 12; transposed flags: −1, −2 Column values after applying flags: [5, -5] | [-5, 10] Resolve negatives/carries → quotient 44, remainder 72

542. **Q = 225, R = 7** — Check: 11 × 225 + 7 = 2,482.

543. **Q = 46, R = 224** — Divisor 998 = 1,000 − 2; write complement digits 002 Split dividend: 46 | 132 Column totals after multiplying by the complement: [4, 6] | [1, 11, 14] Resolve carries and remainder → quotient 46, remainder 224

544. **Q = 38, R = 74** — Divisor 111 = 100 + 11; transposed flags: −1, −1 Column values after applying flags: [4, -2] | [7, 4] Resolve negatives/carries → quotient 38, remainder 74

545. **Q = 49, R = 83** — Divisor 98 = 100 − 2; write complement digits 02 Split dividend: 48 | 85 Column totals after multiplying by the complement: [4, 8] | [16, 21] Resolve carries and remainder → quotient 49, remainder 83

546. **Q = 63, R = 3** — Divisor 13 = 10 + 3; transposed flags: −3 Column values after applying flags: [8, -22] | [68] Resolve negatives/carries → quotient 63, remainder 3

547. **Q = 40, R = 70** — Divisor 89 = 100 − 11; write complement digits 11 Split dividend: 36 | 30 Column totals after multiplying by the complement: [3, 9] | [15, 9] Resolve carries and remainder → quotient 40, remainder 70

548. **Q = 89, R = 0** — Divisor 12 = 10 + 2; transposed flags: −2 Column values after applying flags: [1, -2, 10] | [-12] Resolve negatives/carries → quotient 89, remainder 0

549. **Q = 19, R = 6** — Divisor 88 = 100 − 12; write complement digits 12 Split dividend: 16 | 78 Column totals after multiplying by the complement: [1, 7] | [16, 22] Resolve carries and remainder → quotient 19, remainder 6

550. **Q = 53, R = 10** — Divisor 11 = 10 + 1; transposed flags: −1 Column values after applying flags: [5, 4] | [-1] Resolve negatives/carries → quotient 53, remainder 10

551. **0.(052631578947368421)** — Ekadhika (one more than 1) = 2 Start with 1 on the right; repeatedly multiply by 2, write the unit digit, carry the rest Recurring block (18 digits): 052631578947368421

552. **Q = 41, R = 46** — Divisor 97 = 100 − 3; write complement digits 03 Split dividend: 40 | 23 Column totals after multiplying by the complement: [4, 0] | [14, 3] Resolve carries and remainder → quotient 41, remainder 46

553. **x = 13** — Transpose 19 (change sign): 5x = 84 − 19 = 65; divide by 5: x = 13.

554. **$x^2 - 5x - 24$** — Vertical (left): 1 × 1 = 1. Crosswise: 1 × (3) + 1 × (−8) = −5. Vertical (right): (−8) × (3) = −24.

555. **x = −1** — Transpose −20 (change sign): 2x = −22 + 20 = −2; divide by 2: x = −1.

556. **$3x^2 - 21x - 24$** — Vertical (left): 3 × 1 = 3. Crosswise: 3 × (−8) + 1 × (3) = −21. Vertical (right): (3) × (−8) = −24.

557. **x = −2** — Transpose −15 (change sign): 7x = −29 + 15 = −14; divide by 7: x = −2.

558. **$2x^2 - 7x - 72$** — Vertical (left): 2 × 1 = 2. Crosswise: 2 × (−8) + 1 × (9) = −7. Vertical (right): (9) × (−8) = −72.

559. **x = −4** — Transpose 9 (change sign): 3x = −3 − 9 = −12; divide by 3: x = −4.

560. **$6x^2 - 3x - 3$** — Vertical (left): 3 × 2 = 6. Crosswise: 3 × (1) + 2 × (−3) = −3. Vertical (right): (−3) × (1) = −3.

561. **x = −6** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (−17 − 1)/(12 − 9) = −18/3 = −6.

562. **x = −3, y = 7** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-56 - 40)/32 = -3$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (200 - -24)/32 = 7$.

563. **(x − 6)(x + 5)** — Find two numbers with product −30 and sum −1: −6 and 5. Check with Gunita-samuccaya (put x = 1): (1 − 6)(1 + 5) = −30 and 1 − 1 − 30 = −30.

564. **3,608** — $a^2 - b^2 = (a + b)(a - b) = 82 \times 44 = 3{,}608$.

565. **729** — $(30 - 3)^2 = 900 - 2{\times}30{\times}3 + 9 = 729$.

566. **x = 0** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (5 − 5)/(11 − 8) = 0/3 = 0.

567. **x = 0, y = 2** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (36 - 36)/-28 = 0$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (16 - 72)/-28 = 2$.

568. **(x + 2)(x + 9)** — Find two numbers with product 18 and sum 11: 2 and 9. Check with Gunita-samuccaya (put x = 1): (1 + 2)(1 + 9) = 30 and 1 + 11 + 18 = 30.

569. **893** — $a^2 - b^2 = (a + b)(a - b) = 47 \times 19 = 893$.

570. **5,041** — $(70 + 1)^2 = 4900 + 2{\times}70{\times}1 + 1 = 5{,}041$.

**571.** **x = –2** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (–12 – –10)/(8 – 7) = –2/1 = –2.

**572.** **x = 3, y = –4** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-16 - -7)/-3 = 3$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (-4 - -16)/-3 = -4$.

**573.** **(x + 5)(x + 5)** — Find two numbers with product 25 and sum 10: 5 and 5. Check with Gunita-samuccaya (put x = 1): (1 + 5)(1 + 5) = 36 and 1 + 10 + 25 = 36.

**574.** **5,400** — $a^2 - b^2 = (a + b)(a - b) = 90 \times 60 = 5{,}400$.

**575.** **6,889** — $(80 + 3)^2 = 6400 + 2{\times}80{\times}3 + 9 = 6{,}889$.

**576.** **x = –7** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (22 – 15)/(3 – 4) = 7/-1 = –7.

**577.** **x = –3, y = 1** — Add: 65(x + y) = –130 → x + y = –2. Subtract: 7(x – y) = –28 → x – y = –4. Solve: x = –3, y = 1.

**578.** **x = –7 1/2** — x = (cd – ab)/(a + b – c – d) = (24 – 84)/(19 – 11).

**579.** **x = –5, –4, –3** — $(x + 4)^3 = x^3 + 12x^2 + 48x + 64$. The given cubic = $(x + 4)^3 - (x + 4)$. Let y = x + 4: $y^3 - y = 0 \rightarrow y = 0, \pm 1$.

**580.** **x = 8 or 5** — Differential: 2x – 13. Discriminant = 169 – 160 = 9, √ = 3. 2x – 13 = ±3.

**581.** **x = 5, y = –2** — Add: 89(x + y) = 267 → x + y = 3. Subtract: 17(x – y) = 119 → x – y = 7. Solve: x = 5, y = –2.

**582.** **x = –33** — x = (cd – ab)/(a + b – c – d) = (12 – 45)/(14 – 13).

**583.** **x = 6 or 3** — Differential: 2x – 9. Discriminant = 81 – 72 = 9, √ = 3. 2x – 9 = ±3.

**584.** **x = 4, y = 1** — Add: 63(x + y) = 315 → x + y = 5. Subtract: 19(x – y) = 57 → x – y = 3. Solve: x = 4, y = 1.

**585.** **x = –7 1/2** — x = (cd – ab)/(a + b – c – d) = (33 – 48)/(16 – 14).

**586.** **x = 6 or –5** — Differential: 2x – 1. Discriminant = 1 – -120 = 121, √ = 11. 2x – 1 = ±11.

**587.** **x = 7, y = 8** — Add: 46(x + y) = 690 → x + y = 15. Subtract: -8(x – y) = 8 → x – y = –1. Solve: x = 7, y = 8.

**588.** **x = –5** — x = (cd – ab)/(a + b – c – d) = (15 – 20)/(9 – 8).

**589.** **A ₹1,758, B ₹2,344** — Total parts 7; one part = 4,102 ÷ 7 = 586.

**590.** **53** — Assume 54. Deviations: –6, –5, +8, –7, +4, +0; total –6; ÷ 6 = –1. Average = 53.

**591.** **₹9,000** — SI = PRT/100 = 25,000 × 6 × 6 / 100 = 9,000.

**592.** **90 km; 10 m/s** — D = S × T = 36 × 2 1/2 = 90 km. km/h → m/s: × 5/18 → 10 m/s.

**593.** **A ₹764, B ₹1,719** — Total parts 13; one part = 2,483 ÷ 13 = 191.

**594.** **54** — Assume 54. Deviations: +7, +4, +1, –5, +6, –1, –12; total +0; ÷ 7 = 0. Average = 54.

**595.** **₹1,200** — SI = PRT/100 = 10,000 × 4 × 3 / 100 = 1,200.

**596.** **30 km; 16 2/3 m/s** — D = S × T = 60 × 1/2 = 30 km. km/h → m/s: × 5/18 → 16 2/3 m/s.

**597.** **A ₹152, B ₹912** — Total parts 7; one part = 1,064 ÷ 7 = 152.

**598.** **57** — Assume 57. Deviations: –7, +8, +2, +4, –7; total +0; ÷ 5 = 0. Average = 57.

**599.** **₹7,500** — SI = PRT/100 = 15,000 × 10 × 5 / 100 = 7,500.

**600.** **18 km; 10 m/s** — D = S × T = 36 × 1/2 = 18 km. km/h → m/s: × 5/18 → 10 m/s.

**601.** **59.84** — (11×42 + 38×65) ÷ 49 = 2932 ÷ 49.

**602.** **12% profit** — Profit = 48. % = 48/400 × 100 = 12%.

**603.** **40%** — Net = a + b – ab/100 = 25 + 20 – 500/100 = 40%.

**604.** **₹3,152.5** — $A = P(1 + R/100)^n = 20{,}000 \times (105/100)^3 = ₹23{,}152.5$. CI = A – P = ₹3,152.5.

**605.** **30 hours** — Net rate = 1/12 – 1/20 = 1/30 tank/hour, so time = 30 hours.

**606.** **11 11/15 s** — Relative speed = 135 km/h = 37 1/2 m/s. Distance = 240 + 200 = 440 m. Time = 11 11/15 s.

**607.** **Boat 13 km/h, stream 4 km/h** — Boat = (down + up)/2 = (17 + 9)/2 = 13. Stream = (down – up)/2 = 4.

**608.** **₹13,470** — Ratio = 30,000×12 : 25,000×12 = 6 : 5. A's share = 6/11 × 24,695 = 13,470.

**609.** **Son 9, father 36** — Let son = x, father = 4x. (4x + 8) + (x + 8) = 61 → 5x = 45 → x = 9.

**610.** **6** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 290 mod 2 = 2 → 6.

**611.** **56** — Order does not matter: $^nC_r = 8!/(3!{\cdot}5!) = 56$.

**612.** **1/8** — $C(n, k)/2^n = 1/8 = 1/8$.

**613.** **91989 1/3 cm³** — $V = 4/3\ \pi r^3 = 91989\ 1/3$.

**614.** **69.81** — (10×44 + 17×85) ÷ 27 = 1885 ÷ 27.

**615.** **10% loss** — Loss = 120. % = 120/1200 × 100 = 10%.

**616.** **28%** — Net = a + b − ab/100 = 20 + 10 − 200/100 = 28%.

**617.** **₹4,400** — $A = P(1 + R/100)^n = 10{,}000 \times (120/100)^2 = ₹14{,}400$. CI = A − P = ₹4,400.

**618.** **9 days** — Total work = LCM(12,36) = 36 units. A does 3/day, B does 1/day; together 4/day. Time = 36/4 = 9 days.

**619.** **12 hours** — Net rate = 1/6 − 1/12 = 1/12 tank/hour, so time = 12 hours.

**620.** **128 s** — Relative speed = 9 km/h = 2 1/2 m/s. Distance = 120 + 200 = 320 m. Time = 128 s.

**621.** **Boat 15 km/h, stream 6 km/h** — Boat = (down + up)/2 = (21 + 9)/2 = 15. Stream = (down − up)/2 = 6.

**622.** **₹21,056** — Ratio = 30,000×12 : 45,000×9 = 8 : 9. A's share = 8/17 × 44,744 = 21,056.

**623.** **Son 15, father 60** — Let son = x, father = 4x. (4x + 7) + (x + 7) = 89 → 5x = 75 → x = 15.

**624.** **9** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 274 mod 4 = 2 → 9.

**625.** **90** — Order matters: ${}^nP_r = 10 \times 9 \times \ldots$ (2 factors) = 90.

**626.** **3/8** — C(n, k)/$2^n$ = 3/8 = 3/8.

**627.** **Area 2,464 cm², circumference 176 cm** — $A = \pi r^2 = 22/7 \times 28^2 = 2{,}464$; $C = 2\pi r = 176$.

**628.** **68.52** — (21×45 + 23×90) ÷ 44 = 3015 ÷ 44.

**629.** **8% profit** — Profit = 32. % = 32/400 × 100 = 8%.

**630.** **37%** — Net = a + b − ab/100 = 30 + 10 − 300/100 = 37%.

**631.** **8 days** — Total work = LCM(12,24) = 24 units. A does 2/day, B does 1/day; together 3/day. Time = 24/3 = 8 days.

**632.** **30 hours** — Net rate = 1/10 − 1/15 = 1/30 tank/hour, so time = 30 hours.

**633.** **13 3/5 s** — Relative speed = 90 km/h = 25 m/s. Distance = 240 + 100 = 340 m. Time = 13 3/5 s.

**634.** **Boat 20 km/h, stream 1 km/h** — Boat = (down + up)/2 = (21 + 19)/2 = 20. Stream = (down − up)/2 = 1.

**635.** **3 : 22** — Alligation cross: (80 − 74) : (74 − 30) = 6 : 44 = 3 : 22.

**636.** **Thursday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**637.** **13 : 1** — Alligation cross: (90 − 25) : (25 − 20) = 65 : 5 = 13 : 1.

**638.** **Tuesday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**639.** **4 : 1** — Alligation cross: (90 − 58) : (58 − 50) = 32 : 8 = 4 : 1.

**640.** **Thursday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**641.** **51 : 19** — Alligation cross: (90 − 39) : (39 − 20) = 51 : 19 = 51 : 19.

**642.** **Sunday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**643.** **11 : 9** — Alligation cross: (80 − 47) : (47 − 20) = 33 : 27 = 11 : 9.

**644.** **Wednesday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**645.** **2,00,000** — The digit is 2; lakhs = 1,00,000; 2 × 1,00,000 = 2,00,000.

**646.** **−33** — Subtracting (−2) is the same as adding 2. −28 + 2 + −7 = −33.

**647.** **49** — All from 9, last from 10: 9−5=4, 10−1=9 → 49.

**648.** **2/3** — HCF of 6 and 9 is 3. Divide both by 3.

**649.** **12,167** — 23 × 23 × 23 = 12,167.

**650.** **$2^2 \times 3 \times 7 \times 11$** — Divide repeatedly by the smallest prime that works (factor tree).

**651.** **68,200** — Write first digit 6 and last digit 0 In between, add neighbours: 6+2=8, 2+0=2, 0+0=0 Carry where a sum ≥ 10 Answer = 68,200

**652.** **1,111** — 101 = 100 + 1, so write 11 twice (padded): 11 × 101 = 1,111.

**653.** **624** — Same leading part 2; last digits 6 + 4 = 10 ✓ Left: 2 × 3 = 6 Right: 6 × 4 = 24 (always two digits) Answer = 624

**654.** **1,54,054** — Working base 400 = 4 × 100. Deviations −9, −6. Left: (391 −6) × 4 = 1540. Right: −9 × −6 = 54. 1540 × 100 + 54 = 1,54,054.

**655.** **2,12,521** — Duplexes from left: 16, 48, 44, 12, 1 Combine with carries (right to left) → 2,12,521

**656.** **Q = 36, R = 14** — Divisor 89 = 100 − 11; write complement digits 11 Split dividend: 32 | 18 Column totals after multiplying by the complement: [3, 5] | [9, 13] Resolve carries and remainder → quotient 36, remainder 14

**657.** **318,430,128** — International grouping uses threes from the right: thousand, million, billion.

**658.** **9: 1, 2, 4, 5, 10, 20, 25, 50, 100** — Pair factors from the outside in: 1×100, 2×50, 4×25, 5×20, 10×10.

**659.** **6,323** — Subtract 1,160 instead: 7,480 – 1,160 = 6,320; add back 3 → 6,323.

**660.** **33 1/3%** — 1/3 × 100 = 33 1/3%.

**661.** **1.682** — Multiply without points: 29 × 58 = 1682. Decimal places: 1 + 2 = 3, so the answer is 1.682.

**662.** **25** — Last digit 5 → root ends in 5. Leading part 6 lies between $2^2$ and $3^2$, so tens digit is 2. Check $25^2 = 625$ to choose → 25.

**663.** **1/4** — A negative exponent means reciprocal: $1/2^2$.

**664.** **1,050** — 25 = 100 / 4 → 42 × 100 ÷ 4 = 1,050.

**665.** **8,811** — Base = 100 Deviations: 99 → –1, 89 → –11 Left part (cross): 99 – 11 = 88 Right part: (–1) × (–11) = 11 Right part padded to 2 digit(s): 11 Answer = 8,811

**666.** **11,025** — Digits before 5: 10 10 × (one more = 11) = 110 Attach 25 → 11,025

**667.** **42,480** — Column 1 (from right): 7×0 = 0 → write 0, carry 0 Column 2 (from right): 7×4 + 7×0 = 28 → write 8, carry 2 Column 3 (from right): 7×2 + 7×4 + 1×0 = 42, + carry 2 = 44 → write 4, carry 4 Column 4 (from right): 7×2 + 1×4 = 18, + carry 4 = 22 → write 2, carry 2 Column 5 (from right): 1×2 = 2, + carry 2 = 4 → write 4, carry 0 Answer = 42,480

**668.** **Q = 56, R = 13** — Check: 15 × 56 + 13 = 853.

**669.** **Prime** — √251 ≈ 15. No prime up to 15 divides 251, so it is prime.

**670.** **750** — Hundreds 600, tens 140, units 10: 600 + 140 + 10 = 750.

**671.** **24,726** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 3.

**672.** **600** — Split 12.5% into easy parts (10%, 5%, 1% …) or use fractions: 12.5% = 1/8. Result 600.

**673.** **195** — 37 1/2% = 3/8. 3/8 × 520 = 195.

**674.** **2** — 54,553 = 7 × 7,793 + 2.

**675.** **4,34,565** — Left: 435 – 1 = 434 Right: 999 – 434 = 565 Answer = 4,34,565

**676.** **–486** — Same signs → positive; different signs → negative. 27 × 18 = 486.

**677.** **9,25,444** — Base 1,000, deviation –38 Left: 962 – 38 = 924 Right: $38^2 = 1444$ (needs 3 digit(s), carry 1) Answer = 9,25,444

**678.** **Q = 931, R = 5** — Running digit sums (quotient columns): 8, 11, 19 Last running sum + last digit = remainder column 23 Adjust carries/remainder ≥ 9 → quotient 931, remainder 5

**679.** **x = 6, y = 9** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (156 - 144)/2 = 6$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (96 - 78)/2 = 9$.

**680.** **x = 6, y = 0** — Add: 73(x + y) = 438 → x + y = 6. Subtract: 43(x – y) = 258 → x – y = 6. Solve: x = 6, y = 0.

**681.** **No** — Rule for 4: last two digits divisible by 4. 54,086 ÷ 4 leaves remainder 2.

**682.** **2,437** — 98 = 100 – 2. 2,339 + 100 = 2,439, then – 2 = 2,437.

**683.** **≈ 90,000 (exact 87,685)** — 95 ≈ 100, 923 ≈ 900; 100 × 900 = 90,000. This is an estimate, not the exact product.

**684.** **2 11/28** — LCM of denominators = 28. Convert, then add numerators and simplify: 67/28.

**685.** **HCF 3, LCM 132** — HCF × LCM = product: 3 × 132 = 396 = 33 × 12.

**686.** **≈ 14.57 (true value 14.560)** — Nearest perfect square 225 = $15^2$. $\sqrt{n} \approx 15 + (212 - 225) \div (2 \times 15) = 14.567$.

**687.** **195** — (30 + 9) × 5 = 150 + 45 = 195.

**688.** **6,586** — Column 1 (from right): 9×4 = 36 → write 6, carry 3 Column 2 (from right): 9×7 + 8×4 = 95, + carry 3 = 98 → write 8, carry 9 Column 3 (from right): 8×7 = 56, + carry 9 = 65 → write 5, carry 6 Answer = 6,586

**689.** **9,345** — Base = 100 Deviations: 105 → +5, 89 → –11 Left part (cross): 105 – 11 = 94 Right part: (+5) × (–11) = –55 Right part is negative: borrow 1 from left → 94 × 100 – 55 Answer = 9,345

**690.** **230.92** — ÷ 25 = × 4 ÷ 100: 5,773 × 4 = 23,092, ÷ 100 = 230.92.

**691.** **x = 6** — Transpose –15 (change sign): 4x = 9 + 15 = 24; divide by 4: x = 6.

**692.** **(x + 1)(x + 4)** — Find two numbers with product 4 and sum 5: 1 and 4. Check with Gunita-samuccaya (put x = 1): (1 + 1)(1 + 4) = 10 and 1 + 5 + 4 = 10.

**693.** **4,675** — Write first digit 4 and last digit 5 In between, add neighbours: 4+2=6, 2+5=7 Carry where a sum ≥ 10 Answer = 4,675

**694.** **65** — All from 9, last from 10: 9–3=6, 10–5=5 → 65.

**695.** **4,226** — 199 = 200 – 1. 4,027 + 200 = 4,227, then – 1 = 4,226.

**696.** **1,590** — 5 = 10 / 2 → 318 × 10 ÷ 2 = 1,590.

**697.** **24,025** — Same leading part 15; last digits 5 + 5 = 10 ✓ Left: 15 × 16 = 240 Right: 5 × 5 = 25 (always two digits) Answer = 24,025

**698.** **9,93,010** — Base = 1,000 Deviations: 998 → –2, 995 → –5 Left part (cross): 998 – 5 = 993 Right part: (–2) × (–5) = 10 Right part padded to 3 digit(s): 010 Answer = 9,93,010

**699.** **120** — Split 5% into easy parts (10%, 5%, 1% …) or use fractions: 5% = 1/20. Result 120.

**700.** **40** — 4 groups of 10 = 40.

**701.** **138.88** — ÷ 50 = × 2 ÷ 100: 6,944 × 2 = 13,888, ÷ 100 = 138.88.

**702.** **7,227** — Left: 73 – 1 = 72 Right: 99 – 72 = 27 Answer = 7,227

**703.** **146** — 11 1/9% = 1/9. 1/9 × 1,314 = 146.

**704.** **1,060** — Hundreds 900, tens 160, units 0: 900 + 160 + 0 = 1060.

**705.** **5,220** — $a^2 - b^2 = (a + b)(a - b) = 90 \times 58 = 5{,}220$.

**706.** **25,245** — Write first digit 2 and last digit 5 In between, add neighbours: 2+2=4, 2+9=11, 9+5=14 Carry where a sum ≥ 10 Answer = 25,245

**707.** **225** — Digits before 5: 1 1 × (one more = 2) = 2 Attach 25 → 225

**708.** **54,863** — All from 9, last from 10: 9–4=5 9–5=4 9–1=8 9–3=6, 10–7=3 → 54863.

**709.** **430** — 39 = 40 – 1. 391 + 40 = 431, then – 1 = 430.

**710.** **91,875** — 125 = 1000 / 8 → 735 × 1000 ÷ 8 = 91,875.

**711.** **15,609** — Same leading part 12; last digits 9 + 1 = 10 ✓ Left: 12 × 13 = 156 Right: 9 × 1 = 09 (always two digits) Answer = 15,609

**712.** **9,49,494** — Base = 1,000 Deviations: 962 → –38, 987 → –13 Left part (cross): 962 – 13 = 949 Right part: (–38) × (–13) = 494 Right part padded to 3 digit(s): 494 Answer = 9,49,494

**713.** **408** — All from 9, last from 10: 9–5=4 9–9=0, 10–2=8 → 408.

**714.** **3,567** — 999 = 1000 – 1. 2,568 + 1000 = 3,568, then – 1 = 3,567.

**715.** **3,425** — 25 = 100 / 4 → 137 × 100 ÷ 4 = 3,425.

**716.** **21,021** — Same leading part 14; last digits 3 + 7 = 10 ✓ Left: 14 × 15 = 210 Right: 3 × 7 = 21 (always two digits) Answer = 21,021

**717.** **8,463** — Base = 100 Deviations: 93 → –7, 91 → –9 Left part (cross): 93 – 9 = 84 Right part: (–7) × (–9) = 63 Right part padded to 2 digit(s): 63 Answer = 8,463

**718.** **720** — Split 30% into easy parts (10%, 5%, 1% …) or use fractions: 30% = 3/10. Result 720.

**719.** **88** — 8 groups of 11 = 88.

**720.** **209.8** — ÷ 5 = × 2 ÷ 10: 1,049 × 2 = 2,098, ÷ 10 = 209.8.

**721.** **484** — 22 × 22 = 484.

**722.** **8,217** — Left: 83 – 1 = 82 Right: 99 – 82 = 17 Answer = 8,217

**723.** **62** — 66 2/3% = 2/3. 2/3 × 93 = 62.

**724.** **1,344** — Hundreds 1300, tens 30, units 14: 1300 + 30 + 14 = 1344.

**725.** **1,025** — $a^2 - b^2 = (a + b)(a - b) = 41 \times 25 = 1{,}025$.

**726.** **34,815** — Write first digit 3 and last digit 5 In between, add neighbours: 3+1=4, 1+6=7, 6+5=11 Carry where a sum ≥ 10 Answer = 34,815

**727.** **3,025** — Digits before 5: 5 5 × (one more = 6) = 30 Attach 25 → 3,025

**728.** **9** — All from 9, last from 10: 9–9=0, 10–1=9 → 09.

**729.** **887** — 197 = 200 – 3. 690 + 200 = 890, then – 3 = 887.

**730.** **2,225** — 25 = 100 / 4 → 89 × 100 ÷ 4 = 2,225.

**731.** **3,024** — Same leading part 5; last digits 6 + 4 = 10 ✓ Left: 5 × 6 = 30 Right: 6 × 4 = 24 (always two digits) Answer = 3,024

**732.** **8,820** — Base = 100 Deviations: 98 → –2, 90 → –10 Left part (cross): 98 – 10 = 88 Right part: (–2) × (–10) = 20 Right part padded to 2 digit(s): 20 Answer = 8,820

**733.** **6,384** — All from 9, last from 10: 9–3=6 9–6=3 9–1=8, 10–6=4 → 6384.

**734.** **9,522** — 39 = 40 – 1. 9,483 + 40 = 9,523, then – 1 = 9,522.

**735.** **20,750** — 50 = 100 / 2 → 415 × 100 ÷ 2 = 20,750.

**736.** **18,221** — Same leading part 13; last digits 7 + 3 = 10 ✓ Left: 13 × 14 = 182 Right: 7 × 3 = 21 (always two digits) Answer = 18,221

**737.** **8,190** — Base = 100 Deviations: 91 → –9, 90 → –10 Left part (cross): 91 – 10 = 81 Right part: (–9) × (–10) = 90 Right part padded to 2 digit(s): 90 Answer = 8,190

**738.** **84** — Split 35% into easy parts (10%, 5%, 1% …) or use fractions: 35% = 7/20. Result 84.

**739.** **120** — 10 groups of 12 = 120.

**740.** **4.4** — ÷ 50 = × 2 ÷ 100: 220 × 2 = 440, ÷ 100 = 4.4.

**741.** **6,534** — Left: 66 – 1 = 65 Right: 99 – 65 = 34 Answer = 6,534

**742.** **84** — 75% = 3/4. 3/4 × 112 = 84.

**743.** **666** — Hundreds 600, tens 60, units 6: 600 + 60 + 6 = 666.

**744.** **7,296** — $a^2 - b^2 = (a + b)(a - b) = 96 \times 76 = 7{,}296$.

**745.** **781** — Write first digit 7 and last digit 1 In between, add neighbours: 7+1=8 Carry where a sum ≥ 10 Answer = 781

**746.** **5,625** — Digits before 5: 7 7 × (one more = 8) = 56 Attach 25 → 5,625

**747.** **4,305** — All from 9, last from 10: 9–5=4 9–6=3 9–9=0, 10–5=5 → 4305.

**748.** **1,514** — 299 = 300 – 1. 1,215 + 300 = 1,515, then – 1 = 1,514.

**749.** **16,750** — 50 = 100 / 2 → 335 × 100 ÷ 2 = 16,750.

**750.** **9,409** — Base = 100 Deviations: 97 → –3, 97 → –3 Left part (cross): 97 – 3 = 94 Right part: (–3) × (–3) = 9 Right part padded to 2 digit(s): 09 Answer = 9,409

**751.** **360** — Split 5% into easy parts (10%, 5%, 1% …) or use fractions: 5% = 1/20. Result 360.

**752.** **12** — 4 groups of 3 = 12.

**753.** **7,288** — 39 = 40 – 1. 7,249 + 40 = 7,289, then – 1 = 7,288.

**754.** **43,625** — 125 = 1000 / 8 → 349 × 1000 ÷ 8 = 43,625.

**755.** **18,225** — Same leading part 13; last digits 5 + 5 = 10 ✓ Left: 13 × 14 = 182 Right: 5 × 5 = 25 (always two digits) Answer = 18,225

**756.** **9,207** — Base = 100 Deviations: 99 → –1, 93 → –7 Left part (cross): 99 – 7 = 92 Right part: (–1) × (–7) = 7 Right part padded to 2 digit(s): 07 Answer = 9,207

**757.** **2,160** — Split 30% into easy parts (10%, 5%, 1% …) or use fractions: 30% = 3/10. Result 2,160.

**758.** **99** — 9 groups of 11 = 99.

**759.** **28.36** — ÷ 25 = × 4 ÷ 100: 709 × 4 = 2,836, ÷ 100 = 28.36.

**760.** **5,69,430** — Left: 570 – 1 = 569 Right: 999 – 569 = 430 Answer = 5,69,430

**761.** **86** — 25% = 1/4. 1/4 × 344 = 86.

**762.** **786** — Hundreds 600, tens 170, units 16: 600 + 170 + 16 = 786.

**763.** **1,836** — $a^2 - b^2 = (a + b)(a - b) = 54 \times 34 = 1{,}836$.

**764.** **42,383** — Write first digit 3 and last digit 3 In between, add neighbours: 3+8=11, 8+5=13, 5+3=8 Carry where a sum ≥ 10 Answer = 42,383

**765.** **24,025** — Digits before 5: 15 15 × (one more = 16) = 240 Attach 25 → 24,025

**766.** **198** — All from 9, last from 10: 9–8=1 9–0=9, 10–2=8 → 198.

**767.** **627** — 9 = 10 – 1. 618 + 10 = 628, then – 1 = 627.

**768.** **1,12,750** — 125 = 1000 / 8 → 902 × 1000 ÷ 8 = 1,12,750.

**769.** **11,024** — Same leading part 10; last digits 4 + 6 = 10 ✓ Left: 10 × 11 = 110 Right: 4 × 6 = 24 (always two digits) Answer = 11,024

**770.** **9,58,405** — Base = 1,000 Deviations: 985 → –15, 973 → –27 Left part (cross): 985 – 27 = 958 Right part: (–15) × (–27) = 405 Right part padded to 3 digit(s): 405 Answer = 9,58,405

**771.** **240** — Split 30% into easy parts (10%, 5%, 1% …) or use fractions: 30% = 3/10. Result 240.

**772.** **56** — 8 groups of 7 = 56.

**773.** **≈ 18.17 (true value 18.166)** — Nearest perfect square 324 = $18^2$. $\sqrt{n} \approx 18 + (330 - 324) \div (2 \times 18) = 18.167$.

**774.** **9,434** — Base = 100 Deviations: 106 → +6, 89 → –11 Left part (cross): 106 – 11 = 95 Right part: (+6) × (–11) = –66 Right part is negative: borrow 1 from left → 95 × 100 – 66 Answer = 9,434

**775.** **2,352** — Working base 50 = 5 × 10. Deviations +6, –8. Left: (56 –8) × 5 = 240. Right: 6 × –8 = –48. 240 × 10 + –48 = 2,352.

**776.** **2,18,530** — Column 1 (from right): 3×0 = 0 → write 0, carry 0 Column 2 (from right): 3×1 + 3×0 = 3 → write 3, carry 0 Column 3 (from right): 3×4 + 3×1 + 5×0 = 15 → write 5, carry 1 Column 4 (from right): 3×4 + 5×1 = 17, + carry 1 = 18 → write 8, carry 1 Column 5 (from right): 5×4 = 20, + carry 1 = 21 → write 1, carry 2 Answer = 2,18,530

**777.** **9,48,676** — Duplexes from left: 81, 126, 121, 56, 16 Combine with carries (right to left) → 9,48,676

**778.** **Q = 73, R = 2** — Divisor 8 = 10 – 2; write complement digits 2 Split dividend: 58 | 6 Column totals after multiplying by the complement: [5, 18] | [42] Resolve carries and remainder → quotient 73, remainder 2

**779.** **Q = 45, R = 2** — Divisor 13 = 10 + 3; transposed flags: –3 Column values after applying flags: [5, -7] | [28] Resolve negatives/carries → quotient 45, remainder 2

**780.** **x = 3, y = –1** — Add: 107(x + y) = 214 → x + y = 2. Subtract: -9(x – y) = –36 → x – y = 4. Solve: x = 3, y = –1.

**781.** **x = –16** — x = (cd – ab)/(a + b – c – d) = (44 – 60)/(16 – 15).

**782.** **x = –2, –1, 0** — $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$. The given cubic = $(x + 1)^3 – (x + 1)$. Let y = x + 1: $y^3 – y = 0$ → y = 0, ±1.

**783.** **x = 6 or –3** — Differential: 2x – 3. Discriminant = 9 – -72 = 81, √ = 9. 2x – 3 = ±9.

**784.** **18 : 7** — Alligation cross: (90 – 54) : (54 – 40) = 36 : 14 = 18 : 7.

**785.** **Sunday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**786.** **≈ 18.66 (true value 18.655)** — Nearest perfect square 361 = $19^2$. √n ≈ 19 + (348 – 361) ÷ (2 × 19) = 18.658.

**787.** **10,101** — Base = 100 Deviations: 111 → +11, 91 → –9 Left part (cross): 111 – 9 = 102 Right part: (+11) × (–9) = –99 Right part is negative: borrow 1 from left → 102 × 100 – 99 Answer = 10,101

**788.** **3,135** — Working base 50 = 5 × 10. Deviations +7, +5. Left: (57 +5) × 5 = 310. Right: 7 × 5 = 35. 310 × 10 + 35 = 3,135.

**789.** **76,200** — Column 1 (from right): 0×5 = 0 → write 0, carry 0 Column 2 (from right): 0×3 + 2×5 = 10 → write 0, carry 1 Column 3 (from right): 0×6 + 2×3 + 1×5 = 11, + carry 1 = 12 → write 2, carry 1 Column 4 (from right): 2×6 + 1×3 = 15, + carry 1 = 16 → write 6, carry 1 Column 5 (from right): 1×6 = 6, + carry 1 = 7 → write 7, carry 0 Answer = 76,200

**790.** **1,06,929** — Duplexes from left: 9, 12, 46, 28, 49 Combine with carries (right to left) → 1,06,929

**791.** **Q = 18, R = 7** — Divisor 89 = 100 – 11; write complement digits 11 Split dividend: 16 | 09 Column totals after multiplying by the complement: [1, 7] | [8, 16] Resolve carries and remainder → quotient 18, remainder 7

**792.** **Q = 78, R = 50** — Divisor 113 = 100 + 13; transposed flags: –1, –3 Column values after applying flags: [8, 0] | [-18, 4] Resolve negatives/carries → quotient 78, remainder 50

# Part XIII

## Solved Practice Sets

Seventy practice sets in seven categories, with ten sets in each. Every set shows a difficulty level and a suggested time. Answers and explanations follow each category.

# Chapter 52: Seventy Practice Sets

### Structure of every category

- Sets 1–3: **Beginner** (about 45 seconds per question)
- Sets 4–7: **Intermediate** (about 75 seconds per question)
- Sets 8–10: **Advanced** (about 2 minutes per question)

The suggested time is a target, not a rule. Accuracy comes first.

| # | Category | Sets |
|---|---|---|
| 1 | Foundational arithmetic | 1–10 |
| 2 | Multiplication and division | 11–20 |
| 3 | Squares, cubes and roots | 21–30 |
| 4 | Algebra and factorisation | 31–40 |
| 5 | Percentages, ratios and averages | 41–50 |
| 6 | Government-exam mixed aptitude | 51–60 |
| 7 | Advanced mental-math challenge | 61–70 |

### After each set

Write your score, time taken, and the topic of every mistake in the tracker. Once your accuracy on a category reaches 90%, move to its next difficulty level.

# Foundational arithmetic — 10 sets

## ✎ Set 1: Foundational arithmetic · Beginner · suggested time 6 min

**793.** Add mentally from left to right: 141 + 303

**794.** 5,075 + 29

**795.** 10,000 − 6,447

**796.** 2,008 − 677

**797.** Calculate 18 − 17 + (−15).

**798.** Simplify 66/30.

**799.** Find 15% of 640.

**800.** Is 99,256 divisible by 8? Use the divisibility rule.

*Answers: see the answer key (questions 793–800).*

## ✎ Set 2: Foundational arithmetic · Beginner · suggested time 6 min

**801.** 2,585 + 99

**802.** 1,00,000 − 95,784

**803.** 6,035 − 4,278

**804.** Calculate 13 − 16 + (−1).

**805.** Simplify 90/63.

**806.** Find 25% of 2,400.

**807.** Is 35,012 divisible by 11? Use the divisibility rule.

**808.** Add mentally from left to right: 802 + 490

*Answers: see the answer key (questions 801–808).*

## ✎ Set 3: Foundational arithmetic · Beginner · suggested time 6 min

**809.** 10,000 − 1,982

**810.** 7,000 − 1,939

**811.** Calculate 6 − 18 + (−8).

**812.** Simplify 30/25.

**813.** Find 150% of 640.

**814.** Is 11,388 divisible by 3? Use the divisibility rule.

**815.** Add mentally from left to right: 148 + 895

**816.** 6,670 + 199

*Answers: see the answer key (questions 809–816).*

## ✎ Set 4: Foundational arithmetic · Intermediate · suggested time 10 min

**817.** Find the HCF and LCM of 18 and 12.

**818.** Estimate 29 × 822 by rounding (approximate answer). *(approximate)*

**819.** Find the remainder when 15,053 is divided by 13.

**820.** $5^4 \times 5^5$ (give the value)

**821.** Add: 2,233 + 3,279 + 4,105 + 2,438 + 8,299 + 3,424

**822.** 3/11 + 8/5

**823.** 3.9 × 0.53

**824.** Find the HCF and LCM of 8 and 22.

*Answers: see the answer key (questions 817–824).*

## ✎ Set 5: Foundational arithmetic · Intermediate · suggested time 10 min

**825.** Estimate 70 × 551 by rounding (approximate answer). *(approximate)*

**826.** Find the remainder when 29,048 is divided by 9.

**827.** $2^2 \times 2^5$ (give the value)

**828.** Add: 1,640 + 9,946 + 2,305 + 6,886

**829.** 5/4 − 5/12

**830.** 7.1 × 1.7

**831.** Find the HCF and LCM of 20 and 120.

**832.** Estimate 97 × 616 by rounding (approximate answer). *(approximate)*

*Answers: see the answer key (questions 825–832).*

## ✎ Set 6: Foundational arithmetic · Intermediate · suggested time 10 min

**833.** Find the remainder when 59,711 is divided by 11.

**834.** $(5^3)^3$

**835.** Add: 5,801 + 5,597 + 7,715 + 8,769 + 9,456 + 3,055

**836.** 1/3 + 6/7

**837.** 2.2 × 0.59

**838.** Find the HCF and LCM of 90 and 18.

**839.** Estimate 64 × 296 by rounding (approximate answer). *(approximate)*

**840.** Find the remainder when 51,194 is divided by 11.

*Answers: see the answer key (questions 833–840).*

## ✎ Set 7: Foundational arithmetic · Intermediate · suggested time 10 min

**841.** $3^{-3}$ as a fraction

**842.** Add: 3,768 + 4,689 + 8,797 + 7,293 + 5,033 + 5,381

**843.** 5/12 + 1

**844.** 2.7 × 0.69

**845.** Find the HCF and LCM of 105 and 165.

**846.** Estimate 56 × 860 by rounding (approximate answer). *(approximate)*

**847.** Find the remainder when 45,994 is divided by 11.

**848.** $(3^1)^2$

*Answers: see the answer key (questions 841–848).*

## ✎ Set 8: Foundational arithmetic · Advanced · suggested time 16 min

**849.** Add: 8,834 + 9,542 + 6,302 + 6,877

**850.** 8/9 + 8/3

**851.** 1.3 × 0.95

**852.** Find the HCF and LCM of 108 and 144.

**853.** Estimate 68 × 396 by rounding (approximate answer). *(approximate)*

**854.** Find the remainder when 67,585 is divided by 11.

**855.** $5^{-3}$ as a fraction

**856.** Add: 3,747 + 3,227 + 522 + 4,712 + 2,643 + 1,337

*Answers: see the answer key (questions 849–856).*

## ✎ Set 9: Foundational arithmetic · Advanced · suggested time 16 min

**857.** 5/7 − 3/2

**858.** 7.2 × 6.2

**859.** Find the HCF and LCM of 18 and 66.

**860.** Estimate 21 × 838 by rounding (approximate answer). *(approximate)*

**861.** Find the remainder when 97,136 is divided by 9.

**862.** $5^{5} \times 5^{1}$ (give the value)

**863.** Add: 8,981 + 5,106 + 533 + 677

**864.** 1/3 + 4/3

*Answers: see the answer key (questions 857–864).*

## ✎ Set 10: Foundational arithmetic · Advanced · suggested time 16 min

**865.** 9.4 × 6.1

**866.** Find the HCF and LCM of 48 and 20.

**867.** Estimate 32 × 491 by rounding (approximate answer). *(approximate)*

**868.** Find the remainder when 40,561 is divided by 7.

**869.** Add: 3,957 + 2,687 + 6,972 + 6,502 + 4,330 + 8,852

**870.** 2/11 + 2/11

**871.** 6.4 × 1.9

**872.** Find the HCF and LCM of 40 and 55.

*Answers: see the answer key (questions 865–872).*

## Answers and explanations — Foundational arithmetic

**793.** **444** — Hundreds 400, tens 40, units 4: 400 + 40 + 4 = 444.

**794.** **5,104** — 29 = 30 − 1. 5,075 + 30 = 5,105, then − 1 = 5,104.

**795.** **3,553** — All from 9, last from 10: 9−6=3 9−4=5 9−4=5, 10−7=3 → 3553.

**796.** **1,331** — Subtract 680 instead: 2,008 − 680 = 1,328; add back 3 → 1,331.

**797.** **−14** — Subtracting 17 is the same as adding −17. 18 + −17 + −15 = −14.

**798.** **11/5** — HCF of 66 and 30 is 6. Divide both by 6.

**799.** **96** — Split 15% into easy parts (10%, 5%, 1% …) or use fractions: 15% = 3/20. Result 96.

**800.** **Yes** — Rule for 8: last three digits divisible by 8. 99,256 ÷ 8 leaves remainder 0.

**801.** **2,684** — 99 = 100 − 1. 2,585 + 100 = 2,685, then − 1 = 2,684.

**802.** **4,216** — All from 9, last from 10: 9−9=0 9−5=4 9−7=2 9−8=1, 10−4=6 → 04216.

**803.** **1,757** — Subtract 4,280 instead: 6,035 − 4,280 = 1,755; add back 2 → 1,757.

**804.** **−4** — Subtracting 16 is the same as adding −16. 13 + −16 + −1 = −4.

**805.** **10/7** — HCF of 90 and 63 is 9. Divide both by 9.

**806.** **600** — Split 25% into easy parts (10%, 5%, 1% …) or use fractions: 25% = 1/4. Result 600.

**807.** **No** — Rule for 11: (sum of odd-place digits − sum of even-place digits) divisible by 11. 35,012 ÷ 11 leaves remainder 10.

**808.** **1,292** — Hundreds 1200, tens 90, units 2: 1200 + 90 + 2 = 1292.

**809.** **8,018** — All from 9, last from 10: 9−1=8 9−9=0 9−8=1, 10−2=8 → 8018.

**810.** **5,061** — Subtract 1,940 instead: 7,000 − 1,940 = 5,060; add back 1 → 5,061.

**811.** **−20** — Subtracting 18 is the same as adding −18. 6 + −18 + −8 = −20.

**812.** **6/5** — HCF of 30 and 25 is 5. Divide both by 5.

**813.** **960** — Split 150% into easy parts (10%, 5%, 1% …) or use fractions: 150% = 3/2. Result 960.

**814.** **Yes** — Rule for 3: digit sum divisible by 3. 11,388 ÷ 3 leaves remainder 0.

**815.** **1,043** — Hundreds 900, tens 130, units 13: 900 + 130 + 13 = 1043.

**816.** **6,869** — 199 = 200 − 1. 6,670 + 200 = 6,870, then − 1 = 6,869.

**817.** **HCF 6, LCM 36** — HCF × LCM = product: 6 × 36 = 216 = 18 × 12.

**818.** **≈ 24,000 (exact 23,838)** — 29 ≈ 30, 822 ≈ 800; 30 × 800 = 24,000. This is an estimate, not the exact product.

**819.** **12** — 15,053 = 13 × 1,157 + 12.

**820.** **19,53,125** — Add exponents: $5^9$ = 19,53,125.

**821.** **23,778** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 9.

**822.** **1 48/55** — LCM of denominators = 55. Convert, then add numerators and simplify: 103/55.

**823.** **2.067** — Multiply without points: 39 × 53 = 2067. Decimal places: 1 + 2 = 3, so the answer is 2.067.

**824.** **HCF 2, LCM 88** — HCF × LCM = product: 2 × 88 = 176 = 8 × 22.

**825.** **≈ 42,000 (exact 38,570)** — 70 ≈ 70, 551 ≈ 600; 70 × 600 = 42,000. This is an estimate, not the exact product.

**826.** **5** — 29,048 = 9 × 3,227 + 5. Shortcut: digit sum mod 9.

**827.** **128** — Add exponents: $2^7$ = 128.

**828.** **20,777** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 5.

**829.** **5/6** — LCM of denominators = 12. Convert, then subtract numerators and simplify: 5/6.

**830.** **12.07** — Multiply without points: 71 × 17 = 1207. Decimal places: 1 + 1 = 2, so the answer is 12.07.

**831.** **HCF 20, LCM 120** — HCF × LCM = product: 20 × 120 = 2400 = 20 × 120.

**832.** **≈ 60,000 (exact 59,752)** — 97 ≈ 100, 616 ≈ 600; 100 × 600 = 60,000. This is an estimate, not the exact product.

**833.** **3** — 59,711 = 11 × 5,428 + 3.

**834.** **19,53,125** — Multiply exponents: $5^9$.

**835.** **40,393** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 1.

**836.** **1 4/21** — LCM of denominators = 21. Convert, then add numerators and simplify: 25/21.

**837.** **1.298** — Multiply without points: 22 × 59 = 1298. Decimal places: 1 + 2 = 3, so the answer is 1.298.

**838.** **HCF 18, LCM 90** — HCF × LCM = product: 18 × 90 = 1620 = 90 × 18.

**839.** **≈ 18,000 (exact 18,944)** — 64 ≈ 60, 296 ≈ 300; 60 × 300 = 18,000. This is an estimate, not the exact product.

**840.** **0** — 51,194 = 11 × 4,654 + 0.

**841.** **1/27** — A negative exponent means reciprocal: $1/3^3$.

**842.** **34,961** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 5.

**843.** **1 5/12** — LCM of denominators = 12. Convert, then add numerators and simplify: 17/12.

**844.** **1.863** — Multiply without points: 27 × 69 = 1863. Decimal places: 1 + 2 = 3, so the answer is 1.863.

**845.** **HCF 15, LCM 1155** — HCF × LCM = product: 15 × 1155 = 17325 = 105 × 165.

**846.** **≈ 54,000 (exact 48,160)** — 56 ≈ 60, 860 ≈ 900; 60 × 900 = 54,000. This is an estimate, not the exact product.

**847.** **3** — 45,994 = 11 × 4,181 + 3.

**848.** **9** — Multiply exponents: $3^2$.

**849.** **31,555** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 1.

**850.** **3 5/9** — LCM of denominators = 9. Convert, then add numerators and simplify: 32/9.

**851.** **1.235** — Multiply without points: 13 × 95 = 1235. Decimal places: 1 + 2 = 3, so the answer is 1.235.

**852.** **HCF 36, LCM 432** — HCF × LCM = product: 36 × 432 = 15552 = 108 × 144.

**853.** **≈ 28,000 (exact 26,928)** — 68 ≈ 70, 396 ≈ 400; 70 × 400 = 28,000. This is an estimate, not the exact product.

**854.** **1** — 67,585 = 11 × 6,144 + 1.

**855.** **1/125** — A negative exponent means reciprocal: $1/5^3$.

**856.** **16,188** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 6.

**857.** **–11/14** — LCM of denominators = 14. Convert, then subtract numerators and simplify: –11/14.

**858.** **44.64** — Multiply without points: 72 × 62 = 4464. Decimal places: 1 + 1 = 2, so the answer is 44.64.

**859.** **HCF 6, LCM 198** — HCF × LCM = product: 6 × 198 = 1188 = 18 × 66.

**860.** **≈ 16,000 (exact 17,598)** — 21 ≈ 20, 838 ≈ 800; 20 × 800 = 16,000. This is an estimate, not the exact product.

**861.** **8** — 97,136 = 9 × 10,792 + 8. Shortcut: digit sum mod 9.

**862.** **15,625** — Add exponents: $5^6$ = 15,625.

**863.** **15,297** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 6.

**864.** **1 2/3** — LCM of denominators = 3. Convert, then add numerators and simplify: 5/3.

**865.** **57.34** — Multiply without points: 94 × 61 = 5734. Decimal places: 1 + 1 = 2, so the answer is 57.34.

**866.** **HCF 4, LCM 240** — HCF × LCM = product: 4 × 240 = 960 = 48 × 20.

**867.** **≈ 14,700 (exact 15,712)** — 32 ≈ 30, 491 ≈ 490; 30 × 490 = 14,700. This is an estimate, not the exact product.

**868.** **3** — 40,561 = 7 × 5,794 + 3.

**869.** **33,300** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 9.

**870.** **4/11** — LCM of denominators = 11. Convert, then add numerators and simplify: 4/11.

**871.** **12.16** — Multiply without points: 64 × 19 = 1216. Decimal places: 1 + 1 = 2, so the answer is 12.16.

**872.** **HCF 5, LCM 440** — HCF × LCM = product: 5 × 440 = 2200 = 40 × 55.

## Multiplication and division — 10 sets

### ✎ Set 11: Multiplication and division · Beginner · suggested time 6 min

**873.** 2,004 × 11

**874.** 903 × 50

**875.** 780 × 999

**876.** 8,597 ÷ 25

**877.** 1,140 ÷ 50 (as a decimal)

**878.** 3,052 ÷ 9 (quotient and remainder)

**879.** 1,973 × 11

**880.** 404 × 25

*Answers: see the answer key (questions 873–880).*

### ✎ Set 12: Multiplication and division · Beginner · suggested time 6 min

**881.** 867 × 50

**882.** 47 × 99

**883.** 1,107 ÷ 7

**884.** 8,929 ÷ 50 (as a decimal)

**885.** 7,402 ÷ 9 (quotient and remainder)

**886.** 9,825 × 11

**887.** 309 × 25

**888.** 137 × 999

*Answers: see the answer key (questions 881–888).*

### ✎ Set 13: Multiplication and division · Beginner · suggested time 6 min

**889.** 198 × 999

**890.** 8,551 ÷ 29

**891.** 4,718 ÷ 5 (as a decimal)

**892.** 23,790 ÷ 9 (quotient and remainder)

**893.** 1,055 × 11

**894.** 744 × 50

**895.** 403 × 999

**896.** 1,902 ÷ 7

*Answers: see the answer key (questions 889–896).*

## ✎ Set 14: Multiplication and division · Intermediate · suggested time 10 min

**897.** 62 × 68

**898.** 776 × 1,001

**899.** 966 × 980 (Nikhilam, base 1,000)

**900.** 105 × 112 (base 100)

**901.** 78 × 52 (vertically and crosswise)

**902.** 21 × 29

**903.** 91 × 101

**904.** 89 × 98 (Nikhilam, base 100)

*Answers: see the answer key (questions 897–904).*

## ✎ Set 15: Multiplication and division · Intermediate · suggested time 10 min

**905.** 938 × 1,001

**906.** 99 × 98 (Nikhilam, base 100)

**907.** 108 × 106 (base 100)

**908.** 96 × 79 (vertically and crosswise)

**909.** 29 × 21

**910.** 892 × 1,001

**911.** 1,019 × 1,036 (base 1,000)

**912.** 74 × 28 (vertically and crosswise)

*Answers: see the answer key (questions 905–912).*

## ✎ Set 16: Multiplication and division · Intermediate · suggested time 10 min

**913.** 96 × 91 (Nikhilam, base 100)

**914.** 105 × 101 (base 100)

**915.** 88 × 55 (vertically and crosswise)

**916.** 105 × 105

**917.** 591 × 1,001

**918.** 961 × 972 (Nikhilam, base 1,000)

**919.** 103 × 112 (base 100)

**920.** 82 × 16 (vertically and crosswise)

*Answers: see the answer key (questions 913–920).*

## ✎ Set 17: Multiplication and division · Intermediate · suggested time 10 min

**921.** 1,027 × 1,057 (base 1,000)

**922.** 48 × 78 (vertically and crosswise)

**923.** 81 × 89

**924.** 305 × 1,001

**925.** 89 × 92 (Nikhilam, base 100)

**926.** 110 × 104 (base 100)

**927.** 43 × 58 (vertically and crosswise)

**928.** 107 × 103

*Answers: see the answer key (questions 921–928).*

## ✎ Set 18: Multiplication and division · Advanced · suggested time 16 min

**929.** 47,000 ÷ 996

**930.** 1,123 ÷ 13

**931.** 42 × 45 (working base 50)

**932.** 292 × 576

**933.** 4,213 ÷ 88

**934.** 540 ÷ 12

**935.** 299 × 299 (working base 300)

**936.** 108 × 90

*Answers: see the answer key (questions 929–936).*

## ✎ Set 19: Multiplication and division · Advanced · suggested time 16 min

**937.** 66,477 ÷ 1012

**938.** 304 × 308 (working base 300)

**939.** 111 × 95

**940.** 808 × 587

**941.** 552 ÷ 7

**942.** 746 ÷ 14

**943.** 301 × 295 (working base 300)

**944.** 108 × 96

*Answers: see the answer key (questions 937–944).*

## ✎ Set 20: Multiplication and division · Advanced · suggested time 16 min

**945.** 209 × 192 (working base 200)

**946.** 107 × 94

**947.** 691 × 236

**948.** 2,194 ÷ 98

**949.** 1,079 ÷ 14

**950.** 202 × 196 (working base 200)

**951.** 102 × 93

**952.** 636 × 436

*Answers: see the answer key (questions 945–952).*

## Answers and explanations — Multiplication and division

**873.** **22,044** — Write first digit 2 and last digit 4 In between, add neighbours: 2+0=2, 0+0=0, 0+4=4 Carry where a sum ≥ 10 Answer = 22,044

**874.** **45,150** — 50 = 100 / 2 → 903 × 100 ÷ 2 = 45,150.

**875.** **7,79,220** — Left: 780 − 1 = 779 Right: 999 − 779 = 220 Answer = 7,79,220

**876.** **Q = 343, R = 22** — Check: 25 × 343 + 22 = 8,597.

**877.** **22.8** — ÷ 50 = × 2 ÷ 100: 1,140 × 2 = 2,280, ÷ 100 = 22.8.

**878.** **Q = 339, R = 1** — Running digit sums (quotient columns): 3, 3, 8 Last running sum + last digit = remainder column 10 Adjust carries/remainder ≥ 9 → quotient 339, remainder 1

**879.** **21,703** — Write first digit 1 and last digit 3 In between, add neighbours: 1+9=10, 9+7=16, 7+3=10 Carry where a sum ≥ 10 Answer = 21,703

**880.** **10,100** — 25 = 100 / 4 → 404 × 100 ÷ 4 = 10,100.

**881.** **43,350** — 50 = 100 / 2 → 867 × 100 ÷ 2 = 43,350.

**882.** **4,653** — Left: 47 − 1 = 46 Right: 99 − 46 = 53 Answer = 4,653

**883.** **Q = 158, R = 1** — Check: 7 × 158 + 1 = 1,107.

**884.** **178.58** — ÷ 50 = × 2 ÷ 100: 8,929 × 2 = 17,858, ÷ 100 = 178.58.

**885.** **Q = 822, R = 4** — Running digit sums (quotient columns): 7, 11, 11 Last running sum + last digit = remainder column 13 Adjust carries/remainder ≥ 9 → quotient 822, remainder 4

**886.** **1,08,075** — Write first digit 9 and last digit 5 In between, add neighbours: 9+8=17, 8+2=10, 2+5=7 Carry where a sum ≥ 10 Answer = 1,08,075

**887.** **7,725** — 25 = 100 / 4 → 309 × 100 ÷ 4 = 7,725.

**888.** **1,36,863** — Left: 137 − 1 = 136 Right: 999 − 136 = 863 Answer = 1,36,863

**889.** **1,97,802** — Left: 198 − 1 = 197 Right: 999 − 197 = 802 Answer = 1,97,802

**890.** **Q = 294, R = 25** — Check: 29 × 294 + 25 = 8,551.

**891.** **943.6** — ÷ 5 = × 2 ÷ 10: 4,718 × 2 = 9,436, ÷ 10 = 943.6.

**892.** **Q = 2,643, R = 3** — Running digit sums (quotient columns): 2, 5, 12, 21 Last running sum + last digit = remainder column 21 Adjust carries/remainder ≥ 9 → quotient 2,643, remainder 3

**893.** **11,605** — Write first digit 1 and last digit 5 In between, add neighbours: 1+0=1, 0+5=5, 5+5=10 Carry where a sum ≥ 10 Answer = 11,605

**894.** **37,200** — 50 = 100 / 2 → 744 × 100 ÷ 2 = 37,200.

**895.** **4,02,597** — Left: 403 − 1 = 402 Right: 999 − 402 = 597 Answer = 4,02,597

**896.** **Q = 271, R = 5** — Check: 7 × 271 + 5 = 1,902.

**897.** **4,216** — Same leading part 6; last digits 2 + 8 = 10 ✓ Left: 6 × 7 = 42 Right: 2 × 8 = 16 (always two digits) Answer = 4,216

**898.** **7,76,776** — 1001 = 1000 + 1, so write 776 twice (padded): 776 × 1001 = 7,76,776.

**899.** **9,46,680** — Base = 1,000 Deviations: 966 → −34, 980 → −20 Left part (cross): 966 − 20 = 946 Right part: (−34) × (−20) = 680 Right part padded to 3 digit(s): 680 Answer = 9,46,680

**900.** **11,760** — Base = 100 Deviations: 105 → +5, 112 → +12 Left part (cross): 105 + 12 = 117 Right part: (+5) × (+12) = 60 Right part padded to 2 digit(s): 60 Answer = 11,760

**901.** **4,056** — Column 1 (from right): 8×2 = 16 → write 6, carry 1 Column 2 (from right): 8×5 + 7×2 = 54, + carry 1 = 55 → write 5, carry 5 Column 3 (from right): 7×5 = 35, + carry 5 = 40 → write 0, carry 4 Answer = 4,056

**902.** **609** — Same leading part 2; last digits 1 + 9 = 10 ✓ Left: 2 × 3 = 6 Right: 1 × 9 = 09 (always two digits) Answer = 609

**903.** **9,191** — 101 = 100 + 1, so write 91 twice (padded): 91 × 101 = 9,191.

**904.** **8,722** — Base = 100 Deviations: 89 → −11, 98 → −2 Left part (cross): 89 − 2 = 87 Right part: (−11) × (−2) = 22 Right part padded to 2 digit(s): 22 Answer = 8,722

**905.** **9,38,938** — 1001 = 1000 + 1, so write 938 twice (padded): 938 × 1001 = 9,38,938.

**906.** **9,702** — Base = 100 Deviations: 99 → −1, 98 → −2 Left part (cross): 99 − 2 = 97 Right part: (−1) × (−2) = 2 Right part padded to 2 digit(s): 02 Answer = 9,702

**907.** **11,448** — Base = 100 Deviations: 108 → +8, 106 → +6 Left part (cross): 108 + 6 = 114 Right part: (+8) × (+6) = 48 Right part padded to 2 digit(s): 48 Answer = 11,448

**908.** **7,584** — Column 1 (from right): 6×9 = 54 → write 4, carry 5 Column 2 (from right): 6×7 + 9×9 = 123, + carry 5 = 128 → write 8, carry 12 Column 3 (from right): 9×7 = 63, + carry 12 = 75 → write 5, carry 7 Answer = 7,584

**909.** **609** — Same leading part 2; last digits 9 + 1 = 10 ✓ Left: 2 × 3 = 6 Right: 9 × 1 = 09 (always two digits) Answer = 609

**910.** **8,92,892** — 1001 = 1000 + 1, so write 892 twice (padded): 892 × 1001 = 8,92,892.

**911.** **10,55,684** — Base = 1,000 Deviations: 1019 → +19, 1036 → +36 Left part (cross): 1019 + 36 = 1055 Right part: (+19) × (+36) = 684 Right part padded to 3 digit(s): 684 Answer = 10,55,684

**912.** **2,072** — Column 1 (from right): 4×8 = 32 → write 2, carry 3 Column 2 (from right): 4×2 + 7×8 = 64, + carry 3 = 67 → write 7, carry 6 Column 3 (from right): 7×2 = 14, + carry 6 = 20 → write 0, carry 2 Answer = 2,072

**913.** **8,736** — Base = 100 Deviations: 96 → −4, 91 → −9 Left part (cross): 96 − 9 = 87 Right part: (−4) × (−9) = 36 Right part padded to 2 digit(s): 36 Answer = 8,736

**914.** **10,605** — Base = 100 Deviations: 105 → +5, 101 → +1 Left part (cross): 105 + 1 = 106 Right part: (+5) × (+1) = 5 Right part padded to 2 digit(s): 05 Answer = 10,605

**915.** **4,840** — Column 1 (from right): 8×5 = 40 → write 0, carry 4 Column 2 (from right): 8×5 + 8×5 = 80, + carry 4 = 84 → write 4, carry 8 Column 3 (from right): 8×5 = 40, + carry 8 = 48 → write 8, carry 4 Answer = 4,840

**916.** **11,025** — Same leading part 10; last digits 5 + 5 = 10 ✓ Left: 10 × 11 = 110 Right: 5 × 5 = 25 (always two digits) Answer = 11,025

**917.** **5,91,591** — 1001 = 1000 + 1, so write 591 twice (padded): 591 × 1001 = 5,91,591.

**918.** **9,34,092** — Base = 1,000 Deviations: 961 → −39, 972 → −28 Left part (cross): 961 − 28 = 933 Right part: (−39) × (−28) = 1092 Right part has more than 3 digit(s): carry 1 to the left Answer = 9,34,092

**919.** **11,536** — Base = 100 Deviations: 103 → +3, 112 → +12 Left part (cross): 103 + 12 = 115 Right part: (+3) × (+12) = 36 Right part padded to 2 digit(s): 36 Answer = 11,536

**920.** **1,312** — Column 1 (from right): 2×6 = 12 → write 2, carry 1 Column 2 (from right): 2×1 + 8×6 = 50, + carry 1 = 51 → write 1, carry 5 Column 3 (from right): 8×1 = 8, + carry 5 = 13 → write 3, carry 1 Answer = 1,312

**921.** **10,85,539** — Base = 1,000 Deviations: 1027 → +27, 1057 → +57 Left part (cross): 1027 + 57 = 1084 Right part: (+27) × (+57) = 1539 Right part has more than 3 digit(s): carry 1 to the left Answer = 10,85,539

**922.** **3,744** — Column 1 (from right): 8×8 = 64 → write 4, carry 6 Column 2 (from right): 8×7 + 4×8 = 88, + carry 6 = 94 → write 4, carry 9 Column 3 (from right): 4×7 = 28, + carry 9 = 37 → write 7, carry 3 Answer = 3,744

**923.** **7,209** — Same leading part 8; last digits 1 + 9 = 10 ✓ Left: 8 × 9 = 72 Right: 1 × 9 = 09 (always two digits) Answer = 7,209

**924.** **3,05,305** — 1001 = 1000 + 1, so write 305 twice (padded): 305 × 1001 = 3,05,305.

**925.** **8,188** — Base = 100 Deviations: 89 → −11, 92 → −8 Left part (cross): 89 − 8 = 81 Right part: (−11) × (−8) = 88 Right part padded to 2 digit(s): 88 Answer = 8,188

**926.** **11,440** — Base = 100 Deviations: 110 → +10, 104 → +4 Left part (cross): 110 + 4 = 114 Right part: (+10) × (+4) = 40 Right part padded to 2 digit(s): 40 Answer = 11,440

**927.** **2,494** — Column 1 (from right): 3×8 = 24 → write 4, carry 2 Column 2 (from right): 3×5 + 4×8 = 47, + carry 2 = 49 → write 9, carry 4 Column 3 (from right): 4×5 = 20, + carry 4 = 24 → write 4, carry 2 Answer = 2,494

**928.** **11,021** — Same leading part 10; last digits 7 + 3 = 10 ✓ Left: 10 × 11 = 110 Right: 7 × 3 = 21 (always two digits) Answer = 11,021

**929.** **Q = 47, R = 188** — Divisor 996 = 1,000 − 4; write complement digits 004 Split dividend: 47 | 000 Column totals after multiplying by the complement: [4, 7] | [0, 16, 28] Resolve carries and remainder → quotient 47, remainder 188

**930.** **Q = 86, R = 5** — Divisor 13 = 10 + 3; transposed flags: −3 Column values after applying flags: [1, -2, 8] | [-21] Resolve negatives/carries → quotient 86, remainder 5

**931.** **1,890** — Working base 50 = 5 × 10. Deviations −8, −5. Left: (42 −5) × 5 = 185. Right: −8 × −5 = 40. 185 × 10 + 40 = 1,890.

**932.** **1,68,192** — Column 1 (from right): 2×6 = 12 → write 2, carry 1 Column 2 (from right): 2×7 + 9×6 = 68, + carry 1 = 69 → write 9, carry 6 Column 3 (from right): 2×5 + 9×7 + 2×6 = 85, + carry 6 = 91 → write 1, carry 9 Column 4 (from right): 9×5 + 2×7 = 59, + carry 9 = 68 → write 8, carry 6 Column 5 (from right): 2×5 = 10, + carry 6 = 16 → write 6, carry 1 Answer = 1,68,192

**933.** **Q = 47, R = 77** — Divisor 88 = 100 − 12; write complement digits 12 Split dividend: 42 | 13 Column totals after multiplying by the complement: [4, 6] | [15, 15] Resolve carries and remainder → quotient 47, remainder 77

**934.** **Q = 45, R = 0** — Divisor 12 = 10 + 2; transposed flags: –2 Column values after applying flags: [5, -6] | [12] Resolve negatives/carries → quotient 45, remainder 0

**935.** **89,401** — Working base 300 = 3 × 100. Deviations –1, –1. Left: (299 –1) × 3 = 894. Right: –1 × –1 = 1. 894 × 100 + 1 = 89,401.

**936.** **9,720** — Base = 100 Deviations: 108 → +8, 90 → –10 Left part (cross): 108 – 10 = 98 Right part: (+8) × (–10) = –80 Right part is negative: borrow 1 from left → 98 × 100 – 80 Answer = 9,720

**937.** **Q = 65, R = 697** — Divisor 1012 = 1,000 + 12; transposed flags: –0, –1, –2 Column values after applying flags: [6, 6] | [-2, -11, -5] Resolve negatives/carries → quotient 65, remainder 697

**938.** **93,632** — Working base 300 = 3 × 100. Deviations +4, +8. Left: (304 +8) × 3 = 936. Right: 4 × 8 = 32. 936 × 100 + 32 = 93,632.

**939.** **10,545** — Base = 100 Deviations: 111 → +11, 95 → –5 Left part (cross): 111 – 5 = 106 Right part: (+11) × (–5) = –55 Right part is negative: borrow 1 from left → 106 × 100 – 55 Answer = 10,545

**940.** **4,74,296** — Column 1 (from right): 8×7 = 56 → write 6, carry 5 Column 2 (from right): 8×8 + 0×7 = 64, + carry 5 = 69 → write 9, carry 6 Column 3 (from right): 8×5 + 0×8 + 8×7 = 96, + carry 6 = 102 → write 2, carry 10 Column 4 (from right): 0×5 + 8×8 = 64, + carry 10 = 74 → write 4, carry 7 Column 5 (from right): 8×5 = 40, + carry 7 = 47 → write 7, carry 4 Answer = 4,74,296

**941.** **Q = 78, R = 6** — Divisor 7 = 10 – 3; write complement digits 3 Split dividend: 55 | 2 Column totals after multiplying by the complement: [5, 20] | [62] Resolve carries and remainder → quotient 78, remainder 6

**942.** **Q = 53, R = 4** — Divisor 14 = 10 + 4; transposed flags: –4 Column values after applying flags: [7, -24] | [102] Resolve negatives/carries → quotient 53, remainder 4

**943.** **88,795** — Working base 300 = 3 × 100. Deviations +1, –5. Left: (301 –5) × 3 = 888. Right: 1 × –5 = –5. 888 × 100 + –5 = 88,795.

**944.** **10,368** — Base = 100 Deviations: 108 → +8, 96 → –4 Left part (cross): 108 – 4 = 104 Right part: (+8) × (–4) = –32 Right part is negative: borrow 1 from left → 104 × 100 – 32 Answer = 10,368

**945.** **40,128** — Working base 200 = 2 × 100. Deviations +9, –8. Left: (209 –8) × 2 = 402. Right: 9 × –8 = –72. 402 × 100 + –72 = 40,128.

**946.** **10,058** — Base = 100 Deviations: 107 → +7, 94 → –6 Left part (cross): 107 – 6 = 101 Right part: (+7) × (–6) = –42 Right part is negative: borrow 1 from left → 101 × 100 – 42 Answer = 10,058

**947.** **1,63,076** — Column 1 (from right): 1×6 = 6 → write 6, carry 0 Column 2 (from right): 1×3 + 9×6 = 57 → write 7, carry 5 Column 3 (from right): 1×2 + 9×3 + 6×6 = 65, + carry 5 = 70 → write 0, carry 7 Column 4 (from right): 9×2 + 6×3 = 36, + carry 7 = 43 → write 3, carry 4 Column 5 (from right): 6×2 = 12, + carry 4 = 16 → write 6, carry 1 Answer = 1,63,076

**948.** **Q = 22, R = 38** — Divisor 98 = 100 – 2; write complement digits 02 Split dividend: 21 | 94 Column totals after multiplying by the complement: [2, 1] | [13, 6] Resolve carries and remainder → quotient 22, remainder 38

**949.** **Q = 77, R = 1** — Divisor 14 = 10 + 4; transposed flags: –4 Column values after applying flags: [1, -4, 23] | [-83] Resolve negatives/carries → quotient 77, remainder 1

**950.** **39,592** — Working base 200 = 2 × 100. Deviations +2, –4. Left: (202 –4) × 2 = 396. Right: 2 × –4 = –8. 396 × 100 + –8 = 39,592.

**951.** **9,486** — Base = 100 Deviations: 102 → +2, 93 → –7 Left part (cross): 102 – 7 = 95 Right part: (+2) × (–7) = –14 Right part is negative: borrow 1 from left → 95 × 100 – 14 Answer = 9,486

**952.** **2,77,296** — Column 1 (from right): 6×6 = 36 → write 6, carry 3 Column 2 (from right): 6×3 + 3×6 = 36, + carry 3 = 39 → write 9, carry 3 Column 3 (from right): 6×4 + 3×3 + 6×6 = 69, + carry 3 = 72 → write 2, carry 7 Column 4 (from right): 3×4 + 6×3 = 30, + carry 7 = 37 → write 7, carry 3 Column 5 (from right): 6×4 = 24, + carry 3 = 27 → write 7, carry 2 Answer = 2,77,296

## Squares, cubes and roots — 10 sets

### ✎ Set 21: Squares, cubes and roots · Beginner · suggested time 6 min

**953.** $27^2$

**954.** $17^2$

**955.** $115^2$

**956.** $47^2$

**957.** $43^2$

**958.** $145^2$

**959.** $59^2$

**960.** $175^2$

*Answers: see the answer key (questions 953–960).*

### ✎ Set 22: Squares, cubes and roots · Beginner · suggested time 6 min

**961.** $12^2$

**962.** $39^2$

**963.** $195^2$

**964.** $14^2$

**965.** $95^2$

**966.** $60^2$

**967.** $32^2$

**968.** $11^2$

*Answers: see the answer key (questions 961–968).*

### ✎ Set 23: Squares, cubes and roots · Beginner · suggested time 6 min

**969.** $37^2$

**970.** $65^2$

**971.** $24^2$

**972.** $35^2$

**973.** $33^2$

**974.** $165^2$

**975.** $38^2$

**976.** $56^2$

*Answers: see the answer key (questions 969–976).*

## ✎ Set 24: Squares, cubes and roots · Intermediate · suggested time 10 min

**977.** $\sqrt{441}$

**978.** $\sqrt[3]{2,05,379}$

**979.** Use $(a \pm b)^2$ to find $31^2$

**980.** $103^2$

**981.** $\sqrt{961}$

**982.** $\sqrt[3]{6,58,503}$

**983.** $18^3$

**984.** $1,034^2$

*Answers: see the answer key (questions 977–984).*

## ✎ Set 25: Squares, cubes and roots · Intermediate · suggested time 10 min

**985.** $\sqrt[3]{9,41,192}$

**986.** Use $(a \pm b)^2$ to find $301^2$

**987.** $20^3$

**988.** $106^2$

**989.** $\sqrt{7,744}$

**990.** $\sqrt[3]{5,31,441}$

**991.** $22^3$

**992.** $\sqrt{529}$

*Answers: see the answer key (questions 985–992).*

## ✎ Set 26: Squares, cubes and roots · Intermediate · suggested time 10 min

**993.** Use $(a \pm b)^2$ to find $202^2$

**994.** $17^3$

**995.** $48^2$

**996.** $\sqrt{3,721}$

**997.** $\sqrt[3]{3,375}$

**998.** Use $(a \pm b)^2$ to find $18^2$

**999.** $13^3$

**1000.** $1,009^2$

*Answers: see the answer key (questions 993–1000).*

## ✎ Set 27: Squares, cubes and roots · Intermediate · suggested time 10 min

**1001.** $19^3$

**1002.** $\sqrt{7,056}$

**1003.** $\sqrt[3]{91,125}$

**1004.** Use $(a \pm b)^2$ to find $62^2$

**1005.** $986^2$

**1006.** $\sqrt[3]{6,81,472}$

**1007.** Use $(a \pm b)^2$ to find $87^2$

**1008.** $\sqrt{9,216}$

*Answers: see the answer key (questions 1001–1008).*

## ✎ Set 28: Squares, cubes and roots · Advanced · suggested time 16 min

**1009.** Estimate √346 to 2 decimal places (approximate). *(approximate)*

**1010.** $997^2$ (duplex method)

**1011.** Estimate √80 to 2 decimal places (approximate). *(approximate)*

**1012.** $206^2$ (duplex method)

**1013.** Estimate √374 to 2 decimal places (approximate). *(approximate)*

**1014.** $358^2$ (duplex method)

**1015.** Estimate √83 to 2 decimal places (approximate). *(approximate)*

**1016.** $976^2$ (duplex method)

*Answers: see the answer key (questions 1009–1016).*

## ✎ Set 29: Squares, cubes and roots · Advanced · suggested time 16 min

**1017.** $57^2$ (duplex method)

**1018.** Estimate √140 to 2 decimal places (approximate). *(approximate)*

**1019.** $412^2$ (duplex method)

**1020.** Estimate √220 to 2 decimal places (approximate). *(approximate)*

**1021.** $838^2$ (duplex method)

**1022.** Estimate √118 to 2 decimal places (approximate). *(approximate)*

**1023.** $561^2$ (duplex method)

**1024.** Estimate √383 to 2 decimal places (approximate). *(approximate)*

*Answers: see the answer key (questions 1017–1024).*

## ✎ Set 30: Squares, cubes and roots · Advanced · suggested time 16 min

**1025.** Estimate √294 to 2 decimal places (approximate). *(approximate)*

**1026.** $550^2$ (duplex method)

**1027.** Estimate √291 to 2 decimal places (approximate). *(approximate)*

**1028.** $810^2$ (duplex method)

**1029.** Estimate √253 to 2 decimal places (approximate). *(approximate)*

**1030.** $851^2$ (duplex method)

**1031.** Estimate √240 to 2 decimal places (approximate). *(approximate)*

**1032.** $912^2$ (duplex method)

*Answers: see the answer key (questions 1025–1032).*

## Answers and explanations — Squares, cubes and roots

**953.** **729** — 27 × 27 = 729.

**954.** **289** — 17 × 17 = 289.

**955.** **13,225** — Digits before 5: 11 11 × (one more = 12) = 132 Attach 25 → 13,225

**956.** **2,209** — 47 × 47 = 2209.

**957.** **1,849** — 43 × 43 = 1849.

**958.** **21,025** — Digits before 5: 14 14 × (one more = 15) = 210 Attach 25 → 21,025

**959.** **3,481** — 59 × 59 = 3481.

**960.** **30,625** — Digits before 5: 17 17 × (one more = 18) = 306 Attach 25 → 30,625

**961.** **144** — 12 × 12 = 144.

**962.** **1,521** — 39 × 39 = 1521.

**963.** **38,025** — Digits before 5: 19 19 × (one more = 20) = 380 Attach 25 → 38,025

**964.** **196** — 14 × 14 = 196.

**965.** **9,025** — Digits before 5: 9 9 × (one more = 10) = 90 Attach 25 → 9,025

**966.** **3,600** — 60 × 60 = 3600.

**967.** **1,024** — 32 × 32 = 1024.

**968.** **121** — 11 × 11 = 121.

**969.** **1,369** — 37 × 37 = 1369.

**970.** **4,225** — Digits before 5: 6 6 × (one more = 7) = 42 Attach 25 → 4,225

**971.** **576** — 24 × 24 = 576.

**972.** **1,225** — Digits before 5: 3 3 × (one more = 4) = 12 Attach 25 → 1,225

**973.** **1,089** — 33 × 33 = 1089.

**974.** **27,225** — Digits before 5: 16 16 × (one more = 17) = 272 Attach 25 → 27,225

**975.** **1,444** — 38 × 38 = 1444.

**976.** **3,136** — 56 × 56 = 3136.

**977.** **21** — Last digit 1 → root ends in 1 or 9. Leading part 4 lies between $2^2$ and $3^2$, so tens digit is 2. Check $25^2 = 625$ to choose → 21.

**978.** **59** — Last digit 9 → cube root ends in 9 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 205 lies between $5^3$ and $6^3$ → tens digit 5.

**979.** **961** — $(30 + 1)^2 = 900 + 2\times30\times1 + 1 = 961$.

**980.** **10,609** — Base 100, deviation +3 Left: 103 + 3 = 106 Right: $3^2 = 9$ (needs 2 digit(s)) Answer = 10,609

**981.** **31** — Last digit 1 → root ends in 1 or 9. Leading part 9 lies between $3^2$ and $4^2$, so tens digit is 3. Check $35^2 = 1225$ to choose → 31.

**982.** **87** — Last digit 3 → cube root ends in 7 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 658 lies between $8^3$ and $9^3$ → tens digit 8.

**983.** **5,832** — 18 × 18 × 18 = 5,832.

**984.** **10,69,156** — Base 1,000, deviation +34 Left: 1034 + 34 = 1068 Right: $34^2 = 1156$ (needs 3 digit(s), carry 1) Answer = 10,69,156

**985.** **98** — Last digit 2 → cube root ends in 8 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 941 lies between $9^3$ and $10^3$ → tens digit 9.

**986.** **90,601** — $(300 + 1)^2 = 90000 + 2\times300\times1 + 1 = 90,601$.

**987.** **8,000** — 20 × 20 × 20 = 8,000.

**988.** **11,236** — Base 100, deviation +6 Left: 106 + 6 = 112 Right: $6^2 = 36$ (needs 2 digit(s)) Answer = 11,236

**989.** **88** — Last digit 4 → root ends in 2 or 8. Leading part 77 lies between $8^2$ and $9^2$, so tens digit is 8. Check $85^2 = 7225$ to choose → 88.

**990.** **81** — Last digit 1 → cube root ends in 1 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 531 lies between $8^3$ and $9^3$ → tens digit 8.

**991.** **10,648** — 22 × 22 × 22 = 10,648.

**992.** **23** — Last digit 9 → root ends in 3 or 7. Leading part 5 lies between $2^2$ and $3^2$, so tens digit is 2. Check $25^2 = 625$ to choose → 23.

**993.** **40,804** — $(200 + 2)^2 = 40000 + 2\times200\times2 + 4 = 40{,}804$.

**994.** **4,913** — $17 \times 17 \times 17 = 4{,}913$.

**995.** **2,304** — Deviation from 50: −2 Left: 25 − 2 = 23 Right: $2^2 = 04$ Answer = 2,304

**996.** **61** — Last digit 1 → root ends in 1 or 9. Leading part 37 lies between $6^2$ and $7^2$, so tens digit is 6. Check $65^2 = 4225$ to choose → 61.

**997.** **15** — Last digit 5 → cube root ends in 5 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 3 lies between $1^3$ and $2^3$ → tens digit 1.

**998.** **324** — $(20 - 2)^2 = 400 - 2\times20\times2 + 4 = 324$.

**999.** **2,197** — $13 \times 13 \times 13 = 2{,}197$.

**1000.** **10,18,081** — Base 1,000, deviation +9 Left: 1009 + 9 = 1018 Right: $9^2 = 81$ (needs 3 digit(s)) Answer = 10,18,081

**1001.** **6,859** — $19 \times 19 \times 19 = 6{,}859$.

**1002.** **84** — Last digit 6 → root ends in 4 or 6. Leading part 70 lies between $8^2$ and $9^2$, so tens digit is 8. Check $85^2 = 7225$ to choose → 84.

**1003.** **45** — Last digit 5 → cube root ends in 5 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 91 lies between $4^3$ and $5^3$ → tens digit 4.

**1004.** **3,844** — $(60 + 2)^2 = 3600 + 2\times60\times2 + 4 = 3{,}844$.

**1005.** **9,72,196** — Base 1,000, deviation −14 Left: 986 − 14 = 972 Right: $14^2 = 196$ (needs 3 digit(s)) Answer = 9,72,196

**1006.** **88** — Last digit 2 → cube root ends in 8 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 681 lies between $8^3$ and $9^3$ → tens digit 8.

**1007.** **7,569** — $(90 - 3)^2 = 8100 - 2\times90\times3 + 9 = 7{,}569$.

**1008.** **96** — Last digit 6 → root ends in 4 or 6. Leading part 92 lies between $9^2$ and $10^2$, so tens digit is 9. Check $95^2 = 9025$ to choose → 96.

**1009.** **≈ 18.61 (true value 18.601)** — Nearest perfect square 361 = $19^2$. $\sqrt{n} \approx 19 + (346 - 361) \div (2 \times 19) = 18.605$.

**1010.** **9,94,009** — Duplexes from left: 81, 162, 207, 126, 49 Combine with carries (right to left) → 9,94,009

**1011.** **≈ 8.94 (true value 8.944)** — Nearest perfect square 81 = $9^2$. $\sqrt{n} \approx 9 + (80 - 81) \div (2 \times 9) = 8.944$.

**1012.** **42,436** — Duplexes from left: 4, 0, 24, 0, 36 Combine with carries (right to left) → 42,436

**1013.** **≈ 19.34 (true value 19.339)** — Nearest perfect square 361 = $19^2$. $\sqrt{n} \approx 19 + (374 - 361) \div (2 \times 19) = 19.342$.

**1014.** **1,28,164** — Duplexes from left: 9, 30, 73, 80, 64 Combine with carries (right to left) → 1,28,164

**1015.** **≈ 9.11 (true value 9.110)** — Nearest perfect square 81 = $9^2$. $\sqrt{n} \approx 9 + (83 - 81) \div (2 \times 9) = 9.111$.

**1016.** **9,52,576** — Duplexes from left: 81, 126, 157, 84, 36 Combine with carries (right to left) → 9,52,576

**1017.** **3,249** — Duplexes from left: 25, 70, 49 Combine with carries (right to left) → 3,249

**1018.** **≈ 11.83 (true value 11.832)** — Nearest perfect square 144 = $12^2$. $\sqrt{n} \approx 12 + (140 - 144) \div (2 \times 12) = 11.833$.

**1019.** **1,69,744** — Duplexes from left: 16, 8, 17, 4, 4 Combine with carries (right to left) → 1,69,744

**1020.** **≈ 14.83 (true value 14.832)** — Nearest perfect square 225 = $15^2$. $\sqrt{n} \approx 15 + (220 - 225) \div (2 \times 15) = 14.833$.

**1021.** **7,02,244** — Duplexes from left: 64, 48, 137, 48, 64 Combine with carries (right to left) → 7,02,244

**1022.** **≈ 10.86 (true value 10.863)** — Nearest perfect square 121 = $11^2$. $\sqrt{n} \approx 11 + (118 - 121) \div (2 \times 11) = 10.864$.

**1023.** **3,14,721** — Duplexes from left: 25, 60, 46, 12, 1 Combine with carries (right to left) → 3,14,721

**1024.** **≈ 19.57 (true value 19.570)** — Nearest perfect square 400 = $20^2$. $\sqrt{n} \approx 20 + (383 - 400) \div (2 \times 20) = 19.575$.

**1025.** **≈ 17.15 (true value 17.146)** — Nearest perfect square 289 = $17^2$. $\sqrt{n} \approx 17 + (294 - 289) \div (2 \times 17) = 17.147$.

**1026.** **3,02,500** — Duplexes from left: 25, 50, 25, 0, 0 Combine with carries (right to left) → 3,02,500

**1027.** **≈ 17.06 (true value 17.059)** — Nearest perfect square 289 = $17^2$. $\sqrt{n} \approx 17 + (291 - 289) \div (2 \times 17) = 17.059$.

**1028.** **6,56,100** — Duplexes from left: 64, 16, 1, 0, 0 Combine with carries (right to left) → 6,56,100

**1029.** **≈ 15.91 (true value 15.906)** — Nearest perfect square 256 = $16^2$. $\sqrt{n} \approx 16 + (253 - 256) \div (2 \times 16) = 15.906$.

**1030.** **7,24,201** — Duplexes from left: 64, 80, 41, 10, 1 Combine with carries (right to left) → 7,24,201

**1031.** **≈ 15.50 (true value 15.492)** — Nearest perfect square 225 = $15^2$. $\sqrt{n} \approx 15 + (240 - 225) \div (2 \times 15) = 15.500$.

**1032.** **8,31,744** — Duplexes from left: 81, 18, 37, 4, 4 Combine with carries (right to left) → 8,31,744

# Algebra and factorisation — 10 sets

## ✎ Set 31: Algebra and factorisation · Beginner · suggested time 6 min

**1033.** Solve $4x - 8 = -44$

**1034.** Expand $(4x - 8)(x + 9)$

**1035.** Solve $6x - 11 = -71$

**1036.** Expand $(2x + 5)(2x - 3)$

**1037.** Solve $6x - 1 = -49$

**1038.** Expand $(4x - 6)(x + 7)$

**1039.** Solve $2x - 6 = 8$

**1040.** Expand $(2x - 4)(3x - 1)$

*Answers: see the answer key (questions 1033–1040).*

## ✎ Set 32: Algebra and factorisation · Beginner · suggested time 6 min

**1041.** Expand $(4x + 3)(2x - 7)$

**1042.** Solve $2x - 14 = -12$

**1043.** Expand $(3x + 2)(2x + 7)$

**1044.** Solve $2x + 10 = 2$

**1045.** Expand $(4x + 7)(3x + 7)$

**1046.** Solve $2x - 19 = -15$

**1047.** Expand $(4x + 7)(3x - 9)$

**1048.** Solve $7x + 14 = 105$

*Answers: see the answer key (questions 1041–1048).*

## ✎ Set 33: Algebra and factorisation · Beginner · suggested time 6 min

**1049.** Solve $6x + 2 = -34$

**1050.** Expand $(2x + 6)(3x + 1)$

**1051.** Solve $3x + 10 = 31$

**1052.** Expand $(x + 1)(x - 1)$

**1053.** Solve $7x + 9 = -26$

**1054.** Expand $(3x - 9)(3x + 1)$

**1055.** Solve $9x - 9 = -36$

**1056.** Expand $(4x + 8)(3x - 1)$

*Answers: see the answer key (questions 1049–1056).*

## ✎ Set 34: Algebra and factorisation · Intermediate · suggested time 10 min

**1057.** $62^2 - 24^2$

**1058.** Solve $12x - 2 = 1x + 108$

**1059.** Solve: $4x + 9y = -46$, $5x + 1y = 4$

**1060.** Factorise $x^2 - 12x + 35$

**1061.** $30^2 - 5^2$

**1062.** Use $(a \pm b)^2$ to find $17^2$

**1063.** Solve $3x - 10 = 5x - 24$

**1064.** Solve: $4x + 6y = -16$, $2x + 6y = -8$

*Answers: see the answer key (questions 1057–1064).*

## ✎ Set 35: Algebra and factorisation · Intermediate · suggested time 10 min

**1065.** Use $(a \pm b)^2$ to find $41^2$

**1066.** Solve $12x - 8 = 1x - 52$

**1067.** Solve: $6x + 2y = 20$, $8x + 1y = 35$

**1068.** Factorise $x^2 - 4x - 12$

**1069.** $63^2 - 18^2$

**1070.** Use $(a \pm b)^2$ to find $67^2$

**1071.** Solve $6x + 15 = 1x + 45$

**1072.** Solve: $1x + 5y = -36$, $5x + 8y = -78$

*Answers: see the answer key (questions 1065–1072).*

## ✎ Set 36: Algebra and factorisation · Intermediate · suggested time 10 min

**1073.** Solve $7x - 1 = 1x + 23$

**1074.** Solve: $6x + 1y = -16$, $3x + 1y = -10$

**1075.** Factorise $x^2 + 2x - 24$

**1076.** $83^2 - 20^2$

**1077.** Use $(a \pm b)^2$ to find $103^2$

**1078.** Solve $9x + 6 = 3x + 0$

**1079.** Solve: $8x + 4y = -48$, $7x + 3y = -39$

**1080.** Factorise $x^2 - 6x - 7$

*Answers: see the answer key (questions 1073–1080).*

## ✎ Set 37: Algebra and factorisation · Intermediate · suggested time 10 min

**1081.** Solve: $4x + 5y = 11$, $3x + 3y = 6$

**1082.** Factorise $x^2 - 8x + 12$

**1083.** $88^2 - 17^2$

**1084.** Solve $12x - 2 = 4x - 82$

**1085.** Solve: $3x + 5y = 24$, $5x + 8y = 38$

**1086.** Factorise $x^2 + 2x - 8$

**1087.** $48^2 - 18^2$

**1088.** Use $(a \pm b)^2$ to find $22^2$

*Answers: see the answer key (questions 1081–1088).*

## ✎ Set 38: Algebra and factorisation · Advanced · suggested time 16 min

**1089.** Solve $x^2 - 4x + 3 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1090.** Solve: $34x + 54y = 208$, $54x + 34y = -32$

**1091.** Solve $(x + 6)(x + 1) = (x + 2)(x + 9)$

**1092.** Solve $x^2 + 8x + 12 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1093.** Solve: $20x + 14y = 230$, $14x + 20y = 212$

**1094.** Solve $(x + 11)(x + 12) = (x + 10)(x + 10)$

**1095.** Solve $x^2 + 4x + 3 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1096.** Solve: $50x + 49y = 47$, $49x + 50y = 52$

*Answers: see the answer key (questions 1089–1096).*

## ✎ Set 39: Algebra and factorisation · Advanced · suggested time 16 min

**1097.** Solve: $16x + 47y = 47$, $47x + 16y = 16$

**1098.** Solve $(x + 10)(x + 4) = (x + 1)(x + 11)$

**1099.** Solve $x^2 - 13x + 42 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1100.** Solve: $48x + 49y = -48$, $49x + 48y = -49$

**1101.** Solve $(x + 2)(x + 8) = (x + 4)(x + 3)$

**1102.** Solve $x^2 + 0x - 36 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1103.** Solve: $51x + 46y = -301$, $46x + 51y = -281$

**1104.** Solve $(x + 5)(x + 12) = (x + 12)(x + 4)$

*Answers: see the answer key (questions 1097–1104).*

## ✎ Set 40: Algebra and factorisation · Advanced · suggested time 16 min

**1105.** Solve $(x + 9)(x + 2) = (x + 7)(x + 12)$

**1106.** Solve $x^2 + 9x + 18 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1107.** Solve: $58x + 11y = 36$, $11x + 58y = -105$

**1108.** Solve $(x + 10)(x + 4) = (x + 8)(x + 3)$

**1109.** Solve $x^2 - 8x + 0 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1110.** Solve: $50x + 28y = 344$, $28x + 50y = 124$

**1111.** Solve $(x + 7)(x + 11) = (x + 4)(x + 7)$

**1112.** Solve $x^2 - 7x + 10 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

*Answers: see the answer key (questions 1105–1112).*

# Answers and explanations — Algebra and factorisation

**1033.** **x = −9** — Transpose −8 (change sign): 4x = −44 + 8 = −36; divide by 4: x = −9.

**1034.** **4x² + 28x − 72** — Vertical (left): 4 × 1 = 4. Crosswise: 4 × (9) + 1 × (−8) = 28. Vertical (right): (−8) × (9) = −72.

**1035.** **x = −10** — Transpose −11 (change sign): 6x = −71 + 11 = −60; divide by 6: x = −10.

**1036.** **4x² + 4x − 15** — Vertical (left): 2 × 2 = 4. Crosswise: 2 × (−3) + 2 × (5) = 4. Vertical (right): (5) × (−3) = −15.

**1037.** **x = −8** — Transpose −1 (change sign): 6x = −49 + 1 = −48; divide by 6: x = −8.

**1038.** **4x² + 22x − 42** — Vertical (left): 4 × 1 = 4. Crosswise: 4 × (7) + 1 × (−6) = 22. Vertical (right): (−6) × (7) = −42.

**1039.** **x = 7** — Transpose −6 (change sign): 2x = 8 + 6 = 14; divide by 2: x = 7.

**1040.** **6x² − 14x + 4** — Vertical (left): 2 × 3 = 6. Crosswise: 2 × (−1) + 3 × (−4) = −14. Vertical (right): (−4) × (−1) = 4.

**1041.** **8x² − 22x − 21** — Vertical (left): 4 × 2 = 8. Crosswise: 4 × (−7) + 2 × (3) = −22. Vertical (right): (3) × (−7) = −21.

**1042.** **x = 1** — Transpose −14 (change sign): 2x = −12 + 14 = 2; divide by 2: x = 1.

**1043.** **6x² + 25x + 14** — Vertical (left): 3 × 2 = 6. Crosswise: 3 × (7) + 2 × (2) = 25. Vertical (right): (2) × (7) = 14.

**1044.** **x = −4** — Transpose 10 (change sign): 2x = 2 − 10 = −8; divide by 2: x = −4.

**1045.** **12x² + 49x + 49** — Vertical (left): 4 × 3 = 12. Crosswise: 4 × (7) + 3 × (7) = 49. Vertical (right): (7) × (7) = 49.

**1046.** **x = 2** — Transpose −19 (change sign): 2x = −15 + 19 = 4; divide by 2: x = 2.

**1047.** **12x² − 15x − 63** — Vertical (left): 4 × 3 = 12. Crosswise: 4 × (−9) + 3 × (7) = −15. Vertical (right): (7) × (−9) = −63.

**1048.** **x = 13** — Transpose 14 (change sign): 7x = 105 − 14 = 91; divide by 7: x = 13.

**1049.** **x = −6** — Transpose 2 (change sign): 6x = −34 − 2 = −36; divide by 6: x = −6.

**1050.** **6x² + 20x + 6** — Vertical (left): 2 × 3 = 6. Crosswise: 2 × (1) + 3 × (6) = 20. Vertical (right): (6) × (1) = 6.

**1051.** **x = 7** — Transpose 10 (change sign): 3x = 31 − 10 = 21; divide by 3: x = 7.

**1052.** **x² − 1** — Vertical (left): 1 × 1 = 1. Crosswise: 1 × (−1) + 1 × (1) = 0. Vertical (right): (1) × (−1) = −1.

**1053.** **x = −5** — Transpose 9 (change sign): 7x = −26 − 9 = −35; divide by 7: x = −5.

**1054.** **9x² − 24x − 9** — Vertical (left): 3 × 3 = 9. Crosswise: 3 × (1) + 3 × (−9) = −24. Vertical (right): (−9) × (1) = −9.

**1055.** **x = −3** — Transpose −9 (change sign): 9x = −36 + 9 = −27; divide by 9: x = −3.

**1056.** **12x² + 20x − 8** — Vertical (left): 4 × 3 = 12. Crosswise: 4 × (−1) + 3 × (8) = 20. Vertical (right): (8) × (−1) = −8.

**1057.** **3,268** — $a^2 - b^2 = (a + b)(a - b) = 86 \times 38 = 3{,}268$.

**1058.** **x = 10** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (108 − −2)/(12 − 1) = 110/11 = 10.

**1059.** **x = 2, y = −6** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-46 - 36)/-41 = 2$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (16 - -230)/-41 = -6$.

**1060.** **(x − 7)(x − 5)** — Find two numbers with product 35 and sum −12: −7 and −5. Check with Gunita-samuccaya (put x = 1): (1 − 7)(1 − 5) = 24 and 1 − 12 + 35 = 24.

**1061.** **875** — $a^2 - b^2 = (a + b)(a - b) = 35 \times 25 = 875$.

**1062.** **289** — $(20 - 3)^2 = 400 - 2{\times}20{\times}3 + 9 = 289$.

**1063.** **x = 7** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (−24 − −10)/(3 − 5) = −14/-2 = 7.

**1064.** **x = −4, y = 0** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-96 - -48)/12 = -4$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (-32 - -32)/12 = 0$.

**1065.** **1,681** — $(40 + 1)^2 = 1600 + 2{\times}40{\times}1 + 1 = 1{,}681$.

**1066.** **x = −4** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (−52 − −8)/(12 − 1) = −44/11 = −4.

**1067.** **x = 5, y = −5** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (20 - 70)/-10 = 5$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (210 - 160)/-10 = -5$.

**1068.** **(x − 6)(x + 2)** — Find two numbers with product −12 and sum −4: −6 and 2. Check with Gunita-samuccaya (put x = 1): (1 − 6)(1 + 2) = −15 and 1 − 4 − 12 = −15.

**1069.** **3,645** — $a^2 - b^2 = (a + b)(a - b) = 81 \times 45 = 3{,}645$.

**1070.** **4,489** — $(70 - 3)^2 = 4900 - 2{\times}70{\times}3 + 9 = 4{,}489$.

**1071.** **x = 6** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (45 − 15)/(6 − 1) = 30/5 = 6.

**1072.** **x = −6, y = −6** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-288 - -390)/-17 = -6$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (-78 - -180)/-17 = -6$.

**1073.** **x = 4** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (23 − −1)/(7 − 1) = 24/6 = 4.

**1074.** **x = –2, y = –4** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-16 - -10)/3 = -2$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (-60 - -48)/3 = -4$.

**1075.** **(x + 6)(x – 4)** — Find two numbers with product –24 and sum 2: 6 and –4. Check with Gunita-samuccaya (put x = 1): (1 + 6)(1 – 4) = –21 and 1 + 2 – 24 = –21.

**1076.** **6,489** — $a^2 - b^2 = (a + b)(a - b) = 103 \times 63 = 6{,}489$.

**1077.** **10,609** — $(100 + 3)^2 = 10000 + 2{\times}100{\times}3 + 9 = 10{,}609$.

**1078.** **x = –1** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (0 – 6)/(9 – 3) = –6/6 = –1.

**1079.** **x = –3, y = –6** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-144 - -156)/-4 = -3$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (-312 - -336)/-4 = -6$.

**1080.** **(x – 7)(x + 1)** — Find two numbers with product –7 and sum –6: –7 and 1. Check with Gunita-samuccaya (put x = 1): (1 – 7)(1 + 1) = –12 and 1 – 6 – 7 = –12.

**1081.** **x = –1, y = 3** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (33 - 30)/-3 = -1$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (24 - 33)/-3 = 3$.

**1082.** **(x – 2)(x – 6)** — Find two numbers with product 12 and sum –8: –2 and –6. Check with Gunita-samuccaya (put x = 1): (1 – 2)(1 – 6) = 5 and 1 – 8 + 12 = 5.

**1083.** **7,455** — $a^2 - b^2 = (a + b)(a - b) = 105 \times 71 = 7{,}455$.

**1084.** **x = –10** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (–82 – –2)/(12 – 4) = –80/8 = –10.

**1085.** **x = –2, y = 6** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (192 - 190)/-1 = -2$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (114 - 120)/-1 = 6$.

**1086.** **(x + 4)(x – 2)** — Find two numbers with product –8 and sum 2: 4 and –2. Check with Gunita-samuccaya (put x = 1): (1 + 4)(1 – 2) = –5 and 1 + 2 – 8 = –5.

**1087.** **1,980** — $a^2 - b^2 = (a + b)(a - b) = 66 \times 30 = 1{,}980$.

**1088.** **484** — $(20 + 2)^2 = 400 + 2{\times}20{\times}2 + 4 = 484$.

**1089.** **x = 3 or 1** — Differential: 2x – 4. Discriminant = 16 – 12 = 4, √ = 2. 2x – 4 = ±2.

**1090.** **x = –5, y = 7** — Add: 88(x + y) = 176 → x + y = 2. Subtract: -20(x – y) = 240 → x – y = –12. Solve: x = –5, y = 7.

**1091.** **x = –3** — x = (cd – ab)/(a + b – c – d) = (18 – 6)/(7 – 11).

**1092.** **x = –2 or –6** — Differential: 2x + 8. Discriminant = 64 – 48 = 16, √ = 4. 2x + 8 = ±4.

**1093.** **x = 8, y = 5** — Add: 34(x + y) = 442 → x + y = 13. Subtract: 6(x – y) = 18 → x – y = 3. Solve: x = 8, y = 5.

**1094.** **x = –10 2/3** — x = (cd – ab)/(a + b – c – d) = (100 – 132)/(23 – 20).

**1095.** **x = –1 or –3** — Differential: 2x + 4. Discriminant = 16 – 12 = 4, √ = 2. 2x + 4 = ±2.

**1096.** **x = –2, y = 3** — Add: 99(x + y) = 99 → x + y = 1. Subtract: 1(x – y) = –5 → x – y = –5. Solve: x = –2, y = 3.

**1097.** **x = 0, y = 1** — Add: 63(x + y) = 63 → x + y = 1. Subtract: -31(x – y) = 31 → x – y = –1. Solve: x = 0, y = 1.

**1098.** **x = –14 1/2** — x = (cd – ab)/(a + b – c – d) = (11 – 40)/(14 – 12).

**1099.** **x = 7 or 6** — Differential: 2x – 13. Discriminant = 169 – 168 = 1, √ = 1. 2x – 13 = ±1.

**1100.** **x = –1, y = 0** — Add: 97(x + y) = –97 → x + y = –1. Subtract: -1(x – y) = 1 → x – y = –1. Solve: x = –1, y = 0.

**1101.** **x = –1 1/3** — x = (cd – ab)/(a + b – c – d) = (12 – 16)/(10 – 7).

**1102.** **x = 6 or –6** — Differential: 2x + 0. Discriminant = 0 – -144 = 144, √ = 12. 2x + 0 = ±12.

**1103.** **x = –5, y = –1** — Add: 97(x + y) = –582 → x + y = –6. Subtract: 5(x – y) = –20 → x – y = –4. Solve: x = –5, y = –1.

**1104.** **x = –12** — x = (cd – ab)/(a + b – c – d) = (48 – 60)/(17 – 16).

**1105.** **x = –8 1/4** — x = (cd – ab)/(a + b – c – d) = (84 – 18)/(11 – 19).

**1106.** **x = –3 or –6** — Differential: 2x + 9. Discriminant = 81 – 72 = 9, √ = 3. 2x + 9 = ±3.

**1107.** **x = 1, y = –2** — Add: 69(x + y) = –69 → x + y = –1. Subtract: 47(x – y) = 141 → x – y = 3. Solve: x = 1, y = –2.

**1108.** **x = –5 1/3** — x = (cd – ab)/(a + b – c – d) = (24 – 40)/(14 – 11).

**1109.** **x = 8 or 0** — Differential: 2x – 8. Discriminant = 64 – 0 = 64, √ = 8. 2x – 8 = ±8.

**1110.** **x = 8, y = –2** — Add: 78(x + y) = 468 → x + y = 6. Subtract: 22(x – y) = 220 → x – y = 10. Solve: x = 8, y = –2.

**1111.** **x = –7** — x = (cd – ab)/(a + b – c – d) = (28 – 77)/(18 – 11).

**1112.** **x = 5 or 2** — Differential: 2x – 7. Discriminant = 49 – 40 = 9, √ = 3. 2x – 7 = ±3.

## Percentages, ratios and averages — 10 sets

### ✎ Set 41: Percentages, ratios and averages · Beginner · suggested time 6 min

**1113.** Find 35% of 480.

**1114.** Write 1/6 as a percentage.

**1115.** Find 12.5% of 608 using a fraction.

**1116.** Divide ₹5,600 between A and B in the ratio 2 : 5.

**1117.** Find the average of 79, 73, 81, 75, 79, 69.

**1118.** Find 60% of 160.

**1119.** Find 75% of 104 using a fraction.

**1120.** Divide ₹1,197 between A and B in the ratio 2 : 5.

*Answers: see the answer key (questions 1113–1120).*

### ✎ Set 42: Percentages, ratios and averages · Beginner · suggested time 6 min

**1121.** Write 7/8 as a percentage.

**1122.** Find 25% of 284 using a fraction.

**1123.** Divide ₹2,765 between A and B in the ratio 1 : 4.

**1124.** Find the average of 43, 34, 41, 48, 43, 34, 44.

**1125.** Find 5% of 4,800.

**1126.** Write 1/4 as a percentage.

**1127.** Find 75% of 440 using a fraction.

**1128.** Divide ₹2,285 between A and B in the ratio 2 : 3.

*Answers: see the answer key (questions 1121–1128).*

### ✎ Set 43: Percentages, ratios and averages · Beginner · suggested time 6 min

**1129.** Find 16 2/3% of 438 using a fraction.

**1130.** Divide ₹7,712 between A and B in the ratio 7 : 9.

**1131.** Find the average of 43, 42, 47, 48, 45.

**1132.** Write 2/3 as a percentage.

**1133.** Find 25% of 236 using a fraction.

**1134.** Divide ₹1,881 between A and B in the ratio 2 : 7.

**1135.** Find the average of 46, 36, 38, 35, 39, 39, 47.

**1136.** Find 150% of 160.

*Answers: see the answer key (questions 1129–1136).*

## ✎ Set 44: Percentages, ratios and averages · Intermediate · suggested time 10 min

**1137.** Find the single discount equal to successive discounts of 30% and 5%.

**1138.** A invests ₹40,000 for 12 months and B invests ₹25,000 for 12 months. Profit is ₹27,092. Find A's share.

**1139.** A price changes from ₹400 to ₹320. Find the percentage decrease.

**1140.** Section A has 36 students with average 62; section B has 34 students with average 70. Find the combined average.

**1141.** An item costing ₹1,500 is sold for ₹1,800. Find the profit percentage.

**1142.** A invests ₹50,000 for 12 months and B invests ₹35,000 for 9 months. Profit is ₹1,62,748. Find A's share.

**1143.** A price changes from ₹500 to ₹460. Find the percentage decrease.

**1144.** Section A has 33 students with average 41; section B has 17 students with average 66. Find the combined average.

*Answers: see the answer key (questions 1137–1144).*

## ✎ Set 45: Percentages, ratios and averages · Intermediate · suggested time 10 min

**1145.** A invests ₹40,000 for 12 months and B invests ₹60,000 for 6 months. Profit is ₹6,377. Find A's share.

**1146.** A price changes from ₹4,000 to ₹3,600. Find the percentage decrease.

**1147.** Section A has 30 students with average 59; section B has 32 students with average 81. Find the combined average.

**1148.** An item costing ₹800 is sold for ₹640. Find the loss percentage.

**1149.** A invests ₹20,000 for 12 months and B invests ₹35,000 for 6 months. Profit is ₹23,370. Find A's share.

**1150.** A price changes from ₹400 to ₹340. Find the percentage decrease.

**1151.** Section A has 21 students with average 63; section B has 15 students with average 55. Find the combined average.

**1152.** An item costing ₹250 is sold for ₹220. Find the loss percentage.

*Answers: see the answer key (questions 1145–1152).*

## ✎ Set 46: Percentages, ratios and averages · Intermediate · suggested time 10 min

**1153.** A price changes from ₹1,600 to ₹1,440. Find the percentage decrease.

**1154.** Section A has 11 students with average 48; section B has 30 students with average 79. Find the combined average.

**1155.** An item costing ₹400 is sold for ₹300. Find the loss percentage.

**1156.** Find the single discount equal to successive discounts of 20% and 15%.

**1157.** A invests ₹40,000 for 12 months and B invests ₹35,000 for 10 months. Profit is ₹49,136. Find A's share.

**1158.** Section A has 19 students with average 60; section B has 13 students with average 84. Find the combined average.

**1159.** An item costing ₹2,400 is sold for ₹2,520. Find the profit percentage.

**1160.** A invests ₹50,000 for 12 months and B invests ₹35,000 for 6 months. Profit is ₹34,776. Find A's share.

*Answers: see the answer key (questions 1153–1160).*

## ✎ Set 47: Percentages, ratios and averages · Intermediate · suggested time 10 min

**1161.** Section A has 24 students with average 49; section B has 34 students with average 55. Find the combined average.

**1162.** An item costing ₹500 is sold for ₹440. Find the loss percentage.

**1163.** Find the single discount equal to successive discounts of 20% and 5%.

**1164.** A invests ₹20,000 for 12 months and B invests ₹45,000 for 6 months. Profit is ₹19,788. Find A's share.

**1165.** Section A has 27 students with average 54; section B has 25 students with average 77. Find the combined average.

**1166.** An item costing ₹1,200 is sold for ₹1,140. Find the loss percentage.

**1167.** A invests ₹30,000 for 12 months and B invests ₹60,000 for 9 months. Profit is ₹8,700. Find A's share.

**1168.** A price changes from ₹800 to ₹680. Find the percentage decrease.

*Answers: see the answer key (questions 1161–1168).*

## ✎ Set 48: Percentages, ratios and averages · Advanced · suggested time 16 min

**1169.** In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹56/kg?

**1170.** In what ratio should rice at ₹30/kg be mixed with rice at ₹80/kg to get a mixture worth ₹49/kg?

**1171.** In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹50/kg?

**1172.** In what ratio should rice at ₹30/kg be mixed with rice at ₹80/kg to get a mixture worth ₹73/kg?

**1173.** In what ratio should rice at ₹50/kg be mixed with rice at ₹60/kg to get a mixture worth ₹53/kg?

**1174.** In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹62/kg?

**1175.** In what ratio should rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹46/kg?

**1176.** In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹59/kg?

*Answers: see the answer key (questions 1169–1176).*

## ✎ Set 49: Percentages, ratios and averages · Advanced · suggested time 16 min

**1177.** In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹44/kg?

**1178.** In what ratio should rice at ₹20/kg be mixed with rice at ₹60/kg to get a mixture worth ₹51/kg?

**1179.** In what ratio should rice at ₹20/kg be mixed with rice at ₹90/kg to get a mixture worth ₹37/kg?

**1180.** In what ratio should rice at ₹40/kg be mixed with rice at ₹80/kg to get a mixture worth ₹43/kg?

**1181.** In what ratio should rice at ₹50/kg be mixed with rice at ₹60/kg to get a mixture worth ₹55/kg?

**1182.** In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹61/kg?

**1183.** In what ratio should rice at ₹50/kg be mixed with rice at ₹90/kg to get a mixture worth ₹56/kg?

**1184.** In what ratio should rice at ₹40/kg be mixed with rice at ₹90/kg to get a mixture worth ₹43/kg?

*Answers: see the answer key (questions 1177–1184).*

## ✎ Set 50: Percentages, ratios and averages - Advanced - suggested time 16 min

**1185.** In what ratio should rice at ₹40/kg be mixed with rice at ₹80/kg to get a mixture worth ₹65/kg?

**1186.** In what ratio should rice at ₹20/kg be mixed with rice at ₹60/kg to get a mixture worth ₹49/kg?

**1187.** In what ratio should rice at ₹20/kg be mixed with rice at ₹90/kg to get a mixture worth ₹76/kg?

**1188.** In what ratio should rice at ₹50/kg be mixed with rice at ₹80/kg to get a mixture worth ₹55/kg?

**1189.** In what ratio should rice at ₹20/kg be mixed with rice at ₹70/kg to get a mixture worth ₹42/kg?

**1190.** In what ratio should rice at ₹40/kg be mixed with rice at ₹80/kg to get a mixture worth ₹72/kg?

**1191.** In what ratio should rice at ₹20/kg be mixed with rice at ₹90/kg to get a mixture worth ₹49/kg?

**1192.** In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹43/kg?

*Answers: see the answer key (questions 1185–1192).*

## Answers and explanations — Percentages, ratios and averages

**1113.** **168** — Split 35% into easy parts (10%, 5%, 1% …) or use fractions: 35% = 7/20. Result 168.

**1114.** **16 2/3%** — 1/6 × 100 = 16 2/3%.

**1115.** **76** — 12.5% = 1/8. 1/8 × 608 = 76.

**1116.** **A ₹1,600, B ₹4,000** — Total parts 7; one part = 5,600 ÷ 7 = 800.

**1117.** **76** — Assume 77. Deviations: +2, −4, +4, −2, +2, −8; total −6; ÷ 6 = −1. Average = 76.

**1118.** **96** — Split 60% into easy parts (10%, 5%, 1% …) or use fractions: 60% = 3/5. Result 96.

**1119.** **78** — 75% = 3/4. 3/4 × 104 = 78.

**1120.** **A ₹342, B ₹855** — Total parts 7; one part = 1,197 ÷ 7 = 171.

**1121.** **87 1/2%** — 7/8 × 100 = 87 1/2%.

**1122.** **71** — 25% = 1/4. 1/4 × 284 = 71.

**1123.** **A ₹553, B ₹2,212** — Total parts 5; one part = 2,765 ÷ 5 = 553.

**1124.** **41** — Assume 41. Deviations: +2, −7, +0, +7, +2, −7, +3; total +0; ÷ 7 = 0. Average = 41.

**1125.** **240** — Split 5% into easy parts (10%, 5%, 1% …) or use fractions: 5% = 1/20. Result 240.

**1126.** **25%** — 1/4 × 100 = 25%.

**1127.** **330** — 75% = 3/4. 3/4 × 440 = 330.

**1128.** **A ₹914, B ₹1,371** — Total parts 5; one part = 2,285 ÷ 5 = 457.

**1129.** **73** — 16 2/3% = 1/6. 1/6 × 438 = 73.

**1130.** **A ₹3,374, B ₹4,338** — Total parts 16; one part = 7,712 ÷ 16 = 482.

**1131.** **45** — Assume 47. Deviations: −4, −5, +0, +1, −2; total −10; ÷ 5 = −2. Average = 45.

**1132.** **66 2/3%** — 2/3 × 100 = 66 2/3%.

**1133.** **59** — 25% = 1/4. 1/4 × 236 = 59.

**1134.** **A ₹418, B ₹1,463** — Total parts 9; one part = 1,881 ÷ 9 = 209.

**1135.** **40** — Assume 44. Deviations: +2, −8, −6, −9, −5, −5, +3; total −28; ÷ 7 = −4. Average = 40.

**1136.** **240** — Split 150% into easy parts (10%, 5%, 1% …) or use fractions: 150% = 3/2. Result 240.

**1137.** **33 1/2%** — Net = a + b − ab/100 = 30 + 5 − 150/100 = 33 1/2%.

**1138.** **₹16,672** — Ratio = 40,000×12 : 25,000×12 = 8 : 5. A's share = 8/13 × 27,092 = 16,672.

**1139.** **20%** — Change = 80. % change = change ÷ original × 100 = 80/400 × 100 = 20%.

**1140.** **65.89** — (36×62 + 34×70) ÷ 70 = 4612 ÷ 70.

**1141.** **20% profit** — Profit = 300. % = 300/1500 × 100 = 20%.

**1142.** **₹1,06,720** — Ratio = 50,000×12 : 35,000×9 = 40 : 21. A's share = 40/61 × 1,62,748 = 1,06,720.

**1143.** **8%** — Change = 40. % change = change ÷ original × 100 = 40/500 × 100 = 8%.

**1144.** **49.50** — (33×41 + 17×66) ÷ 50 = 2475 ÷ 50.

**1145.** **₹3,644** — Ratio = 40,000×12 : 60,000×6 = 4 : 3. A's share = 4/7 × 6,377 = 3,644.

**1146.** **10%** — Change = 400. % change = change ÷ original × 100 = 400/4000 × 100 = 10%.

**1147.** **70.35** — (30×59 + 32×81) ÷ 62 = 4362 ÷ 62.

**1148.** **20% loss** — Loss = 160. % = 160/800 × 100 = 20%.

**1149.** **₹12,464** — Ratio = 20,000×12 : 35,000×6 = 8 : 7. A's share = 8/15 × 23,370 = 12,464.

**1150.** **15%** — Change = 60. % change = change ÷ original × 100 = 60/400 × 100 = 15%.

**1151.** **59.67** — (21×63 + 15×55) ÷ 36 = 2148 ÷ 36.

**1152.** **12% loss** — Loss = 30. % = 30/250 × 100 = 12%.

**1153.** **10%** — Change = 160. % change = change ÷ original × 100 = 160/1600 × 100 = 10%.

**1154.** **70.68** — (11×48 + 30×79) ÷ 41 = 2898 ÷ 41.

**1155.** **25% loss** — Loss = 100. % = 100/400 × 100 = 25%.

**1156.** **32%** — Net = a + b − ab/100 = 20 + 15 − 300/100 = 32%.

**1157.** **₹28,416** — Ratio = 40,000×12 : 35,000×10 = 48 : 35. A's share = 48/83 × 49,136 = 28,416.

**1158.** **69.75** — (19×60 + 13×84) ÷ 32 = 2232 ÷ 32.

**1159.** **5% profit** — Profit = 120. % = 120/2400 × 100 = 5%.

**1160.** **₹25,760** — Ratio = 50,000×12 : 35,000×6 = 20 : 7. A's share = 20/27 × 34,776 = 25,760.

**1161.** **52.52** — (24×49 + 34×55) ÷ 58 = 3046 ÷ 58.

**1162.** **12% loss** — Loss = 60. % = 60/500 × 100 = 12%.

**1163.** **24%** — Net = a + b − ab/100 = 20 + 5 − 100/100 = 24%.

**1164.** **₹9,312** — Ratio = 20,000×12 : 45,000×6 = 8 : 9. A's share = 8/17 × 19,788 = 9,312.

**1165.** **65.06** — (27×54 + 25×77) ÷ 52 = 3383 ÷ 52.

**1166.** **5% loss** — Loss = 60. % = 60/1200 × 100 = 5%.

**1167.** **₹3,480** — Ratio = 30,000×12 : 60,000×9 = 2 : 3. A's share = 2/5 × 8,700 = 3,480.

**1168.** **15%** — Change = 120. % change = change ÷ original × 100 = 120/800 × 100 = 15%.

**1169.** **7 : 3** — Alligation cross: (70 − 56) : (56 − 50) = 14 : 6 = 7 : 3.

**1170.** **31 : 19** — Alligation cross: (80 − 49) : (49 − 30) = 31 : 19 = 31 : 19.

**1171.** **1 : 2** — Alligation cross: (60 − 50) : (50 − 30) = 10 : 20 = 1 : 2.

**1172.** **7 : 43** — Alligation cross: (80 − 73) : (73 − 30) = 7 : 43 = 7 : 43.

**1173.** **7 : 3** — Alligation cross: (60 − 53) : (53 − 50) = 7 : 3 = 7 : 3.

**1174.** **2 : 3** — Alligation cross: (70 − 62) : (62 − 50) = 8 : 12 = 2 : 3.

**1175.** **7 : 3** — Alligation cross: (60 − 46) : (46 − 40) = 14 : 6 = 7 : 3.

**1176.** **11 : 19** — Alligation cross: (70 − 59) : (59 − 40) = 11 : 19 = 11 : 19.

**1177.** **13 : 2** — Alligation cross: (70 − 44) : (44 − 40) = 26 : 4 = 13 : 2.

**1178.** **9 : 31** — Alligation cross: (60 − 51) : (51 − 20) = 9 : 31 = 9 : 31.

**1179.** **53 : 17** — Alligation cross: (90 − 37) : (37 − 20) = 53 : 17 = 53 : 17.

**1180.** **37 : 3** — Alligation cross: (80 − 43) : (43 − 40) = 37 : 3 = 37 : 3.

**1181.** **1 : 1** — Alligation cross: (60 − 55) : (55 − 50) = 5 : 5 = 1 : 1.

**1182.** **3 : 7** — Alligation cross: (70 − 61) : (61 − 40) = 9 : 21 = 3 : 7.

**1183.** **17 : 3** — Alligation cross: (90 − 56) : (56 − 50) = 34 : 6 = 17 : 3.

**1184.** **47 : 3** — Alligation cross: (90 − 43) : (43 − 40) = 47 : 3 = 47 : 3.

**1185.** **3 : 5** — Alligation cross: (80 − 65) : (65 − 40) = 15 : 25 = 3 : 5.

**1186.** **11 : 29** — Alligation cross: (60 − 49) : (49 − 20) = 11 : 29 = 11 : 29.

**1187.** **1 : 4** — Alligation cross: (90 − 76) : (76 − 20) = 14 : 56 = 1 : 4.

**1188.** **5 : 1** — Alligation cross: (80 − 55) : (55 − 50) = 25 : 5 = 5 : 1.

**1189.** **14 : 11** — Alligation cross: (70 − 42) : (42 − 20) = 28 : 22 = 14 : 11.

**1190.** **1 : 4** — Alligation cross: (80 − 72) : (72 − 40) = 8 : 32 = 1 : 4.

**1191.** **41 : 29** — Alligation cross: (90 − 49) : (49 − 20) = 41 : 29 = 41 : 29.

**1192.** **17 : 13** — Alligation cross: (60 − 43) : (43 − 30) = 17 : 13 = 17 : 13.

# Government-exam mixed aptitude — 10 sets

## ✎ Set 51: Government-exam mixed aptitude · Beginner · suggested time 6 min

**1193.** Find the simple interest on ₹25,000 at 8% per annum for 2 years.

**1194.** A car travels at 54 km/h for 3/4 hours. Find the distance, and the speed in m/s.

**1195.** Divide ₹5,841 between A and B in the ratio 6 : 5.

**1196.** A car travels at 36 km/h for 1 1/2 hours. Find the distance, and the speed in m/s.

**1197.** Divide ₹11,674 between A and B in the ratio 5 : 8.

**1198.** Find the simple interest on ₹40,000 at 5% per annum for 4 years.

**1199.** A car travels at 36 km/h for 3/4 hours. Find the distance, and the speed in m/s.

**1200.** Divide ₹8,624 between A and B in the ratio 2 : 9.

*Answers: see the answer key (questions 1193–1200).*

## ✎ Set 52: Government-exam mixed aptitude · Beginner · suggested time 6 min

**1201.** A car travels at 90 km/h for 2 hours. Find the distance, and the speed in m/s.

**1202.** Divide ₹4,192 between A and B in the ratio 9 : 7.

**1203.** Find the simple interest on ₹25,000 at 6% per annum for 4 years.

**1204.** Divide ₹2,598 between A and B in the ratio 2 : 1.

**1205.** Find the simple interest on ₹25,000 at 6% per annum for 2 years.

**1206.** A car travels at 72 km/h for 1 1/2 hours. Find the distance, and the speed in m/s.

**1207.** Divide ₹6,831 between A and B in the ratio 4 : 5.

**1208.** Find the simple interest on ₹10,000 at 12% per annum for 3 years.

*Answers: see the answer key (questions 1201–1208).*

## ✎ Set 53: Government-exam mixed aptitude · Beginner · suggested time 6 min

**1209.** Divide ₹1,370 between A and B in the ratio 2 : 3.

**1210.** Find the simple interest on ₹5,000 at 5% per annum for 3 years.

**1211.** A car travels at 90 km/h for 1 1/2 hours. Find the distance, and the speed in m/s.

**1212.** Divide ₹5,271 between A and B in the ratio 6 : 1.

**1213.** Find the simple interest on ₹25,000 at 4% per annum for 3 years.

**1214.** A car travels at 54 km/h for 2 1/2 hours. Find the distance, and the speed in m/s.

**1215.** Divide ₹3,975 between A and B in the ratio 7 : 8.

**1216.** Find the simple interest on ₹5,000 at 4% per annum for 2 years.

*Answers: see the answer key (questions 1209–1216).*

## ✎ Set 54: Government-exam mixed aptitude · Intermediate · suggested time 10 min

**1217.** Find the compound interest on ₹40,000 at 10% per annum, compounded yearly, for 2 years.

**1218.** Two trains 180 m and 80 m long run at 54 km/h and 54 km/h in opposite directions. How long do they take to cross each other?

**1219.** A boat goes downstream at 20 km/h and upstream at 12 km/h. Find the speed of the boat in still water and of the stream.

**1220.** A father is 4 times as old as his son. In 5 years the sum of their ages will be 35. Find their present ages.

**1221.** Find the area and circumference of a circle of radius 14 cm (π = 22/7).

**1222.** In how many ways can 2 prizes (1st, 2nd, …) be given to 12 students, one prize each?

**1223.** Find the last digit of $22^{799}$.

**1224.** Find the remainder when 31,522 is divided by 13.

*Answers: see the answer key (questions 1217–1224).*

## ✎ Set 55: Government-exam mixed aptitude · Intermediate · suggested time 10 min

**1225.** A can finish a job in 15 days and B in 10 days. How long will they take together?

**1226.** A boat goes downstream at 18 km/h and upstream at 8 km/h. Find the speed of the boat in still water and of the stream.

**1227.** A father is 3 times as old as his son. In 5 years the sum of their ages will be 86. Find their present ages.

**1228.** A card is drawn from a well-shuffled deck of 52. Probability of a heart?

**1229.** In how many ways can a committee of 3 be chosen from 10 people?

**1230.** Find the last digit of $8^{745}$.

**1231.** Find the remainder when 60,683 is divided by 9.

**1232.** Simplify: 20 + 40 ÷ 4 × 10 − 6

*Answers: see the answer key (questions 1225–1232).*

## ✎ Set 56: Government-exam mixed aptitude · Intermediate · suggested time 10 min

**1233.** Two trains 100 m and 130 m long run at 90 km/h and 36 km/h in opposite directions. How long do they take to cross each other?

**1234.** A boat goes downstream at 24 km/h and upstream at 16 km/h. Find the speed of the boat in still water and of the stream.

**1235.** A father is 2 times as old as his son. In 6 years the sum of their ages will be 45. Find their present ages.

**1236.** Volume of a sphere of radius 21 cm (π = 22/7)?

**1237.** Two dice are thrown. What is the probability that the sum is 7?

**1238.** In how many ways can 2 prizes (1st, 2nd, …) be given to 9 students, one prize each?

**1239.** Find the last digit of $2^{131}$.

**1240.** Find the remainder when 84,942 is divided by 11.

*Answers: see the answer key (questions 1233–1240).*

## ✎ Set 57: Government-exam mixed aptitude · Intermediate · suggested time 10 min

**1241.** Two trains 240 m and 160 m long run at 72 km/h and 63 km/h in the same direction. How long do they take to cross each other?

**1242.** A boat goes downstream at 16 km/h and upstream at 12 km/h. Find the speed of the boat in still water and of the stream.

**1243.** A father is 3 times as old as his son. In 5 years the sum of their ages will be 74. Find their present ages.

**1244.** A rectangular field is 14 m by 23 m. Find its area and perimeter.

**1245.** Two dice are thrown. What is the probability that the sum is 6?

**1246.** Find the last digit of $4^{443}$.

**1247.** Find the remainder when 97,115 is divided by 7.

**1248.** Simplify: 18 + 27 ÷ 3 × 9 − 3

*Answers: see the answer key (questions 1241–1248).*

## ✎ Set 58: Government-exam mixed aptitude · Advanced · suggested time 16 min

**1249.** In what ratio should rice at ₹50/kg be mixed with rice at ₹80/kg to get a mixture worth ₹69/kg?

**1250.** Approximate: 875.02 × 49.02 ÷ 10 + 215.1 *(approximate)*

**1251.** Approximate: 558.98 × 12.01 ÷ 10 + 398.1 *(approximate)*

**1252.** In what ratio should rice at ₹40/kg be mixed with rice at ₹90/kg to get a mixture worth ₹77/kg?

**1253.** Approximate: 789.03 × 47.02 ÷ 10 + 644.1 *(approximate)*

**1254.** In what ratio should rice at ₹30/kg be mixed with rice at ₹80/kg to get a mixture worth ₹37/kg?

**1255.** Approximate: 590.97 × 49.01 ÷ 10 + 220.9 *(approximate)*

**1256.** In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹57/kg?

*Answers: see the answer key (questions 1249–1256).*

## ✎ Set 59: Government-exam mixed aptitude · Advanced · suggested time 16 min

**1257.** Approximate: 488.02 × 37.01 ÷ 10 + 376.9 *(approximate)*

**1258.** In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹56/kg?

**1259.** Approximate: 862.98 × 34.02 ÷ 10 + 598.9 *(approximate)*

**1260.** In what ratio should rice at ₹30/kg be mixed with rice at ₹70/kg to get a mixture worth ₹38/kg?

**1261.** Approximate: 366.03 × 25.02 ÷ 10 + 176.1 *(approximate)*

**1262.** In what ratio should rice at ₹20/kg be mixed with rice at ₹80/kg to get a mixture worth ₹32/kg?

**1263.** Approximate: 595.03 × 44.98 ÷ 10 + 384.1 *(approximate)*

**1264.** Approximate: 797.97 × 35.01 ÷ 10 + 455.9 *(approximate)*

*Answers: see the answer key (questions 1257–1264).*

## ✎ Set 60: Government-exam mixed aptitude - Advanced - suggested time 16 min

**1265.** In what ratio should rice at ₹20/kg be mixed with rice at ₹70/kg to get a mixture worth ₹60/kg?

**1266.** Approximate: 627.02 × 33.98 ÷ 10 + 353.9 *(approximate)*

**1267.** In what ratio should rice at ₹20/kg be mixed with rice at ₹60/kg to get a mixture worth ₹31/kg?

**1268.** Approximate: 453.03 × 35.01 ÷ 10 + 309.1 *(approximate)*

**1269.** In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹55/kg?

**1270.** Approximate: 460.98 × 12.01 ÷ 10 + 223.9 *(approximate)*

**1271.** In what ratio should rice at ₹40/kg be mixed with rice at ₹90/kg to get a mixture worth ₹85/kg?

**1272.** Approximate: 297.98 × 38.02 ÷ 10 + 157.9 *(approximate)*

*Answers: see the answer key (questions 1265–1272).*

## Answers and explanations — Government-exam mixed aptitude

**1193.** **₹4,000** — SI = PRT/100 = 25,000 × 8 × 2 / 100 = 4,000.

**1194.** **40 1/2 km; 15 m/s** — D = S × T = 54 × 3/4 = 40 1/2 km. km/h → m/s: × 5/18 → 15 m/s.

**1195.** **A ₹3,186, B ₹2,655** — Total parts 11; one part = 5,841 ÷ 11 = 531.

**1196.** **54 km; 10 m/s** — D = S × T = 36 × 1 1/2 = 54 km. km/h → m/s: × 5/18 → 10 m/s.

**1197.** **A ₹4,490, B ₹7,184** — Total parts 13; one part = 11,674 ÷ 13 = 898.

**1198.** **₹8,000** — SI = PRT/100 = 40,000 × 5 × 4 / 100 = 8,000.

**1199.** **27 km; 10 m/s** — D = S × T = 36 × 3/4 = 27 km. km/h → m/s: × 5/18 → 10 m/s.

**1200.** **A ₹1,568, B ₹7,056** — Total parts 11; one part = 8,624 ÷ 11 = 784.

**1201.** **180 km; 25 m/s** — D = S × T = 90 × 2 = 180 km. km/h → m/s: × 5/18 → 25 m/s.

**1202.** **A ₹2,358, B ₹1,834** — Total parts 16; one part = 4,192 ÷ 16 = 262.

**1203.** **₹6,000** — SI = PRT/100 = 25,000 × 6 × 4 / 100 = 6,000.

**1204.** **A ₹1,732, B ₹866** — Total parts 3; one part = 2,598 ÷ 3 = 866.

**1205.** **₹3,000** — SI = PRT/100 = 25,000 × 6 × 2 / 100 = 3,000.

**1206.** **108 km; 20 m/s** — D = S × T = 72 × 1 1/2 = 108 km. km/h → m/s: × 5/18 → 20 m/s.

**1207.** **A ₹3,036, B ₹3,795** — Total parts 9; one part = 6,831 ÷ 9 = 759.

**1208.** **₹3,600** — SI = PRT/100 = 10,000 × 12 × 3 / 100 = 3,600.

**1209.** **A ₹548, B ₹822** — Total parts 5; one part = 1,370 ÷ 5 = 274.

**1210.** **₹750** — SI = PRT/100 = 5,000 × 5 × 3 / 100 = 750.

**1211.** **135 km; 25 m/s** — D = S × T = 90 × 1 1/2 = 135 km. km/h → m/s: × 5/18 → 25 m/s.

**1212.** **A ₹4,518, B ₹753** — Total parts 7; one part = 5,271 ÷ 7 = 753.

**1213.** **₹3,000** — SI = PRT/100 = 25,000 × 4 × 3 / 100 = 3,000.

**1214.** **135 km; 15 m/s** — D = S × T = 54 × 2 1/2 = 135 km. km/h → m/s: × 5/18 → 15 m/s.

**1215.** **A ₹1,855, B ₹2,120** — Total parts 15; one part = 3,975 ÷ 15 = 265.

**1216.** **₹400** — SI = PRT/100 = 5,000 × 4 × 2 / 100 = 400.

**1217.** **₹8,400** — $A = P(1 + R/100)^n$ = 40,000 × $(110/100)^2$ = ₹48,400. CI = A − P = ₹8,400.

**1218.** **8 2/3 s** — Relative speed = 108 km/h = 30 m/s. Distance = 180 + 80 = 260 m. Time = 8 2/3 s.

**1219.** **Boat 16 km/h, stream 4 km/h** — Boat = (down + up)/2 = (20 + 12)/2 = 16. Stream = (down − up)/2 = 4.

**1220.** **Son 5, father 20** — Let son = x, father = 4x. (4x + 5) + (x + 5) = 35 → 5x = 25 → x = 5.

**1221.** **Area 616 cm², circumference 88 cm** — $A = \pi r^2 = 22/7 \times 14^2 = 616$; $C = 2\pi r = 88$.

**1222.** **132** — Order matters: ${}^nP_r$ = 12 × 11 × … (2 factors) = 132.

**1223.** **8** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 799 mod 4 = 3 → 8.

**1224.** **10** — 31,522 = 13 × 2,424 + 10.

**1225.** **6 days** — Total work = LCM(15,10) = 30 units. A does 2/day, B does 3/day; together 5/day. Time = 30/5 = 6 days.

**1226.** **Boat 13 km/h, stream 5 km/h** — Boat = (down + up)/2 = (18 + 8)/2 = 13. Stream = (down − up)/2 = 5.

**1227.** **Son 19, father 57** — Let son = x, father = 3x. (3x + 5) + (x + 5) = 86 → 4x = 76 → x = 19.

**1228.** **1/4** — 13/52 = 1/4.

**1229.** **120** — Order does not matter: ${}^nC_r = 10!/(3!\cdot 7!) = 120$.

**1230.** **8** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 745 mod 4 = 1 → 8.

**1231.** **5** — 60,683 = 9 × 6,742 + 5. Shortcut: digit sum mod 9.

**1232.** **114** — Division and multiplication left to right first: 40 ÷ 4 = 10; × 10 = 100. Then 20 + 100 − 6 = 114.

**1233.** **6 4/7 s** — Relative speed = 126 km/h = 35 m/s. Distance = 100 + 130 = 230 m. Time = 6 4/7 s.

**1234.** **Boat 20 km/h, stream 4 km/h** — Boat = (down + up)/2 = (24 + 16)/2 = 20. Stream = (down − up)/2 = 4.

**1235.** **Son 11, father 22** — Let son = x, father = 2x. (2x + 6) + (x + 6) = 45 → 3x = 33 → x = 11.

**1236.** **38,808 cm³** — $V = 4/3\ \pi r^3$ = 38,808.

**1237.** **1/6** — 6 favourable outcomes out of 36 → 6/36 = 1/6.

**1238.** **72** — Order matters: ${}^{n}P_{r} = 9 \times 8 \times \ldots$ (2 factors) = 72.

**1239.** **8** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 131 mod 4 = 3 → 8.

**1240.** **0** — 84,942 = 11 × 7,722 + 0.

**1241.** **160 s** — Relative speed = 9 km/h = 2 1/2 m/s. Distance = 240 + 160 = 400 m. Time = 160 s.

**1242.** **Boat 14 km/h, stream 2 km/h** — Boat = (down + up)/2 = (16 + 12)/2 = 14. Stream = (down − up)/2 = 2.

**1243.** **Son 16, father 48** — Let son = x, father = 3x. (3x + 5) + (x + 5) = 74 → 4x = 64 → x = 16.

**1244.** **Area 322 m², perimeter 74 m** — A = l × w; P = 2(l + w).

**1245.** **5/36** — 5 favourable outcomes out of 36 → 5/36 = 5/36.

**1246.** **4** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 443 mod 2 = 1 → 4.

**1247.** **4** — 97,115 = 7 × 13,873 + 4.

**1248.** **96** — Division and multiplication left to right first: 27 ÷ 3 = 9; × 9 = 81. Then 18 + 81 − 3 = 96.

**1249.** **11 : 19** — Alligation cross: (80 − 69) : (69 − 50) = 11 : 19 = 11 : 19.

**1250.** **≈ 4502 (exact 4504.45)** — Round each number: 875 × 49 ÷ 10 + 215 ≈ 4502.5.

**1251.** **≈ 1069 (exact 1069.43)** — Round each number: 559 × 12 ÷ 10 + 398 ≈ 1068.8.

**1252.** **13 : 37** — Alligation cross: (90 − 77) : (77 − 40) = 13 : 37 = 13 : 37.

**1253.** **≈ 4352 (exact 4354.12)** — Round each number: 789 × 47 ÷ 10 + 644 ≈ 4352.3.

**1254.** **43 : 7** — Alligation cross: (80 − 37) : (37 − 30) = 43 : 7 = 43 : 7.

**1255.** **≈ 3117 (exact 3117.24)** — Round each number: 591 × 49 ÷ 10 + 221 ≈ 3116.9.

**1256.** **1 : 9** — Alligation cross: (60 − 57) : (57 − 30) = 3 : 27 = 1 : 9.

**1257.** **≈ 2183 (exact 2183.06)** — Round each number: 488 × 37 ÷ 10 + 377 ≈ 2182.6.

**1258.** **2 : 13** — Alligation cross: (60 − 56) : (56 − 30) = 4 : 26 = 2 : 13.

**1259.** **≈ 3533 (exact 3534.76)** — Round each number: 863 × 34 ÷ 10 + 599 ≈ 3533.2.

**1260.** **4 : 1** — Alligation cross: (70 − 38) : (38 − 30) = 32 : 8 = 4 : 1.

**1261.** **≈ 1091 (exact 1091.91)** — Round each number: 366 × 25 ÷ 10 + 176 ≈ 1091.0.

**1262.** **4 : 1** — Alligation cross: (80 − 32) : (32 − 20) = 48 : 12 = 4 : 1.

**1263.** **≈ 3062 (exact 3060.54)** — Round each number: 595 × 45 ÷ 10 + 384 ≈ 3061.5.

**1264.** **≈ 3249 (exact 3249.59)** — Round each number: 798 × 35 ÷ 10 + 456 ≈ 3249.0.

**1265.** **1 : 4** — Alligation cross: (70 − 60) : (60 − 20) = 10 : 40 = 1 : 4.

**1266.** **≈ 2486 (exact 2484.51)** — Round each number: 627 × 34 ÷ 10 + 354 ≈ 2485.8.

**1267.** **29 : 11** — Alligation cross: (60 − 31) : (31 − 20) = 29 : 11 = 29 : 11.

**1268.** **≈ 1894 (exact 1895.16)** — Round each number: 453 × 35 ÷ 10 + 309 ≈ 1894.5.

**1269.** **3 : 1** — Alligation cross: (70 − 55) : (55 − 50) = 15 : 5 = 3 : 1.

**1270.** **≈ 777 (exact 777.54)** — Round each number: 461 × 12 ÷ 10 + 224 ≈ 777.2.

**1271.** **1 : 9** — Alligation cross: (90 − 85) : (85 − 40) = 5 : 45 = 1 : 9.

**1272.** **≈ 1290 (exact 1290.82)** — Round each number: 298 × 38 ÷ 10 + 158 ≈ 1290.4.

# Advanced mental-math challenge — 10 sets

## ✎ Set 61: Advanced mental-math challenge · Beginner · suggested time 6 min

**1273.** Find the last digit of $2^{598}$.

**1274.** $1{,}032^2$

**1275.** Find the last digit of $33^{38}$.

**1276.** $1{,}004^2$

**1277.** Find the last digit of $2^{578}$.

**1278.** $92^2$

**1279.** Find the last digit of $2^{363}$.

**1280.** $88^2$

*Answers: see the answer key (questions 1273–1280).*

## ✎ Set 62: Advanced mental-math challenge · Beginner · suggested time 6 min

**1281.** Find the last digit of $27^{248}$.

**1282.** $1{,}008^2$

**1283.** Find the last digit of $7^{852}$.

**1284.** Find the last digit of $4^{218}$.

**1285.** $90^2$

**1286.** Find the last digit of $4^{237}$.

**1287.** $102^2$

**1288.** Find the last digit of $22^{25}$.

*Answers: see the answer key (questions 1281–1288).*

## ✎ Set 63: Advanced mental-math challenge · Beginner · suggested time 6 min

**1289.** Find the last digit of $13^{261}$.

**1290.** Find the last digit of $4^{532}$.

**1291.** $104^2$

**1292.** Find the last digit of $4^{337}$.

**1293.** Find the last digit of $8^{368}$.

**1294.** Find the last digit of $17^{954}$.

**1295.** $91^2$

**1296.** Find the last digit of $3^{880}$.

*Answers: see the answer key (questions 1289–1296).*

## ✎ Set 64: Advanced mental-math challenge · Intermediate · suggested time 10 min

**1297.** Find the last digit of $13^{61}$.

**1298.** $93^2$

**1299.** Find the last digit of $27^{894}$.

**1300.** $114^2$

**1301.** Find the last digit of $8^{777}$.

**1302.** Find the last digit of $7^{621}$.

**1303.** $1{,}036^2$

**1304.** Find the last digit of $8^{417}$.

*Answers: see the answer key (questions 1297–1304).*

## ✎ Set 65: Advanced mental-math challenge · Intermediate · suggested time 10 min

**1305.** Find the last digit of $17^{842}$.

**1306.** $94^2$

**1307.** Find the last digit of $9^{573}$.

**1308.** Find the last digit of $4^{54}$.

**1309.** $99^2$

**1310.** Find the last digit of $9^{22}$.

**1311.** Find the last digit of $17^{474}$.

**1312.** $1{,}021^2$

*Answers: see the answer key (questions 1305–1312).*

## ✎ Set 66: Advanced mental-math challenge · Intermediate · suggested time 10 min

**1313.** Find the last digit of $22^{330}$.

**1314.** $1{,}006^2$

**1315.** Find the last digit of $27^{88}$.

**1316.** $109^2$

**1317.** Find the last digit of $33^{596}$.

**1318.** Find the last digit of $33^{208}$.

**1319.** $970^2$

**1320.** Find the last digit of $8^{565}$.

*Answers: see the answer key (questions 1313–1320).*

## ✎ Set 67: Advanced mental-math challenge · Intermediate · suggested time 10 min

**1321.** Find the last digit of $7^{600}$.

**1322.** Find the last digit of $2^{707}$.

**1323.** $978^2$

**1324.** Find the last digit of $13^{912}$.

**1325.** Find the last digit of $13^{597}$.

**1326.** $1{,}020^2$

**1327.** Find the last digit of $17^{182}$.

**1328.** Find the last digit of $17^{902}$.

*Answers: see the answer key (questions 1321–1328).*

## ✎ Set 68: Advanced mental-math challenge · Advanced · suggested time 16 min

**1329.** Solve: 14x + 45y = 298, 45x + 14y = 174

**1330.** Estimate √68 to 2 decimal places (approximate). *(approximate)*

**1331.** Solve $x^2 - 1x - 56 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1332.** 555 × 215

**1333.** $378^2$ (duplex method)

**1334.** 4,338 ÷ 113

**1335.** 34,649 ÷ 998

**1336.** 54 × 51 (working base 50)

*Answers: see the answer key (questions 1329–1336).*

## ✎ Set 69: Advanced mental-math challenge · Advanced · suggested time 16 min

**1337.** Estimate √391 to 2 decimal places (approximate). *(approximate)*

**1338.** Solve $x^2 + 0x - 16 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1339.** 225 × 459

**1340.** $219^2$ (duplex method)

**1341.** 3,759 ÷ 111

**1342.** 3,490 ÷ 98

**1343.** 196 × 205 (working base 200)

**1344.** What day of the week was/is 17 September 1966?

*Answers: see the answer key (questions 1337–1344).*

## ✎ Set 70: Advanced mental-math challenge · Advanced · suggested time 16 min

**1345.** Solve $x^2 + 8x + 7 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

**1346.** 114 × 516

**1347.** $781^2$ (duplex method)

**1348.** 5,611 ÷ 112

**1349.** 644 ÷ 8

**1350.** 494 × 506 (working base 500)

**1351.** What day of the week was/is 18 August 2005?

**1352.** Solve: 11x + 38y = 250, 38x + 11y = 142

*Answers: see the answer key (questions 1345–1352).*

# Answers and explanations — Advanced mental-math challenge

**1273.** **4** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 598 mod 4 = 2 → 4.

**1274.** **10,65,024** — Base 1,000, deviation +32 Left: 1032 + 32 = 1064 Right: $32^2$ = 1024 (needs 3 digit(s), carry 1) Answer = 10,65,024

**1275.** **9** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 38 mod 4 = 2 → 9.

**1276.** **10,08,016** — Base 1,000, deviation +4 Left: 1004 + 4 = 1008 Right: $4^2$ = 16 (needs 3 digit(s)) Answer = 10,08,016

**1277.** **4** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 578 mod 4 = 2 → 4.

**1278.** **8,464** — Base 100, deviation −8 Left: 92 − 8 = 84 Right: $8^2$ = 64 (needs 2 digit(s)) Answer = 8,464

**1279.** **8** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 363 mod 4 = 3 → 8.

**1280.** **7,744** — Base 100, deviation −12 Left: 88 − 12 = 76 Right: $12^2$ = 144 (needs 2 digit(s), carry 1) Answer = 7,744

**1281.** **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 248 mod 4 = 4 → 1.

**1282.** **10,16,064** — Base 1,000, deviation +8 Left: 1008 + 8 = 1016 Right: $8^2$ = 64 (needs 3 digit(s)) Answer = 10,16,064

**1283.** **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 852 mod 4 = 4 → 1.

**1284.** **6** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 218 mod 2 = 2 → 6.

**1285.** **8,100** — Base 100, deviation −10 Left: 90 − 10 = 80 Right: $10^2$ = 100 (needs 2 digit(s), carry 1) Answer = 8,100

**1286.** **4** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 237 mod 2 = 1 → 4.

**1287.** **10,404** — Base 100, deviation +2 Left: 102 + 2 = 104 Right: $2^2$ = 4 (needs 2 digit(s)) Answer = 10,404

**1288.** **2** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 25 mod 4 = 1 → 2.

**1289.** **3** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 261 mod 4 = 1 → 3.

**1290.** **6** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 532 mod 2 = 2 → 6.

**1291.** **10,816** — Base 100, deviation +4 Left: 104 + 4 = 108 Right: $4^2$ = 16 (needs 2 digit(s)) Answer = 10,816

**1292.** **4** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 337 mod 2 = 1 → 4.

**1293.** **6** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 368 mod 4 = 4 → 6.

**1294.** **9** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 954 mod 4 = 2 → 9.

**1295.** **8,281** — Base 100, deviation −9 Left: 91 − 9 = 82 Right: $9^2$ = 81 (needs 2 digit(s)) Answer = 8,281

**1296.** **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 880 mod 4 = 4 → 1.

**1297.** **3** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 61 mod 4 = 1 → 3.

**1298.** **8,649** — Base 100, deviation −7 Left: 93 − 7 = 86 Right: $7^2$ = 49 (needs 2 digit(s)) Answer = 8,649

**1299.** **9** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 894 mod 4 = 2 → 9.

**1300.** **12,996** — Base 100, deviation +14 Left: 114 + 14 = 128 Right: $14^2$ = 196 (needs 2 digit(s), carry 1) Answer = 12,996

**1301.** **8** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 777 mod 4 = 1 → 8.

**1302.** **7** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 621 mod 4 = 1 → 7.

**1303.** **10,73,296** — Base 1,000, deviation +36 Left: 1036 + 36 = 1072 Right: $36^2$ = 1296 (needs 3 digit(s), carry 1) Answer = 10,73,296

**1304.** **8** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 417 mod 4 = 1 → 8.

**1305.** **9** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 842 mod 4 = 2 → 9.

**1306.** **8,836** — Base 100, deviation −6 Left: 94 − 6 = 88 Right: $6^2$ = 36 (needs 2 digit(s)) Answer = 8,836

**1307.** **9** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 573 mod 2 = 1 → 9.

**1308.** **6** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 54 mod 2 = 2 → 6.

**1309.** **9,801** — Base 100, deviation −1 Left: 99 − 1 = 98 Right: $1^2$ = 1 (needs 2 digit(s)) Answer = 9,801

**1310.** **1** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 22 mod 2 = 2 → 1.

**1311.** **9** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 474 mod 4 = 2 → 9.

**1312.** **10,42,441** — Base 1,000, deviation +21 Left: 1021 + 21 = 1042 Right: $21^2$ = 441 (needs 3 digit(s)) Answer = 10,42,441

**1313.** **4** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 330 mod 4 = 2 → 4.

**1314.** **10,12,036** — Base 1,000, deviation +6 Left: 1006 + 6 = 1012 Right: $6^2$ = 36 (needs 3 digit(s)) Answer = 10,12,036

**1315.** **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 88 mod 4 = 4 → 1.

**1316.** **11,881** — Base 100, deviation +9 Left: 109 + 9 = 118 Right: $9^2$ = 81 (needs 2 digit(s)) Answer = 11,881

**1317.** **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 596 mod 4 = 4 → 1.

**1318.** **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 208 mod 4 = 4 → 1.

**1319.** **9,40,900** — Base 1,000, deviation −30 Left: 970 − 30 = 940 Right: $30^2$ = 900 (needs 3 digit(s)) Answer = 9,40,900

**1320.** **8** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 565 mod 4 = 1 → 8.

**1321.** **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 600 mod 4 = 4 → 1.

**1322.** **8** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 707 mod 4 = 3 → 8.

**1323.** **9,56,484** — Base 1,000, deviation −22 Left: 978 − 22 = 956 Right: $22^2$ = 484 (needs 3 digit(s)) Answer = 9,56,484

**1324.** **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 912 mod 4 = 4 → 1.

**1325.** **3** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 597 mod 4 = 1 → 3.

**1326.** **10,40,400** — Base 1,000, deviation +20 Left: 1020 + 20 = 1040 Right: $20^2$ = 400 (needs 3 digit(s)) Answer = 10,40,400

**1327.** **9** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 182 mod 4 = 2 → 9.

**1328.** **9** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 902 mod 4 = 2 → 9.

**1329.** **x = 2, y = 6** — Add: 59(x + y) = 472 → x + y = 8. Subtract: -31(x − y) = 124 → x − y = −4. Solve: x = 2, y = 6.

**1330.** **≈ 8.25 (true value 8.246)** — Nearest perfect square 64 = $8^2$. $\sqrt{n} \approx 8 + (68 - 64) \div (2 \times 8) = 8.250$.

**1331.** **x = 8 or −7** — Differential: 2x − 1. Discriminant = 1 − -224 = 225, $\sqrt{}$ = 15. 2x − 1 = ±15.

**1332.** **1,19,325** — Column 1 (from right): 5×5 = 25 → write 5, carry 2 Column 2 (from right): 5×1 + 5×5 = 30, + carry 2 = 32 → write 2, carry 3 Column 3 (from right): 5×2 + 5×1 + 5×5 = 40, + carry 3 = 43 → write 3, carry 4 Column 4 (from right): 5×2 + 5×1 = 15, + carry 4 = 19 → write 9, carry 1 Column 5 (from right): 5×2 = 10, + carry 1 = 11 → write 1, carry 1 Answer = 1,19,325

**1333.** **1,42,884** — Duplexes from left: 9, 42, 97, 112, 64 Combine with carries (right to left) → 1,42,884

**1334.** **Q = 38, R = 44** — Divisor 113 = 100 + 13; transposed flags: −1, −3 Column values after applying flags: [4, -1] | [-8, 11] Resolve negatives/carries → quotient 38, remainder 44

**1335.** **Q = 34, R = 717** — Divisor 998 = 1,000 − 2; write complement digits 002 Split dividend: 34 | 649 Column totals after multiplying by the complement: [3, 4] | [6, 10, 17] Resolve carries and remainder → quotient 34, remainder 717

**1336.** **2,754** — Working base 50 = 5 × 10. Deviations +4, +1. Left: (54 +1) × 5 = 275. Right: 4 × 1 = 4. 275 × 10 + 4 = 2,754.

**1337.** **≈ 19.77 (true value 19.774)** — Nearest perfect square 400 = $20^2$. $\sqrt{n} \approx 20 + (391 - 400) \div (2 \times 20) = 19.775$.

**1338.** **x = 4 or −4** — Differential: 2x + 0. Discriminant = 0 − -64 = 64, $\sqrt{}$ = 8. 2x + 0 = ±8.

**1339.** **1,03,275** — Column 1 (from right): 5×9 = 45 → write 5, carry 4 Column 2 (from right): 5×5 + 2×9 = 43, + carry 4 = 47 → write 7, carry 4 Column 3 (from right): 5×4 + 2×5 + 2×9 = 48, + carry 4 = 52 → write 2, carry 5 Column 4 (from right): 2×4 + 2×5 = 18, + carry 5 = 23 → write 3, carry 2 Column 5 (from right): 2×4 = 8, + carry 2 = 10 → write 0, carry 1 Answer = 1,03,275

**1340.** **47,961** — Duplexes from left: 4, 4, 37, 18, 81 Combine with carries (right to left) → 47,961

**1341.** **Q = 33, R = 96** — Divisor 111 = 100 + 11; transposed flags: −1, −1 Column values after applying flags: [3, 4] | [-2, 5] Resolve negatives/carries → quotient 33, remainder 96

**1342.** **Q = 35, R = 60** — Divisor 98 = 100 − 2; write complement digits 02 Split dividend: 34 | 90 Column totals after multiplying by the complement: [3, 4] | [15, 8] Resolve carries and remainder → quotient 35, remainder 60

**1343.** **40,180** — Working base 200 = 2 × 100. Deviations −4, +5. Left: (196 +5) × 2 = 402. Right: −4 × 5 = −20. 402 × 100 + −20 = 40,180.

**1344.** **Saturday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**1345.** **x = −1 or −7** — Differential: 2x + 8. Discriminant = 64 − 28 = 36, $\sqrt{}$ = 6. 2x + 8 = ±6.

**1346.** **58,824** — Column 1 (from right): 4×6 = 24 → write 4, carry 2 Column 2 (from right): 4×1 + 1×6 = 10, + carry 2 = 12 → write 2, carry 1 Column 3 (from right): 4×5 + 1×1 + 1×6 = 27, + carry 1 = 28 → write 8, carry 2 Column 4 (from right): 1×5 + 1×1 = 6, + carry 2 = 8 → write 8, carry 0 Column 5 (from right): 1×5 = 5 → write 5, carry 0 Answer = 58,824

**1347.** **6,09,961** — Duplexes from left: 49, 112, 78, 16, 1 Combine with carries (right to left) → 6,09,961

**1348.** **Q = 50, R = 11** — Divisor 112 = 100 + 12; transposed flags: –1, –2 Column values after applying flags: [5, 1] | [-10, -1] Resolve negatives/carries → quotient 50, remainder 11

**1349.** **Q = 80, R = 4** — Divisor 8 = 10 – 2; write complement digits 2 Split dividend: 64 | 4 Column totals after multiplying by the complement: [6, 16] | [36] Resolve carries and remainder → quotient 80, remainder 4

**1350.** **2,49,964** — Working base 500 = 5 × 100. Deviations –6, +6. Left: (494 +6) × 5 = 2500. Right: –6 × 6 = –36. 2500 × 100 + –36 = 2,49,964.

**1351.** **Thursday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

**1352.** **x = 2, y = 6** — Add: 49(x + y) = 392 → x + y = 8. Subtract: -27(x – y) = 108 → x – y = –4. Solve: x = 2, y = 6.

# Part XIV

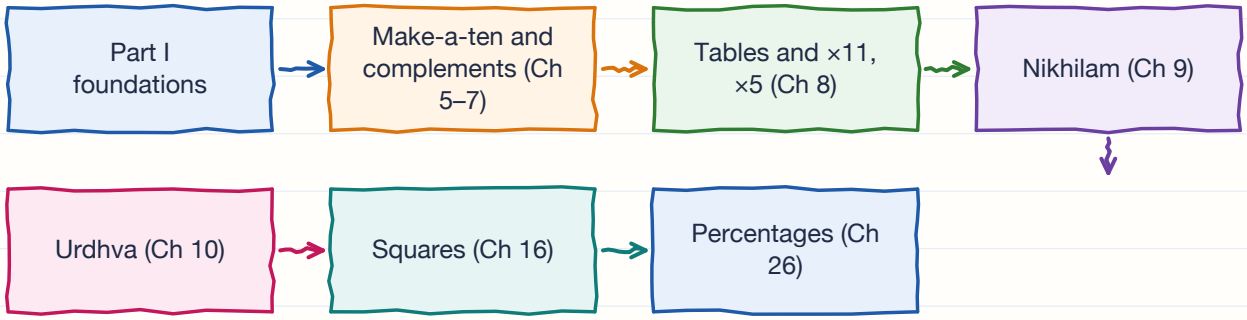
## Learning Roadmaps

Choose the roadmap that fits you. Every roadmap uses the chapters of this book in a sensible order and has checkpoints, so you know when to move on.

> **No guarantees**
> These plans describe what regular practice typically achieves. Results depend on your starting level, consistency and the quality of your practice. No plan can guarantee a score or a selection.

# Chapter 53: Five Roadmaps

## 53.1 Roadmap A — Complete beginner (adult)




**Goal:** comfortable mental arithmetic for daily life and as a base for exams. **Time:** 20–30 minutes a day for 60 days.

## 53.2 Roadmap B — School student (ages 11–16)

Foundations → mental add/sub → multiplication methods → squares/roots → algebra (Part VII) → fractions and percentages. Use the methods to **check** homework, not to skip showing working where the teacher requires it.

## 53.3 Roadmap C — Government-exam aspirant

Foundations review (1 week) → calculation speed (Parts III–VI, 3 weeks) → arithmetic aptitude (Part VIII, 4 weeks) → exam strategy and DI (Part XI) → practice sets and mocks (Part XIII). Follow the 90-day plan below.

## 53.4 Roadmap D — Parent teaching a child

Read Part X first. Follow the level that matches the child's age, with 10–15 minutes a day. Use the Children's Worksheets PDF for printable practice.

## 53.5 Roadmap E — Advanced learner

Duplex squaring, square roots by the long-division method, Paravartya and Nikhilam division, recurring decimals, vinculum, polynomial multiplication and the algebra sutras. Use the Advanced practice sets (61–70).

# Chapter 54: 30, 60 and 90-Day Plans

## 54.1 30-day plan — calculation speed

| Week | Daily focus (25 min) | Checkpoint at the end of the week |
|---|---|---|
| 1 | Complements, add/sub left to right, subtraction from powers of 10 | 20 additions of 3-digit numbers in 5 minutes with 90% accuracy |
| 2 | ×11, ×5/25/125, ×9s, Nikhilam near 10 and 100 | 15 near-base products in 5 minutes |
| 3 | Urdhva 2×2, squares ending in 5, near-base squares | any 2-digit × 2-digit in under 20 seconds |
| 4 | Division by 9, 5/25/125, fraction–percentage chart | Timed tests 1 and 2 of Part XII with ≥ 16/20 |

## 54.2 60-day plan — calculation plus aptitude

- **Days 1–30:** the 30-day plan above.
- **Days 31–40:** ratio, averages, percentages, profit and loss (Ch 25–27).
- **Days 41–50:** interest, work, pipes, speed, trains, boats (Ch 28–30).
- **Days 51–60:** alligation, ages, number system, probability, mensuration (Ch 31–35), plus one practice set each day.

## 54.3 90-day plan — government exams

| Phase | Days | Focus | Weekly mock |
|---|---|---|---|
| Foundation | 1–15 | Parts I–III; tables to 20; squares to 30; cubes to 15 | — |
| Speed | 16–35 | Parts IV–VI; 1 timed test every day | Timed test from Part XII |
| Aptitude | 36–65 | Part VIII, two chapters a week, plus practice sets | 1 sectional mock |
| Exam skills | 66–80 | Part XI; DI and approximation every day | 2 sectional mocks |
| Revision | 81–90 | Part XV formula sheets; repeat circled mistakes | 3 full mocks and their analysis |

# Chapter 55: Daily Routines and Weekly Revision

## 55.1 A 30-minute daily routine

| Minutes | Activity |
|---|---|
| 0–5 | Warm-up: squares, cubes, tables or bonds (alternate days) |
| 5–15 | Learn or revise one technique from the book |
| 15–25 | 10–15 practice questions on that technique |
| 25–30 | Note mistakes in the tracker; read one recap box |

## 55.2 Weekly revision (Sunday, 45 minutes)

1. Re-do every circled mistake from the week.
2. Take one timed test or practice set.
3. Read the recap boxes of the chapters you studied.
4. Update the tracker: accuracy and average time per topic.

# Chapter 56: Tracking Progress and Checkpoints

## 56.1 Topic mastery tracker

| Topic | Accuracy ≥ 90% | Time target met | Date mastered |
|---|---|---|---|
| Complements and add/sub | | | |
| ×11, ×5, ×25, ×9s | | | |
| Nikhilam | | | |
| Urdhva | | | |
| Squares and square roots | | | |
| Cubes and cube roots | | | |
| Division methods | | | |
| Fraction–percentage chart | | | |
| Algebra identities and factorising | | | |
| Arithmetic aptitude | | | |
| Data interpretation | | | |

## 56.2 Checkpoint tests

| Checkpoint | When | Test | Pass mark |
|---|---|---|---|
| 1 | After Part III | Part XII section B, Intermediate | 85% |
| 2 | After Part IV | Part XII section E, Beginner and Intermediate | 85% |
| 3 | After Part VI | Practice sets 21–25 | 80% |
| 4 | After Part VIII | Practice sets 41–45 and 51–55 | 75% |
| 5 | Before the exam | Practice sets 56–60 and 61–65 | 70% (these are hard) |

### If you fail a checkpoint

Do not move on. Re-read the recap boxes, redo that chapter's practice questions, and take the checkpoint again after three days.

### Part XIV summary

- Pick a roadmap, then follow a 30, 60 or 90-day plan.
- Spend 20–30 minutes a day, with a weekly revision session.
- Track accuracy first, then speed.
- Pass each checkpoint before moving on.

# Part XV

## Revision and Formula Sheets

Use these pages for last-minute revision. Each sheet fits a page or two, so you can print the separate **Formula Sheets** PDF and keep it on your desk.

# Chapter 57: The Sutra Chart

| # | Sutra | Meaning | Use it for | Example |
|---|---|---|---|---|
| 1 | Ekadhikena Purvena | One more than the previous | squares ending in 5; 1/19 | $65^2 \to 6 \times 7 \mid 25 \to 4{,}225$ |
| 2 | Nikhilam | All from 9, last from 10 | $10^n - n$; near-base × and ÷ | $1{,}000 - 357 \to 643$ |
| 3 | Urdhva-Tiryag bhyam | Vertically and crosswise | any multiplication | $23 \times 41 \to 943$ |
| 4 | Paravartya Yojayet | Transpose and adjust | divisors just above a base; equations | $1{,}234 \div 12 \to Q\ 102, R\ 10$ |
| 5 | Shunyam Saa myasamuccay e | Same total → zero | special equations | $(x + 7)(x + 9) = (x + 3)(x + 21)$: $7 \times 9$ and $3 \times 21$ are both $63 \to x = 0$ |
| 6 | Anurupye Shunyamanyat | One in ratio, other zero | simultaneous equations | $2x + 3y = 6$, $7x + 9y = 18 \to x = 0$ |
| 7 | Sankalana-Vya vakalanabhya m | Add and subtract | swapped coefficients | add and subtract the equations |
| 8 | Puranapurana bhyam | Completion | completing the square | $x^2 + 6x \to (x + 3)^2 - 9$ |
| 9 | Chalana-Kalan abhyam | Differential | quadratic roots | $2x + b = \pm\sqrt{(b^2 - 4ac)}$ |
| 10 | Yavadunam | By the deficiency | squares near a base | $96^2 \to 92 \mid 16 \to 9{,}216$ |
| 11 | Vyashtisamans tih | Part and whole | splitting, averages | $49 \times 51 = 50^2 - 1$ |
| 12 | Shesanyanken a Charamena | Remainders by last digit | decimals of 1/7, 1/13 | $1/7 = 0.142857\ldots$ |
| 13 | Sopantyadvay amantyam | Last + twice penultimate | special fraction equations | see Part II |
| 14 | Ekanyunena Purvena | One less than the previous | × 9, 99, 999 | $47 \times 99 \to 46 \mid 53 \to 4{,}653$ |
| 15 | Gunitasamucc ayah | Sum-product check | checking products | digit sums |
| 16 | Gunakasamuc cayah | Factors of the sum | checking factorisations | $x = 1$ check |

# Chapter 58: Formula Sheets

## 58.1 Multiplication shortcuts

| Situation | Method |
|---|---|
| × 5, 25, 125 | × 10/2, × 100/4, × 1000/8 |
| × 9, 99, 999 (same number of digits) | (n − 1) \| (complement of n) |
| × 11 | write the outer digits, add neighbours in between |
| × 101 | write the number twice (for 2 digits) |
| Both near a base | $(a - d_2) \mid d_1 \times d_2$, with signs |
| Same tens, units add to 10 | tens × (tens + 1) \| units × units |
| General 2-digit | right: vertical; middle: crosswise sum; left: vertical |
| Squares ending in 5 | n(n + 1) \| 25 |
| Squares near 50 | $(25 \pm d) \mid d^2$ |
| Squares near 100 | $(n \pm d) \mid d^2$ |
| Any square | duplex method |

## 58.2 Squares and cubes

| n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ | n | $n^2$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 7 | 49 | 13 | 169 | 19 | 361 | 25 | 625 |
| 2 | 4 | 8 | 64 | 14 | 196 | 20 | 400 | 26 | 676 |
| 3 | 9 | 9 | 81 | 15 | 225 | 21 | 441 | 27 | 729 |
| 4 | 16 | 10 | 100 | 16 | 256 | 22 | 484 | 28 | 784 |
| 5 | 25 | 11 | 121 | 17 | 289 | 23 | 529 | 29 | 841 |
| 6 | 36 | 12 | 144 | 18 | 324 | 24 | 576 | 30 | 900 |

| n | $n^3$ | n | $n^3$ | n | $n^3$ |
|---|---|---|---|---|---|
| 1 | 1 | 6 | 216 | 11 | 1,331 |
| 2 | 8 | 7 | 343 | 12 | 1,728 |
| 3 | 27 | 8 | 512 | 13 | 2,197 |
| 4 | 64 | 9 | 729 | 14 | 2,744 |
| 5 | 125 | 10 | 1,000 | 15 | 3,375 |

## 58.3 Fractions and percentages

| Fraction | Percentage | Decimal |
|---|---|---|
| 1/2 | 50% | 0.5 |
| 1/3 | 33 1/3% | 0.3333… (recurring) |
| 2/3 | 66 2/3% | 0.6667… (recurring) |
| 1/4 | 25% | 0.25 |

| Fraction | Percentage | Decimal |
|---|---|---|
| 3/4 | 75% | 0.75 |
| 1/5 | 20% | 0.2 |
| 2/5 | 40% | 0.4 |
| 3/5 | 60% | 0.6 |
| 4/5 | 80% | 0.8 |
| 1/6 | 16 2/3% | 0.1667… (recurring) |
| 5/6 | 83 1/3% | 0.8333… (recurring) |
| 1/7 | 14 2/7% | 0.1429… (recurring) |
| 1/8 | 12 1/2% | 0.125 |
| 3/8 | 37 1/2% | 0.375 |
| 5/8 | 62 1/2% | 0.625 |
| 7/8 | 87 1/2% | 0.875 |
| 1/9 | 11 1/9% | 0.1111… (recurring) |
| 1/10 | 10% | 0.1 |
| 1/11 | 9 1/11% | 0.0909… (recurring) |
| 1/12 | 8 1/3% | 0.0833… (recurring) |
| 1/15 | 6 2/3% | 0.0667… (recurring) |
| 1/16 | 6 1/4% | 0.0625 |
| 1/20 | 5% | 0.05 |
| 1/25 | 4% | 0.04 |
| 1/40 | 2 1/2% | 0.025 |
| 1/50 | 2% | 0.02 |

## 58.4 Powers of 2 and unit-digit cycles

| n | $2^n$ | $3^n$ | $5^n$ |
|---|---|---|---|
| 1 | 2 | 3 | 5 |
| 2 | 4 | 9 | 25 |
| 3 | 8 | 27 | 125 |
| 4 | 16 | 81 | 625 |
| 5 | 32 | 243 | 3,125 |
| 6 | 64 | 729 | 15,625 |
| 7 | 128 | 2,187 | 78,125 |
| 8 | 256 | 6,561 | 3,90,625 |
| 9 | 512 | 19,683 | |
| 10 | 1,024 | 59,049 | |
| 11 | 2,048 | | |
| 12 | 4,096 | | |

| Unit digit | Last digits of powers 1, 2, 3, 4 … | Cycle length |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 1 |
| 2 | 2, 4, 8, 6 | 4 |
| 3 | 3, 9, 7, 1 | 4 |
| 4 | 4, 6 | 2 |
| 5 | 5 | 1 |
| 6 | 6 | 1 |
| 7 | 7, 9, 3, 1 | 4 |
| 8 | 8, 4, 2, 6 | 4 |
| 9 | 9, 1 | 2 |

## 58.5 Square and cube root endings

| Number ends in | Square ends in | Cube ends in |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 1 | 1 |
| 2 | 4 | 8 |
| 3 | 9 | 7 |
| 4 | 6 | 4 |
| 5 | 5 | 5 |
| 6 | 6 | 6 |
| 7 | 9 | 3 |
| 8 | 4 | 2 |
| 9 | 1 | 9 |

## 58.6 Primes and divisibility

| Primes up to 100 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 5 | 7 | 11 | 13 | 17 | 19 | 23 | 29 |
| 31 | 37 | 41 | 43 | 47 | 53 | 59 | 61 | 67 | 71 |
| 73 | 79 | 83 | 89 | 97 | | | | | |

| Divisor | Rule | Example |
|---|---|---|
| 2 | Last digit even (0, 2, 4, 6, 8) | 3,574 ✓ |
| 3 | Digit sum divisible by 3 | 4,512 → 12 ✓ |
| 4 | Last two digits divisible by 4 | 7,316 → 16 ✓ |
| 5 | Last digit 0 or 5 | 8,435 ✓ |
| 6 | Divisible by 2 and by 3 | 2,154 ✓ |
| 7 | Double the last digit, subtract from the rest; repeat | 203 → 20 − 6 = 14 ✓ |
| 8 | Last three digits divisible by 8 | 51,128 → 128 ✓ |

| Divisor | Rule | Example |
|---|---|---|
| 9 | Digit sum divisible by 9 | 6,723 → 18 ✓ |
| 10 | Last digit 0 | 4,560 ✓ |
| 11 | Alternating digit sum divisible by 11 | 9,537 → (9+3) – (5+7) = 0 ✓ |
| 12 | Divisible by 3 and by 4 | 1,524 ✓ |
| 13 | Add 4 × last digit to the rest; repeat | 247 → 24 + 28 = 52 ✓ |
| 25 | Last two digits 00, 25, 50, 75 | 3,475 ✓ |

## 58.7 Algebraic identities

### Identities

$(a + b)^2 = a^2 + 2ab + b^2$ · $(a - b)^2 = a^2 - 2ab + b^2$ · $a^2 - b^2 = (a + b)(a - b)$ $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ · $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ · $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ $(x + a)(x + b) = x^2 + (a + b)x + ab$ Quadratic roots: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$

## 58.8 Aptitude formulas

### Percentages, profit and interest

Successive change: $a + b + ab/100$ · r% more → $r/(100 + r)$ less SP = $CP(100 + P\%)/100$ · SP = $MP(100 - D\%)/100$ · Profit % on CP SI = $PRT/100$ · $A = P(1 + R/100)^T$ · 2-year CI – SI = $P(R/100)^2$

### Ratio, averages and mixtures

Share = total × part/(sum of parts) · Weighted average = $\Sigma n_i a_i / \Sigma n_i$ Alligation: cheaper : dearer = (dearer – mean) : (mean – cheaper) Replacement: $x(1 - y/x)^n$ · Partnership: capital × time

### Time, work, speed

Work: total = LCM; outlets negative · $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$ D = S × T · km/h × 5/18 = m/s · Average speed (equal distances) = $2ab/(a + b)$ Trains: add lengths; relative speed adds (opposite) or subtracts (same) Boats: boat = (down + up)/2, stream = (down – up)/2

### Counting and probability

${}^nP_r = n!/(n - r)!$ · ${}^nC_r = n!/(r!(n - r)!)$ · P(not A) = 1 – P(A) P(A or B) = P(A) + P(B) – P(A and B)

### Mensuration

Rectangle lb, 2(l + b) · Triangle ½bh, Heron $\sqrt{(s(s-a)(s-b)(s-c))}$ · Circle $\pi r^2$, $2\pi r$ Cuboid lbh, 2(lb + bh + hl) · Cube $a^3$, $6a^2$ · Cylinder $\pi r^2 h$, $2\pi rh$ · Cone $\frac{1}{3}\pi r^2 h$, $\pi rl$ · Sphere 4/3 $\pi r^3$, $4\pi r^2$

### Calendar

Odd days: ordinary year 1, leap year 2; 100 yrs 5, 200 yrs 3, 300 yrs 1, 400 yrs 0 Months: Jan 3, Feb 0/1, Mar 3, Apr 2, May 3, Jun 2, Jul 3, Aug 3, Sep 2, Oct 3, Nov 2, Dec 3

# Chapter 59: Common Mistakes, Exam-Day Strategy and Checklists

## 59.1 Twenty common mistakes

| # | Mistake | Correct idea |
|---|---|---|
| 1 | Forgetting to carry in Urdhva | carry everything except one digit from each column |
| 2 | Wrong number of digits on the right in Nikhilam | right part has as many digits as the base has zeros |
| 3 | Sign errors with mixed deviations | one above and one below → right part is negative |
| 4 | "All from 9" applied to the last digit | the last non-zero digit is from 10 |
| 5 | Ekanyunena with unequal digit counts | pad with zeros or use a base method |
| 6 | Squares ending in 5 for 3-digit numbers | $n(n + 1)$ still works: $115^2 \rightarrow 11 \times 12 \mid 25$ |
| 7 | Treating $a^2 + b^2$ as $(a + b)^2$ | remember $2ab$ |
| 8 | Profit % on SP | profit % is on CP |
| 9 | Averaging speeds | use $2ab/(a + b)$ |
| 10 | Adding successive percentages | $a + b + ab/100$ |
| 11 | Wrong units (km/h with metres) | convert first |
| 12 | Leap-year rule for 1900 | 1900 is not a leap year |
| 13 | Digit-sum check "proves" an answer | it only rejects wrong answers |
| 14 | Over-rounding in approximation | round to the nearest easy number |
| 15 | Missing the overlap in "or" probability | subtract P(A and B) |
| 16 | Remainder taken as negative | add the divisor to make it positive |
| 17 | $-(a - b)$ written as $-a - b$ | $-(a - b) = -a + b$ |
| 18 | Misreading the scale in DI | read title, units and scale first |
| 19 | Using a shortcut that does not fit | check the condition (same tens? near a base?) |
| 20 | Spending too long on one question | use the two-pass method |

## 59.2 Exam-day strategy

1. Sleep well the night before. A tired brain is slower than any shortcut can make up for.
2. Warm up for 5 minutes: a few squares, tables and percentages.
3. Read each question fully before calculating.
4. First pass: easy questions only. Second pass: marked questions.
5. Estimate before choosing; check the units digit and the size of the answer.
6. With negative marking, avoid pure guesses. Guess only after eliminating at least two options.

## 59.3 Revision checklists

**One week before the exam**

- Revise all recap boxes.
- Take 3 full mocks and analyse each one.
- Re-do all circled mistakes.

**One day before the exam**

- Read this chapter's formula sheets once.
- Do one light timed test. Do not start anything new.

**Skills checklist**

- [ ] Squares to 30 and cubes to 15 from memory
- [ ] Fraction–percentage chart from memory
- [ ] Nikhilam and Urdhva without writing intermediate steps
- [ ] Divide by 9 and by numbers near 10/100
- [ ] All aptitude formulas in this chapter
- [ ] Two-pass method practised in at least 5 mocks

> **Final words**
>
> Vedic Maths is a toolbox, not magic. Understand why each method works, choose the right tool for each problem, check your answers, and practise a little every day. Good luck on your journey from zero to hero!